Foundations of Autonomous Robotics

Mathematics, Physics, Algorithms, and Engineering

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Preface

This book is my collection of learning notes for understanding intelligence, especially intelligence embodied in agents that perceive their environment, predict what may happen, choose actions, and act in the physical world. Writing the notes is part of the learning process: the aim is to make each idea precise enough that I can explain it, derive it, implement it, and test it.

My main algorithmic interests are **reinforcement learning** and **world models**. Reinforcement learning studies how an agent can improve its decisions through interaction and feedback. A world model is an internal model that predicts how the environment or system may evolve, allowing an agent to evaluate possible actions before or while taking them. These algorithms depend on more fundamental ideas from mathematics, dynamics, control, estimation, planning, perception, and software engineering, so the notes build those foundations first.

Unified Notation and Conventions

This note is the canonical notation registry for the project. Chapters may introduce local symbols, but they must not redefine symbols listed here. When a textbook uses different notation, translate it into these conventions.

Core conventions

1. Scalars use italic lowercase letters, vectors use bold lowercase letters, and matrices use bold uppercase letters.
2. Vectors are column vectors unless stated otherwise.
3. SI units are used in calculations. Angles are expressed in radians.
4. Coordinate frames are right-handed. A leading left superscript identifies the frame in which coordinates are expressed.
5. Continuous time uses t; sampled time uses the integer index k.
6. A wide hat over a twist, as in $\widehat{\boldsymbol{\xi}}$, denotes the declared Lie-algebra hat operator and is not an estimate.
7. Bold symbols and scalar symbols with the same letter are distinct; for example, $\mathbf{R}$ is a rotation matrix whereas R may be a signed turning radius.
8. A symbol may have a different local meaning only in a clearly separated mathematical domain where its type and meaning are explicitly redeclared; for example, m may denote a matrix dimension in a generic linear-algebra statement and robot mass in a dynamics model.

Part I: Motion, Mechanics, and Control

Notation introduced in Chapter 1

Symbol	Meaning	Units or convention
a, α	Scalar quantity	Depends on the quantity
m, n, i, j	Integer dimensions or indices	Declared locally
$\mathbb{R}$	Set of real numbers	—
$\mathbb{R}^n, \mathbb{R}^{m \times n}$	Real coordinate columns and real matrices	Dimensions shown by the superscripts

Symbol	Meaning	Units or convention
$\mathbb{R}_{\geq 0}, \mathbb{R}_{>0}$	Nonnegative and strictly positive real numbers	—
$\mathbf{v} \in \mathbb{R}^n$	Column vector with n components	Depends on the quantity
$\mathbf{A} \in \mathbb{R}^{m \times n}$	Matrix with m rows and n columns	Depends on the map
$\mathbf{I}_n$	$n \times n$ identity matrix	Dimensionless
$\mathbf{0}_n$	Zero vector with n components	Depends on the quantity
$(\cdot)^\mathsf{T}$	Transpose	—
$(\cdot)^{-1}$	Inverse, when it exists	—
$(\cdot)^{-\mathsf{T}}$	Inverse transpose, $(\mathbf{A}^{-1})^\mathsf{T} = (\mathbf{A}^\mathsf{T})^{-1}$ for nonsingular $\mathbf{A}$	—
$\\|\mathbf{v}\\|_2$	Euclidean norm of $\mathbf{v}$	Same units as $\mathbf{v}$
$\det(\mathbf{A})$	Determinant of a square matrix	Depends on the matrix units
$\operatorname{rank}(\mathbf{A})$	Dimension of the column space of $\mathbf{A}$	Nonnegative integer
$\operatorname{nullity}(\mathbf{A})$	Dimension of the null space of $\mathbf{A}$	Nonnegative integer
$\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$	Set of all linear combinations of the listed vectors	Same vector space as the vectors
$\operatorname{Col}(\mathbf{A}), \mathcal{N}(\mathbf{A})$	Column space and null space of $\mathbf{A}$	Subspaces of the output and input spaces
$\mathbf{u} \times \mathbf{v}$	Three-dimensional cross product of vectors expressed in the same right-handed orthonormal frame	Product of the input units
$O(n), SO(n)$	Orthogonal group and special orthogonal group of real $n \times n$ matrices	$\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_n$; $SO(n)$ additionally requires $\det \mathbf{Q} = 1$

Notation introduced in Chapter 2

Symbol	Meaning	Units or convention
$\{a\}$	Coordinate frame named a	Right-handed
O_a	Origin of frame $\{a\}$	—
$\mathbf{e}_{x,a}, \mathbf{e}_{y,a}, \mathbf{e}_{z,a}$	Unit basis vectors of frame $\{a\}$	Dimensionless

Symbol	Meaning	Units or convention
${}^{a}\mathbf{v}$	Coordinates of geometric vector $\mathbf{v}$ expressed in frame $\{a\}$	Depends on $\mathbf{v}$
${}^{a}\mathbf{p}_P$	Position of point P relative to O_a, expressed in frame $\{a\}$	m
${}^{a}\mathbf{r}_{OP}$	Relative-position vector from point O to point P, expressed in frame $\{a\}$	m
$\mathbf{R}_{ab}$	Rotation mapping coordinates from frame $\{b\}$ to frame $\{a\}$	$\mathbf{R}_{ab} \in SO(2)$ or $SO(3)$
${}^{a}\mathbf{p}_b$	Position of origin O_b relative to O_a, expressed in frame $\{a\}$	m
$\boldsymbol{\omega}$	Three-dimensional angular-velocity vector	$\mathrm{rad\,s^{-1}}$

Notation introduced in Chapter 3

Symbol	Meaning	Units or convention
t	Continuous time	s
$k \in \mathbb{N}_0$	Sample index, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$	Dimensionless integer
t_k	Time at sample k	s
$\Delta t = t_{k+1} - t_k$	Sampling or numerical-integration interval	s
$\dot{\mathbf{x}}, \ddot{\mathbf{x}}$	First and second derivatives with respect to time	Quantity per s and per s^2
$\{w\}, \{b\}$	World and robot body frames	—
O_w, O_b	Origins of the world and body frames	—
$\mathbf{q} = [x\ y\ \theta]^\mathsf{T}$	Planar robot pose coordinates in the world frame	$\mathrm{m, m, rad}$
θ	Counter-clockwise heading of $\{b\}$ relative to $\{w\}$	rad
$\mathbf{u} = [v\ \omega]^\mathsf{T}$	Planar body-velocity input	$\mathrm{m\,s^{-1}, rad\,s^{-1}}$
v, ω	Signed body-forward speed and counter-clockwise yaw rate	$\mathrm{m\,s^{-1}, rad\,s^{-1}}$
ϕ_L, ϕ_R	Left and right wheel angles	rad
$\dot{\phi}_L, \dot{\phi}_R$	Left and right wheel angular speeds	$\mathrm{rad\,s^{-1}}$
v_L, v_R	Signed longitudinal ground speeds of the left and right wheels	$\mathrm{m\,s^{-1}}$

Symbol	Meaning	Units or convention
r, L	Effective wheel radius and track width between wheel contact lines	m
C	Instantaneous centre of curvature for planar motion	Geometric point
ρ	Nonnegative radial distance declared in the local geometric context	m
R	Signed turning radius; positive for an ICC on the robot's left	m; distinct from rotation matrix $\mathbf{R}$
$\Delta\ell, \Delta\theta$	Signed arc-length and heading changes over an interval	m, rad
$\mathbf{q}_k^{\mathrm{E}}$	Forward Euler approximation of the pose at sample k	$\mathrm{m}, \mathrm{m}, \mathrm{rad}$
$E(\Delta t)$	Declared numerical-integration error magnitude at step size Δt	Depends on the error definition
e_p	One-step position-error magnitude in the declared integration comparison	m
$O(h^p)$	Quantity bounded in magnitude by a constant times h^p as $h \to 0$	Units follow the bounded quantity
$\operatorname{sinc}(z)$	Unnormalised sinc function, $\sin z/z$ for $z \neq 0$ and 1 for $z = 0$	Dimensionless argument and value

Notation introduced in Chapter 6

Symbol	Meaning	Units or convention
${}^a\overline{\mathbf{p}}_P$	Homogeneous point coordinate obtained by appending 1 to ${}^a\mathbf{p}_P$	Position entries in m; final entry dimensionless
$SO(2)$	Special orthogonal group of planar rotation matrices	$\mathbf{R}^{\mathsf{T}}\mathbf{R} = \mathbf{I}_2$, $\det \mathbf{R} = 1$
$SE(2)$	Special Euclidean group of planar rigid transformations	Planar rotation and translation represented together
$\mathbf{T}_{ab}$	Homogeneous transform mapping coordinates from frame $\{b\}$ to frame $\{a\}$	$\mathbf{T}_{ab} \in SE(2)$
$T_{\mathbf{I}_2}SO(2)$	Tangent space of $SO(2)$ at the identity rotation $\mathbf{I}_2$	Linear space of rotation-curve derivatives through $\mathbf{I}_2$
$T_{\mathbf{I}_3}SE(2)$	Tangent space of $SE(2)$ at the identity transformation $\mathbf{I}_3$	Linear space of transformation-curve derivatives through $\mathbf{I}_3$

Symbol	Meaning	Units or convention
$\mathbf{S}_2$	Planar skew generator $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	Dimensionless
$\mathfrak{so}(2)$	Lie algebra and identity tangent space of $SO(2)$	2×2 real skew-symmetric matrices
$\mathfrak{se}(2)$	Lie algebra and identity tangent space of $SE(2)$	Structured 3×3 infinitesimal planar rigid-motion matrices
$\boldsymbol{\xi} = [\boldsymbol{\nu}^\mathsf{T}\ \omega]^\mathsf{T}$	Planar twist coordinate column with linear part $\boldsymbol{\nu} \in \mathbb{R}^2$ and angular part $\omega \in \mathbb{R}$	$\mathrm{m\,s^{-1}}$ and $\mathrm{rad\,s^{-1}}$
$\widehat{\boldsymbol{\xi}}$	Hat-operator matrix representing $\boldsymbol{\xi}$ in $\mathfrak{se}(2)$	Upper-left block in $\mathrm{s^{-1}}$; translation column in $\mathrm{m\,s^{-1}}$
$(\cdot)^\vee$	Vee operator, inverse of the declared hat operator	Returns the twist coordinate column
$\boldsymbol{\xi}_b, \boldsymbol{\xi}_w$	Body and spatial twist coordinates for the same instantaneous motion	Body axes/body origin and world axes/world origin conventions
$\boldsymbol{\nu}_b, \boldsymbol{\nu}_w$	Linear parts of the body and spatial twists	$\mathrm{m\,s^{-1}}$
$\mathrm{Ad}_{\mathbf{T}}$	Adjoint mapping that changes twist coordinates according to rigid transform $\mathbf{T}$	Matrix dimensions and mixed units follow the twist convention
ℓ	Declared characteristic length used to compare linear and angular twist components	m

Notation introduced in Chapter 7

Symbol	Meaning	Units or convention
$\mathcal{B}$	Set of labelled material points belonging to the rigid body	Abstract set
$\chi : \mathcal{B} \to \mathbb{R}^d$	Placement map assigning a world position to every material point in a d-dimensional model	Output coordinates measured in m
$\mathcal{C}$	Configuration space: the set of all configurations allowed by the model	May be curved and need not be a vector space
$\chi \in \mathcal{C}$	One complete physical configuration represented as an admissible placement map	Abstract element of $\mathcal{C}$

Symbol	Meaning	Units or convention
$\mathbf{q}$	Chosen column of generalised coordinates describing a configuration	Components and units depend on the system
n_{dof}	Number of configuration degrees of freedom	Nonnegative integer
n_v, n_c	Numbers of scalar variables and independent regular equality constraints in a local count	Nonnegative integers
$\mathbf{z} \in \mathbb{R}^{n_v}$	Ambient coordinate column used to express equality constraints	Components and units depend on the system
$\mathbf{g} : \mathbb{R}^{n_v} \to \mathbb{R}^{n_c}$	Equality-constraint mapping, with valid coordinates satisfying $\mathbf{g}(\mathbf{z}) = \mathbf{0}_{n_c}$	Output units depend on the constraints
$\mathbf{J}_{\mathbf{g}}(\mathbf{z})$	Constraint Jacobian with entries $\partial g_i/\partial z_j$	$n_c \times n_v$; row units depend on g_i and z_j
$SO(3)$	Special orthogonal group of spatial rotation matrices	$\mathbf{R}^{\mathsf{T}}\mathbf{R} = \mathbf{I}_3$, $\det \mathbf{R} = 1$
$\mathbf{R}_x(\theta), \mathbf{R}_y(\theta), \mathbf{R}_z(\theta)$	Active right-hand-rule rotations about the declared coordinate axes	Elements of $SO(3)$
$\mathbf{Q}_w, \mathbf{Q}_b$	Finite orientation increments expressed along world or body axes	Elements of $SO(3)$; premultiply or postmultiply accordingly
$\boldsymbol{\omega}_w, \boldsymbol{\omega}_b$	Same world-relative angular velocity resolved along world or body axes	$\mathrm{rad\,s^{-1}}$
$SE(3)$	Special Euclidean group of spatial rigid transformations	Spatial rotation and translation represented together
$\mathbf{T}_{wb} \in SE(3)$	Spatial pose of frame $\{b\}$ relative to frame $\{w\}$ and transform from body to world point coordinates	Rotation block dimensionless; translation block in m
$[\mathbf{v}]_\times$	Cross-product matrix satisfying $[\mathbf{v}]_\times \mathbf{y} = \mathbf{v} \times \mathbf{y}$	Same units as $\mathbf{v}$
$\mathfrak{so}(3)$	Lie algebra and identity tangent space of $SO(3)$	3×3 real skew-symmetric matrices
$\mathbf{a}_w, \alpha, \boldsymbol{\varphi}_w = \alpha \mathbf{a}_w$	Unit rotation axis, signed angle, and rotation-vector coordinates expressed along world axes	Dimensionless axis; angle and rotation vector labelled in rad
$\exp(\mathbf{A})$	Matrix exponential $\sum_{j=0}^{\infty} \mathbf{A}^j/j!$	Dimensionless output when $\mathbf{A}$ is dimensionless
$\mathsf{q} = (q_w, \mathbf{q}_v)$	Quaternion written as an ordered scalar–vector pair	Dimensionless components
$\otimes$	Hamilton product of two quaternions	Order follows rotation-composition order

Symbol	Meaning	Units or convention
$\mathsf{q}^* = (q_w, -\mathbf{q}_v)$	Quaternion conjugate; the inverse when q has unit norm	Dimensionless
$\mathfrak{se}(3)$	Lie algebra and identity tangent space of $SE(3)$	Structured 4×4 infinitesimal spatial rigid-motion matrices
$\boldsymbol{\xi} = [\boldsymbol{\nu}^\mathsf{T}\ \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}$	Spatial twist coordinate column in the project's linear-first order	Three entries in $\mathrm{m\,s^{-1}}$ followed by three in $\mathrm{rad\,s^{-1}}$
$\|\mathbf{M}\|_\mathrm{F}$	Frobenius norm of a real matrix	Units follow the matrix entries
$e_\mathrm{orth}, e_\mathrm{det}, e_q$	Orthogonality, determinant, and unit-quaternion residuals	Dimensionless
$\|\boldsymbol{\xi}\|_\ell$	Twist comparison magnitude after choosing characteristic length ℓ	$\mathrm{m\,s^{-1}}$; application-dependent
$\boldsymbol{\Omega}_k = [\boldsymbol{\omega}_{b,k}]_\times$	Body-expressed angular-velocity matrix at sample k	$\mathrm{s^{-1}}$
$\widetilde{\mathbf{R}}_{k+1}$	Forward Euler candidate for the next orientation matrix	Dimensionless; not necessarily in $SO(3)$

Part I

Motion, Mechanics, and Control

1 Linear Algebra Foundations

Why linear algebra matters

Autonomous robots represent directions, velocities, forces, sensor measurements, model parameters, and control commands with vectors. Matrices transform those vectors, combine coordinate descriptions, solve systems of equations, and expose whether information has been preserved or lost. Linear algebra provides the language needed to reason about these operations before they are used in geometry, kinematics, estimation, optimisation, and control.

This chapter develops that language from first principles. It distinguishes geometric vectors from coordinate columns, explains matrices as linear maps, connects null spaces and rank to invertibility, defines determinants as signed volume scaling, and derives the properties of orthogonal matrices used later for rotations.

Prerequisites and assumptions

Prerequisites

The reader should be comfortable with:

- arithmetic and elementary algebra;
- Cartesian coordinates;
- solving simple equations;
- sine and cosine; and
- the project notation conventions.

Assumptions

- Scalars are real numbers unless stated otherwise.
- Vectors are written as columns.
- Matrix and vector dimensions are declared when symbols are introduced.
- Coordinate bases used for Euclidean lengths and angles are orthonormal.
- Physical units are carried by the represented quantity; abstract basis and transformation matrices are dimensionless unless stated otherwise.

Dependency map

```
Vectors and linear combinations
        –
        v
Bases and coordinate columns
        –
        v
Matrices as linear maps
        –
        v
Null spaces, rank, and inverses
        –
        v
Determinants and orthogonal matrices
        –
        v
Coordinate frames, rotations, kinematics, estimation, and control
```

1.1 Vectors: quantities with magnitude and direction

1.1.1 Geometric vector versus coordinate column

A geometric vector is independent of any coordinate frame. A coordinate column is the numerical description of that vector in a chosen basis.

Let $\{a\}$ denote a right-handed coordinate frame with orthonormal basis vectors $\mathbf{e}_{x,a}$, $\mathbf{e}_{y,a}$, and $\mathbf{e}_{z,a}$. For a geometric vector $\mathbf{v}$, the symbol ${}^{a}\mathbf{v}$ denotes its coordinate column expressed in frame $\{a\}$. In three-dimensional space,

$$ {}^{a}\mathbf{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}. $$

The entries mean that

$$ \mathbf{v} = v_x \mathbf{e}_{x,a} + v_y \mathbf{e}_{y,a} + v_z \mathbf{e}_{z,a}. $$

The vector $\mathbf{v}$ is the physical or geometric object, whereas the column ${}^{a}\mathbf{v}$ depends on frame $\{a\}$. Omitting the frame label is safe only when the surrounding context makes the frame unambiguous.

1.1.2 Addition and scalar multiplication

For vectors expressed in the same frame,

$$
{}^{a}\mathbf{u} + {}^{a}\mathbf{v} = \begin{bmatrix} u_x + v_x \\ u_y + v_y \\ u_z + v_z \end{bmatrix}.
$$

For a scalar α,

$$
\alpha\, {}^{a}\mathbf{v} = \begin{bmatrix} \alpha v_x \\ \alpha v_y \\ \alpha v_z \end{bmatrix}.
$$

The algebra is valid only when the quantities are physically compatible. For example, adding two positions expressed in different frames is not meaningful until one has been transformed. Adding a velocity in metres per second to a position in metres is dimensionally invalid even if both are three-element columns.

1.1.3 Length and unit direction

The Euclidean norm is

$$
\|{}^{a}\mathbf{v}\|_2 = \sqrt{v_x^2 + v_y^2 + v_z^2}.
$$

If $\mathbf{v} \neq \mathbf{0}$, let ${}^{a}\mathbf{e}_v$ denote its unit direction expressed in frame $\{a\}$:

$$
{}^{a}\mathbf{e}_v = \frac{{}^{a}\mathbf{v}}{\|{}^{a}\mathbf{v}\|_2}.
$$

The norm has the same physical units as the vector, while the unit vector is dimensionless because the units cancel.

1.1.4 Dot product and angle

Let two vectors be expressed as coordinate columns in the same orthonormal frame:

$$
{}^{a}\mathbf{u} = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix}, \qquad {}^{a}\mathbf{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}.
$$

The superscript T denotes the **transpose** operation. It turns the column ${}^{a}\mathbf{u} \in \mathbb{R}^{3\times 1}$ into the row

$$\left({}^{a}\mathbf{u}\right)^{\mathsf{T}} = {}^{a}\mathbf{u}^{\mathsf{T}} = \begin{bmatrix} u_x & u_y & u_z \end{bmatrix} \in \mathbb{R}^{1\times 3}.$$

Here, T is an operation on the coordinate column; it is not a power. Multiplying this row by the column ${}^{a}\mathbf{v}$ produces a scalar:

$${}^{a}\mathbf{u}^{\mathsf{T}}\,{}^{a}\mathbf{v} = \begin{bmatrix} u_x & u_y & u_z \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = u_x v_x + u_y v_y + u_z v_z.$$

This scalar is the **dot product**, or Euclidean inner product, of the two vectors:

$$\mathbf{u} \cdot \mathbf{v} := {}^{a}\mathbf{u}^{\mathsf{T}}\,{}^{a}\mathbf{v}.$$

The norm introduced in Section 1.1.3 can equivalently be defined from the dot product of a vector with itself:

$$\begin{aligned} \left\|{}^{a}\mathbf{u}\right\|_2 &:= \sqrt{{}^{a}\mathbf{u}^{\mathsf{T}}\,{}^{a}\mathbf{u}} \\ &= \sqrt{u_x^2 + u_y^2 + u_z^2}. \end{aligned}$$

Equivalently, the squared norm is

$$\left\|{}^{a}\mathbf{u}\right\|_2^2 = {}^{a}\mathbf{u}^{\mathsf{T}}\,{}^{a}\mathbf{u}.$$

Thus, dotting a vector with itself gives its squared length, and taking the nonnegative square root gives its length. Because frame $\{a\}$ is orthonormal, this coordinate norm equals the geometric length $\|\mathbf{u}\|_2$.

The coordinate formula above has a geometric interpretation. To establish it, we first derive the law of cosines.

1.1.4.1 Law of cosines

Consider a triangle with two sides of lengths r and s and included angle θ. Place its vertices at

$$\mathbf{p}_0 = (0,0), \qquad \mathbf{p}_1 = (r,0), \qquad \mathbf{p}_2 = (s\cos\theta, s\sin\theta).$$

Let c be the length of the side from $\mathbf{p}_1$ to $\mathbf{p}_2$. The horizontal and vertical displacements along that side are $s\cos\theta - r$ and $s\sin\theta$. By the Pythagorean theorem,

$$\begin{aligned} c^2 &= (s\cos\theta - r)^2 + (s\sin\theta)^2 \\ &= s^2\cos^2\theta - 2rs\cos\theta + r^2 + s^2\sin^2\theta \\ &= r^2 + s^2\left(\cos^2\theta + \sin^2\theta\right) - 2rs\cos\theta \\ &= r^2 + s^2 - 2rs\cos\theta. \end{aligned}$$

Therefore, the law of cosines is

$$\boxed{c^2 = r^2 + s^2 - 2rs\cos\theta}.$$

1.1.4.2 Deriving the dot-product angle formula

For nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, place their tails at the same point and let θ be the angle between them. The vectors form a triangle whose side lengths are

$$r = \|\mathbf{u}\|_2, \qquad s = \|\mathbf{v}\|_2, \qquad c = \|\mathbf{u} - \mathbf{v}\|_2.$$

The law of cosines therefore gives

$$\|\mathbf{u} - \mathbf{v}\|_2^2 = \|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 - 2\|\mathbf{u}\|_2\|\mathbf{v}\|_2\cos\theta.$$

Because frame $\{a\}$ is orthonormal, Euclidean lengths are preserved by its coordinate representation. Expanding the same squared length using coordinate columns gives

$$\begin{aligned} \|\mathbf{u} - \mathbf{v}\|_2^2 &= ({}^a\mathbf{u} - {}^a\mathbf{v})^\mathsf{T}({}^a\mathbf{u} - {}^a\mathbf{v}) \\ &= \|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 - 2\,{}^a\mathbf{u}^\mathsf{T}\,{}^a\mathbf{v}. \end{aligned}$$

Equating the two expressions for $\|\mathbf{u} - \mathbf{v}\|_2^2$ and cancelling their common terms gives

$${}^a\mathbf{u}^\mathsf{T}\,{}^a\mathbf{v} = \|\mathbf{u}\|_2\|\mathbf{v}\|_2\cos\theta,$$

which is the dot-product angle formula. Solving it for the angle gives

$$\theta = \cos^{-1}\left(\frac{{}^a\mathbf{u}^\mathsf{T}\,{}^a\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2}\right),$$

provided neither vector is zero.

The dot product is useful for projection, alignment, and checking orthogonality. If

$${}^a\mathbf{u}^\mathsf{T}\,{}^a\mathbf{v} = 0,$$

then nonzero $\mathbf{u}$ and $\mathbf{v}$ are perpendicular.

1.1.5 Cross product and useful identities

The dot product returns a scalar that measures alignment. Three-dimensional geometry also needs an operation that constructs a vector perpendicular to two given directions. Let $\mathbf{u} = [u_1\ u_2\ u_3]^\mathsf{T} \in \mathbb{R}^3$ and $\mathbf{v} = [v_1\ v_2\ v_3]^\mathsf{T} \in \mathbb{R}^3$ be coordinate columns expressed in the same right-handed orthonormal frame. Their **cross product** is the coordinate column

$$\boxed{\mathbf{u} \times \mathbf{v} := \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix} \in \mathbb{R}^3}.$$

The output is perpendicular to both inputs because direct substitution gives

$$\mathbf{u}^\mathsf{T}(\mathbf{u} \times \mathbf{v}) = 0, \qquad \mathbf{v}^\mathsf{T}(\mathbf{u} \times \mathbf{v}) = 0.$$

Its direction follows the right-hand rule. Expanding its squared norm gives

$$\begin{aligned} \|\mathbf{u} \times \mathbf{v}\|_2^2 &= (u_2 v_3 - u_3 v_2)^2 + (u_3 v_1 - u_1 v_3)^2 + (u_1 v_2 - u_2 v_1)^2 \\ &= \|\mathbf{u}\|_2^2 \|\mathbf{v}\|_2^2 - (\mathbf{u}^\mathsf{T}\mathbf{v})^2 \\ &= \|\mathbf{u}\|_2^2 \|\mathbf{v}\|_2^2 \sin^2\theta, \end{aligned}$$

where $\theta \in [0, \pi]$ is the angle between the nonzero inputs and the last equality uses $\mathbf{u}^\mathsf{T}\mathbf{v} = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \cos\theta$. Taking the nonnegative square root gives

$$\boxed{\|\mathbf{u} \times \mathbf{v}\|_2 = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \sin\theta}.$$

If the input components have physical units, the cross-product components have the product of those units. The coordinate definition also gives the useful identities

$$\boxed{\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}, \qquad \mathbf{u} \times \mathbf{u} = \mathbf{0}_3}.$$

Cross products are not associative: $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ and $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$ are generally different. For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$ expressed in the same right-handed orthonormal frame, the **vector triple-product identity** is

$$\boxed{\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a}^\mathsf{T}\mathbf{c}) - \mathbf{c}(\mathbf{a}^\mathsf{T}\mathbf{b})}.$$

To verify the identity from the cross-product definition, set $\mathbf{d} := \mathbf{b} \times \mathbf{c}$. Substituting the components of $\mathbf{d}$ into $\mathbf{a} \times \mathbf{d}$ and regrouping by the components of $\mathbf{b}$ and $\mathbf{c}$ gives

$$
\begin{aligned}
\mathbf{a} \times \mathbf{d} &= \begin{bmatrix} a_2(b_1c_2 - b_2c_1) - a_3(b_3c_1 - b_1c_3) \\ a_3(b_2c_3 - b_3c_2) - a_1(b_1c_2 - b_2c_1) \\ a_1(b_3c_1 - b_1c_3) - a_2(b_2c_3 - b_3c_2) \end{bmatrix} \\
&= \begin{bmatrix} b_1(\mathbf{a}^\mathsf{T}\mathbf{c}) - c_1(\mathbf{a}^\mathsf{T}\mathbf{b}) \\ b_2(\mathbf{a}^\mathsf{T}\mathbf{c}) - c_2(\mathbf{a}^\mathsf{T}\mathbf{b}) \\ b_3(\mathbf{a}^\mathsf{T}\mathbf{c}) - c_3(\mathbf{a}^\mathsf{T}\mathbf{b}) \end{bmatrix} \\
&= \mathbf{b}(\mathbf{a}^\mathsf{T}\mathbf{c}) - \mathbf{c}(\mathbf{a}^\mathsf{T}\mathbf{b}).
\end{aligned}
$$

A case used later for rotations sets $\mathbf{b} = \mathbf{a}$ and $\mathbf{c} = \mathbf{y}$:

$$\boxed{\mathbf{a} \times (\mathbf{a} \times \mathbf{y}) = \mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}) - (\mathbf{a}^\mathsf{T}\mathbf{a})\mathbf{y}}.$$

If $\mathbf{a}$ is a unit vector, then $\mathbf{a}^\mathsf{T}\mathbf{a} = 1$.

1.1.6 Scalar triple product

A cross product supplies a direction perpendicular to two vectors and a magnitude equal to their parallelogram area. Taking its dot product with a third vector converts this area vector into a signed volume. For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$ expressed in the same right-handed orthonormal frame, the **scalar triple product** is

$$\boxed{\mathbf{a}^\mathsf{T}(\mathbf{b} \times \mathbf{c}) \in \mathbb{R}}.$$

The word *scalar* records that the outer dot product returns one real number, while *triple* records that the expression uses three vectors. Substituting the cross-product components gives the explicit formula

$$
\begin{aligned}
\mathbf{a}^\mathsf{T}(\mathbf{b} \times \mathbf{c}) = a_1(b_2c_3 - b_3c_2) &+ a_2(b_3c_1 - b_1c_3) \\
&+ a_3(b_1c_2 - b_2c_1).
\end{aligned}
$$

Geometrically, $\|\mathbf{b} \times \mathbf{c}\|_2$ is the area of the parallelogram spanned by $\mathbf{b}$ and $\mathbf{c}$. If $\mathbf{b} \times \mathbf{c} \neq \mathbf{0}_3$, define its unit normal by $\mathbf{n} := (\mathbf{b} \times \mathbf{c})/\|\mathbf{b} \times \mathbf{c}\|_2$. Then

$$\mathbf{a}^\mathsf{T}(\mathbf{b} \times \mathbf{c}) = \underbrace{\mathbf{a}^\mathsf{T}\mathbf{n}}_{\text{signed height}} \underbrace{\|\mathbf{b} \times \mathbf{c}\|_2}_{\text{base area}}.$$

Its absolute value is therefore the volume of the parallelepiped spanned by the three vectors. Its sign states whether the ordered directions $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ have positive or negative orientation, and it is zero when the three vectors are coplanar. If the input vectors carry physical units, the scalar triple product carries the product of all three units.

Direct component expansion also gives the cyclic identities

$$\boxed{\mathbf{a}^\mathsf{T}(\mathbf{b} \times \mathbf{c}) = \mathbf{b}^\mathsf{T}(\mathbf{c} \times \mathbf{a}) = \mathbf{c}^\mathsf{T}(\mathbf{a} \times \mathbf{b})}.$$

Cyclically moving the three vectors does not change the value, but swapping any two vectors reverses its sign. This scalar-valued expression differs from the vector triple product in Section 1.1.5, whose output is a vector. The determinant discussion in Section 1.3 will show that the scalar triple product is also the determinant of the matrix whose columns are $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ in that order.

1.1.7 Linear combinations, span, and basis

Let $\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ and let $\alpha_1, \ldots, \alpha_k \in \mathbb{R}$. A **linear combination** of these vectors is any vector of the form

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k.$$

The **span** of $\mathbf{v}_1, \ldots, \mathbf{v}_k$ is the set of every linear combination that can be formed from them:

$$\operatorname{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} := \left\{ \sum_{j=1}^{k} \alpha_j \mathbf{v}_j : \alpha_1, \ldots, \alpha_k \in \mathbb{R} \right\}.$$

The vectors are **linearly independent** if

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_k \mathbf{v}_k = \mathbf{0}_n$$

implies $\alpha_1 = \cdots = \alpha_k = 0$, where $\mathbf{0}_n \in \mathbb{R}^n$ is the zero vector. If a nonzero set of coefficients produces $\mathbf{0}_n$, the vectors are linearly dependent.

A **basis** of $\mathbb{R}^n$ is a set of n linearly independent vectors that spans $\mathbb{R}^n$. If $\mathbf{e}_1, \ldots, \mathbf{e}_n$ is a basis, every $\mathbf{x} \in \mathbb{R}^n$ has a unique expansion

$$\mathbf{x} = x_1 \mathbf{e}_1 + \cdots + x_n \mathbf{e}_n.$$

The scalar coefficients $x_1, \ldots, x_n \in \mathbb{R}$ form the coordinate column of $\mathbf{x}$ in that basis. Uniqueness is essential: if the basis vectors were dependent, the same vector could have more than one coefficient description.

The **standard basis** of $\mathbb{R}^n$ is $\mathbf{e}_1, \ldots, \mathbf{e}_n$, where $\mathbf{e}_i$ has entry 1 in position i and entry 0 in every other position. In the standard basis,

$$\mathbf{x} = x_1 \mathbf{e}_1 + \cdots + x_n \mathbf{e}_n \quad \Longleftrightarrow \quad \mathbf{x} = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}^\mathsf{T}.$$

Quick test A: vectors and bases

Complete this test without looking back at the section.

1. **Recall:** What is the difference between a geometric vector $\mathbf{v}$ and a coordinate column ${}^{a}\mathbf{v}$?
2. **Explanation:** Why must coordinate columns be associated with a basis?
3. **Derivation:** Use the law of cosines and an algebraic expansion of $\|\mathbf{u}-\mathbf{v}\|_2^2$ to derive ${}^{a}\mathbf{u}^{\mathsf{T}}\,{}^{a}\mathbf{v} = \|\mathbf{u}\|_2\|\mathbf{v}\|_2\cos\theta$.
4. **Application:** Find the norm and unit direction of $\mathbf{x} = [3 \;\; 4]^{\mathsf{T}}$.
5. **Limitations:** Why is the angle formula undefined when either vector is zero?
6. **Basis:** Explain why two parallel nonzero vectors cannot form a basis of $\mathbb{R}^2$.
7. **Cross product:** For a unit vector $\mathbf{a} \in \mathbb{R}^3$, use the vector triple-product identity to simplify $\mathbf{a} \times (\mathbf{a} \times \mathbf{y})$.
8. **Scalar triple product:** Evaluate $\mathbf{e}_1^{\mathsf{T}}(\mathbf{e}_2 \times \mathbf{e}_3)$, then predict the value after swapping $\mathbf{e}_2$ and $\mathbf{e}_3$. What geometric change causes the sign change?

Answers and hints

1. $\mathbf{v}$ is the coordinate-independent geometric object; ${}^{a}\mathbf{v}$ is its numerical representation in basis or frame $\{a\}$.
2. The entries are coefficients multiplying particular basis vectors. Without the basis, the numbers do not determine physical or geometric directions.
3. Equate the law-of-cosines expression for $\|\mathbf{u}-\mathbf{v}\|_2^2$ with its dot-product expansion and cancel the common norm terms.
4. $\|\mathbf{x}\|_2 = 5$ and the unit direction is $[0.6 \;\; 0.8]^{\mathsf{T}}$.
5. The denominator $\|\mathbf{u}\|_2\|\mathbf{v}\|_2$ becomes zero, and a zero vector has no direction.
6. One vector is a scalar multiple of the other, so their span is only a line and their coefficient representation is not unique.
7. $\mathbf{a} \times (\mathbf{a} \times \mathbf{y}) = \mathbf{a}(\mathbf{a}^{\mathsf{T}}\mathbf{y}) - \mathbf{y}$ because $\mathbf{a}^{\mathsf{T}}\mathbf{a} = 1$.
8. Since $\mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1$, the first value is 1. Swapping the two cross-product inputs gives $\mathbf{e}_3 \times \mathbf{e}_2 = -\mathbf{e}_1$, so the second value is -1. The swap reverses the orientation of the ordered directions while preserving the volume magnitude.

1.2 Matrices: linear maps and basis changes

1.2.1 Matrix dimensions

A matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$ maps an n-element coordinate vector to an m-element coordinate vector:

$$\mathbf{y} = \mathbf{A}\mathbf{x}$$

where $\mathbf{y} \in \mathbb{R}^m$ and $\mathbf{x} \in \mathbb{R}^n$.

If

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

then

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix}.$$

Equivalently, if $\mathbf{a}_1$ and $\mathbf{a}_2$ are the columns of $\mathbf{A}$,

$$\mathbf{A}\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2.$$

This column interpretation is central to rotation matrices: their columns are the axes of one frame expressed in another frame.

1.2.2 Identity, transpose, null space, rank, and inverse

Let $\mathbf{I}_n$ denote the $n \times n$ identity matrix. It leaves compatible coordinates unchanged:

$$\mathbf{I}_n\mathbf{x} = \mathbf{x}.$$

The transpose swaps rows and columns:

$$\left(\mathbf{A}^\mathsf{T}\right)_{ij} = a_{ji}.$$

Let $\mathbf{A} \in \mathbb{R}^{m\times n}$. Its **column space**, denoted $\mathrm{Col}(\mathbf{A})$, is the set of all linear combinations of its columns. The **rank**, denoted $\mathrm{rank}(\mathbf{A})$, is the number of linearly independent columns, equivalently the dimension of $\mathrm{Col}(\mathbf{A})$.

The **null space** of $\mathbf{A}$ is

$$\mathcal{N}(\mathbf{A}) := \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{0}_m\},$$

where $\mathbf{0}_m \in \mathbb{R}^m$ is the zero vector. The null space contains all input directions that the mapping sends to zero.

1.2.2.1 Rank-nullity theorem

The **nullity** of $\mathbf{A}$ is the dimension of its null space:

$$\operatorname{nullity}(\mathbf{A}) := \dim \mathcal{N}(\mathbf{A}).$$

For every matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$, the **rank-nullity theorem** states

$$\boxed{\operatorname{rank}(\mathbf{A}) + \operatorname{nullity}(\mathbf{A}) = n}.$$

The number n is the dimension of the input space $\mathbb{R}^n$. The theorem says that its independent input directions divide into two counts: $\operatorname{nullity}(\mathbf{A})$ directions that the mapping sends to zero and $\operatorname{rank}(\mathbf{A})$ directions that produce independent output directions.

To prove the theorem, let

$$k := \operatorname{nullity}(\mathbf{A}),$$

and choose a basis $\mathbf{n}_1, \dots, \mathbf{n}_k \in \mathbb{R}^n$ for $\mathcal{N}(\mathbf{A})$. Because these vectors are linearly independent, they can be extended to a basis of the whole input space by repeatedly adding vectors outside their current span. Write the resulting basis as

$$\mathbf{n}_1, \dots, \mathbf{n}_k, \mathbf{v}_1, \dots, \mathbf{v}_{n-k}.$$

The output vectors

$$\mathbf{A}\mathbf{v}_1, \dots, \mathbf{A}\mathbf{v}_{n-k}$$

span $\operatorname{Col}(\mathbf{A})$. To see this, express any input $\mathbf{x} \in \mathbb{R}^n$ in the chosen input basis:

$$\mathbf{x} = \sum_{i=1}^{k} a_i \mathbf{n}_i + \sum_{j=1}^{n-k} b_j \mathbf{v}_j.$$

Because every $\mathbf{n}_i$ belongs to the null space, $\mathbf{A}\mathbf{n}_i = \mathbf{0}_m$. Applying $\mathbf{A}$ therefore gives

$$\mathbf{A}\mathbf{x} = \sum_{j=1}^{n-k} b_j \mathbf{A}\mathbf{v}_j,$$

so every possible output is a linear combination of the listed output vectors.

These output vectors are also linearly independent. If

$$\sum_{j=1}^{n-k} c_j \mathbf{A}\mathbf{v}_j = \mathbf{0}_m,$$

then

$$\mathbf{A}\left(\sum_{j=1}^{n-k} c_j \mathbf{v}_j\right) = \mathbf{0}_m.$$

Thus, $\sum_j c_j \mathbf{v}_j$ belongs to $\mathcal{N}(\mathbf{A})$ and can also be written as a linear combination of $\mathbf{n}_1, \ldots, \mathbf{n}_k$. The complete list $\mathbf{n}_1, \ldots, \mathbf{n}_k, \mathbf{v}_1, \ldots, \mathbf{v}_{n-k}$ is a basis and is therefore linearly independent, so this is possible only when every $c_j = 0$. Hence, the $n-k$ vectors $\mathbf{A}\mathbf{v}_j$ form a basis of $\operatorname{Col}(\mathbf{A})$, and

$$\operatorname{rank}(\mathbf{A}) = n - k.$$

Since $k = \operatorname{nullity}(\mathbf{A})$,

$$\operatorname{rank}(\mathbf{A}) + \operatorname{nullity}(\mathbf{A}) = (n-k) + k = n,$$

which proves the theorem.

For example, let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \in \mathbb{R}^{2\times 3}.$$

The first two columns are independent and the third is their sum, so $\operatorname{rank}(\mathbf{A}) = 2$. Solving $\mathbf{A}\mathbf{x} = \mathbf{0}_2$ gives

$$\mathcal{N}(\mathbf{A}) = \left\{ \alpha \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \;\middle|\; \alpha \in \mathbb{R} \right\},$$

so $\operatorname{nullity}(\mathbf{A}) = 1$. The input dimension is therefore partitioned as

$$\underbrace{3}_{\text{input dimension}} = \underbrace{2}_{\text{rank}} + \underbrace{1}_{\text{nullity}}.$$

1.2.2.2 Injective mappings and unique recovery

Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ define the linear map

$$T_{\mathbf{A}} : \mathbb{R}^n \to \mathbb{R}^m, \qquad T_{\mathbf{A}}(\mathbf{x}) = \mathbf{A}\mathbf{x}.$$

The map $T_{\mathbf{A}}$ is **injective**, or **one-to-one**, if equal outputs can occur only when the inputs are equal. Formally, for every $\mathbf{x}^{(1)}, \mathbf{x}^{(2)} \in \mathbb{R}^n$, where the parenthesised superscripts (1) and (2) label two inputs and are not exponents,

$$\mathbf{A}\mathbf{x}^{(1)} = \mathbf{A}\mathbf{x}^{(2)} \quad \Longrightarrow \quad \mathbf{x}^{(1)} = \mathbf{x}^{(2)}.$$

Equivalently, distinct inputs must produce distinct outputs:

$$\mathbf{x}^{(1)} \neq \mathbf{x}^{(2)} \quad \Longrightarrow \quad \mathbf{A}\mathbf{x}^{(1)} \neq \mathbf{A}\mathbf{x}^{(2)}.$$

The map is **not injective** if at least one pair of distinct inputs produces the same output:

$$\exists\, \mathbf{x}^{(1)}, \mathbf{x}^{(2)} \in \mathbb{R}^n : \quad \mathbf{x}^{(1)} \neq \mathbf{x}^{(2)}, \qquad \mathbf{A}\mathbf{x}^{(1)} = \mathbf{A}\mathbf{x}^{(2)}.$$

To connect injectivity to the null space, suppose two inputs have the same output. Subtracting the outputs gives

$$\begin{aligned} \mathbf{A}\mathbf{x}^{(1)} &= \mathbf{A}\mathbf{x}^{(2)} \\ \Longrightarrow \quad \mathbf{A}(\mathbf{x}^{(1)} - \mathbf{x}^{(2)}) &= \mathbf{0}_m. \end{aligned}$$

If $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}_n\}$, then $\mathbf{x}^{(1)} - \mathbf{x}^{(2)} = \mathbf{0}_n$, so the inputs must be equal. Conversely, if a nonzero vector $\mathbf{z} \in \mathcal{N}(\mathbf{A})$ exists, then for any $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{A}(\mathbf{x} + \mathbf{z}) = \mathbf{A}\mathbf{x} + \mathbf{A}\mathbf{z} = \mathbf{A}\mathbf{x}.$$

Because $\mathbf{z} \neq \mathbf{0}_n$, the inputs $\mathbf{x}$ and $\mathbf{x}+\mathbf{z}$ are different but produce the same output. Therefore,

$$\boxed{T_{\mathbf{A}} \text{ is injective} \quad \Longleftrightarrow \quad \mathcal{N}(\mathbf{A}) = \{\mathbf{0}_n\} \quad \Longleftrightarrow \quad \mathbf{A} \text{ has full column rank}}.$$

For a square matrix, these conditions are also equivalent to nonsingularity and invertibility. If the map is injective, its output uniquely identifies its input; if it is not injective, some input information cannot be recovered from the output.

Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be a square matrix with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n \in \mathbb{R}^n$, and let $\mathbf{x} = [x_1 \ \cdots \ x_n]^{\mathsf{T}} \in \mathbb{R}^n$. Matrix-vector multiplication forms the linear combination

$$\mathbf{A}\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \sum_{j=1}^{n} x_j\mathbf{a}_j.$$

For every matrix, $\mathbf{x} = \mathbf{0}$ implies $\mathbf{A}\mathbf{x} = \mathbf{0}$. The square matrix $\mathbf{A}$ is **nonsingular** when the converse is also true:

$$\mathbf{A}\mathbf{x} = \mathbf{0} \quad \Longleftrightarrow \quad \mathbf{x} = \mathbf{0}.$$

Equivalently,

$$\mathbf{x} \neq \mathbf{0} \quad \Longrightarrow \quad \mathbf{A}\mathbf{x} \neq \mathbf{0}.$$

Because $\mathbf{A}\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{a}_j$, nonsingularity means that

$$x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$$

only when $x_1 = \cdots = x_n = 0$. This is precisely the definition that the columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ are linearly independent.

A square matrix that is not nonsingular is called **singular**. For a singular matrix, at least one nonzero $\mathbf{x}$ satisfies $\mathbf{A}\mathbf{x} = \mathbf{0}$, so the mapping loses information about that input direction.

More generally, if $\mathbf{A} \in \mathbb{R}^{m\times n}$, then its columns are $\mathbf{a}_1, \ldots, \mathbf{a}_n \in \mathbb{R}^m$ and $\mathbf{A}\mathbf{x} = \sum_{j=1}^{n} x_j\mathbf{a}_j$. The condition $\mathbf{A}\mathbf{x} = \mathbf{0}_m \Rightarrow \mathbf{x} = \mathbf{0}_n$ means that $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}_n\}$ and that all n columns are linearly independent. In that case $\operatorname{rank}(\mathbf{A}) = n$, which is called **full column rank**. The term *nonsingular* is normally reserved for the square case $m = n$.

Only a nonsingular square matrix has an inverse $\mathbf{A}^{-1} \in \mathbb{R}^{n\times n}$. The inverse reverses the action of $\mathbf{A}$ and satisfies

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n.$$

1.2.3 Multiplication order

Let n be a positive integer. Let $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{n\times n}$ be three matrices, and let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ be coordinate vectors. Define the intermediate vector $\mathbf{z}$ and output vector $\mathbf{y}$ by

$$\mathbf{z} = \mathbf{B}\mathbf{x} \qquad \text{and} \qquad \mathbf{y} = \mathbf{A}\mathbf{z}.$$

Substituting the expression for $\mathbf{z}$ into the expression for $\mathbf{y}$ gives

$$\mathbf{y} = \mathbf{A}(\mathbf{B}\mathbf{x}) = (\mathbf{A}\mathbf{B})\mathbf{x}.$$

Thus, the rightmost matrix acts first: $\mathbf{B}$ maps $\mathbf{x}$ to $\mathbf{z}$, and then $\mathbf{A}$ maps $\mathbf{z}$ to $\mathbf{y}$.

The third matrix $\mathbf{C}$ allows three transformations to be composed. Matrix multiplication is **associative**, meaning that changing the grouping does not change the product:

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}).$$

Both sides are matrices in $\mathbb{R}^{n\times n}$. However, matrix multiplication is generally **not commutative**, meaning that changing the order can change the result:

$$\mathbf{AB} \neq \mathbf{BA}.$$

This order becomes critical when several coordinate transformations are composed.

Quick test B: matrices, rank, and inverses

1. **Recall:** For $\mathbf{A} \in \mathbb{R}^{m\times n}$, what are the dimensions of an input $\mathbf{x}$ and output $\mathbf{y} = \mathbf{Ax}$?
2. **Explanation:** What does column j of $\mathbf{A}$ tell us about the mapping?
3. **Derivation:** Explain why $\mathbf{Ax} = \mathbf{0}$ has only the zero solution exactly when the columns of $\mathbf{A}$ are linearly independent.
4. **Application:** Determine the null space and rank of $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.
5. **Limitations:** Why can a nonsquare matrix not have a two-sided inverse of the same kind as a nonsingular square matrix?
6. **Composition:** If $\mathbf{z} = \mathbf{Bx}$ and $\mathbf{y} = \mathbf{Az}$, which matrix acts first in $\mathbf{y} = \mathbf{ABx}$?
7. **Injectivity:** Define an injective matrix mapping and explain why it is equivalent to $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$.
8. **Rank-nullity:** State the rank-nullity theorem for $\mathbf{A} \in \mathbb{R}^{m\times n}$ and verify it for the matrix in Question 4.

Answers and hints

1. $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^m$.
2. It is the output produced by applying $\mathbf{A}$ to the corresponding standard basis input; general outputs are linear combinations of the columns.
3. The equation is the linear combination $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$. Only zero coefficients solve it precisely when the columns are independent.
4. The second column is twice the first, so the rank is 1. Solving $x_1 + 2x_2 = 0$ gives $\mathcal{N}(\mathbf{A}) = \{t[-2 \;\; 1]^\mathsf{T} : t \in \mathbb{R}\}$.
5. A two-sided inverse would have to undo mappings in both domain and codomain, which requires equal dimensions and no lost directions.
6. $\mathbf{B}$ acts first, followed by $\mathbf{A}$.

7. A map is injective if $\mathbf{A}\mathbf{x}^{(1)} = \mathbf{A}\mathbf{x}^{(2)}$ implies $\mathbf{x}^{(1)} = \mathbf{x}^{(2)}$. If $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$, subtracting equal outputs gives $\mathbf{A}(\mathbf{x}^{(1)} - \mathbf{x}^{(2)}) = \mathbf{0}$, so $\mathbf{x}^{(1)} - \mathbf{x}^{(2)} = \mathbf{0}$. Conversely, a nonzero null vector $\mathbf{z}$ makes $\mathbf{x}$ and $\mathbf{x} + \mathbf{z}$ distinct inputs with the same output.
8. The theorem is $\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n$. The matrix in Question 4 has two input columns, rank one, and a one-dimensional null space, so $1 + 1 = 2$.

1.3 Determinants and signed volume

1.3.1 What the determinant measures

The determinant is defined for a square matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ and returns a scalar:

$$\det : \mathbb{R}^{n\times n} \to \mathbb{R}.$$

Geometrically, $|\det(\mathbf{A})|$ is the factor by which the linear map $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ scales n-dimensional volume. The sign records whether the ordered basis orientation is preserved or reversed.

For $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{R}^{2\times 2}$,

$$\det(\mathbf{A}) = ad - bc.$$

Its magnitude is the area of the parallelogram formed by the columns of $\mathbf{A}$. For $\mathbf{A} = [\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3] \in \mathbb{R}^{3\times 3}$, the determinant is the scalar triple product defined in Section 1.1.6:

$$\det(\mathbf{A}) = \mathbf{a}_1^\mathsf{T}(\mathbf{a}_2 \times \mathbf{a}_3).$$

Its magnitude is the volume of the parallelepiped formed by the columns.

1.3.1.1 Determinant sign, the right-hand rule, and reflection

The columns of $\mathbf{A} = [\mathbf{a}_1\ \cdots\ \mathbf{a}_n]$ are the images of the standard basis vectors because $\mathbf{A}\mathbf{e}_i = \mathbf{a}_i$. The determinant sign therefore compares the orientation of the ordered transformed columns with the orientation of the ordered standard basis.

In three dimensions, the standard basis is right-handed:

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3.$$

For $\mathbf{A} = [\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3] \in \mathbb{R}^{3\times 3}$, the determinant can be written as

$$\det(\mathbf{A}) = (\mathbf{a}_1 \times \mathbf{a}_2)^\mathsf{T}\mathbf{a}_3.$$

The cross product $\mathbf{a}_1 \times \mathbf{a}_2$ points in the right-hand-rule normal direction determined by the first two columns. Consequently:

- If $\det(\mathbf{A}) > 0$, then $\mathbf{a}_3$ has a positive component in that right-hand-rule direction. The ordered columns $(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$ are right-handed, so the transformation preserves orientation.
- If $\det(\mathbf{A}) < 0$, then $\mathbf{a}_3$ points to the opposite side of the plane spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$. The ordered columns are left-handed, so the transformation reverses orientation.
- If $\det(\mathbf{A}) = 0$, then the three columns are coplanar or otherwise linearly dependent. The three-dimensional volume collapses to zero, so a full three-dimensional orientation is no longer defined.

A reflection reverses orientation. For example, reflection across the y-z plane is represented by

$$\mathbf{S} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \det(\mathbf{S}) = -1.$$

This transformation sends $\mathbf{e}_1$ to $-\mathbf{e}_1$ while leaving $\mathbf{e}_2$ and $\mathbf{e}_3$ unchanged. The transformed basis is left-handed because

$$(-\mathbf{e}_1) \times \mathbf{e}_2 = -\mathbf{e}_3,$$

whereas its third basis vector is $+\mathbf{e}_3$. Thus, the negative determinant detects the orientation reversal introduced by the reflection.

A matrix with negative determinant need not be a pure reflection: it may also stretch, scale, or shear. The negative sign means that its total action contains an orientation-reversing, or reflection-like, component. By contrast, a proper rotation preserves right-handedness and has determinant $+1$.

1.3.2 Algebraic definition

The determinant is the unique scalar-valued function of the columns of a square matrix with three defining properties:

1. **Multilinearity:** it is linear in each column while the other columns are held fixed.
2. **Alternation:** swapping two columns changes the sign; consequently, a matrix with repeated columns has determinant zero.
3. **Normalisation:** $\det(\mathbf{I}_n) = 1$.

1.3.2.1 Multilinearity in one column

Write the columns of $\mathbf{A} \in \mathbb{R}^{n\times n}$ as $\mathbf{A} = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$, where each $\mathbf{a}_i \in \mathbb{R}^n$. Choose a column index $j \in \{1, \ldots, n\}$ and hold every column except column j fixed. For any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and scalars $\alpha, \beta \in \mathbb{R}$, linearity in column j means

$$\begin{aligned}
&\det(\mathbf{a}_1, \ldots, \mathbf{a}_{j-1}, \alpha\mathbf{u} + \beta\mathbf{v}, \mathbf{a}_{j+1}, \ldots, \mathbf{a}_n) \\
&\quad = \alpha \det(\mathbf{a}_1, \ldots, \mathbf{a}_{j-1}, \mathbf{u}, \mathbf{a}_{j+1}, \ldots, \mathbf{a}_n) \\
&\qquad + \beta \det(\mathbf{a}_1, \ldots, \mathbf{a}_{j-1}, \mathbf{v}, \mathbf{a}_{j+1}, \ldots, \mathbf{a}_n).
\end{aligned}$$

Thus, a sum placed in one column can be separated into a sum of determinants, and a scalar multiplying one column can be moved outside the determinant. The word **multilinear** means that this linearity property holds separately for every column. It does not mean that the determinant is linear in the whole matrix at once; in general,

$$\det(\mathbf{A} + \mathbf{B}) \neq \det(\mathbf{A}) + \det(\mathbf{B}).$$

For a two-dimensional check, let $\mathbf{u} = [u_1 \ u_2]^\mathsf{T}$, $\mathbf{v} = [v_1 \ v_2]^\mathsf{T}$, and $\mathbf{w} = [w_1 \ w_2]^\mathsf{T}$. Linearity in the first column follows directly from

$$\begin{aligned}
\det\begin{bmatrix} \alpha u_1 + \beta v_1 & w_1 \\ \alpha u_2 + \beta v_2 & w_2 \end{bmatrix} &= (\alpha u_1 + \beta v_1)w_2 - w_1(\alpha u_2 + \beta v_2) \\
&= \alpha(u_1 w_2 - w_1 u_2) + \beta(v_1 w_2 - w_1 v_2) \\
&= \alpha \det[\mathbf{u} \ \mathbf{w}] + \beta \det[\mathbf{v} \ \mathbf{w}].
\end{aligned}$$

1.3.2.2 Alternation and repeated columns

Let $\mathbf{A}'$ be obtained from $\mathbf{A}$ by swapping columns i and j, where $i \neq j$. Alternation means

$$\det(\mathbf{A}') = -\det(\mathbf{A}).$$

A single swap reverses the orientation of the ordered columns, so it reverses the sign of the signed volume without changing its magnitude.

Now suppose columns i and j are equal to the same vector $\mathbf{c} \in \mathbb{R}^n$. Let the determinant of this matrix be D. Swapping the two equal columns leaves the matrix numerically unchanged, so its determinant is still D. Alternation simultaneously requires the determinant to become $-D$. Therefore,

$$D = -D \quad \Longrightarrow \quad 2D = 0 \quad \Longrightarrow \quad D = 0.$$

Hence, every matrix with two identical columns has determinant zero. Geometrically, two identical column directions cannot span an n-dimensional parallelepiped, so its n-dimensional volume collapses to zero.

Equivalently, if $\mathbf{A} = [a_{ij}] \in \mathbb{R}^{n\times n}$, then

$$\det(\mathbf{A}) = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^{n} a_{\sigma(j),j}.$$

Here, S_n is the set of all permutations of the indices $1,\dots,n$. The sign $\operatorname{sgn}(\sigma)$ equals $+1$ when the permutation can be produced by an even number of pairwise swaps and -1 when it requires an odd number.

If the columns of $\mathbf{A}$ are linearly dependent, one column can be written as a linear combination of the others. Multilinearity and alternation then force $\det(\mathbf{A}) = 0$. Conversely, $\det(\mathbf{A}) = 0$ means that the signed volume has collapsed, so the columns do not span $\mathbb{R}^n$. Therefore,

$$\boxed{\det(\mathbf{A}) \neq 0 \quad \Longleftrightarrow \quad \mathbf{A} \text{ is nonsingular}}.$$

1.3.2.3 Example: a singular matrix loses one input component

Consider the linear map $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$ from $\mathbb{R}^2$ to $\mathbb{R}^2$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Its determinant is

$$\det(\mathbf{A}) = 0.$$

For an input coordinate column

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2,$$

the output is

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} \in \mathbb{R}^2.$$

The output retains x_1 but completely removes x_2. For example,

$$\mathbf{A}\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \qquad \mathbf{A}\begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}.$$

The two inputs are different, but their outputs are identical. From the output $[3\ 0]^\mathsf{T}$, we can recover $x_1 = 3$, but we cannot determine whether x_2 was 1, 7, or any other real number. In fact,

$$\mathcal{N}(\mathbf{A}) = \left\{ t \begin{bmatrix} 0 \\ 1 \end{bmatrix} : t \in \mathbb{R} \right\},$$

so the entire input direction $[0\ 1]^\mathsf{T}$ is lost.

Geometrically, $\mathbf{A}$ maps the entire two-dimensional plane onto the one-dimensional x-axis:

$$\mathbb{R}^2 \longrightarrow \left\{ \begin{bmatrix} y_1 \\ 0 \end{bmatrix} : y_1 \in \mathbb{R} \right\}.$$

Every two-dimensional region is flattened into a line segment, whose two-dimensional area is zero. This agrees with $\det(\mathbf{A}) = 0$.

Recall that **not injective** means that there exist at least two distinct inputs $\mathbf{x}^{(1)}, \mathbf{x}^{(2)} \in \mathbb{R}^n$ with the same output:

$$\mathbf{x}^{(1)} \neq \mathbf{x}^{(2)}, \qquad \mathbf{A}\mathbf{x}^{(1)} = \mathbf{A}\mathbf{x}^{(2)}.$$

The statement does not mean that every pair of inputs has the same output; it means that at least one distinct pair cannot be distinguished from the output.

If $\mathbf{A}$ is singular, then there exists $\mathbf{z} \in \mathcal{N}(\mathbf{A})$ such that $\mathbf{A}\mathbf{z} = \mathbf{0}$. This implies $\mathbf{A}(\mathbf{x} + \mathbf{z}) = \mathbf{A}\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. Hence, a singular matrix is not injective and different inputs can produce the same output.

For a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, the general equivalences are

$$\boxed{\begin{aligned} \det(\mathbf{A}) = 0 &\iff \mathbf{A} \text{ is singular} \\ &\iff \mathcal{N}(\mathbf{A}) \neq \{\mathbf{0}\} \\ &\iff \mathbf{A} \text{ is not injective} \\ &\iff \text{different inputs can produce the same output} \\ &\iff \text{the input cannot be uniquely recovered.} \end{aligned}}$$

1.3.3 Why determinants multiply

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$. The product $\mathbf{AB}$ represents the composition

$$\mathbf{x} \mapsto \mathbf{Bx} \mapsto \mathbf{ABx}.$$

Geometrically, $\mathbf{B}$ first scales signed volume by $\det(\mathbf{B})$, and $\mathbf{A}$ then scales the result by $\det(\mathbf{A})$. This suggests

$$\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B}).$$

A formal proof follows from the defining properties of the determinant. Hold $\mathbf{A}$ fixed and define

$$F : \mathbb{R}^{n\times n} \to \mathbb{R}, \qquad F(\mathbf{B}) := \det(\mathbf{AB}).$$

Write $\mathbf{B} = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_n]$, where each $\mathbf{b}_j \in \mathbb{R}^n$. Then

$$\mathbf{AB} = \begin{bmatrix} \mathbf{Ab}_1 & \cdots & \mathbf{Ab}_n \end{bmatrix}.$$

To check linearity in column j, let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Because multiplication by $\mathbf{A}$ is linear,

$$\mathbf{A}(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha\mathbf{Au} + \beta\mathbf{Av}.$$

The determinant is linear in column j, so

$$\begin{aligned} &F(\mathbf{b}_1, \ldots, \alpha\mathbf{u} + \beta\mathbf{v}, \ldots, \mathbf{b}_n) \\ &= \alpha F(\mathbf{b}_1, \ldots, \mathbf{u}, \ldots, \mathbf{b}_n) + \beta F(\mathbf{b}_1, \ldots, \mathbf{v}, \ldots, \mathbf{b}_n). \end{aligned}$$

Thus, F is multilinear in the columns of $\mathbf{B}$. If two columns of $\mathbf{B}$ are equal, their images under $\mathbf{A}$ are equal, so $\mathbf{AB}$ has repeated columns and $F(\mathbf{B}) = 0$. Swapping two columns of $\mathbf{B}$ swaps the corresponding columns of $\mathbf{AB}$ and changes the determinant sign.

Hence, F is an alternating multilinear scalar-valued function of the columns of $\mathbf{B}$. We claim that

$$F(\mathbf{B}) = c\det(\mathbf{B}), \qquad c := F(\mathbf{e}_1, \ldots, \mathbf{e}_n).$$

The following expansion proves the claim.

Let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the standard basis of $\mathbb{R}^n$. Because it is a basis, every column $\mathbf{b}_j$ of $\mathbf{B}$ has the unique expansion

$$\mathbf{b}_j = \sum_{i=1}^{n} b_{ij}\mathbf{e}_i.$$

Using multilinearity once for each of the n columns gives

$$F(\mathbf{B}) = \sum_{i_1=1}^{n} \cdots \sum_{i_n=1}^{n} \left(\prod_{j=1}^{n} b_{i_j,j} \right) F(\mathbf{e}_{i_1}, \ldots, \mathbf{e}_{i_n}).$$

This sum initially contains n^n terms. If two indices are equal, such as $i_p = i_q$, then the corresponding argument list contains the same basis vector twice. Alternation makes that term zero:

$$F(\ldots, \mathbf{e}_{i_p}, \ldots, \mathbf{e}_{i_p}, \ldots) = 0.$$

A term can therefore survive only when $i_1, \ldots, i_n$ are all different. Because every i_j belongs to $\{1, \ldots, n\}$, the surviving index list must contain every index exactly once. Thus, it has the form

$$(i_1, \ldots, i_n) = (\sigma(1), \ldots, \sigma(n))$$

for some permutation $\sigma \in S_n$.

Alternation determines the value of every surviving basis term relative to the ordered basis term. Reordering $(\mathbf{e}_{\sigma(1)}, \ldots, \mathbf{e}_{\sigma(n)})$ into $(\mathbf{e}_1, \ldots, \mathbf{e}_n)$ requires an even or odd number of swaps, so

$$F(\mathbf{e}_{\sigma(1)}, \ldots, \mathbf{e}_{\sigma(n)}) = \operatorname{sgn}(\sigma) F(\mathbf{e}_1, \ldots, \mathbf{e}_n).$$

Substituting this result into the expansion for $F(\mathbf{B})$ leaves only the permutation terms:

$$F(\mathbf{B}) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^{n} b_{\sigma(j),j} F(\mathbf{e}_1, \ldots, \mathbf{e}_n)$$

The scalar appearing in the claim is

$$c := F(\mathbf{e}_1, \ldots, \mathbf{e}_n).$$

The columns of $\mathbf{I}_n$ are the ordered standard basis vectors, so the same reference value can be written as

$$c = F(\mathbf{e}_1, \dots, \mathbf{e}_n) = F(\mathbf{I}_n).$$

Substituting the surviving terms into the expansion gives

$$\begin{aligned} F(\mathbf{B}) &= c \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^{n} b_{\sigma(j),j} \\ &= c \det(\mathbf{B}). \end{aligned}$$

A two-dimensional example makes the one-free-value idea explicit. In this example, $F(\mathbf{u}, \mathbf{v})$ is shorthand for $F([\mathbf{u}\ \mathbf{v}])$, where $\mathbf{u}$ and $\mathbf{v}$ are the two columns of the input matrix. Let $\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$ and $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$. Using multilinearity in both columns gives

$$\begin{aligned} F(\mathbf{u}, \mathbf{v}) = u_1 v_1 F(\mathbf{e}_1, \mathbf{e}_1) + u_1 v_2 F(\mathbf{e}_1, \mathbf{e}_2) \\ + u_2 v_1 F(\mathbf{e}_2, \mathbf{e}_1) + u_2 v_2 F(\mathbf{e}_2, \mathbf{e}_2). \end{aligned}$$

The repeated-basis terms are zero, $F(\mathbf{e}_1, \mathbf{e}_2) = c$, and $F(\mathbf{e}_2, \mathbf{e}_1) = -c$. Therefore,

$$\begin{aligned} F(\mathbf{u}, \mathbf{v}) &= u_1 v_2 c - u_2 v_1 c \\ &= c(u_1 v_2 - u_2 v_1) \\ &= c \det[\mathbf{u}\ \mathbf{v}]. \end{aligned}$$

After c is known, this equation fixes $F(\mathbf{u}, \mathbf{v})$ for every pair $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$.

For the particular function $F(\mathbf{B}) = \det(\mathbf{AB})$,

$$\begin{aligned} c &= F(\mathbf{I}_n) \\ &= \det(\mathbf{A}\mathbf{I}_n) \\ &= \det(\mathbf{A}). \end{aligned}$$

Consequently,

$$\boxed{\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})}.$$

1.3.3.1 Why transposition does not change the determinant

The related identity

$$\boxed{\det(\mathbf{A}^\mathsf{T}) = \det(\mathbf{A})}$$

does not follow from multiplicativity alone. It follows directly from the permutation definition of the determinant.

Let $\mathbf{A} = [a_{ij}] \in \mathbb{R}^{n\times n}$. Because $(\mathbf{A}^\mathsf{T})_{ij} = a_{ji}$,

$$\begin{aligned}\det(\mathbf{A}^\mathsf{T}) &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^{n} (\mathbf{A}^\mathsf{T})_{\sigma(j),j} \\ &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^{n} a_{j,\sigma(j)}.\end{aligned}$$

A permutation $\sigma \in S_n$ is a bijection from the index set $\{1, \dots, n\}$ to itself. Because it is bijective, it has a unique **inverse permutation**, denoted by σ^{-1}, that reverses the mapping performed by σ. Define

$$\tau := \sigma^{-1}.$$

The defining identities are

$$\tau(\sigma(j)) = j, \qquad \sigma(\tau(k)) = k.$$

Equivalently,

$$k = \sigma(j) \quad\Longleftrightarrow\quad j = \tau(k).$$

For example, suppose $n = 3$ and

$$\sigma(1) = 2, \qquad \sigma(2) = 3, \qquad \sigma(3) = 1.$$

The inverse permutation reverses these assignments:

$$\tau(1) = 3, \qquad \tau(2) = 1, \qquad \tau(3) = 2.$$

For instance, $\sigma(1) = 2$ is reversed by $\tau(2) = 1$.

As σ runs through all permutations in S_n, $\tau = \sigma^{-1}$ also runs through all permutations in S_n. This is because inversion pairs every σ with exactly one τ, and the pairing is reversible:

$\sigma = \tau^{-1}$. Therefore, replacing the summation index σ by τ neither loses nor duplicates a permutation.

The inverse permutation also has the same sign as the original permutation. Suppose σ can be produced by r pairwise swaps. To undo σ, perform those same r swaps in reverse order. Thus, σ^{-1} uses the same number of swaps as σ. They therefore have the same even-or-odd parity, so

$$\operatorname{sgn}(\tau) = \operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma).$$

Now consider the product

$$\prod_{j=1}^{n} a_{j,\sigma(j)}.$$

Introduce the new index

$$k := \sigma(j).$$

Because σ is a permutation, as j visits $1, \ldots, n$, the index $k = \sigma(j)$ also visits every value $1, \ldots, n$ exactly once. From $\tau = \sigma^{-1}$, the relation $k = \sigma(j)$ implies $j = \tau(k)$. Each factor can therefore be rewritten as

$$a_{j,\sigma(j)} = a_{\tau(k),k}.$$

Consequently,

$$\prod_{j=1}^{n} a_{j,\sigma(j)} = \prod_{k=1}^{n} a_{\tau(k),k}.$$

The two products contain the same scalar factors, possibly in a different order. Scalar multiplication is commutative, so their order does not affect the product. In the three-dimensional example above,

$$\prod_{j=1}^{3} a_{j,\sigma(j)} = a_{1,2}a_{2,3}a_{3,1},$$
$$\prod_{k=1}^{3} a_{\tau(k),k} = a_{3,1}a_{1,2}a_{2,3}.$$

These products are equal because they contain exactly the same three factors.

Therefore,

$$\begin{aligned}\det(\mathbf{A}^\mathsf{T}) &= \sum_{\tau \in S_n} \operatorname{sgn}(\tau) \prod_{k=1}^{n} a_{\tau(k),k} \\ &= \det(\mathbf{A}).\end{aligned}$$

Thus, exchanging the rows and columns changes how the same determinant terms are indexed but does not change their values or signs.

1.3.3.2 Determinant of an inverse

For nonsingular $\mathbf{A}$, multiplicativity does give the inverse identity. Taking determinants of $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$ yields

$$\begin{aligned}1 &= \det(\mathbf{I}_n) \\ &= \det(\mathbf{A}^{-1}\mathbf{A}) \\ &= \det(\mathbf{A}^{-1})\det(\mathbf{A}).\end{aligned}$$

Since $\det(\mathbf{A}) \neq 0$,

$$\boxed{\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}}.$$

1.3.4 How an invertible linear map transforms a cross product

The cross product depends on lengths, angles, and the handedness of the coordinate system. A general linear map may stretch those lengths, change the angles, or reverse the handedness, so it does not usually pass through a cross product unchanged.

Let $\mathbf{A} \in \mathbb{R}^{3\times 3}$ be nonsingular, and let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ be expressed in the same right-handed orthonormal coordinates. The transformed cross product is

$$\boxed{(\mathbf{A}\mathbf{a}) \times (\mathbf{A}\mathbf{b}) = \det(\mathbf{A})\mathbf{A}^{-\mathsf{T}}(\mathbf{a} \times \mathbf{b})},$$

where $\mathbf{A}^{-\mathsf{T}} := (\mathbf{A}^{-1})^\mathsf{T} = (\mathbf{A}^\mathsf{T})^{-1}$ is the **inverse transpose** of $\mathbf{A}$. Nonsingularity is required because $\mathbf{A}^{-1}$ must exist.

To prove the identity, compare the dot product of both sides with an arbitrary test column $\mathbf{c} \in \mathbb{R}^3$. The scalar-triple-product formula from Section 1.1.6 gives

$$\begin{aligned}
\mathbf{c}^\mathsf{T}\left[(\mathbf{A}\mathbf{a}) \times (\mathbf{A}\mathbf{b})\right] &= \det\begin{bmatrix}\mathbf{c} & \mathbf{A}\mathbf{a} & \mathbf{A}\mathbf{b}\end{bmatrix} \\
&= \det\left(\mathbf{A}\begin{bmatrix}\mathbf{A}^{-1}\mathbf{c} & \mathbf{a} & \mathbf{b}\end{bmatrix}\right) \\
&= \det(\mathbf{A})\det\begin{bmatrix}\mathbf{A}^{-1}\mathbf{c} & \mathbf{a} & \mathbf{b}\end{bmatrix} \\
&= \det(\mathbf{A})(\mathbf{A}^{-1}\mathbf{c})^\mathsf{T}(\mathbf{a}\times\mathbf{b}) \\
&= \mathbf{c}^\mathsf{T}\left[\det(\mathbf{A})\mathbf{A}^{-\mathsf{T}}(\mathbf{a}\times\mathbf{b})\right].
\end{aligned}$$

The second equality uses $\mathbf{c} = \mathbf{A}(\mathbf{A}^{-1}\mathbf{c})$ to factor $\mathbf{A}$ from all three columns. The third uses determinant multiplicativity. The final equality uses $(\mathbf{A}^{-1}\mathbf{c})^\mathsf{T} = \mathbf{c}^\mathsf{T}\mathbf{A}^{-\mathsf{T}}$. Because both candidate vectors have the same dot product with every $\mathbf{c}$, they are equal.

For an orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{3\times3}$, $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_3$ implies $\mathbf{Q}^{-\mathsf{T}} = \mathbf{Q}$. The general identity then becomes

$$(\mathbf{Q}\mathbf{a}) \times (\mathbf{Q}\mathbf{b}) = \det(\mathbf{Q})\mathbf{Q}(\mathbf{a}\times\mathbf{b}).$$

If $\det(\mathbf{Q}) = 1$, then $\mathbf{Q}$ is a proper rotation and preserves the cross-product direction. If $\det(\mathbf{Q}) = -1$, then $\mathbf{Q}$ contains a reflection and reverses that direction. This determinant sign is why orthogonality alone is insufficient for a rotation to preserve cross products.

Quick test C: determinants

1. **Recall:** What does $|\det(\mathbf{A})|$ measure geometrically?
2. **Explanation:** What does the sign of a determinant record?
3. **Derivation:** Explain why repeated columns force the determinant to be zero.
4. **Application:** Compute $\det\begin{bmatrix}2 & 1\\ 3 & 4\end{bmatrix}$ and interpret its magnitude.
5. **Limitations:** Why is the determinant not defined in the same way for a rectangular matrix?
6. **Composition:** Explain geometrically why $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$.
7. **Reindexing:** If $\sigma(1) = 2$, $\sigma(2) = 3$, and $\sigma(3) = 1$, find $\tau = \sigma^{-1}$ and verify $\prod_{j=1}^{3} a_{j,\sigma(j)} = \prod_{k=1}^{3} a_{\tau(k),k}$.
8. **Cross products under linear maps:** State how a nonsingular $\mathbf{A} \in \mathbb{R}^{3\times3}$ transforms a cross product. Why does a proper rotation preserve the cross product while a reflection reverses it?

Answers and hints

1. The factor by which the associated linear map scales n-dimensional volume.

2. A positive determinant preserves the ordered right-handed orientation; a negative determinant reverses it to a left-handed orientation and therefore indicates a reflection component; a zero determinant collapses the volume, so no full-dimensional orientation remains.
3. Swapping identical columns leaves the matrix unchanged but must negate the determinant, so the determinant equals its own negative and is zero.
4. The determinant is $2(4) - 1(3) = 5$, so areas are scaled by a factor of 5 and orientation is preserved.
5. A rectangular map changes dimension, so it does not map an n-dimensional volume to another volume in the same n-dimensional space.
6. $\mathbf{B}$ scales signed volume first and $\mathbf{A}$ scales the resulting signed volume second, so the factors multiply.
7. The inverse is $\tau(1) = 3$, $\tau(2) = 1$, and $\tau(3) = 2$. The left product is $a_{1,2}a_{2,3}a_{3,1}$, while the reindexed product is $a_{3,1}a_{1,2}a_{2,3}$; these are equal because scalar factors commute.
8. $(\mathbf{Aa}) \times (\mathbf{Ab}) = \det(\mathbf{A})\mathbf{A}^{-\mathsf{T}}(\mathbf{a} \times \mathbf{b})$. For orthogonal $\mathbf{Q}$, $\mathbf{Q}^{-\mathsf{T}} = \mathbf{Q}$. A proper rotation has determinant $+1$ and therefore preserves the cross product, whereas an orthogonal reflection has determinant -1 and introduces a minus sign.

1.4 Orthogonal matrices

Let $\mathbf{Q} \in \mathbb{R}^{n\times n}$. The matrix $\mathbf{Q}$ is **orthogonal** if

$$\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_n.$$

The entries of $\mathbf{Q}^{\mathsf{T}}\mathbf{Q}$ are pairwise dot products between columns of $\mathbf{Q}$. Therefore, this equation is equivalent to saying that the columns form an orthonormal basis of $\mathbb{R}^n$.

Because $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_n$, the transpose satisfies the defining equation for the inverse:

$$\boxed{\mathbf{Q}^{-1} = \mathbf{Q}^{\mathsf{T}}}.$$

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$(\mathbf{Qx})^{\mathsf{T}}(\mathbf{Qy}) = \mathbf{x}^{\mathsf{T}}\mathbf{Q}^{\mathsf{T}}\mathbf{Q}\mathbf{y} = \mathbf{x}^{\mathsf{T}}\mathbf{y}.$$

Thus, orthogonal matrices preserve dot products, lengths, and angles.

Taking determinants of $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_n$ gives

$$\det(\mathbf{Q})^2 = 1,$$

so

$$\det(\mathbf{Q}) = +1 \qquad \text{or} \qquad \det(\mathbf{Q}) = -1.$$

The **orthogonal group** is

$$O(n) := \left\{\mathbf{Q} \in \mathbb{R}^{n\times n} : \mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_n\right\}.$$

The **special orthogonal group** is

$$SO(n) := \{\mathbf{Q} \in O(n) : \det(\mathbf{Q}) = 1\}.$$

Matrices in $SO(n)$ preserve lengths, angles, and orientation. Orthogonal matrices with determinant -1 preserve lengths and angles but reverse orientation through a reflection. Rotation matrices used for right-handed coordinate frames belong to $SO(2)$ or $SO(3)$.

Quick test D: orthogonal matrices

1. **Recall:** What equation defines an orthogonal matrix?
2. **Explanation:** Why is the inverse of an orthogonal matrix its transpose?
3. **Derivation:** Show that $\|\mathbf{Qx}\|_2 = \|\mathbf{x}\|_2$.
4. **Application:** Determine whether $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ belongs to $SO(2)$.
5. **Limitations:** Why is $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_n$ insufficient by itself to identify a proper rotation?

Answers and hints

1. $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_n$.
2. The transpose already satisfies the defining inverse equation.
3. Square both norms and use $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_n$.
4. It is orthogonal but has determinant -1, so it is a reflection in $O(2)$ and does not belong to $SO(2)$.
5. Orthogonality permits determinant -1 reflections as well as determinant $+1$ rotations.

Cumulative chapter test

1. Define a basis and explain why coordinate columns are unique in a basis.
2. Given $\mathbf{A} \in \mathbb{R}^{3\times 2}$ and $\mathbf{x} \in \mathbb{R}^2$, state the dimension of $\mathbf{Ax}$ and interpret it using the columns of $\mathbf{A}$.
3. Explain the relationship between $\mathcal{N}(\mathbf{A})$, independent columns, and full column rank.
4. Prove that a nonsingular square matrix cannot map a nonzero vector to zero.

5. Define the determinant using multilinearity, alternation, and normalisation.
6. Explain why $\det(\mathbf{A}) = 0$ indicates lost information.
7. Derive $\det(\mathbf{A}^{-1}) = 1/\det(\mathbf{A})$.
8. Distinguish $O(3)$ from $SO(3)$ and explain which set contains rigid-body rotation matrices.
9. State the rank-nullity theorem and explain what its two terms count.
10. State the vector triple-product identity and explain why the order and parentheses in a nested cross product matter.

Cumulative answers and hints

1. A basis is a linearly independent spanning set. Independence prevents multiple coefficient descriptions, while spanning guarantees that a description exists.

2. $\mathbf{Ax} \in \mathbb{R}^3$ and equals $x_1\mathbf{a}_1 + x_2\mathbf{a}_2$.

3. $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$ exactly when the columns are independent; for n columns this gives $\operatorname{rank}(\mathbf{A}) = n$.

4. Nonsingularity is equivalent to $\mathbf{Ax} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$; the contrapositive is $\mathbf{x} \neq \mathbf{0} \Rightarrow \mathbf{Ax} \neq \mathbf{0}$.

5. The determinant is the unique scalar function that is linear in each column, changes sign under a column swap, and equals 1 on $\mathbf{I}_n$.

6. For $\mathbf{A} \in \mathbb{R}^{n\times n}$, $\det(\mathbf{A}) = 0$ means that $\mathbf{A}$ is singular, so its null space contains a nonzero vector $\mathbf{z} \in \mathbb{R}^n$ satisfying $\mathbf{Az} = \mathbf{0}$. For any $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{A}(\mathbf{x} + \mathbf{z}) = \mathbf{Ax} + \mathbf{Az} = \mathbf{Ax}.$$

   Therefore, the distinct inputs $\mathbf{x}$ and $\mathbf{x} + \mathbf{z}$ produce the same output. The output cannot reveal the input component in the null-space direction $\mathbf{z}$, so the input cannot be uniquely recovered. This is the meaning of **lost information** here: an independent input direction or degree of freedom has disappeared, not that Shannon information has been calculated. Geometrically, the map collapses n-dimensional volume into a lower-dimensional set, which is why its volume-scaling factor is zero.

7. Take determinants of $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$ and use multiplicativity.

8. $O(3)$ contains all orthogonal 3×3 matrices with determinant ± 1; $SO(3)$ contains the determinant-1 members and therefore the proper rigid-body rotations.

9. For $\mathbf{A} \in \mathbb{R}^{m\times n}$, $\operatorname{rank}(\mathbf{A}) + \operatorname{nullity}(\mathbf{A}) = n$. Rank counts independent output directions produced by the map, while nullity counts independent input directions sent to zero.

10. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a}^\mathsf{T}\mathbf{c}) - \mathbf{c}(\mathbf{a}^\mathsf{T}\mathbf{b})$. The cross product is not associative, so changing the parentheses generally changes the result.

References

- Richard Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed., Appendix A, especially the discussion of orthonormal matrices, rank, singular values, and matrix factorisations.
- Kevin M. Lynch and Frank C. Park, *Modern Robotics*, Chapter 3, especially Section 3.2.1 on rotation matrices and the distinction between orthogonality and right-handed orientation.

2 Geometry and Coordinate Frames

Why coordinate frames matter

A robot represents positions, directions, velocities, measurements, and commands with numbers. Those numbers acquire physical meaning only when their coordinate frame and units are known. A controller that combines quantities expressed in incompatible frames can perform a mathematically valid calculation and produce a physically incorrect command.

Coordinate-frame methods allow us to:

- represent a robot's pose relative to a world frame;
- distinguish world-frame and body-frame quantities;
- transform sensor measurements into a shared frame;
- express target and tracking errors consistently;
- convert body-relative motion into world-relative motion; and
- detect invalid transformations before they reach a controller.

The central rule is simple: interpret every coordinate column together with the frame in which it is expressed.

Prerequisites and assumptions

Prerequisites

The reader should understand the material in Chapter 1, especially:

- geometric vectors and coordinate columns;
- dot products, norms, and orthogonality;
- matrices as linear maps and compositions;
- linear independence, rank, and inverses;
- determinants and signed orientation; and
- orthogonal matrices.

The reader should also be comfortable with sine, cosine, and angles measured in radians.

Assumptions

- Physical space is modelled as Euclidean space.
- Robot links and sensor mounts are treated as rigid bodies.
- Coordinate frames are right-handed and use orthonormal basis vectors.
- Length is measured in metres, time in seconds, and angles in radians.
- Coordinate vectors are written as columns.
- Numerical examples ignore measurement noise unless stated otherwise.

These idealisations isolate the geometry. A physical robot also introduces measurement noise, calibration error, structural compliance, and model mismatch.

Dependency map

```
Linear algebra foundations
        –
        v
Cross products and rigid-body velocity
        –
        v
Coordinate frames and units
        –
        v
Rotation matrices
        –
        v
Point transformations
        –
        v
Differential-drive motion and feedback control
```

2.1 Cross product and rigid-body velocity

Let $\mathbf{u}$ and $\mathbf{v}$ be geometric vectors with coordinate columns ${}^a\mathbf{u}, {}^a\mathbf{v} \in \mathbb{R}^3$ expressed in the same right-handed orthonormal frame $\{a\}$. Their cross product, expressed in frame $\{a\}$, is defined by

$$
{}^a\mathbf{u} \times {}^a\mathbf{v} = \begin{bmatrix} u_y v_z - u_z v_y \\ u_z v_x - u_x v_z \\ u_x v_y - u_y v_x \end{bmatrix} \in \mathbb{R}^3.
$$

Its direction follows the right-hand rule. If both vectors are nonzero, let $\theta \in [0, \pi]$ be the smaller angle between them. The cross-product magnitude is

$$\|{}^a\mathbf{u} \times {}^a\mathbf{v}\|_2 = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \sin\theta.$$

The cross product is perpendicular to both input vectors because

$${}^a\mathbf{u}^\mathsf{T} \left({}^a\mathbf{u} \times {}^a\mathbf{v}\right) = 0, \qquad {}^a\mathbf{v}^\mathsf{T} \left({}^a\mathbf{u} \times {}^a\mathbf{v}\right) = 0.$$

2.1.1 Example: velocity of a point on a rotating rigid body

Let O and P be two points fixed to the same rigid body. Point O is the chosen reference point; it need not be stationary, and it need not be the body's centre. The vector ${}^a\mathbf{r}_{OP}$ runs from O to P. Although this vector is fixed relative to the body, its coordinates generally change as the body rotates. Express every quantity in the same orthonormal frame $\{a\}$:

- ${}^a\boldsymbol{\omega}$ as the body's angular-velocity vector, measured in $\mathrm{rad\,s^{-1}}$;
- ${}^a\mathbf{r}_{OP}$ as the position of P relative to O, measured in m;
- ${}^a\mathbf{v}_O$ as the linear velocity of O, measured in $\mathrm{m\,s^{-1}}$; and
- ${}^a\mathbf{v}_P$ as the linear velocity of P, measured in $\mathrm{m\,s^{-1}}$.

At any instant, the motion of P can be understood as the sum of two simultaneous contributions:

1. **Translation with O.** If the body were not rotating, P would share the reference-point velocity ${}^a\mathbf{v}_O$.
2. **Motion relative to O caused by rotation.** Rotation adds a velocity whose direction and magnitude depend on ${}^a\boldsymbol{\omega}$ and ${}^a\mathbf{r}_{OP}$.

The angular-velocity vector points along the instantaneous rotation axis according to the right-hand rule, and its magnitude is the angular speed. Define the rotational contribution to the velocity of P by

$${}^a\mathbf{v}_{P/O} := {}^a\boldsymbol{\omega} \times {}^a\mathbf{r}_{OP}.$$

The cross product determines both the direction and the magnitude of this contribution. In particular,

$${}^a\boldsymbol{\omega}^\mathsf{T}\,{}^a\mathbf{v}_{P/O} = 0, \qquad {}^a\mathbf{r}_{OP}^\mathsf{T}\,{}^a\mathbf{v}_{P/O} = 0.$$

Therefore, ${}^a\mathbf{v}_{P/O}$ is perpendicular to the rotation axis and to ${}^a\mathbf{r}_{OP}$. It points tangent to the instantaneous circular path generated by the rotation. Adding the translational and rotational contributions gives the rigid-body velocity relation

$$\boxed{{}^a\mathbf{v}_P = {}^a\mathbf{v}_O + {}^a\boldsymbol{\omega} \times {}^a\mathbf{r}_{OP}}.$$

2.1.1.1 Velocity-vector construction

The construction in Figure 2.1 shows this addition for one instant:

- The green arrow ${}^a\mathbf{r}_{OP}$ locates P relative to O.
- The magenta arrow is the reference-point velocity ${}^a\mathbf{v}_O$. It is deliberately not parallel to ${}^a\mathbf{r}_{OP}$.
- The orange arrow is the rotational contribution ${}^a\boldsymbol{\omega} \times {}^a\mathbf{r}_{OP} = {}^a\mathbf{v}_{P/O}$. The right-angle marker shows that it is perpendicular to ${}^a\mathbf{r}_{OP}$.
- The dashed arrows are translated copies of the two velocity contributions. Moving a vector without rotating or resizing it does not change the vector, so the copies can be arranged head to tail.
- The blue arrow joins the beginning of the first contribution to the end of the second. It is therefore the resultant velocity ${}^a\mathbf{v}_P$.

In this planar view, ${}^a\boldsymbol{\omega}$ points along $+z_a$, out of the page. The right-hand rule then gives the displayed tangential direction for ${}^a\boldsymbol{\omega} \times {}^a\mathbf{r}_{OP}$. Notice that only the rotational contribution must be perpendicular to ${}^a\mathbf{r}_{OP}$. The total velocity ${}^a\mathbf{v}_P$ generally is not, because ${}^a\mathbf{v}_O$ may point in any direction.

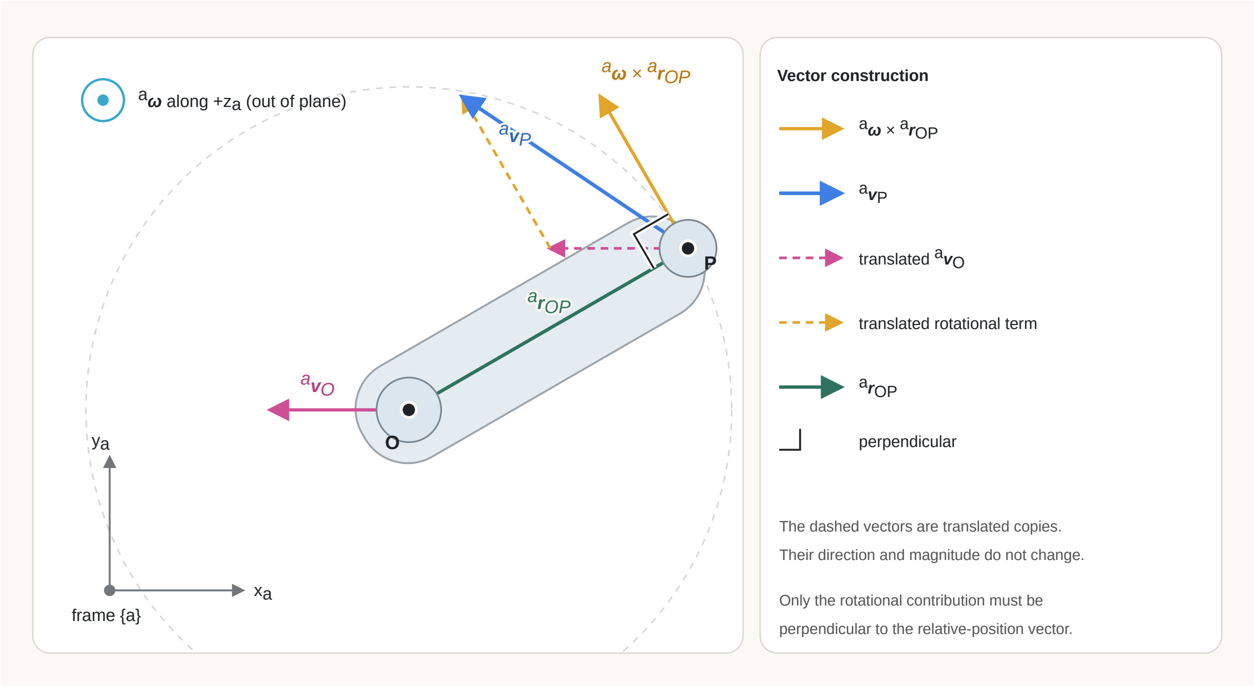

Figure 2.1: Velocity construction for two points on a rigid body. The rotational contribution is perpendicular to the position vector, while the reference-point velocity is independent of that perpendicular relationship.

For pure rotation about a fixed axis through O, the reference point is stationary, so ${}^{a}\mathbf{v}_O = \mathbf{0}$. Therefore,

$$\boxed{{}^{a}\mathbf{v}_P = {}^{a}\boldsymbol{\omega} \times {}^{a}\mathbf{r}_{OP}}.$$

If ϕ is the angle between ${}^{a}\boldsymbol{\omega}$ and ${}^{a}\mathbf{r}_{OP}$, then the relative speed caused by rotation is

$$\left\|{}^{a}\mathbf{v}_{P/O}\right\|_2 = \|{}^{a}\boldsymbol{\omega}\|_2 \, \|{}^{a}\mathbf{r}_{OP}\|_2 \sin\phi = \omega r_\perp,$$

where $r_\perp$ is the perpendicular distance from P to the rotation axis. A point on the axis has $r_\perp = 0$ and therefore has no relative linear velocity due to rotation. Under the pure-rotation assumption, ${}^{a}\mathbf{v}_P = {}^{a}\mathbf{v}_{P/O}$.

For planar counter-clockwise rotation about the positive z-axis,

$${}^{a}\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ \omega \end{bmatrix}, \qquad {}^{a}\mathbf{r}_{OP} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}.$$

The point velocity is

$${}^{a}\mathbf{v}_P = {}^{a}\boldsymbol{\omega} \times {}^{a}\mathbf{r}_{OP} = \begin{bmatrix} -\omega y \\ \omega x \\ 0 \end{bmatrix}.$$

For example, if $\omega = 2\,\mathrm{rad\,s^{-1}}$ and ${}^{a}\mathbf{r}_{OP} = [0.5 \;\; 0 \;\; 0]^\mathsf{T}\,\mathrm{m}$, then

$${}^{a}\mathbf{v}_P = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}.$$

The point lies on the positive x-axis, while its instantaneous velocity points along the positive y-axis, tangent to its circular path.

Quick test A: cross products and rigid-body velocity

Complete this test without looking back at the section.

1. **Recall:** What rule determines the direction of $\mathbf{u} \times \mathbf{v}$?
2. **Explanation:** Why is $\mathbf{u} \times \mathbf{v}$ perpendicular to both input vectors?
3. **Derivation:** Substitute the component formula for ${}^{a}\mathbf{u} \times {}^{a}\mathbf{v}$ into ${}^{a}\mathbf{u}^\mathsf{T}({}^{a}\mathbf{u} \times {}^{a}\mathbf{v})$ and show that the terms cancel.
4. **Application:** Compute $[1 \;\; 2 \;\; 0]^\mathsf{T} \times [0 \;\; 1 \;\; 0]^\mathsf{T}$.

5. **Limitations:** What happens to the cross product when the input vectors are parallel, and why does that prevent it from defining a unique normal direction?
6. **Transfer:** At one instant, a rigid body has ${}^a\boldsymbol{\omega} = [0\ \ 0\ \ 2]^\mathsf{T}\,\mathrm{rad\,s^{-1}}$, ${}^a\mathbf{r}_{OP} = [1\ \ 0\ \ 0]^\mathsf{T}\,\mathrm{m}$, and ${}^a\mathbf{v}_O = [0.5\ \ 0.5\ \ 0]^\mathsf{T}\,\mathrm{m\,s^{-1}}$. Find ${}^a\mathbf{v}_{P/O}$ and ${}^a\mathbf{v}_P$. Which velocity is perpendicular to ${}^a\mathbf{r}_{OP}$?

Answers and hints

1. The right-hand rule, applied in the order from $\mathbf{u}$ to $\mathbf{v}$.
2. Its dot product with either input is zero.
3. Expanding gives $u_x(u_yv_z - u_zv_y) + u_y(u_zv_x - u_xv_z) + u_z(u_xv_y - u_yv_x) = 0$ after pairwise cancellation.
4. The result is $[0\ \ 0\ \ 1]^\mathsf{T}$.
5. Parallel vectors have $\sin\theta = 0$, so their cross product is the zero vector. A zero vector has no direction and therefore cannot specify a unique normal.
6. ${}^a\mathbf{v}_{P/O} = [0\ \ 2\ \ 0]^\mathsf{T}\,\mathrm{m\,s^{-1}}$ and ${}^a\mathbf{v}_P = [0.5\ \ 2.5\ \ 0]^\mathsf{T}\,\mathrm{m\,s^{-1}}$. The rotational contribution is perpendicular to ${}^a\mathbf{r}_{OP}$; the total velocity is not.

2.2 Coordinate frames

2.2.1 What defines a frame?

A three-dimensional coordinate frame $\{a\}$ consists of:

1. an origin O_a, which is the physical point from which positions in frame $\{a\}$ are measured; and
2. three right-handed orthonormal basis vectors $\mathbf{e}_{x,a}$, $\mathbf{e}_{y,a}$, and $\mathbf{e}_{z,a}$, which define the positive x_a-, y_a-, and z_a-axis directions.

The subscript a associates the origin and basis vectors with frame $\{a\}$. Orthonormal means that each basis vector has unit length and that distinct basis vectors are perpendicular. Right-handed means that $\mathbf{e}_{x,a} \times \mathbf{e}_{y,a} = \mathbf{e}_{z,a}$.

A free vector depends only on the basis orientation used to describe it. A point also depends on the frame origin. This distinction explains why changing the coordinate frame of a free vector requires only a rotation, whereas changing the coordinates of a point generally requires a rotation and a translation.

2.2.2 Orientation convention

Let $\{a\}$ and $\{b\}$ be three-dimensional right-handed orthonormal frames. Define $\mathbf{R}_{ab} \in \mathbb{R}^{3\times 3}$ as the rotation matrix that maps coordinate columns from frame $\{b\}$ to frame $\{a\}$. Its columns are the basis vectors of $\{b\}$ expressed in frame $\{a\}$:

$$\mathbf{R}_{ab} = \begin{bmatrix} {}^{a}\mathbf{e}_{x,b} & {}^{a}\mathbf{e}_{y,b} & {}^{a}\mathbf{e}_{z,b} \end{bmatrix},$$

where each column ${}^{a}\mathbf{e}_{i,b} \in \mathbb{R}^3$ is an axis of frame $\{b\}$ written using the basis of frame $\{a\}$, for $i \in \{x, y, z\}$.

The following notation distinguishes a geometric vector from its coordinate representations:

Symbol	Meaning
$\mathbf{v}$	A geometric vector, independent of the coordinate frame used to describe it.
$\mathbf{e}_{x,b}$	The geometric unit vector along the positive x_b-axis of frame $\{b\}$.
$[\mathbf{v}]_b = {}^{b}\mathbf{v} \in \mathbb{R}^3$	The numerical coordinate column of $\mathbf{v}$ in frame $\{b\}$.
$[\mathbf{v}]_a = {}^{a}\mathbf{v} \in \mathbb{R}^3$	The numerical coordinate column of $\mathbf{v}$ in frame $\{a\}$.
$[\mathbf{e}_{x,b}]_a = {}^{a}\mathbf{e}_{x,b} \in \mathbb{R}^3$	The coordinate column of frame $\{b\}$'s geometric x-axis, expressed in frame $\{a\}$.

The bracket operator $[\,\cdot\,]_a$ means "take the coordinate column of the enclosed geometric vector in frame $\{a\}$." The geometric vector $\mathbf{v}$ can be expanded using either frame's basis:

$$\boxed{\mathbf{v} = v_x^{(b)}\mathbf{e}_{x,b} + v_y^{(b)}\mathbf{e}_{y,b} + v_z^{(b)}\mathbf{e}_{z,b} = v_x^{(a)}\mathbf{e}_{x,a} + v_y^{(a)}\mathbf{e}_{y,a} + v_z^{(a)}\mathbf{e}_{z,a}}.$$

Here, $v_i^{(b)} \in \mathbb{R}$ and $v_i^{(a)} \in \mathbb{R}$ are the scalar components of $\mathbf{v}$ along axis $i \in \{x, y, z\}$ of frames $\{b\}$ and $\{a\}$, respectively. The corresponding coordinate columns are

$${}^{b}\mathbf{v} = [\mathbf{v}]_b = \begin{bmatrix} v_x^{(b)} \\ v_y^{(b)} \\ v_z^{(b)} \end{bmatrix}, \qquad {}^{a}\mathbf{v} = [\mathbf{v}]_a = \begin{bmatrix} v_x^{(a)} \\ v_y^{(a)} \\ v_z^{(a)} \end{bmatrix}.$$

Apply the coordinate operator $[\,\cdot\,]_a$ to the expansion of $\mathbf{v}$ in the frame-$\{b\}$ basis. Because taking coordinates is a linear operation,

$$[\mathbf{v}]_a = v_x^{(b)}[\mathbf{e}_{x,b}]_a + v_y^{(b)}[\mathbf{e}_{y,b}]_a + v_z^{(b)}[\mathbf{e}_{z,b}]_a.$$

Using the equivalent superscript notation gives

$$ {}^{a}\mathbf{v} = v_x^{(b)}\,{}^{a}\mathbf{e}_{x,b} + v_y^{(b)}\,{}^{a}\mathbf{e}_{y,b} + v_z^{(b)}\,{}^{a}\mathbf{e}_{z,b}. $$

Writing this linear combination as matrix-vector multiplication gives

$$ {}^{a}\mathbf{v} = \underbrace{\begin{bmatrix} {}^{a}\mathbf{e}_{x,b} & {}^{a}\mathbf{e}_{y,b} & {}^{a}\mathbf{e}_{z,b} \end{bmatrix}}_{\mathbf{R}_{ab}} \underbrace{\begin{bmatrix} v_x^{(b)} \\ v_y^{(b)} \\ v_z^{(b)} \end{bmatrix}}_{{}^{b}\mathbf{v}}. $$

Therefore,

$$ \boxed{{}^{a}\mathbf{v} = \mathbf{R}_{ab}\,{}^{b}\mathbf{v}}. $$

The subscript order can be read as:

> $\mathbf{R}_{ab}$ maps coordinates **from frame** $\{b\}$ **to frame** $\{a\}$.

The linear algebra foundations chapter develops orthogonal matrices, determinants, and inverses in general. Applying those results to the orthonormal, right-handed columns of $\mathbf{R}_{ab}$ gives:

$$ \boxed{\mathbf{R}_{ab}^{\top}\mathbf{R}_{ab} = \mathbf{I}_3, \qquad \det(\mathbf{R}_{ab}) = 1, \qquad \mathbf{R}_{ab}^{-1} = \mathbf{R}_{ab}^{\top} = \mathbf{R}_{ba}}. $$

The following derivations explain each property.

2.2.2.1 Orthonormal columns

Let $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3 \in \mathbb{R}^3$ denote the columns of $\mathbf{R}_{ab}$:

$$ \mathbf{R}_{ab} = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix}, $$

where $\mathbf{r}_1 = {}^{a}\mathbf{e}_{x,b}$, $\mathbf{r}_2 = {}^{a}\mathbf{e}_{y,b}$, and $\mathbf{r}_3 = {}^{a}\mathbf{e}_{z,b}$. Multiplying the transpose by the original matrix forms every pairwise dot product between these columns:

$$ \mathbf{R}_{ab}^{\top}\mathbf{R}_{ab} = \begin{bmatrix} \mathbf{r}_1^{\top}\mathbf{r}_1 & \mathbf{r}_1^{\top}\mathbf{r}_2 & \mathbf{r}_1^{\top}\mathbf{r}_3 \\ \mathbf{r}_2^{\top}\mathbf{r}_1 & \mathbf{r}_2^{\top}\mathbf{r}_2 & \mathbf{r}_2^{\top}\mathbf{r}_3 \\ \mathbf{r}_3^{\top}\mathbf{r}_1 & \mathbf{r}_3^{\top}\mathbf{r}_2 & \mathbf{r}_3^{\top}\mathbf{r}_3 \end{bmatrix}. $$

Because the frame axes are orthonormal, every column has unit length,

$$\mathbf{r}_i^\mathsf{T}\mathbf{r}_i = 1,$$

and distinct columns are perpendicular,

$$\mathbf{r}_i^\mathsf{T}\mathbf{r}_j = 0 \qquad \text{for } i \neq j,$$

where $i, j \in \{1, 2, 3\}$ are column indices. Therefore,

$$\boxed{\mathbf{R}_{ab}^\mathsf{T}\mathbf{R}_{ab} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I}_3}.$$

This identity means that $\mathbf{R}_{ab}$ preserves dot products and therefore preserves vector lengths and angles.

2.2.2.2 Inverse and reverse coordinate mapping

The inverse $\mathbf{R}_{ab}^{-1} \in \mathbb{R}^{3\times3}$ is defined by

$$\mathbf{R}_{ab}^{-1}\mathbf{R}_{ab} = \mathbf{I}_3.$$

Because $\mathbf{R}_{ab}^\mathsf{T}\mathbf{R}_{ab} = \mathbf{I}_3$, the transpose satisfies the defining property of the inverse. Hence,

$$\boxed{\mathbf{R}_{ab}^{-1} = \mathbf{R}_{ab}^\mathsf{T}}.$$

The coordinate transformation from frame $\{b\}$ to frame $\{a\}$ is

$${}^a\mathbf{v} = \mathbf{R}_{ab}\,{}^b\mathbf{v}.$$

Multiplying by the inverse gives

$${}^b\mathbf{v} = \mathbf{R}_{ab}^{-1}\,{}^a\mathbf{v}.$$

Define $\mathbf{R}_{ba} \in \mathbb{R}^{3\times3}$ as the matrix that maps coordinates in the reverse direction, from frame $\{a\}$ to frame $\{b\}$. By definition,

$${}^b\mathbf{v} = \mathbf{R}_{ba}\,{}^a\mathbf{v}.$$

The two reverse mappings must be the same, so

$$\boxed{\mathbf{R}_{ba} = \mathbf{R}_{ab}^{-1} = \mathbf{R}_{ab}^\mathsf{T}}.$$

2.2.2.3 Determinant and orientation

The notation $\det(\mathbf{R})$ denotes the **determinant** of a square matrix $\mathbf{R}$. Geometrically, $|\det(\mathbf{R})|$ is the volume-scaling factor of the corresponding linear transformation. For an orthogonal matrix $\mathbf{R}$, taking determinants of $\mathbf{R}^\top\mathbf{R} = \mathbf{I}_3$ gives

$$\det(\mathbf{R}^\top\mathbf{R}) = \det(\mathbf{I}_3),$$

and therefore by $\det(\mathbf{R}^\top\mathbf{R}) = \det(\mathbf{R}^\top)\det(\mathbf{R})$ and $\det(R^\top) = \det(\mathbf{R})$ gives

$$\det(\mathbf{R})^2 = 1.$$

Thus, an orthogonal matrix can have determinant $+1$ or -1. Here, **orientation** means the handedness of the ordered coordinate axes. A right-handed frame satisfies

$$\mathbf{e}_x \times \mathbf{e}_y = \mathbf{e}_z.$$

The determinant sign has the following interpretation:

- $\det(\mathbf{R}) = +1$ preserves handedness, so a right-handed ordered basis remains right-handed.
- $\det(\mathbf{R}) = -1$ reverses handedness, so the transformation contains a reflection and produces a left-handed ordered basis from a right-handed one.

For example, consider

$$\mathbf{M} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The matrix $\mathbf{M} \in \mathbb{R}^{3\times 3}$ reflects the x-coordinate. Its columns remain orthonormal, so $\mathbf{M}^\top\mathbf{M} = \mathbf{I}_3$, but $\det(\mathbf{M}) = -1$. It reverses handedness and is therefore a reflection rather than a rigid-body rotation.

Because frames $\{a\}$ and $\{b\}$ are both right-handed, their rotation matrix must satisfy

$$\boxed{\mathbf{R}_{ab}^\top\mathbf{R}_{ab} = \mathbf{I}_3 \quad \text{and} \quad \det(\mathbf{R}_{ab}) = 1}.$$

The determinant sign does not indicate whether a rotation is clockwise or counter-clockwise. It indicates whether the transformation preserves or reverses the handedness of space.

2.2.3 Points require translation

Let O_a and O_b be the origins of frames $\{a\}$ and $\{b\}$. Define ${}^a\mathbf{p}_b \in \mathbb{R}^3$ as the position of O_b relative to O_a, expressed in frame $\{a\}$ and measured in metres.

For a physical point P, let ${}^b\mathbf{p}_P \in \mathbb{R}^3$ be the position of P relative to O_b, expressed in frame $\{b\}$, and let ${}^a\mathbf{p}_P \in \mathbb{R}^3$ be the position of P relative to O_a, expressed in frame $\{a\}$. Both position columns are measured in metres. Their relationship is

$$ {}^a\mathbf{p}_P = \mathbf{R}_{ab}\, {}^b\mathbf{p}_P + {}^a\mathbf{p}_b. $$

The rotation $\mathbf{R}_{ab}\, {}^b\mathbf{p}_P$ changes the basis used to express the displacement from O_b to P. The translation ${}^a\mathbf{p}_b$ then accounts for the displacement from O_a to O_b. A homogeneous transformation combines these two operations into one matrix equation.

2.3 Worked frame-change example

Consider two planar frames $\{a\}$ and $\{b\}$ whose axes differ by an angle $\theta \in \mathbb{R}$, measured in radians. A positive θ means that frame $\{b\}$ is rotated counter-clockwise relative to frame $\{a\}$. Define $\mathbf{R}_{ab}(\theta) \in \mathbb{R}^{2\times 2}$ as the rotation matrix that maps two-dimensional coordinate columns from frame $\{b\}$ to frame $\{a\}$:

$$ {}^a\mathbf{v} = \mathbf{R}_{ab}(\theta)\, {}^b\mathbf{v}, $$

where $\mathbf{v}$ is a geometric free vector and ${}^a\mathbf{v}, {}^b\mathbf{v} \in \mathbb{R}^2$ are its coordinate columns in frames $\{a\}$ and $\{b\}$.

Let $\mathbf{e}_{x,a}$ and $\mathbf{e}_{y,a}$ be the geometric unit axes of frame $\{a\}$, and let $\mathbf{e}_{x,b}$ and $\mathbf{e}_{y,b}$ be the geometric unit axes of frame $\{b\}$. By definition, the columns of $\mathbf{R}_{ab}$ are the axes of frame $\{b\}$ expressed in frame $\{a\}$:

$$ \mathbf{R}_{ab} = \begin{bmatrix} {}^a\mathbf{e}_{x,b} & {}^a\mathbf{e}_{y,b} \end{bmatrix}. $$

The x_b-axis makes angle θ with the x_a-axis. Its projection onto $\mathbf{e}_{x,a}$ is $\cos\theta$, and its projection onto $\mathbf{e}_{y,a}$ is $\sin\theta$. Therefore,

$$ {}^a\mathbf{e}_{x,b} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}. $$

Because the planar frames are right-handed and orthonormal, the y_b-axis is obtained by rotating the x_b-axis counter-clockwise by $\pi/2$. Its angle measured from the x_a-axis is therefore $\theta + \pi/2$, so

$$^{a}\mathbf{e}_{y,b} = \begin{bmatrix} \cos(\theta + \pi/2) \\ \sin(\theta + \pi/2) \end{bmatrix}.$$

Using

$$\cos(\theta + \pi/2) = -\sin\theta, \qquad \sin(\theta + \pi/2) = \cos\theta,$$

gives

$$^{a}\mathbf{e}_{y,b} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}.$$

Placing these two coordinate columns side by side produces

$$\boxed{\mathbf{R}_{ab}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}.$$

The column construction also explains why this matrix maps coordinates from frame $\{b\}$ to frame $\{a\}$. If

$$^{b}\mathbf{v} = \begin{bmatrix} v_x^{(b)} \\ v_y^{(b)} \end{bmatrix},$$

where $v_x^{(b)}, v_y^{(b)} \in \mathbb{R}$ are the components of $\mathbf{v}$ along the x_b- and y_b-axes, then

$$\begin{aligned} ^{a}\mathbf{v} &= v_x^{(b)}\, {}^{a}\mathbf{e}_{x,b} + v_y^{(b)}\, {}^{a}\mathbf{e}_{y,b} \\ &= \begin{bmatrix} ^{a}\mathbf{e}_{x,b} & ^{a}\mathbf{e}_{y,b} \end{bmatrix} \begin{bmatrix} v_x^{(b)} \\ v_y^{(b)} \end{bmatrix} \\ &= \mathbf{R}_{ab}(\theta)\, {}^{b}\mathbf{v}. \end{aligned}$$

For the worked example, $\theta = \pi/2\,\mathrm{rad}$. Substitution gives

$$\mathbf{R}_{ab}(\pi/2) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The first column $[0\ 1]^\mathsf{T}$ says that the positive x_b-axis points along the positive y_a-axis. The second column $[-1\ 0]^\mathsf{T}$ says that the positive y_b-axis points along the negative x_a-axis.

Let ${}^{b}\mathbf{v} \in \mathbb{R}^2$ be the coordinate column of a free vector $\mathbf{v}$, expressed in frame $\{b\}$ and measured in metres:

$$
{}^{b}\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \mathrm{m}.
$$

The coordinate column of the same geometric vector in frame $\{a\}$ is ${}^{a}\mathbf{v} \in \mathbb{R}^2$:

$$
{}^{a}\mathbf{v} = \mathbf{R}_{ab}(\theta)\, {}^{b}\mathbf{v} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \mathrm{m}.
$$

The coordinates changed, but the physical length did not:

$$
\left\| {}^{b}\mathbf{v} \right\|_2 = \sqrt{2^2 + 1^2} = \sqrt{5} = \sqrt{(-1)^2 + 2^2} = \left\| {}^{a}\mathbf{v} \right\|_2 .
$$

2.3.1 Interactive frame-rotation graph

Fix the frame-$\{b\}$ coordinate column

$$
{}^{b}\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.
$$

For each angle $\theta \in [-\pi, \pi]$, the corresponding frame-$\{a\}$ coordinate column is

$$
{}^{a}\mathbf{v}(\theta) = \mathbf{R}_{ab}(\theta)\, {}^{b}\mathbf{v} = \begin{bmatrix} 2\cos\theta - \sin\theta \\ 2\sin\theta + \cos\theta \end{bmatrix}.
$$

In the HTML book, move the angle slider to rotate frame $\{b\}$ relative to frame $\{a\}$. The graph keeps ${}^{b}\mathbf{v} = [2\ 1]^{\mathsf{T}}$ fixed, so the geometric vector rotates with frame $\{b\}$ and its coordinates ${}^{a}\mathbf{v}(\theta)$ update continuously.

The PDF shows the default slider state $\theta = \pi/2\,\mathrm{rad}$. Use the HTML book to vary θ interactively.

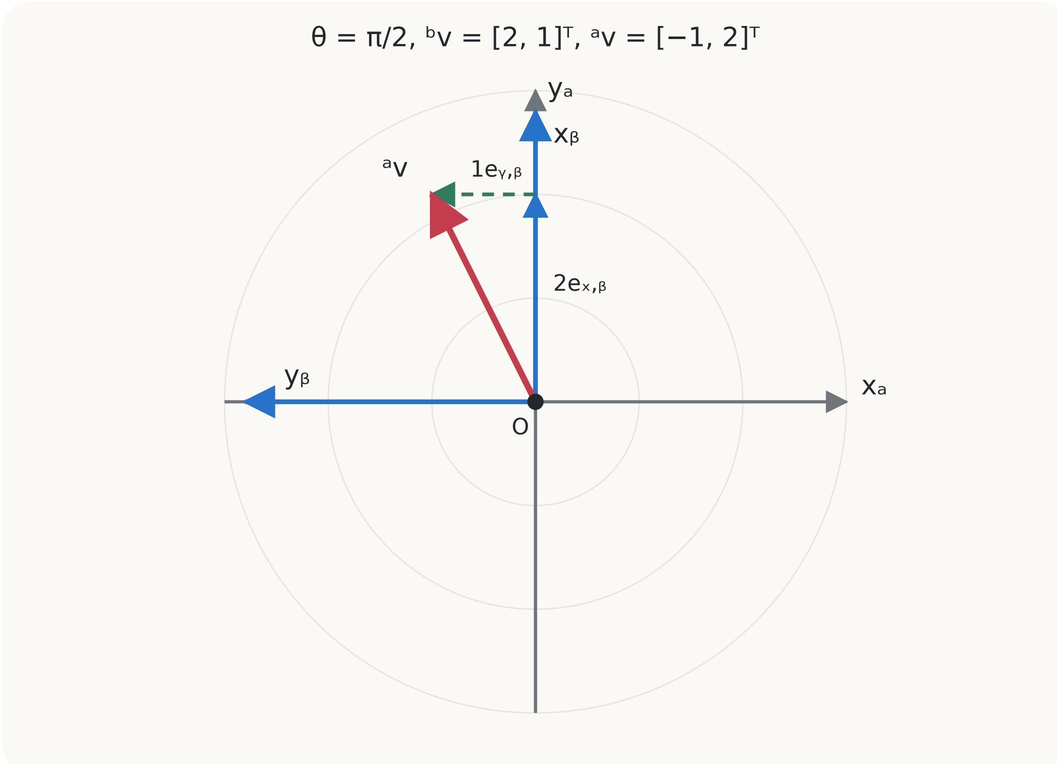

Figure 2.2: Static PDF view of the frame rotation at $\theta = \pi/2$ with ${}^b\mathbf{v} = [2\ 1]^\mathsf{T}$ and ${}^a\mathbf{v} = [-1\ 2]^\mathsf{T}$.

2.3.1.1 Equivalent NumPy calculation

The following NumPy snippet is the direct software translation of ${}^a\mathbf{v} = \mathbf{R}_{ab}(\theta){}^b\mathbf{v}$. The scalar `theta` stores θ in radians; `R_ab` is a NumPy array with shape `(2, 2)` representing $\mathbf{R}_{ab}(\theta) \in \mathbb{R}^{2\times 2}$; `v_b` is a NumPy array with shape `(2,)` representing the fixed coordinate column ${}^b\mathbf{v} = [2\ 1]^\mathsf{T}$; and `v_a` is a NumPy array with shape `(2,)` representing ${}^a\mathbf{v} \in \mathbb{R}^2$. The operator `@` performs matrix-vector multiplication.

```
import numpy as np

theta = np.pi / 2  # radians; replace this value with the selected angle

R_ab = np.array([
    [np.cos(theta), -np.sin(theta)],
    [np.sin(theta),  np.cos(theta)],
])

v_b = np.array([2.0, 1.0])
v_a = R_ab @ v_b
```

For $\theta = \pi/2\,\mathrm{rad}$, the calculation gives `v_a` approximately equal to `[-1.0, 2.0]`. Changing `theta` recomputes the same values displayed by the interactive graph while `v_b` remains fixed.

2.4 Differential-drive interpretation

For a planar mobile robot, let frame $\{w\}$ be fixed to the world and frame $\{b\}$ be attached to the robot. Let $\theta \in \mathbb{R}$ be the robot's heading angle, measured counter-clockwise in radians from the positive x_w-axis to the positive x_b-axis. Let $v \in \mathbb{R}$ be the signed forward speed, measured in $\mathrm{m\,s^{-1}}$; positive v means motion along the positive x_b-axis.

The body-frame velocity ${}^{b}\mathbf{v} \in \mathbb{R}^2$ is

$$
{}^{b}\mathbf{v} = \begin{bmatrix} v \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}.
$$

Define $\mathbf{R}_{wb}(\theta) \in \mathbb{R}^{2\times 2}$ as the rotation matrix that maps planar coordinates from the robot frame $\{b\}$ to the world frame $\{w\}$. The world-frame velocity ${}^{w}\mathbf{v} \in \mathbb{R}^2$ is therefore

$$
{}^{w}\mathbf{v} = \mathbf{R}_{wb}(\theta)\,{}^{b}\mathbf{v} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} v \\ 0 \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \end{bmatrix} \mathrm{m\,s^{-1}}.
$$

This equation is the coordinate-frame foundation of the differential-drive kinematic model. It shows why a controller must distinguish a forward command in the robot frame from motion observed in the world frame.

2.5 Failure modes and engineering checks

2.5.1 Missing frame labels

Two identical columns can describe different physical directions when they are expressed in different frames. Interfaces should name the frame explicitly.

2.5.2 Reversed transformation

Using $\mathbf{R}_{ba}$ where $\mathbf{R}_{ab}$ is required rotates coordinates in the opposite direction. Check the subscript rule before multiplying.

2.5.3 Mixed units

Degrees passed to a function expecting radians produce valid-looking but wrong numbers. Convert at the system boundary and use radians internally.

2.5.4 Point-vector confusion

A point changes under translation; a free vector does not. Applying a translation to velocity or direction data is a modelling error.

2.5.5 Invalid numerical rotation

Let $\mathbf{R} \in \mathbb{R}^{n\times n}$ be a numerically stored matrix intended to represent a rotation, and let $\mathbf{I}_n \in \mathbb{R}^{n\times n}$ be the identity matrix. Rounding, integration drift, or corrupted data can make $\mathbf{R}$ non-orthonormal.

For a matrix $\mathbf{M} \in \mathbb{R}^{n\times n}$ with entries m_{ij}, define the Frobenius norm by

$$\|\mathbf{M}\|_F := \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} m_{ij}^2}.$$

Let $\varepsilon_{\mathrm{orth}} > 0$ and $\varepsilon_{\mathrm{det}} > 0$ be declared numerical tolerances. A robust interface checks

$$\left\|\mathbf{R}^\mathsf{T}\mathbf{R} - \mathbf{I}_n\right\|_F \le \varepsilon_{\mathrm{orth}}$$

and

$$|\det(\mathbf{R}) - 1| \le \varepsilon_{\mathrm{det}}.$$

The first condition tests approximate orthonormality; the second tests approximate preservation of right-handed orientation.

Quick test B: coordinate frames and rotations

1. **Recall:** What do the columns of $\mathbf{R}_{ab}$ represent?
2. **Explanation:** Explain in words why $\mathbf{R}_{ab}^{-1} = \mathbf{R}_{ba}$.
3. **Derivation:** Starting with ${}^a\mathbf{v} = \mathbf{R}_{ab}{}^b\mathbf{v}$, solve for ${}^b\mathbf{v}$.
4. **Application:** A robot has heading $\theta = \pi/2$ and forward speed $v = 0.5\,\mathrm{m\,s^{-1}}$. Find its world-frame velocity.

5. **Limitations:** Give two ways a numerically stored matrix intended to be a rotation matrix can become invalid.
6. **Derivation:** Let $\{a\}$ and $\{b\}$ be right-handed orthonormal planar frames. Frame $\{b\}$ is rotated counter-clockwise by $\theta\,\mathrm{rad}$ relative to frame $\{a\}$, and $\mathbf{R}_{ab}(\theta) \in \mathbb{R}^{2\times 2}$ maps coordinate columns from frame $\{b\}$ to frame $\{a\}$. First express the x_b-axis and y_b-axis as coordinate columns ${}^{a}\mathbf{e}_{x,b}, {}^{a}\mathbf{e}_{y,b} \in \mathbb{R}^2$; then place those columns side by side to derive $\mathbf{R}_{ab}(\theta)$.

Answers and hints

1. The axes of frame $\{b\}$ expressed in coordinates of frame $\{a\}$.

2. Reversing the frame change undoes the original change; for an orthonormal rotation matrix, the inverse is its transpose.

3. Premultiply by $\mathbf{R}_{ab}^{-1} = \mathbf{R}_{ba}$: ${}^{b}\mathbf{v} = \mathbf{R}_{ba}\,{}^{a}\mathbf{v}$.

4. $[0 \;\; 0.5]^{\mathsf{T}}\,\mathrm{m\,s^{-1}}$, up to floating-point rounding.

5. Examples include accumulated integration error, rounded entries, malformed input, unit mistakes in angle construction, and data corruption.

6. Since the x_b-axis makes angle θ with the x_a-axis, its frame-$\{a\}$ coordinate column is ${}^{a}\mathbf{e}_{x,b} = [\cos\theta \;\; \sin\theta]^{\mathsf{T}}$. Right-handed orthonormality places the y_b-axis at angle $\theta + \pi/2$, so ${}^{a}\mathbf{e}_{y,b} = [\cos(\theta+\pi/2) \;\; \sin(\theta+\pi/2)]^{\mathsf{T}} = [-\sin\theta \;\; \cos\theta]^{\mathsf{T}}$. Because these two coordinate columns are the columns of $\mathbf{R}_{ab}(\theta)$,

$$\mathbf{R}_{ab}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

Cumulative chapter test

Complete this test without looking back at the chapter. Show the intermediate steps in every calculation, preserve the frame labels, and state the physical units of every dimensional result.

1. **Recall and meaning:** Define a three-dimensional coordinate frame $\{a\}$. Then state what the columns of $\mathbf{R}_{ab} \in \mathbb{R}^{3\times 3}$ represent and the direction in which this matrix maps coordinate columns.

2. **Derivation, application, and explanation:** Frame $\{b\}$ is rotated counter-clockwise by $\theta = \pi/2\,\mathrm{rad}$ relative to frame $\{a\}$. Derive $\mathbf{R}_{ab}(\pi/2)$ by expressing the three axes of frame $\{b\}$ in frame $\{a\}$. Then let

$$
{}^{b}\mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \text{ m}.
$$

Compute ${}^{a}\mathbf{v}$. Which physical property of the geometric free vector is unchanged, and why?

3. **Point–vector contrast:** Continue from Question 2: use the same frames, with frame $\{b\}$ rotated counter-clockwise by $\theta = \pi/2\,\text{rad}$ relative to frame $\{a\}$, and use the rotation matrix $\mathbf{R}_{ab}(\pi/2)$ derived there. Now let the position of O_b relative to O_a, expressed in frame $\{a\}$, be

$$
{}^{a}\mathbf{p}_{b} = \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} \text{ m},
$$

and let a physical point P have ${}^{b}\mathbf{p}_{P} = [2\ 1\ 0]^{\mathsf{T}}$ m. Compute ${}^{a}\mathbf{p}_{P}$. Then compute the frame-$\{a\}$ coordinates of a free displacement vector whose frame-$\{b\}$ coordinates are also $[2\ 1\ 0]^{\mathsf{T}}$ m. Explain why the two calculations differ.

4. **Derivation:** Starting from $\mathbf{R}_{ab}^{\mathsf{T}}\mathbf{R}_{ab} = \mathbf{I}_3$, derive $\mathbf{R}_{ba} = \mathbf{R}_{ab}^{\mathsf{T}}$. Then prove that changing the coordinate representation of a free vector preserves its Euclidean norm. Finally, show that applying the same point transformation to two physical points P and Q preserves the distance between them.

5. **Judgement:** Consider

$$
\mathbf{M} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.
$$

Determine whether $\mathbf{M}$ is a valid rotation matrix. State which numerical rotation checks it passes or fails and explain the physical meaning of the failure.

6. **Rigid-body application:** At one instant, all of the following coordinate columns are expressed in the same frame $\{a\}$:

$$
{}^{a}\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \text{ rad s}^{-1}, \qquad {}^{a}\mathbf{r}_{OP} = \begin{bmatrix} 0.5 \\ -0.25 \\ 0 \end{bmatrix} \text{ m}, \qquad {}^{a}\mathbf{v}_{O} = \begin{bmatrix} 0.2 \\ -0.1 \\ 0 \end{bmatrix} \text{ m s}^{-1}.
$$

Compute ${}^{a}\mathbf{v}_{P/O}$ and ${}^{a}\mathbf{v}_{P}$. Verify by a dot product which velocity contribution must be perpendicular to ${}^{a}\mathbf{r}_{OP}$, and state why the same conclusion need not hold for the total velocity.

7. **Transfer to differential drive:** A planar robot has heading $\theta = -\pi/2\,\mathrm{rad}$ and signed forward speed $v = 0.6\,\mathrm{m\,s^{-1}}$. Compute its world-frame velocity ${}^{w}\mathbf{v}$. Then use the reverse transformation to recover the original body-frame velocity.

8. **Failure diagnosis:** A software interface documents $\mathbf{R}_{ab}$ as mapping coordinate columns from frame $\{b\}$ to frame $\{a\}$, but its implementation computes `R_ab.transpose() * v_b` and labels the result as frame $\{a\}$. Explain the semantic error. Give one simple basis-axis test that would expose it.

9. **Contrast two situations:** Explain the difference between these statements: (a) one unchanged geometric vector is represented by ${}^{b}\mathbf{v}$ and ${}^{a}\mathbf{v} = \mathbf{R}_{ab}{}^{b}\mathbf{v}$; and (b) the coordinate column ${}^{b}\mathbf{v}$ is held fixed while frame $\{b\}$ physically rotates relative to frame $\{a\}$. What changes physically in each situation?

10. **Limitations and engineering checks:** Name at least four declarations or validation checks required at a coordinate-transformation interface. Then name one real-world source of error that valid frame algebra alone cannot remove.

Cumulative answers and hints

1. A frame $\{a\}$ consists of an origin O_a and three right-handed orthonormal basis vectors $\mathbf{e}_{x,a}$, $\mathbf{e}_{y,a}$, and $\mathbf{e}_{z,a}$. The columns of $\mathbf{R}_{ab}$ are the axes of frame $\{b\}$ expressed in frame $\{a\}$, and $\mathbf{R}_{ab}$ maps coordinate columns from frame $\{b\}$ to frame $\{a\}$.

2. At $\theta = \pi/2$, the frame-$\{b\}$ axes expressed in frame $\{a\}$ are

$$ {}^{a}\mathbf{e}_{x,b} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad {}^{a}\mathbf{e}_{y,b} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \qquad {}^{a}\mathbf{e}_{z,b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. $$

   Placing these axes in the columns gives

$$ \mathbf{R}_{ab}(\pi/2) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}. $$

   The coordinate change is

$$ {}^{a}\mathbf{v} = \mathbf{R}_{ab}{}^{b}\mathbf{v} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \mathrm{m}. $$

   The geometric vector and its length are unchanged; only its coordinate representation changes. Its norm remains $\sqrt{5}\,\mathrm{m}$ because a rotation matrix is orthogonal.

3. A point requires both rotation and translation:

$$
{}^{a}\mathbf{p}_P = \mathbf{R}_{ab}{}^{b}\mathbf{p}_P + {}^{a}\mathbf{p}_b = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \text{ m.}
$$

   A free displacement vector is unaffected by the origin separation, so

$$
{}^{a}\mathbf{v} = \mathbf{R}_{ab}{}^{b}\mathbf{v} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \text{ m.}
$$

   A point is located relative to a frame origin, whereas a free vector has direction and magnitude but no fixed origin.

4. Since $\mathbf{R}_{ab}^{\mathsf{T}}\mathbf{R}_{ab} = \mathbf{I}_3$, the transpose satisfies the defining equation for the inverse, so $\mathbf{R}_{ab}^{-1} = \mathbf{R}_{ab}^{\mathsf{T}}$. The reverse coordinate map is the inverse of the forward map, hence $\mathbf{R}_{ba} = \mathbf{R}_{ab}^{-1} = \mathbf{R}_{ab}^{\mathsf{T}}$. For ${}^{a}\mathbf{v} = \mathbf{R}_{ab}{}^{b}\mathbf{v}$,

$$
\begin{aligned}
\|{}^{a}\mathbf{v}\|_2^2 &= ({}^{a}\mathbf{v})^{\mathsf{T}}{}^{a}\mathbf{v} \\
&= ({}^{b}\mathbf{v})^{\mathsf{T}}\mathbf{R}_{ab}^{\mathsf{T}}\mathbf{R}_{ab}{}^{b}\mathbf{v} \\
&= ({}^{b}\mathbf{v})^{\mathsf{T}}{}^{b}\mathbf{v} \\
&= \left\|{}^{b}\mathbf{v}\right\|_2^2 .
\end{aligned}
$$

   Norms are nonnegative, so equality of the squared norms implies equality of the norms. For two points transformed using the same $\mathbf{R}_{ab}$ and ${}^{a}\mathbf{p}_b$, subtracting their frame-$\{a\}$ coordinates cancels the translation:

$$
\begin{aligned}
{}^{a}\mathbf{p}_P - {}^{a}\mathbf{p}_Q &= \left(\mathbf{R}_{ab}{}^{b}\mathbf{p}_P + {}^{a}\mathbf{p}_b\right) - \left(\mathbf{R}_{ab}{}^{b}\mathbf{p}_Q + {}^{a}\mathbf{p}_b\right) \\
&= \mathbf{R}_{ab}\left({}^{b}\mathbf{p}_P - {}^{b}\mathbf{p}_Q\right) .
\end{aligned}
$$

   This difference is a free displacement vector, so the norm-preservation result gives

$$
\|{}^{a}\mathbf{p}_P - {}^{a}\mathbf{p}_Q\|_2 = \left\|{}^{b}\mathbf{p}_P - {}^{b}\mathbf{p}_Q\right\|_2 .
$$

5. The matrix passes the orthonormality check because $\mathbf{M}^{\mathsf{T}}\mathbf{M} = \mathbf{I}_3$, but it fails the orientation check because $\det(\mathbf{M}) = -1$. It is a reflection across the yz-plane, not a right-handed rigid-body rotation. This is why an orthonormality check alone is insufficient.

6. The rotational contribution is

$$ {}^{a}\mathbf{v}_{P/O} = {}^{a}\boldsymbol{\omega} \times {}^{a}\mathbf{r}_{OP} = \begin{bmatrix} 0.5 \\ 1.0 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}. $$

Therefore,

$$ {}^{a}\mathbf{v}_{P} = {}^{a}\mathbf{v}_{O} + {}^{a}\mathbf{v}_{P/O} = \begin{bmatrix} 0.7 \\ 0.9 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}. $$

The required orthogonality check is

$$ ({}^{a}\mathbf{r}_{OP})^{\mathsf{T}}\,{}^{a}\mathbf{v}_{P/O} = 0.5(0.5) - 0.25(1.0) = 0. $$

The total velocity also contains the independent translation ${}^{a}\mathbf{v}_{O}$, so it need not be perpendicular to ${}^{a}\mathbf{r}_{OP}$; here their dot product is $0.125\,\mathrm{m^2\,s^{-1}}$.

7. Since

$$ \mathbf{R}_{wb}(-\pi/2) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, $$

the world-frame velocity is

$$ {}^{w}\mathbf{v} = \mathbf{R}_{wb}\,{}^{b}\mathbf{v} = \begin{bmatrix} 0 \\ -0.6 \end{bmatrix} \mathrm{m\,s^{-1}}. $$

The reverse map is $\mathbf{R}_{bw} = \mathbf{R}_{wb}^{\mathsf{T}}$, and

$$ {}^{b}\mathbf{v} = \mathbf{R}_{bw}\,{}^{w}\mathbf{v} = \begin{bmatrix} 0.6 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}. $$

8. The transpose is $\mathbf{R}_{ab}^{\mathsf{T}} = \mathbf{R}_{ba}$, whose declared input is a frame-$\{a\}$ coordinate column and whose output is a frame-$\{b\}$ coordinate column. Applying it to ${}^{b}\mathbf{v}$ violates the frame contract, even though the matrix dimensions agree, and the result cannot validly be labelled ${}^{a}\mathbf{v}$. A basis-axis test can provide ${}^{b}\mathbf{e}_{x,b} = [1\ 0\ 0]^{\mathsf{T}}$ and require the output to equal the first column of $\mathbf{R}_{ab}$; the transposed implementation generally fails this test.

9. In situation (a), the geometric vector is unchanged and only its numerical description changes with the coordinate basis. In situation (b), the components relative to frame $\{b\}$ stay fixed while the basis vectors of $\{b\}$ turn relative to frame $\{a\}$, so the geometric vector constructed from those fixed components rotates physically with frame $\{b\}$. The same matrix entries can participate in both calculations, but the physical operation and the meanings of the input and output are different.

10. Suitable declarations and checks include source and destination frame identifiers, point-versus-free-vector semantics, SI units and angle units, matrix and vector dimensions, finite inputs, the intended multiplication direction, approximate orthonormality, determinant near $+1$, and declared numerical tolerances. Valid algebra cannot remove effects such as sensor noise, calibration error, structural compliance, timing error, or model mismatch.

References

- Kevin M. Lynch and Frank C. Park, *Modern Robotics*, Chapter 3, especially Sections 3.1 and 3.2.1 on free vectors, frames, rotation matrices, and rigid-body motions.
- Steven M. LaValle, *Planning Algorithms*, Chapters 3 and 4 on geometric representations, transformations, and configuration spaces.

3 Differential-Drive Kinematics and Numerical Integration: Model and Derivation

Why this chapter matters

Kinematics describes motion without asking which forces or torques produce it. For a differential-drive robot, kinematics connects wheel angular speeds to the time evolution of position and heading. Numerical integration converts this continuous-time model into sampled updates used by simulation, odometry, planning, and control.

This chapter develops that chain in four steps: define planar pose and its derivative, derive the no-sideways-motion model, derive body velocity from wheel speeds, and compare exact constant-input motion with Forward Euler integration.

Prerequisites and assumptions

The reader should understand vectors and matrices from Chapter 1, coordinate frames and rotations from Chapter 2, and elementary derivatives, integrals, sine, and cosine.

The system considered here is a differential-drive ground robot: a rigid chassis propelled by left and right drive wheels that are mounted on a common axle and can be driven independently. The robot operates on a stationary, level, rigid floor, and the model describes its motion as viewed from above. Any additional support hardware does not affect the ideal kinematic model.

Frame $\{w\}$ is fixed to the floor and is called the world frame. Frame $\{b\}$ is rigidly attached to the robot and is called the body frame. Its origin O_b lies midway between the two drive-wheel ground-contact centre lines. The positive x_b-axis points forward, the positive y_b-axis points left, and the positive z_b-axis points upward, so frame $\{b\}$ is right-handed. As the robot moves, frame $\{b\}$ translates and rotates relative to the stationary frame $\{w\}$.

Both drive wheels have the same effective rolling radius $r > 0$, measured in metres. The track width $L > 0$, measured in metres, is the lateral distance between the left and right wheel ground-contact centre lines.

The kinematic model uses the following assumptions:

- The chassis remains rigid, and its pose changes only through horizontal position and rotation about the upward z_b-axis; its height, roll, and pitch remain unchanged.
- Each wheel maintains contact with the floor and rolls without longitudinal slip, so the distance travelled in the wheel's rolling direction equals the arc length swept out at its effective radius.
- The scalar body-frame lateral component ${}^{b}v_y$, also written v_y when the frame is clear, is measured in $\mathrm{m\,s^{-1}}$ and lies along the $+y_b$-direction. The wheel constraints impose ${}^{b}v_y = 0$ at O_b. The robot can still turn because its forward direction changes continuously.
- The effective wheel radius r and track width L remain constant.
- Wheel deformation, tyre slip, backlash, encoder error, actuator dynamics, collisions, and uneven or moving terrain are ignored.
- Length is measured in metres, time in seconds, and angle in radians; corresponding velocities are measured in $\mathrm{m\,s^{-1}}$ and $\mathrm{rad\,s^{-1}}$.

Together, these assumptions define an ideal planar kinematic model rather than a complete physical model. A real robot violates them to some degree, especially during rapid acceleration, braking, collision, wheel slip, or motion over compliant or uneven terrain.

Figure Figure 3.1 summarizes the frames, wheel dimensions, rolling condition, and lateral-velocity constraint used by the model.

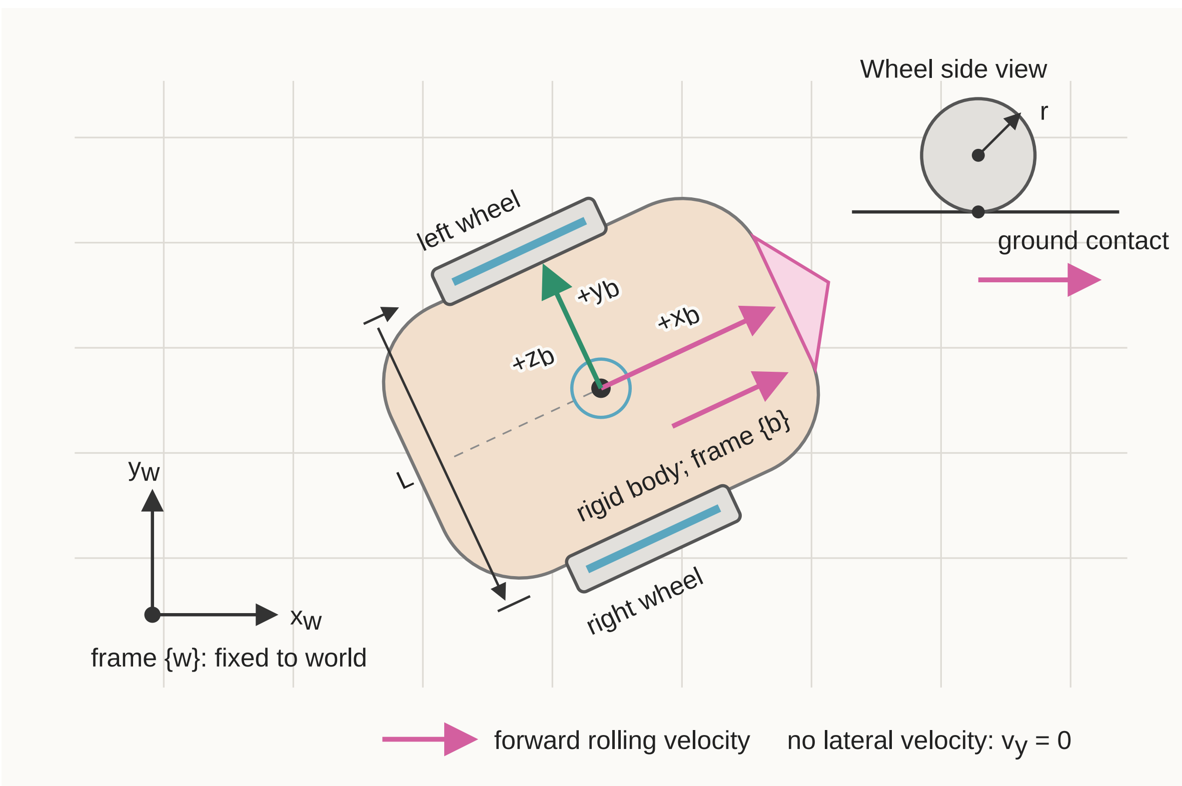

Figure 3.1: Differential-drive robot geometry and ideal rolling constraints.

As a preview of the pure-rolling relation, $v_i = r\dot{\phi}_i$ for $i \in \{L, R\}$, where v_i is the signed longitudinal ground speed of wheel i and $\dot{\phi}_i$ is its signed angular speed. Section 3.2.1 derives how the individual wheel speeds determine the body-forward speed v and yaw rate ω.

3.1 Planar pose and its derivative

Let $t \in \mathbb{R}_{\geq 0}$ denote continuous time, measured in seconds. Define the planar robot pose by

$$\mathbf{q}(t) := \begin{bmatrix} x(t) \\ y(t) \\ \theta(t) \end{bmatrix}.$$

Here, $x(t), y(t) \in \mathbb{R}$ are the world-frame coordinates of body-frame origin O_b, measured in metres, and $\theta(t) \in \mathbb{R}$ is the counter-clockwise heading of frame $\{b\}$ relative to frame $\{w\}$, measured in radians. Angles that differ by an integer multiple of 2π represent the same physical heading.

The pose derivative is

$$\dot{\mathbf{q}}(t) := \frac{d\mathbf{q}(t)}{dt} = \begin{bmatrix} \dot{x}(t) \\ \dot{y}(t) \\ \dot{\theta}(t) \end{bmatrix}.$$

The components $\dot{x}$ and $\dot{y}$ are measured in $\mathrm{m\,s^{-1}}$, while $\dot{\theta}$ is measured in $\mathrm{rad\,s^{-1}}$. Although they appear in one column, their units are not identical.

Define the body-velocity input by

$$\mathbf{u}(t) := \begin{bmatrix} v(t) \\ \omega(t) \end{bmatrix},$$

where $v(t) \in \mathbb{R}$ is signed forward speed along $+x_b$, measured in $\mathrm{m\,s^{-1}}$, and $\omega(t) \in \mathbb{R}$ is signed yaw rate about $+z_b$, measured in $\mathrm{rad\,s^{-1}}$. Positive ω produces counter-clockwise rotation when viewed from above.

3.1.1 Deriving the planar model

Because the ideal robot has no body-frame lateral velocity, its translational velocity expressed in frame $\{b\}$ is

$${}^{b}\mathbf{v} = \begin{bmatrix} v \\ 0 \end{bmatrix} \in \mathbb{R}^2.$$

The rotation matrix $\mathbf{R}_{wb}(\theta) \in \mathbb{R}^{2\times 2}$ maps coordinates from frame $\{b\}$ to frame $\{w\}$. Therefore,

$$\begin{bmatrix}\dot{x}\\ \dot{y}\end{bmatrix} = \underbrace{\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}}_{\mathbf{R}_{wb}(\theta)} \begin{bmatrix}v\\ 0\end{bmatrix} = \begin{bmatrix}v\cos\theta\\ v\sin\theta\end{bmatrix}.$$

Together with $\dot{\theta} = \omega$, this gives

$$\boxed{\dot{\mathbf{q}} = \mathbf{f}(\mathbf{q}, \mathbf{u}) = \begin{bmatrix}v\cos\theta\\ v\sin\theta\\ \omega\end{bmatrix} = \begin{bmatrix}\cos\theta & 0\\ \sin\theta & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}v\\ \omega\end{bmatrix}}.$$

Here, $\mathbf{f}$ is the kinematic state-derivative function: it maps current pose $\mathbf{q}$ and input $\mathbf{u}$ to instantaneous pose derivative $\dot{\mathbf{q}}$. The position rates depend on θ because the body forward direction changes relative to the world frame.

3.1.2 The no-sideways-motion constraint

Define the axle-midpoint linear velocity expressed in the world frame by

$${}^{w}\mathbf{v} := \begin{bmatrix}\dot{x}\\ \dot{y}\end{bmatrix} \in \mathbb{R}^2,$$

where $\dot{x}$ and $\dot{y}$ are its components along $+x_w$ and $+y_w$, respectively, and are measured in $\mathrm{m\,s^{-1}}$.

The geometric unit vector along the body-frame $+y_b$-axis, expressed using world-frame coordinates, is the second column of $\mathbf{R}_{wb}(\theta)$:

$${}^{w}\mathbf{e}_{y,b} = \begin{bmatrix}-\sin\theta\\ \cos\theta\end{bmatrix} \in \mathbb{R}^2.$$

Both ${}^{w}\mathbf{e}_{y,b}$ and ${}^{w}\mathbf{v}$ are expressed in frame $\{w\}$, and ${}^{w}\mathbf{e}_{y,b}$ has unit length. Their dot product therefore gives the signed component of the velocity along the robot's lateral $+y_b$-direction:

$$\begin{aligned} {}^{b}v_y &= ({}^{w}\mathbf{e}_{y,b})^{\top}\,{}^{w}\mathbf{v} \\ &= \begin{bmatrix}-\sin\theta & \cos\theta\end{bmatrix}\begin{bmatrix}\dot{x}\\ \dot{y}\end{bmatrix} \\ &= -\dot{x}\sin\theta + \dot{y}\cos\theta. \end{aligned}$$

The left superscript in ${}^{b}v_y$ makes explicit that this scalar is the y-component of velocity measured along a body-frame axis, even though the projection was calculated using world-frame coordinates.

The matrix $\mathbf{R}_{wb}(\theta)$ maps coordinate columns from frame $\{b\}$ to frame $\{w\}$. Its columns are orthonormal, so its inverse equals its transpose. Define the reverse coordinate map by

$$\begin{aligned}\mathbf{R}_{bw}(\theta) &:= \mathbf{R}_{wb}(\theta)^{-1} \\ &= \mathbf{R}_{wb}(\theta)^{\mathsf{T}} \\ &= \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}.\end{aligned}$$

Therefore, $\mathbf{R}_{bw}(\theta)$ maps coordinate columns from frame $\{w\}$ to frame $\{b\}$. The notation uses the same heading θ because both matrices depend on the robot's heading, but the reverse map applies the opposite rotation: $\mathbf{R}_{bw}(\theta) = \mathbf{R}_{wb}(-\theta)$.

The complete change from world-frame velocity components to body-frame velocity components is

$${}^{b}\mathbf{v} = \begin{bmatrix} {}^{b}v_x \\ {}^{b}v_y \end{bmatrix} = \mathbf{R}_{bw}(\theta)\,{}^{w}\mathbf{v} = \mathbf{R}_{wb}(\theta)^{\mathsf{T}}\,{}^{w}\mathbf{v} = \begin{bmatrix} \dot{x}\cos\theta + \dot{y}\sin\theta \\ -\dot{x}\sin\theta + \dot{y}\cos\theta \end{bmatrix}.$$

Here, ${}^{b}v_x$ and ${}^{b}v_y$ are the signed forward and lateral velocity components along $+x_b$ and $+y_b$, respectively, and both are measured in $\mathrm{m\,s^{-1}}$. Thus, $\dot{x}$ and $\dot{y}$ are world-frame components, whereas ${}^{b}v_x$ and ${}^{b}v_y$ are body-frame components.

The ideal no-sideways-motion assumption is therefore written most clearly as

$$\boxed{{}^{b}v_y = -\dot{x}\sin\theta + \dot{y}\cos\theta = 0}.$$

The signed forward speed v defined earlier equals ${}^{b}v_x$, so the constrained body-frame velocity and its world-frame representation are

$${}^{b}\mathbf{v} = \begin{bmatrix} v \\ 0 \end{bmatrix}, \qquad {}^{w}\mathbf{v} = \mathbf{R}_{wb}(\theta)\begin{bmatrix} v \\ 0 \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \end{bmatrix}.$$

Consequently, zero body-frame lateral velocity does not imply $\dot{y} = 0$. For example, when $\theta = \pi/4$, forward motion gives $\dot{x} = \dot{y} = v/\sqrt{2}$. The angle θ specifies orientation; it does not itself create translation. If $v = 0$ and $\omega \neq 0$, the robot rotates in place about O_b, so $\dot{x} = \dot{y} = 0$ while θ changes.

This equation is a nonholonomic velocity constraint. It restricts the directions in which the robot can move at the current pose, but it does not require the pose variables x, y, and θ to remain on a fixed lower-dimensional surface. In contrast, a holonomic condition such as $y = 0$ would confine every allowable pose to the world x_w-axis.

The distinction concerns instantaneous motion versus accumulated motion. At every instant, the axle midpoint translates only along the robot's current x_b-axis, but turning changes the direction of x_b relative to the world. For example, a robot initially aligned with $+x_w$ can rotate by $\pi/2$, drive forward a distance $d > 0$, and rotate back by $-\pi/2$. Its final heading equals its initial heading, yet it has acquired a world-frame $+y_w$ displacement of d. It never moved along its instantaneous y_b-axis; the net sideways displacement resulted from a sequence of allowed rotations and forward motions.

Quick test A: planar kinematics

1. Define every component of $\mathbf{q} = [x\ y\ \theta]^\mathsf{T}$, including its frame and units.
2. Why do $\dot{x}$ and $\dot{y}$ depend on θ, while v does not?
3. Starting with ${}^b\mathbf{v} = [v\ 0]^\mathsf{T}$, derive $\dot{y} = v \sin\theta$.
4. If $v = 1\,\mathrm{m\,s^{-1}}$ and $\theta = \pi/2\,\mathrm{rad}$, find $\dot{x}$ and $\dot{y}$.
5. Does the no-sideways-motion constraint prevent a net sideways displacement? Explain.

Answers and hints

1. x and y are the world-frame position coordinates of O_b in metres; θ is counter-clockwise body heading relative to the world frame in radians.
2. v is defined along the moving x_b-axis. Its world-frame components depend on the orientation of that axis.
3. The second row of $\mathbf{R}_{wb}(\theta)[v\ 0]^\mathsf{T}$ is $v \sin\theta$.
4. $\dot{x} = 0\,\mathrm{m\,s^{-1}}$ and $\dot{y} = 1\,\mathrm{m\,s^{-1}}$.
5. No. The constraint forbids instantaneous sideways velocity, but combined forward and turning motions can produce a sideways displacement.

3.2 Differential-drive wheel kinematics

Let $\phi_L(t), \phi_R(t) \in \mathbb{R}$ be the rotational positions of the left and right wheels about their axles at time t, measured in radians. The subscripts L and R identify the left and right wheels. These are rolling angles, not steering angles: the wheel planes remain fixed relative to the chassis.

Choose $\phi_L = \phi_R = 0$ at an arbitrary reference position. A change of 2π radians is one complete wheel revolution. Although the marked orientation of a wheel repeats after each revolution, ϕ_L and ϕ_R are treated as unwrapped accumulated angles in $\mathbb{R}$, so they may increase beyond 2π or decrease below zero.

Their time derivatives $\dot{\phi}_L(t)$ and $\dot{\phi}_R(t)$ are the left and right wheel angular speeds, measured in $\mathrm{rad\,s^{-1}}$. Positive angular speed is defined to roll the corresponding wheel and the robot forward.

For a wheel of effective radius r, let $\Delta\phi$ be a signed angular change and let Δs be the corresponding signed arc displacement swept out at the wheel rim. Both r and Δs are measured in metres. By the definition of radian measure,

$$\Delta\phi = \frac{\Delta s}{r}, \qquad \frac{\mathrm{m}}{\mathrm{m}} = 1.$$

The metre units cancel, so a radian is dimensionless in the International System of Units (SI). Nevertheless, angles are labelled in rad and angular speeds in $\mathrm{rad\,s^{-1}}$ so their physical meanings remain clear. Rearranging the radian definition gives the swept arc displacement $\Delta s = r\Delta\phi$. Under rolling without slip, this swept arc displacement equals the signed longitudinal displacement along the ground.

For wheel $i \in \{L, R\}$, let $s_i(t)$ be its signed accumulated longitudinal ground displacement. Over a time interval $\Delta t > 0$, pure rolling gives $\Delta s_i = r\Delta\phi_i$. Because r is constant, dividing by Δt and taking the limit $\Delta t \to 0$ gives

$$v_i := \dot{s}_i = \lim_{\Delta t\to 0} \frac{\Delta s_i}{\Delta t} = r \lim_{\Delta t\to 0} \frac{\Delta \phi_i}{\Delta t} = r\dot{\phi}_i.$$

Applying this relation to the left and right wheels gives

$$v_L = r\dot{\phi}_L, \qquad v_R = r\dot{\phi}_R,$$

where $v_L, v_R \in \mathbb{R}$ are measured in $\mathrm{m\,s^{-1}}$. The unit calculation for either wheel is

$$\begin{aligned} \text{metres} \times \frac{\text{radians}}{\text{second}} &= \mathrm{m} \cdot \frac{\mathrm{rad}}{\mathrm{s}} \\ &= \mathrm{m} \cdot \frac{1}{\mathrm{s}} \\ &= \mathrm{m\,s^{-1}}. \end{aligned}$$

The symbol rad is retained to communicate that ϕ is an angle, even though it contributes no independent physical dimension in the unit calculation.

3.2.1 Relating wheel speeds to body motion

The rigid-body point-velocity relation derived in the geometry and coordinate-frames chapter is

$$^{b}\mathbf{v}_P = {}^{b}\mathbf{v}_O + {}^{b}\boldsymbol{\omega} \times {}^{b}\mathbf{r}_{OP},$$

provided that every vector is expressed in the same orthonormal frame, here the body frame $\{b\}$. The vector $^{b}\mathbf{v}_P$ is the velocity of a rigid-body point P, $^{b}\mathbf{v}_O$ is the velocity of a reference point O, $^{b}\mathbf{r}_{OP}$ is the position of P relative to O, and $^{b}\boldsymbol{\omega}$ is the rigid body angular-velocity vector.

For the differential-drive chassis, choose the reference point as the axle midpoint $O = O_b$. Let C_L and C_R denote the left and right wheel centres on the axle. Both points are fixed relative to the chassis. Under ideal rolling without longitudinal slip, the $+x_b$ components of their translational velocities equal the longitudinal wheel speeds v_L and v_R defined above.

Because $+y_b$ points left, the wheel-centre position vectors are

$$ {}^{b}\mathbf{r}_{O_bC_L} = \begin{bmatrix} 0 \\ L/2 \\ 0 \end{bmatrix}, \qquad {}^{b}\mathbf{r}_{O_bC_R} = \begin{bmatrix} 0 \\ -L/2 \\ 0 \end{bmatrix}, \qquad {}^{b}\mathbf{r}_{O_bC_L}, {}^{b}\mathbf{r}_{O_bC_R} \in \mathbb{R}^3. $$

Their components have units of metres. The axle midpoint moves forward with scalar speed v, and the chassis yaws about $+z_b$ with scalar rate ω, so

$$ {}^{b}\mathbf{v}_{O_b} = \begin{bmatrix} v \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^3, \qquad {}^{b}\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ \omega \end{bmatrix} \in \mathbb{R}^3. $$

The components of ${}^{b}\mathbf{v}_{O_b}$ have units of $\mathrm{m\,s^{-1}}$, and the components of ${}^{b}\boldsymbol{\omega}$ have units of $\mathrm{rad\,s^{-1}}$.

Applying the rigid-body relation to each wheel-centre point gives

$$ {}^{b}\mathbf{v}_{C_L} = {}^{b}\mathbf{v}_{O_b} + {}^{b}\boldsymbol{\omega} \times {}^{b}\mathbf{r}_{O_bC_L}, \qquad {}^{b}\mathbf{v}_{C_R} = {}^{b}\mathbf{v}_{O_b} + {}^{b}\boldsymbol{\omega} \times {}^{b}\mathbf{r}_{O_bC_R}. $$

For the left wheel, the rotational contribution is

$$ {}^{b}\boldsymbol{\omega} \times {}^{b}\mathbf{r}_{O_bC_L} = \begin{bmatrix} 0 \\ 0 \\ \omega \end{bmatrix} \times \begin{bmatrix} 0 \\ L/2 \\ 0 \end{bmatrix} = \begin{bmatrix} -L\omega/2 \\ 0 \\ 0 \end{bmatrix}. $$

For the right wheel, it is

$$ {}^{b}\boldsymbol{\omega} \times {}^{b}\mathbf{r}_{O_bC_R} = \begin{bmatrix} 0 \\ 0 \\ \omega \end{bmatrix} \times \begin{bmatrix} 0 \\ -L/2 \\ 0 \end{bmatrix} = \begin{bmatrix} L\omega/2 \\ 0 \\ 0 \end{bmatrix}. $$

Therefore, the wheel-centre velocity vectors are

$$ {}^{b}\mathbf{v}_{C_L} = \begin{bmatrix} v - L\omega/2 \\ 0 \\ 0 \end{bmatrix}, \qquad {}^{b}\mathbf{v}_{C_R} = \begin{bmatrix} v + L\omega/2 \\ 0 \\ 0 \end{bmatrix}. $$

The longitudinal wheel speeds v_L and v_R are the $+x_b$ components of these vectors. Consequently,

$$\boxed{v_L = v - \frac{L}{2}\omega, \qquad v_R = v + \frac{L}{2}\omega}.$$

The signs follow from the right-hand-rule identities $\mathbf{e}_{z,b} \times \mathbf{e}_{y,b} = -\mathbf{e}_{x,b}$ and $\mathbf{e}_{z,b} \times (-\mathbf{e}_{y,b}) = \mathbf{e}_{x,b}$. For a positive yaw rate, the robot turns left: rotation reduces the left-wheel forward speed and increases the right-wheel forward speed. The product $L\omega$ has units $\mathrm{m\,s^{-1}}$ because radians are dimensionless in dimensional calculations.

Adding the two equations isolates forward speed, while subtracting the left equation from the right equation isolates yaw rate:

$$v = \frac{v_R + v_L}{2}, \qquad \omega = \frac{v_R - v_L}{L}.$$

Substituting the pure-rolling relations produces

$$\boxed{v = \frac{r}{2}\left(\dot{\phi}_R + \dot{\phi}_L\right), \qquad \omega = \frac{r}{L}\left(\dot{\phi}_R - \dot{\phi}_L\right)}.$$

In matrix form,

$$\boxed{\begin{bmatrix} v \\ \omega \end{bmatrix} = \underbrace{\begin{bmatrix} r/2 & r/2 \\ -r/L & r/L \end{bmatrix}}_{\mathbf{G} \in \mathbb{R}^{2\times 2}} \begin{bmatrix} \dot{\phi}_L \\ \dot{\phi}_R \end{bmatrix}}.$$

The matrix $\mathbf{G}$ maps the wheel-speed column $[\dot{\phi}_L\ \dot{\phi}_R]^\top$ to the body-velocity column $[v\ \omega]^\top$. In this chapter, this input-output direction is called the **wheel-to-body velocity mapping**: its inputs are the two wheel angular speeds, and its outputs are the resulting body-forward speed and yaw rate. It is also called the **forward differential-drive velocity kinematics**. Here, *velocity kinematics* means an algebraic relationship among instantaneous velocities without integrating the pose or modelling the forces that produce the motion. The word *forward* describes the calculation from wheel motion to body motion; it does not mean that the robot must travel in the positive forward direction, because the signed wheel speeds and v may be negative.

The two rows of $\mathbf{G}$ have different physical units because the two outputs have different units.

Because $r > 0$ and $L > 0$, the mapping can be inverted. Solving for the wheel speeds gives

$$\boxed{\dot{\phi}_L = \frac{v}{r} - \frac{L\omega}{2r}, \qquad \dot{\phi}_R = \frac{v}{r} + \frac{L\omega}{2r}}.$$

This opposite input-output direction is called the **body-to-wheel velocity mapping**, or the **inverse differential-drive velocity kinematics**. Its inputs are a desired body-forward speed v and yaw rate ω, and its outputs are the ideal left and right wheel angular speeds

$\dot{\phi}_L$ and $\dot{\phi}_R$ required by the no-slip model. The word *inverse* means that this calculation reverses the wheel-to-body mapping; it does not mean that the robot is commanded to drive backwards. In a controller, the body-to-wheel mapping converts a desired chassis motion into wheel-speed commands, whereas the wheel-to-body mapping can convert measured wheel speeds into an estimate of chassis motion.

3.2.2 Characteristic motions

- If $\dot{\phi}_L = \dot{\phi}_R$, then $\omega = 0$ and the robot moves straight.
- If $\dot{\phi}_L = -\dot{\phi}_R$, then $v = 0$ and the robot rotates about O_b.
- If one wheel is stationary, the ideal robot follows a circle about that wheel contact point.
- If the wheel speeds are unequal and not opposite, the robot follows an arc about an instantaneous centre; the signed turning radius is derived next.

3.2.2.1 Instantaneous centre of curvature and signed turning radius

Assume $\omega \neq 0$ at time t. At that instant, the planar rigid-body motion can be regarded as rotation about a point C in the plane whose instantaneous linear velocity is zero. This point is called the instantaneous centre of curvature (ICC). The body origin O_b therefore moves instantaneously along a circular arc centred at C.

The no-sideways-motion constraint makes the velocity of O_b tangent to the arc and parallel to the body x_b-axis. A circle radius is perpendicular to its tangent, so the line joining O_b and C lies along the lateral y_b-axis.

Let ${}^w\mathbf{p}_{O_b}, {}^w\mathbf{p}_C \in \mathbb{R}^2$ be the world-frame position columns of O_b and C, respectively. Define the ordinary nonnegative radius ρ, measured in metres, by

$$\rho := \mathrm{distance}(O_b, C) = \|{}^w\mathbf{p}_C - {}^w\mathbf{p}_{O_b}\|_2 \geq 0.$$

The magnitude ρ records only the distance between the two points. Define the signed turning radius $R \in \mathbb{R}$ by using its sign to record which side of the robot contains the ICC:

$$R := \begin{cases} +\rho, & C \text{ is to the left of } O_b, \\ -\rho, & C \text{ is to the right of } O_b. \end{cases}$$

Consequently, $|R| = \rho$. The relative-position vector from O_b to C, expressed in world-frame coordinates, is

$${}^w\mathbf{r}_{O_bC} := {}^w\mathbf{p}_C - {}^w\mathbf{p}_{O_b} = R\,{}^w\mathbf{e}_{y,b}, \qquad {}^w\mathbf{e}_{y,b} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix} \in \mathbb{R}^2.$$

The unit vector ${}^{w}\mathbf{e}_{y,b}$ points along the robot lateral $+y_b$-axis and is expressed in world coordinates. Multiplying it by $R > 0$ places C to the left of O_b, while multiplying it by $R < 0$ reverses the direction and places C to the right. Figure Figure 3.2 shows both cases. The forward velocity is tangent to the circular path and therefore perpendicular to the radius direction.

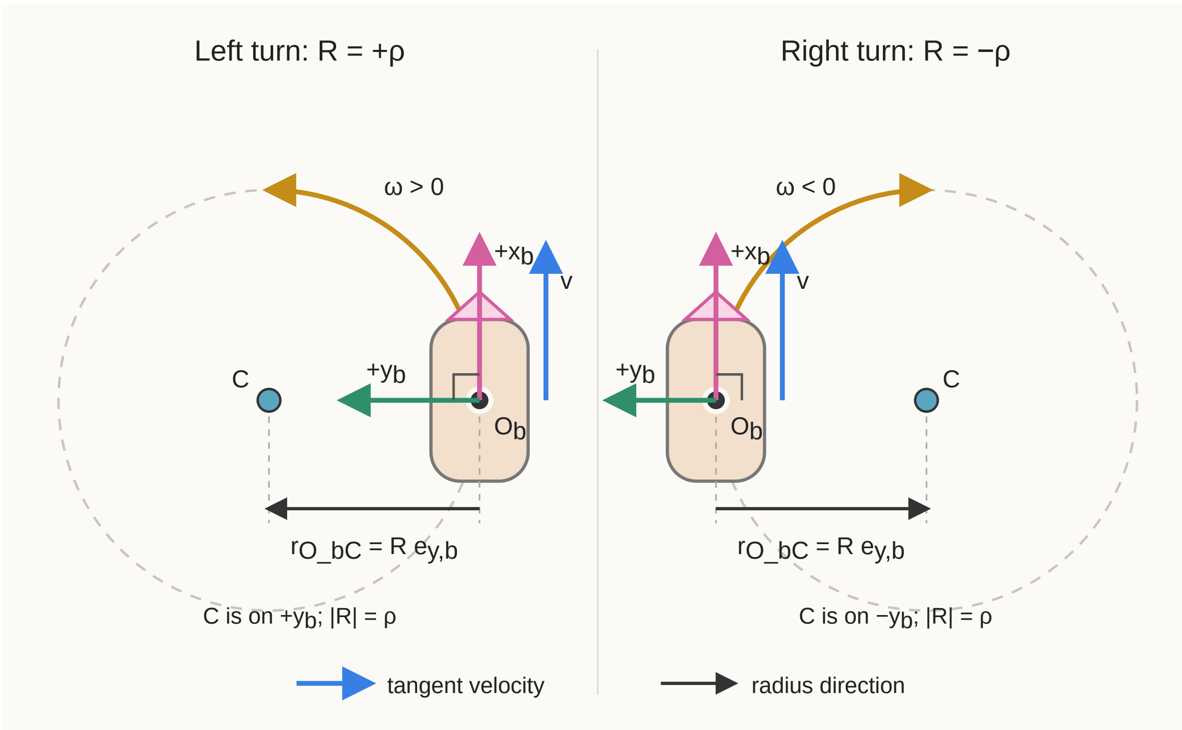

Figure 3.2: Relationship among the body origin, body axes, instantaneous centre of curvature, tangent velocity, and signed turning radius.

To derive the relationship among R, v, and ω, consider a short interval $\Delta t > 0$ during which the instantaneous values are treated as constant. Define the signed heading change by

$$\Delta\theta := \theta(t + \Delta t) - \theta(t),$$

where $\Delta\theta$ is measured in radians. Let $\Delta\ell \in \mathbb{R}$ be the signed distance travelled by O_b along the arc, measured in metres; $\Delta\ell > 0$ denotes travel in the $+x_b$-direction and $\Delta\ell < 0$ denotes travel in the $-x_b$-direction.

The radian definition gives arc length equal to radius times angular change. With the signed quantities just defined, it becomes

$$\boxed{\Delta\ell = R\,\Delta\theta}.$$

For a forward left turn, both R and $\Delta\theta$ are positive. For a forward right turn, both are negative. Their product is therefore the positive forward distance $\Delta\ell$ in either case; the same signed equation also remains valid for backward motion.

Divide the signed arc equation by Δt:

$$\frac{\Delta \ell}{\Delta t} = R\frac{\Delta \theta}{\Delta t}.$$

Taking the limit as $\Delta t \to 0$ gives

$$\frac{d\ell}{dt} = R\frac{d\theta}{dt}.$$

By definition, the signed forward speed is $v = d\ell/dt$, measured in $\mathrm{m\,s^{-1}}$, and the signed yaw rate is $\omega = d\theta/dt$, measured in $\mathrm{rad\,s^{-1}}$. Therefore,

$$\boxed{v = R\omega}.$$

When $\omega \neq 0$, division by ω yields

$$\boxed{R = \frac{v}{\omega}}, \qquad \omega \neq 0.$$

The quotient has units of metres because radians are dimensionless in dimensional calculations. If $\omega = 0$ and $v \neq 0$, the robot moves straight and has no finite ICC; its turning radius is informally described as infinite. If $v = 0$ and $\omega \neq 0$, then $R = 0$ and the robot rotates in place about O_b.

The symbol R in this subsection is a scalar signed turning radius, not a rotation matrix.

Quick test B: wheel-to-body mapping

1. Define ϕ_L, ϕ_R, $\dot{\phi}_L$, $\dot{\phi}_R$, r, and L, including their units, and distinguish wheel angle from angular speed.
2. Why is a radian dimensionless in an SI unit calculation, and why does $r\dot{\phi}$ have units of metres per second?
3. Starting from ${}^b\mathbf{v}_P = {}^b\mathbf{v}_O + {}^b\boldsymbol{\omega} \times {}^b\mathbf{r}_{OP}$, use the two wheel-centre position vectors to derive $v_L = v - L\omega/2$ and $v_R = v + L\omega/2$.
4. Why does equal wheel speed produce $\omega = 0$?
5. Derive $\omega = r(\dot{\phi}_R - \dot{\phi}_L)/L$.
6. Let $r = 0.10\,\mathrm{m}$, $L = 0.50\,\mathrm{m}$, $\dot{\phi}_L = 2\,\mathrm{rad\,s^{-1}}$, and $\dot{\phi}_R = 4\,\mathrm{rad\,s^{-1}}$. Find v and ω.
7. Define the ICC C, ordinary radius ρ, signed radius R, signed distance $\Delta\ell$, and heading change $\Delta\theta$. Starting from $\Delta\ell = R\Delta\theta$, derive $R = v/\omega$ and explain the signs for left and right turns.
8. Name two physical effects that can make predicted and real motion disagree.

Answers and hints

1. ϕ_L and ϕ_R are unwrapped left and right wheel rotational positions in radians; $\dot{\phi}_L$ and $\dot{\phi}_R$ are their angular speeds in $\mathrm{rad\,s^{-1}}$; r is effective wheel radius in metres; and L is track width in metres.
2. Radian measure is an arc-length-to-radius ratio, so its metre units cancel. Therefore, $r\dot{\phi}$ has units $\mathrm{m \cdot rad\,s^{-1}} = \mathrm{m\,s^{-1}}$.
3. In frame $\{b\}$, use ${}^b\mathbf{r}_{O_bC_L} = [0, L/2, 0]^\mathsf{T}$, ${}^b\mathbf{r}_{O_bC_R} = [0, -L/2, 0]^\mathsf{T}$, ${}^b\mathbf{v}_{O_b} = [v, 0, 0]^\mathsf{T}$, and ${}^b\boldsymbol{\omega} = [0, 0, \omega]^\mathsf{T}$. The cross products contribute $[-L\omega/2, 0, 0]^\mathsf{T}$ on the left and $[L\omega/2, 0, 0]^\mathsf{T}$ on the right. Adding ${}^b\mathbf{v}_{O_b}$ and selecting the $+x_b$ components gives the two required equations.
4. Equal wheel speeds have zero difference, so the yaw-rate equation gives $\omega = 0$.
5. Subtract $v_L = v - L\omega/2$ from $v_R = v + L\omega/2$, then substitute $v_R = r\dot{\phi}_R$ and $v_L = r\dot{\phi}_L$.
6. $v = 0.30\,\mathrm{m\,s^{-1}}$ and $\omega = 0.40\,\mathrm{rad\,s^{-1}}$.
7. The ICC C is the point about which the robot is instantaneously rotating. The radius $\rho = \mathrm{distance}(O_b, C) \geq 0$ is an ordinary distance, while $R = +\rho$ for an ICC on the left and $R = -\rho$ for one on the right. The quantity $\Delta\ell$ is signed forward distance and $\Delta\theta$ is signed heading change. Dividing $\Delta\ell = R\Delta\theta$ by Δt and taking the limit gives $v = R\omega$, hence $R = v/\omega$ for $\omega \neq 0$. A forward left turn has $R > 0$ and $\omega > 0$; a forward right turn has $R < 0$ and $\omega < 0$.
8. Examples include wheel slip, unequal wheel radii, tyre deformation, backlash, encoder error, an incorrect track width, and uneven ground.

3.3 From velocity to a finite pose change

The differential equation $\dot{\mathbf{q}} = \mathbf{f}(\mathbf{q}, \mathbf{u})$ specifies instantaneous motion. A simulator or odometry system needs the pose after a finite interval.

Let t_k denote sample time k, where $k \in \{0, 1, 2, \ldots\}$, and define

$$\Delta t := t_{k+1} - t_k > 0.$$

The sampling interval Δt is measured in seconds. Define $\mathbf{q}_k := \mathbf{q}(t_k)$ and assume that v and ω remain constant on $[t_k, t_{k+1}]$. This is a zero-order-hold input assumption.

3.3.1 Exact straight motion

If $\omega = 0$, the heading remains constant. Recall that

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \\ \omega \end{bmatrix}.$$

Integrating the three rate equations gives

$$\boxed{\begin{aligned} x_{k+1} &= x_k + v\Delta t \cos\theta_k, \\ y_{k+1} &= y_k + v\Delta t \sin\theta_k, \\ \theta_{k+1} &= \theta_k. \end{aligned}}$$

3.3.2 Exact constant-curvature motion

If $\omega \neq 0$, let $\tau \in [0, \Delta t]$ denote elapsed time within the interval. Then

$$\theta(t_k + \tau) = \theta_k + \omega\tau.$$

Substituting this heading into the position-rate equations and integrating gives

$$\begin{aligned} x_{k+1} - x_k &= \int_0^{\Delta t} v\cos(\theta_k + \omega\tau)\, d\tau \\ &= \frac{v}{\omega}\left[\sin(\theta_k + \omega\Delta t) - \sin\theta_k\right], \\ y_{k+1} - y_k &= \int_0^{\Delta t} v\sin(\theta_k + \omega\tau)\, d\tau \\ &= \frac{v}{\omega}\left[\cos\theta_k - \cos(\theta_k + \omega\Delta t)\right]. \end{aligned}$$

Therefore,

$$\boxed{\begin{aligned} x_{k+1} &= x_k + \frac{v}{\omega}\left[\sin(\theta_k + \omega\Delta t) - \sin\theta_k\right], \\ y_{k+1} &= y_k + \frac{v}{\omega}\left[\cos\theta_k - \cos(\theta_k + \omega\Delta t)\right], \\ \theta_{k+1} &= \theta_k + \omega\Delta t. \end{aligned}}$$

The signed turning radius is $R = v/\omega$, measured in metres, and the signed angular change is $\Delta\theta = \omega\Delta t$, measured in radians. The robot therefore follows a circular arc.

The formulas contain v/ω and therefore cannot be evaluated directly when $\omega = 0$. To express their continuous limit, define the unnormalised sinc function $\operatorname{sinc} : \mathbb{R} \to \mathbb{R}$ by

$$\operatorname{sinc}(z) := \begin{cases} \dfrac{\sin z}{z}, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Here, $z \in \mathbb{R}$ is a dimensionless scalar, and the value at zero follows from $\lim_{z\to 0} \sin z/z = 1$. This convention must be stated because some software libraries instead define a normalised sinc function containing πz.

Using the trigonometric identities

$$\sin(\theta_k + \Delta\theta) - \sin\theta_k = 2\cos\left(\theta_k + \frac{\Delta\theta}{2}\right)\sin\left(\frac{\Delta\theta}{2}\right)$$

and

$$\cos\theta_k - \cos(\theta_k + \Delta\theta) = 2\sin\left(\theta_k + \frac{\Delta\theta}{2}\right)\sin\left(\frac{\Delta\theta}{2}\right),$$

the common scale factor in both position increments can be rewritten, for $\omega \neq 0$, as

$$\begin{aligned} \frac{v}{\omega}\,2\sin\left(\frac{\Delta\theta}{2}\right) &= v\Delta t\,\frac{\sin(\Delta\theta/2)}{\Delta\theta/2} \\ &= v\Delta t\ \mathrm{sinc}\left(\frac{\Delta\theta}{2}\right), \end{aligned}$$

where $\Delta\theta = \omega\Delta t$ was used in the first equality. The exact update is consequently equivalent to

$$\boxed{\begin{aligned} x_{k+1} &= x_k + v\Delta t\ \mathrm{sinc}\left(\frac{\Delta\theta}{2}\right)\cos\left(\theta_k + \frac{\Delta\theta}{2}\right), \\ y_{k+1} &= y_k + v\Delta t\ \mathrm{sinc}\left(\frac{\Delta\theta}{2}\right)\sin\left(\theta_k + \frac{\Delta\theta}{2}\right), \\ \theta_{k+1} &= \theta_k + \Delta\theta. \end{aligned}}$$

As $\omega \to 0$, the angular change $\Delta\theta \to 0$, $\mathrm{sinc}(\Delta\theta/2) \to 1$, and the midpoint heading $\theta_k + \Delta\theta/2 \to \theta_k$. The position equations therefore reduce smoothly to the straight-motion update

$$x_{k+1} = x_k + v\Delta t\cos\theta_k, \qquad y_{k+1} = y_k + v\Delta t\sin\theta_k.$$

Software may use an explicit straight-motion case when $|\omega|$ is below a declared tolerance or a sinc implementation designed to remain numerically stable near zero. Directly computing $\sin z/z$ at $z = 0$ would still divide by zero despite the function's well-defined limiting value.

Quick test C: exact finite motion

1. What does the zero-order-hold assumption say about v and ω?
2. Why is $\theta(t_k + \tau) = \theta_k + \omega\tau$ when ω is constant?
3. Where does the factor $1/\omega$ in the position integral come from?
4. Starting from $\mathbf{q}_k = [0\ 0\ 0]^\mathsf{T}$, let $v = 1\,\mathrm{m\,s^{-1}}$, $\omega = 1\,\mathrm{rad\,s^{-1}}$, and $\Delta t = 1\,\mathrm{s}$. Find the exact next pose.
5. Define the unnormalised sinc function and explain how it exposes the straight-motion limit.
6. Why should software not directly evaluate either v/ω at $\omega = 0$ or $\sin z/z$ at $z = 0$?

Answers and hints

1. Both inputs remain constant throughout the sample interval.
2. Integrating $\dot{\theta} = \omega$ for elapsed time τ gives heading change $\omega\tau$.
3. With $z = \theta_k + \omega\tau$, $dz = \omega\,d\tau$, so $d\tau = dz/\omega$.
4. $\mathbf{q}_{k+1} = [\sin 1\ \ 1 - \cos 1\ \ 1]^\mathsf{T} \approx [0.8415\ \ 0.4597\ \ 1.0000]^\mathsf{T}$, with position in metres and heading in radians.
5. The unnormalised function is $\operatorname{sinc}(z) = \sin z/z$ for $z \neq 0$ and $\operatorname{sinc}(0) = 1$. Rewriting the update with $\operatorname{sinc}(\Delta\theta/2)$ makes the limit visible: as $\omega \to 0$, $\Delta\theta \to 0$, the sinc factor tends to 1, and the midpoint heading tends to θ_k.
6. Both direct expressions would divide by zero even though their mathematical limits are well defined. Software must select the straight case or evaluate sinc with a numerically stable near-zero implementation.

3.4 Numerical integration with Forward Euler

For a general time-varying input, the exact solution may not have a convenient closed form. The integral form of the state equation over one interval is

$$\mathbf{q}(t_{k+1}) = \mathbf{q}(t_k) + \int_{t_k}^{t_{k+1}} \mathbf{f}(\mathbf{q}(t), \mathbf{u}(t))\,dt.$$

Numerical integration approximates this integral using a finite number of evaluations of $\mathbf{f}$.

3.4.1 Derivation from a Taylor expansion

Assume $\mathbf{q}(t)$ is sufficiently differentiable near t_k. Its Taylor expansion is

$$\mathbf{q}(t_k + \Delta t) = \mathbf{q}(t_k) + \Delta t\,\dot{\mathbf{q}}(t_k) + \frac{\Delta t^2}{2}\,\ddot{\mathbf{q}}(\xi_k),$$

for some $\xi_k \in (t_k, t_{k+1})$. Substituting $\dot{\mathbf{q}}(t_k) = \mathbf{f}(\mathbf{q}_k, \mathbf{u}_k)$ and discarding the term proportional to Δt^2 gives the Forward Euler update

$$\boxed{\mathbf{q}^{\mathrm{E}}_{k+1} = \mathbf{q}^{\mathrm{E}}_k + \Delta t\, \mathbf{f}(\mathbf{q}^{\mathrm{E}}_k, \mathbf{u}_k)}.$$

The superscript E identifies the Euler approximation, and $\mathbf{u}_k := \mathbf{u}(t_k)$ is the sampled input. Forward Euler uses the slope at the beginning of the interval and treats it as constant across the whole interval.

For the planar kinematic model,

$$\boxed{\begin{aligned} x^{\mathrm{E}}_{k+1} &= x^{\mathrm{E}}_k + \Delta t\, v_k \cos\theta^{\mathrm{E}}_k, \\ y^{\mathrm{E}}_{k+1} &= y^{\mathrm{E}}_k + \Delta t\, v_k \sin\theta^{\mathrm{E}}_k, \\ \theta^{\mathrm{E}}_{k+1} &= \theta^{\mathrm{E}}_k + \Delta t\, \omega_k. \end{aligned}}$$

Here, v_k and ω_k are the body-forward speed and yaw rate used for sample k.

3.4.2 Local and accumulated error

Let $E(\Delta t) \geq 0$ denote a chosen error magnitude produced using step size $\Delta t > 0$. For an exponent $p > 0$, the statement

$$E(\Delta t) = O(\Delta t^p) \qquad \text{as} \qquad \Delta t \to 0$$

means that there are constants $C > 0$ and $\Delta t_0 > 0$, independent of Δt, such that

$$E(\Delta t) \leq C \Delta t^p \qquad \text{for every} \qquad 0 < \Delta t \leq \Delta t_0.$$

The exponent p is the convergence order. The big-O statement is a bound on how the error scales for sufficiently small steps, not an equation claiming that the error equals Δt^p. When the leading error term is nonzero and Δt is sufficiently small, one commonly observes the approximate scaling $E(\Delta t) \approx K \Delta t^p$ for a constant $K > 0$ that does not depend on Δt.

The local truncation error is the error introduced in one step when that step begins from the exact state. Forward Euler has local truncation error $O(\Delta t^2)$. Over a fixed total time interval, reducing Δt creates more steps, and their errors accumulate. The resulting global error of Forward Euler is generally $O(\Delta t)$. In the sufficiently-small-step regime where the leading first-order term dominates, halving Δt therefore approximately halves the global error; it does not usually quarter it.

3.4.3 Worked comparison with the exact update

Use the initial pose and constant input

$$\mathbf{q}_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \qquad v = 1\,\mathrm{m\,s^{-1}}, \qquad \omega = 1\,\mathrm{rad\,s^{-1}}, \qquad \Delta t = 1\,\mathrm{s}.$$

The exact update is

$$\mathbf{q}_1 \approx \begin{bmatrix} 0.8415\,\mathrm{m} \\ 0.4597\,\mathrm{m} \\ 1.0000\,\mathrm{rad} \end{bmatrix}.$$

Forward Euler evaluates the direction only at initial heading $\theta_0 = 0$:

$$\mathbf{q}_1^{\mathrm{E}} = \begin{bmatrix} 0 + 1\cos 0 \\ 0 + 1\sin 0 \\ 0 + 1 \end{bmatrix} = \begin{bmatrix} 1\,\mathrm{m} \\ 0\,\mathrm{m} \\ 1\,\mathrm{rad} \end{bmatrix}.$$

Both methods produce the same heading because ω is constant and the heading equation is linear. The Euler position is wrong because the forward direction rotates substantially during the one-second interval. Define one-step position error by

$$e_p := \sqrt{(x_1^{\mathrm{E}} - x_1)^2 + (y_1^{\mathrm{E}} - y_1)^2}.$$

For this example, $e_p \approx 0.4863\,\mathrm{m}$. Smaller steps allow Euler to update the direction more often. When constant-input differential-drive motion is assumed, however, the exact arc update is available and more accurate.

3.4.4 Choosing and checking an integration method

- Use the exact straight-or-arc update when v and ω are held constant and the ideal planar model is appropriate.
- Use a numerical method when the input or model varies within the interval and no convenient exact solution is available.
- Reduce Δt when the state changes too much during one step, while recognising that smaller steps increase computation.
- Compare a numerical trajectory with a known exact case before trusting it on a harder case.
- Compute Δt from the intended time source; do not silently assume perfectly periodic execution.

Quick test D: Forward Euler

1. State the Forward Euler update and define every symbol in it.
2. In plain language, what approximation does Forward Euler make during one step?
3. Suppose one Forward Euler step begins from the exact pose and v and ω remain constant during the step. Why does Forward Euler compute the exact position when $\omega = 0$, but generally not when the robot turns with $\omega \neq 0$?
4. What does the global-error statement $E(\Delta t) = O(\Delta t)$ mean, and what approximate change in global error should occur when a sufficiently small Δt is halved over the same fixed total time interval?
5. Give one exact verification case and one physical limitation for an odometry integrator.

Answers and hints

1. $\mathbf{q}^{\mathrm{E}}_{k+1} = \mathbf{q}^{\mathrm{E}}_k + \Delta t\, \mathbf{f}(\mathbf{q}^{\mathrm{E}}_k, \mathbf{u}_k)$. The terms are the next approximate state, current approximate state, step duration, state-derivative function, and sampled input.
2. It uses the derivative at the start as if that derivative remained constant across the interval.
3. When $\omega = 0$, the heading θ remains constant. Constant v then gives the constant world-frame velocity $[v \cos\theta_k \;\; v \sin\theta_k]^{\mathsf{T}}$, so multiplying the initial velocity by Δt evaluates the position integral exactly. When $\omega \neq 0$ and $v \neq 0$, the heading and therefore the world-frame velocity direction change continuously during the step. Forward Euler holds the initial direction constant and produces a straight tangent displacement instead of the exact arc displacement. The word "generally" allows special cases such as pure rotation with $v = 0$, for which the position remains unchanged and is therefore exact.
4. The statement $E(\Delta t) = O(\Delta t)$ means that constants $C > 0$ and $\Delta t_0 > 0$ exist such that $E(\Delta t) \leq C\Delta t$ whenever $0 < \Delta t \leq \Delta t_0$. It describes first-order scaling as $\Delta t \to 0$, not exact equality. In the regime where the leading first-order error dominates, $E(\Delta t) \approx K\Delta t$, so replacing Δt by $\Delta t/2$ approximately halves the global error over the same fixed total time interval.
5. Exact cases include constant straight motion, pure rotation, and constant-curvature motion. Physical limitations include slip, parameter error, encoder quantisation, and timing error.

3.5 Engineering implications and failure modes

1. **Frame inconsistency:** The scalar v is a body-frame forward speed, but $\dot{x}$ and $\dot{y}$ are world-frame components. Assigning $\dot{x} = v$ is valid only when $\theta = 0$ modulo 2π.

2. **Sign-convention mismatch:** Swapping left and right wheels or using a clockwise-positive yaw convention reverses the predicted turn. Under this chapter convention, $\dot{\phi}_L < \dot{\phi}_R$ must produce positive counter-clockwise yaw.

3. **Unit mismatch:** Wheel speed may arrive in revolutions per minute while the model expects radians per second. Heading may arrive in degrees while trigonometric functions expect radians. Convert units at explicit interface boundaries.

4. **Incorrect or variable sampling time:** Using an assumed Δt when measurements arrive irregularly causes distance and angle errors. An update should validate timestamp differences and define behaviour for zero, negative, missing, or unreasonably large intervals.

5. **Angle representation:** The equations allow θ to grow without bound, but an interface may require $\theta \in (-\pi, \pi]$. Wrapping changes the numerical representative, not the physical orientation. Heading differences require a wrap-aware calculation.

6. **Model error versus integration error:** Reducing Δt reduces integration error but cannot correct an inaccurate physical model. If a wheel slips, even exact integration of the ideal no-slip equations gives the wrong physical trajectory.

7. **Encoder integration and drift:** Wheel encoders measure incremental rotation rather than global position. Integrating these increments estimates pose relative to an initial condition. Small systematic and random errors accumulate, so wheel odometry alone generally drifts and requires external sensing for long-term correction.

Cumulative chapter test

1. Define $\mathbf{q}$, $\mathbf{u}$, $\mathbf{f}$, r, L, $\dot{\phi}_L$, $\dot{\phi}_R$, t_k, and Δt, including dimensions or units.
2. Starting with ${}^b\mathbf{v} = [v\ 0]^\mathsf{T}$, derive $\dot{x} = v\cos\theta$ and $\dot{y} = v\sin\theta$.
3. Derive the no-sideways-motion constraint from the world-frame coordinate column of the y_b-axis.
4. Derive the wheel-to-body equations for v and ω and the body-to-wheel equations for $\dot{\phi}_L$ and $\dot{\phi}_R$.
5. A robot has $r = 0.08\,\mathrm{m}$, $L = 0.40\,\mathrm{m}$, $\dot{\phi}_L = 5\,\mathrm{rad\,s^{-1}}$, and $\dot{\phi}_R = 7\,\mathrm{rad\,s^{-1}}$. Find v and ω and state the turn direction.
6. Derive the exact constant-curvature update by integrating the position rates over one interval.
7. Derive Forward Euler from the first two Taylor terms and explain the discarded term.
8. Explain why exact mathematical integration can still produce inaccurate physical odometry.
9. Design three verification cases that separately reveal a wheel-order error, a yaw-sign error, and excessive integration error.

Cumulative answers and hints

1. $\mathbf{q} = [x\ y\ \theta]^\mathsf{T}$ is planar pose with units $\mathrm{m}, \mathrm{m}, \mathrm{rad}$; $\mathbf{u} = [v\ \omega]^\mathsf{T}$ contains forward speed and yaw rate in $\mathrm{m\,s^{-1}}$ and $\mathrm{rad\,s^{-1}}$; $\mathbf{f}$ maps pose and input to pose derivative; r is wheel radius in metres; L is track width in metres; $\dot{\phi}_L$ and $\dot{\phi}_R$ are wheel angular speeds in

radians per second; t_k is sample time in seconds; and $\Delta t = t_{k+1} - t_k$ is the positive integration interval in seconds.

2. Premultiply ${}^b\mathbf{v}$ by $\mathbf{R}_{wb}(\theta)$ and read the two components.
3. Use ${}^w\mathbf{e}_{y,b} = [-\sin\theta \;\; \cos\theta]^\mathsf{T}$ and set its dot product with $[\dot{x} \;\; \dot{y}]^\mathsf{T}$ equal to zero.
4. Use $r\dot{\phi}_L = v - L\omega/2$ and $r\dot{\phi}_R = v + L\omega/2$. Addition isolates v and subtraction isolates ω; solving the pair for wheel speeds gives the inverse equations.
5. $v = (0.08/2)(5+7) = 0.48\,\mathrm{m\,s^{-1}}$ and $\omega = (0.08/0.40)(7-5) = 0.40\,\mathrm{rad\,s^{-1}}$. The positive yaw rate is a counter-clockwise or left turn.
6. Write $\theta(t_k+\tau) = \theta_k+\omega\tau$, integrate over $\tau \in [0, \Delta t]$, and use antiderivatives $\sin(\theta_k+\omega\tau)/\omega$ and $-\cos(\theta_k + \omega\tau)/\omega$.
7. Taylor expansion gives $\mathbf{q}(t_k + \Delta t) = \mathbf{q}(t_k) + \Delta t\dot{\mathbf{q}}(t_k) + O(\Delta t^2)$. Replacing $\dot{\mathbf{q}}$ with $\mathbf{f}$ and omitting the second-order remainder gives Forward Euler. The discarded term represents within-step change of the derivative.
8. Exact integration eliminates approximation for the stated model and input assumption; it cannot eliminate slip, parameter error, encoder error, timing error, or other model violations.
9. For wheel order, command unequal known wheel speeds and check the turn direction. For yaw sign, command pure positive rotation and check that heading increases counter-clockwise. For integration error, compare with the exact constant-curvature solution at decreasing values of Δt and verify convergence.

References

- Kevin M. Lynch and Frank C. Park, *Modern Robotics: Mechanics, Planning, and Control*, Chapter 13, especially the wheeled-mobile-robot kinematic models and odometry discussion.
- Steven M. LaValle, *Planning Algorithms*, Chapter 13, especially continuous-time differential models, velocity constraints, and wheeled-vehicle models.

4 Differential-Drive Kinematics and Numerical Integration: Implementation

Purpose and scope

This companion translates the differential-drive mathematics derived in Differential-Drive Kinematics and Numerical Integration: Model and Derivation into reusable software contracts, invariants, validation rules, and tests. It records how those equations map into dependable software without repeating the derivations.

4.1 Wheel-to-body velocity mapping

Let $r > 0$ be the common effective wheel radius in metres and let $L > 0$ be the track width in metres. Let $\dot{\phi}_L$ and $\dot{\phi}_R$ be the signed left and right wheel angular speeds in $\mathrm{rad\,s^{-1}}$, where positive angular speed rolls the corresponding wheel forward.

The software implements

$$v = \frac{r}{2}\left(\dot{\phi}_R + \dot{\phi}_L\right), \qquad \omega = \frac{r}{L}\left(\dot{\phi}_R - \dot{\phi}_L\right),$$

where v is signed body-forward speed in $\mathrm{m\,s^{-1}}$ and ω is signed counter-clockwise yaw rate in $\mathrm{rad\,s^{-1}}$.

4.1.1 Software contract

The `wheel_to_body` operation accepts four finite scalar inputs: left wheel angular speed, right wheel angular speed, wheel radius, and track width. The wheel radius and track width must be strictly positive. Invalid inputs raise `ValueError`; the operation does not clamp values or substitute default geometry.

The operation returns an immutable `BodyVelocity` value containing `forward_speed_m_s` and `yaw_rate_rad_s`. It is deterministic, does not mutate its inputs, and does not depend on ROS state, timestamps, configuration files, or hidden global state.

4.1.2 Simplified reference snippet

The following equation-focused snippet shows how the mathematical factors map to Python expressions. It assumes that all inputs are finite and that `wheel_radius_m` and `track_width_m` are strictly positive; the production operation enforces those preconditions before performing the calculation.

```
@dataclass(frozen=True, slots=True)
class BodyVelocity:
    """Reduced planar body velocity in SI units."""

    forward_speed_m_s: float
    yaw_rate_rad_s: float


def wheel_to_body(
    *,
    left_angular_speed_rad_s: float,
    right_angular_speed_rad_s: float,
    wheel_radius_m: float,
    track_width_m: float,
) -> BodyVelocity:
    """Apply the wheel-to-body equations to validated inputs."""
    forward_speed_m_s = (
        wheel_radius_m / 2
        * (right_angular_speed_rad_s + left_angular_speed_rad_s)
    )
    yaw_rate_rad_s = (
        wheel_radius_m / track_width_m
        * (right_angular_speed_rad_s - left_angular_speed_rad_s)
    )

    return BodyVelocity(
        forward_speed_m_s=forward_speed_m_s,
        yaw_rate_rad_s=yaw_rate_rad_s,
    )
```

4.1.3 Invariants

- Equal wheel speeds produce zero yaw rate.
- Equal and opposite wheel speeds produce zero forward speed.
- Swapping the left and right wheel speeds preserves forward speed and reverses yaw rate.
- Two zero wheel speeds produce zero forward speed and zero yaw rate.

- Positive right-wheel speed relative to left-wheel speed produces positive counter-clockwise yaw.

4.2 Body-to-wheel velocity mapping

The `body_to_wheel` operation implements the inverse of the wheel-to-body mapping. Its inputs are signed body-forward speed v in $\mathrm{m\,s^{-1}}$, signed counter-clockwise yaw rate ω in $\mathrm{rad\,s^{-1}}$, wheel radius $r > 0$ in metres, and track width $L > 0$ in metres.

The required wheel angular speeds are

$$\dot{\phi}_L = \frac{v}{r} - \frac{L\omega}{2r}, \qquad \dot{\phi}_R = \frac{v}{r} + \frac{L\omega}{2r},$$

where $\dot{\phi}_L$ and $\dot{\phi}_R$ are the signed left and right wheel angular speeds in $\mathrm{rad\,s^{-1}}$.

4.2.1 Software contract

The operation accepts four finite scalar inputs. The wheel radius and track width must be strictly positive. Invalid inputs raise `ValueError`; the operation does not clamp values or substitute geometry.

The result is an immutable `WheelAngularSpeeds` value containing `left_angular_speed_rad_s` and `right_angular_speed_rad_s`.

4.2.2 Implementation mapping

The calculation separates forward and turning contributions:

```
@dataclass(frozen=True, slots=True)
class WheelAngularSpeeds:
    """Signed left and right wheel angular speeds in radians per second."""

    left_angular_speed_rad_s: float
    right_angular_speed_rad_s: float


def body_to_wheel(
    *,
    forward_speed_m_s: float,
    yaw_rate_rad_s: float,
    wheel_radius_m: float,
    track_width_m: float,
```

```
) -> WheelAngularSpeeds:
    """Apply the body-to-wheel equations to validated inputs."""
    forward_component_rad_s = forward_speed_m_s / wheel_radius_m
    turning_component_rad_s = (
        track_width_m * yaw_rate_rad_s / (2 * wheel_radius_m)
    )
    left_angular_speed_rad_s = (
        forward_component_rad_s - turning_component_rad_s
    )
    right_angular_speed_rad_s = (
        forward_component_rad_s + turning_component_rad_s
    )

    return WheelAngularSpeeds(
        left_angular_speed_rad_s=left_angular_speed_rad_s,
        right_angular_speed_rad_s=right_angular_speed_rad_s,
    )
```

The forward component contributes equally to both wheels. The turning component is subtracted from the left wheel and added to the right wheel. Therefore, a positive yaw rate makes the right wheel turn faster than the left wheel.

4.2.3 Invariants and required tests

- Zero yaw rate produces equal wheel speeds of v/r.
- Zero forward speed produces equal-and-opposite wheel speeds.
- Reversing the yaw-rate sign swaps the left and right wheel speeds.
- Applying `body_to_wheel` and then `wheel_to_body` recovers the original v and ω within the declared floating-point tolerance.
- Non-finite inputs and non-positive geometry raise `ValueError`.

4.3 Exact constant-input pose integration

Let the initial planar pose be (x_k, y_k, θ_k), where x_k and y_k are the world-frame coordinates of the axle midpoint in metres and θ_k is the unwrapped counter-clockwise heading in radians.

Assume that forward speed v in $\mathrm{m\,s^{-1}}$ and yaw rate ω in $\mathrm{rad\,s^{-1}}$ remain constant over a time step $\Delta t > 0$ in seconds. Define the heading change

$$\Delta\theta = \omega \Delta t.$$

The exact update is

$$
\begin{aligned}
x_{k+1} &= x_k + v\Delta t \operatorname{sinc}\left(\frac{\Delta\theta}{2}\right)\cos\left(\theta_k + \frac{\Delta\theta}{2}\right),\\
y_{k+1} &= y_k + v\Delta t \operatorname{sinc}\left(\frac{\Delta\theta}{2}\right)\sin\left(\theta_k + \frac{\Delta\theta}{2}\right),\\
\theta_{k+1} &= \theta_k + \Delta\theta,
\end{aligned}
$$

where the unnormalised sinc function is

$$
\operatorname{sinc}(z) = \begin{cases} \dfrac{\sin z}{z}, & z \neq 0,\\ 1, & z = 0. \end{cases}
$$

The update is exact for the ideal continuous kinematic model only when v and ω remain constant during the time step. It does not account for wheel slip, changing commands, actuator dynamics, or model error.

4.3.1 Software contract

The `integrate_exact` operation accepts an immutable `PlanarPose`, finite forward speed, finite yaw rate, and a finite positive time step. A non-finite input or non-positive time step raises `ValueError`.

The operation returns a new `PlanarPose`. It does not mutate the initial pose or wrap the resulting heading into a fixed angular interval.

4.3.2 Numerically stable implementation mapping

The implementation avoids evaluating v/ω, which would be undefined when $\omega = 0$. Instead, it uses the sinc form:

```
@dataclass(frozen=True, slots=True)
class PlanarPose:
    """Pose of the robot's axle midpoint in the world frame.

    Attributes:
        position_x_m: World-frame x-coordinate in metres.
        position_y_m: World-frame y-coordinate in metres.
        heading_rad: Counter-clockwise heading from the world-frame
            positive x-axis in radians.
    """

    position_x_m: float
```

```
    position_y_m: float
    heading_rad: float


def _sinc(value: float) -> float:
    """Return the unnormalised sinc function, including its value at zero."""
    if value == 0.0:
        return 1.0

    return sin(value) / value


heading_change_rad = yaw_rate_rad_s * time_step_s
half_heading_change_rad = heading_change_rad / 2
midpoint_heading_rad = pose.heading_rad + half_heading_change_rad

signed_displacement_m = (
    forward_speed_m_s
    * time_step_s
    * _sinc(half_heading_change_rad)
)

next_pose = PlanarPose(
    position_x_m=(
        pose.position_x_m
        + signed_displacement_m * cos(midpoint_heading_rad)
    ),
    position_y_m=(
        pose.position_y_m
        + signed_displacement_m * sin(midpoint_heading_rad)
    ),
    heading_rad=pose.heading_rad + heading_change_rad,
)
```

When $\omega = 0$, we have $\Delta\theta = 0$ and $\mathrm{sinc}(0) = 1$. The same implementation therefore reduces smoothly to straight motion without a separate division by zero.

4.3.3 Invariants and required tests

- Zero yaw rate produces the exact straight-motion update.
- Zero forward speed changes only the heading.
- Constant nonzero forward speed and yaw rate produce constant-curvature motion.
- The input pose remains unchanged.
- The output heading remains unwrapped.

- Non-finite inputs raise `ValueError`.
- A zero or negative time step raises `ValueError`.

4.4 Forward Euler pose integration

Forward Euler approximates motion over a time step $\Delta t > 0$ by evaluating the pose derivative at the beginning of the step. For initial pose (x_k, y_k, θ_k), constant forward speed v, and constant yaw rate ω, the update is

$$\begin{aligned} x_{k+1}^{\mathrm{E}} &= x_k + \Delta t\, v \cos\theta_k, \\ y_{k+1}^{\mathrm{E}} &= y_k + \Delta t\, v \sin\theta_k, \\ \theta_{k+1}^{\mathrm{E}} &= \theta_k + \Delta t\, \omega. \end{aligned}$$

The superscript E identifies a Forward Euler approximation. Position is calculated entirely from the beginning-of-step heading θ_k; the changing heading within the step does not affect that step's position increment.

4.4.1 Software contract

The `integrate_forward_euler` operation accepts the same inputs and validation rules as `integrate_exact`: an immutable `PlanarPose`, finite forward speed, finite yaw rate, and a finite positive time step.

The operation returns a new `PlanarPose`, does not mutate the input pose, and leaves the heading unwrapped. Invalid inputs raise `ValueError`.

4.4.2 Implementation mapping

```
next_pose = PlanarPose(
    position_x_m=(
        pose.position_x_m
        + time_step_s * forward_speed_m_s * cos(pose.heading_rad)
    ),
    position_y_m=(
        pose.position_y_m
        + time_step_s * forward_speed_m_s * sin(pose.heading_rad)
    ),
    heading_rad=(
        pose.heading_rad + time_step_s * yaw_rate_rad_s
    ),
)
```

4.4.3 Exact and approximate cases

When $\omega = 0$, the heading remains constant. The world-frame velocity is therefore constant, so one Forward Euler step produces the exact straight-motion position.

When $v = 0$, position remains unchanged and the heading update is exact because $\dot{\theta} = \omega$ is constant.

When both $v \neq 0$ and $\omega \neq 0$, the true path curves during the time step. Forward Euler instead follows the tangent defined by θ_k, so its one-step position is generally approximate.

For repeated integration over a fixed duration, Forward Euler has global error $O(\Delta t)$. In this regime, halving Δt should approximately halve the accumulated error.

4.4.4 Invariants and required tests

- Straight constant-speed motion agrees with the exact update.
- Pure rotation changes only the heading.
- A turning step uses the beginning-of-step heading.
- Smaller time steps reduce accumulated turning-position error.
- Halving the time step approximately halves global error in the first-order regime.
- Non-finite inputs and non-positive time steps raise `ValueError`.

5 ROS 2 Package Development Workflow

Goal

This workflow creates, builds, runs, and tests a small ROS 2 Python package from scratch. It uses `my_robot_package` as an example package name and `my_node` as an example executable name; replace them in a real project.

A **workspace** holds packages developed and built together. A **ROS 2 package** combines source code with the metadata needed for building, installing, and discovery. An **executable** is a program installed by a package and started with `ros2 run`.

The generated `my_node` executable only prints a message. It demonstrates the package workflow but does not yet communicate with other ROS 2 nodes.

5.1 Source the ROS 2 underlay

For Zsh, run:

```
source /opt/ros/jazzy/setup.zsh
```

For Bash, run:

```
source /opt/ros/jazzy/setup.bash
```

Use the script matching the current shell.

What happens: `/opt/ros/jazzy` is the ROS 2 Jazzy **underlay**, meaning the existing installation on which the new workspace depends. Its setup script adds ROS installation prefixes, Python packages, and executables to the current shell's search environment.

The command uses `source` because the environment of the current shell must change. If the script ran in a separate **child process**, meaning a process started by the shell, its environment changes would disappear when that child exited.

Verify the environment:

```
printf '%s\n' "$ROS_DISTRO"
command -v ros2
command -v colcon
```

The expected observations are `jazzy` and executable paths for `ros2` and `colcon`.

5.2 Create a workspace

For a new standalone workspace, run:

```
mkdir -p ~/ros_ws/src
cd ~/ros_ws
```

The workspace develops into:

```
ros_ws/
├── src/        # Editable source
├── build/      # Generated build workspace
├── install/    # Generated runnable workspace image
└── log/        # Generated build and test records
```

Only `src/` exists initially. `colcon` creates the other directories. Edit `src/`, not the generated files in `build/`, `install/`, or `log/`.

5.3 Create a Python package

From the workspace root, run:

```
cd src
ros2 pkg create \
  --build-type ament_python \
  --license Apache-2.0 \
  --node-name my_node \
  my_robot_package
cd ..
```

Run this command only when the package does not already exist.

What happens: `ros2 pkg create` generates `src/my_robot_package`.'

- `ament_python` selects Python packaging through `setuptools`.

- `Apache-2.0` is recorded in `package.xml`, and a matching `LICENSE` file is created.
- `--node-name my_node` creates a minimal Python executable and its `setup.py` entry point.

Omit `--node-name` when creating a library-only package.

For comparison, create a C++ package with:

```
ros2 pkg create \
  --build-type ament_cmake \
  --license Apache-2.0 \
  robot_controller
```

The common build types are `ament_python` for Python with `setuptools` and `ament_cmake` for C++ with CMake.

5.4 Inspect the generated package

```
src/my_robot_package/
├── LICENSE
├── package.xml
├── setup.py
├── setup.cfg
├── resource/
│   └── my_robot_package
├── my_robot_package/
│   ├── __init__.py
│   └── my_node.py
└── test/
    ├── test_copyright.py
    ├── test_flake8.py
    └── test_pep257.py
```

The outer `my_robot_package/` is the ROS package root. The inner directory is the Python import package used by `import my_robot_package`; the repeated name does not indicate nested repositories.

File or directory	Purpose
`package.xml`	Declares package identity, licence, dependencies, and build type
`setup.py`	Selects Python packages, data files, and executables to install

File or directory	Purpose
`setup.cfg`	Selects the ROS-compatible destination for Python executables
`resource/my_robot_package`	Registers the installed package in the ament resource index
`my_robot_package/`	Contains importable Python source
`test/`	Contains checks run by `pytest` and `colcon test`

The resource marker is intentionally empty. Its filename identifies the package.

5.5 Complete the package metadata

Replace placeholder descriptions and maintainer details in `package.xml` and `setup.py`. Keep their package name, version, and licence consistent. The `package.xml` export tells `colcon` which build integration to use:

```
<export>
  <build_type>ament_python</build_type>
</export>
```

A concise `setup.py` for the example is:

```
from setuptools import find_packages, setup

package_name = "my_robot_package"

setup(
    name=package_name,
    version="0.0.0",
    packages=find_packages(exclude=["test"]),
    data_files=[
        (
            "share/ament_index/resource_index/packages",
            [f"resource/{package_name}"],
        ),
        (f"share/{package_name}", ["package.xml"]),
    ],
    install_requires=["setuptools"],
    zip_safe=True,
    maintainer="Your Name",
    maintainer_email="you@example.com",
```

```
    description="Example ROS 2 Python package.",
    license="Apache-2.0",
    entry_points={
        "console_scripts": [
            "my_node = my_robot_package.my_node:main",
        ],
    },
)
```

The important connections are:

- `packages` installs importable Python packages.
- `data_files` installs the resource marker and `package.xml` where ROS expects them.
- `console_scripts` creates an executable that calls a Python function.

The entry point `my_node = my_robot_package.my_node:main` maps the executable name `my_node` to the function `main()` in the Python module `my_robot_package.my_node`.

The matching `setup.cfg` is:

```
[develop]
script_dir=$base/lib/my_robot_package

[install]
install_scripts=$base/lib/my_robot_package
```

Here, `$base` is the package installation prefix. `setup.cfg` specifies that executable scripts belong under `lib/my_robot_package`, where `ros2 run` looks for them. It does not create an executable by itself; the `console_scripts` entry does that.

5.6 Resolve dependencies and run fast tests

Resolve dependencies declared by packages under `src/`:

```
rosdep install \
  --from-paths src \
  --ignore-src \
  --rosdistro jazzy \
  -r \
  -y
```

`rosdep` ignores dependencies supplied by source packages in the workspace, maps the remaining dependency keys to operating-system packages, and installs missing packages.

Run fast tests directly against the source tree:

```
python3 -m pytest \
  src/my_robot_package/test -q
```

Source-level tests provide fast feedback but do not verify ROS metadata or the installed layout. Continue only when they pass; the count and duration depend on the package.

5.7 Build and install the package

From the workspace root, run:

```
colcon build \
  --packages-select my_robot_package \
  --symlink-install
```

What happens: `colcon` discovers packages under `src/`, reads their manifests, orders them by declared dependencies, and selects the appropriate build integration. `ament_python` then invokes `setuptools` using `setup.py` and `setup.cfg`.

The logical installed layout is:

```
install/my_robot_package/
├── lib/
│   ├── python3.12/
│   │   └── site-packages/
│   │       └── my_robot_package/
│   │           ├── __init__.py
│   │           └── my_node.py
│   └── my_robot_package/
│       └── my_node
└── share/
    ├── ament_index/
    │   └── resource_index/
    │       └── packages/
    │           └── my_robot_package
    └── my_robot_package/
        └── package.xml
```

The Python version in the path depends on the system.

- `site-packages/my_robot_package` supplies importable Python code.
- `lib/my_robot_package/my_node` supplies the executable used by `ros2 run`.
- The resource marker registers the package with ROS.
- The installed `package.xml` supplies package metadata.

ROS 2 needs this standard runtime layout to locate packages, executables, metadata, launch files, configuration, Python imports, and dependency environment hooks.

5.7.1 What `--symlink-install` changes

Supported files are linked back to editable source rather than copied:

```
install/.../my_node.py
        ↓ symbolic link
src/my_robot_package/my_robot_package/my_node.py
```

Python packages may use an `.egg-link` instead of the copied directory shown in the logical tree. Consequently, edits to existing linked Python files may appear without rebuilding, but `install/` is still needed for metadata, executables, and environment scripts.

Rebuild after changing `setup.py`, `setup.cfg`, `console_scripts`, installed data, dependencies, CMake configuration, or compiled C++ code. Rebuild after structural changes and before formal verification to avoid stale state.

5.8 Source the overlay and run the package

After a successful build, source the script matching the current shell.

```
source install/setup.zsh
```

```
source install/setup.bash
```

This command does not install anything. It adds the workspace's `install/` tree as an **overlay** on the ROS underlay by updating variables such as `AMENT_PREFIX_PATH`, `PYTHONPATH`, and `PATH`.

Sourcing affects only the current shell. A new terminal must source the ROS underlay and then the workspace overlay again.

Verify package discovery, Python import, and executable discovery:

```
ros2 pkg prefix my_robot_package
python3 -c "import my_robot_package"
ros2 run my_robot_package my_node
```

The final command should print:

```
Hi from my_robot_package.
```

`ros2 pkg prefix` queries the ament resource index. `ros2 run` uses the first argument as the package name and the second as the executable name, then starts `lib/my_robot_package/my_node`.

5.9 Run tests through colcon

```
colcon test --packages-select my_robot_package
colcon test-result --verbose
```

`colcon test` runs the package's configured tests and stores structured results under the build tree. `colcon test-result` aggregates them. A completed command is not necessarily successful; verification passes only when the summary reports no errors or failures.

5.10 Repeat the daily development loop

From the workspace root, repeat:

```
# 1. Edit source.
vim src/my_robot_package/my_robot_package/my_node.py

# 2. Run fast tests.
python3 -m pytest src/my_robot_package/test -q

# 3. Build and install.
colcon build \
  --packages-select my_robot_package \
  --symlink-install

# 4. Reload the overlay.
source install/setup.zsh

# 5. Run the executable.
ros2 run my_robot_package my_node

# 6. Run package-level verification.
colcon test --packages-select my_robot_package
colcon test-result --verbose
```

Use `source install/setup.bash` in Bash.

Common failure checks

Symptom	Check
`ros2: command not found`	Source the ROS underlay in the current shell
Package absent from `colcon list`	Check its location under `src/` and validate `package.xml`
Python package is not installed	Check `__init__.py` and the `packages` argument in `setup.py`
`ros2 pkg prefix` cannot find the package	Build and source the overlay
`ros2 run` cannot find the executable	Check `console_scripts` and `setup.cfg`, then rebuild and source
Test status is unclear	Run `colcon test-result --verbose`

References

- Open Robotics, "Developing a ROS 2 package," *ROS 2 Jazzy Documentation.*
- Open Robotics, "Using Python Packages with ROS 2," *ROS 2 Jazzy Documentation.*
- colcon, "What is a Workspace?" *colcon Documentation.*

6 Planar Rigid-Body Motion: $SE(2)$ and Twists

Why this chapter matters

The differential-drive pose $\mathbf{q} = [x\ y\ \theta]^\mathsf{T}$ is convenient for storing three coordinates, but it does not by itself expose how rigid motions compose, how the same motion is expressed in different frames, or why constant body velocity produces the exact curved-motion update derived previously. This chapter develops the planar rigid-motion structure needed to answer those questions. It begins by distinguishing a pose from its coordinates and representing a planar pose by a homogeneous transformation in the special Euclidean group $SE(2)$.

Prerequisites and assumptions

The reader should understand matrices and matrix multiplication from Chapter 1, coordinate frames and rotation matrices from Chapter 2, and planar pose notation from Chapter 3.

All frames in this learning block are right-handed planar frames embedded in three-dimensional physical space with a common vertical axis. Frame $\{w\}$ is the fixed world frame and frame $\{b\}$ is attached rigidly to the robot body. Distances are measured in metres and angles in radians.

6.1 Pose coordinates and rigid transformations

6.1.1 A pose is a relative geometric relationship

A planar rigid body's **pose** specifies both its position and its orientation relative to a reference frame. The word *relative* is essential: saying that a robot is at position $(2, 1)$ with heading $\pi/2$ is incomplete until the reference frame and sign convention are known.

Let O_w and O_b denote the origins of frames $\{w\}$ and $\{b\}$. The position of O_b relative to O_w, expressed in world-frame coordinates, is

$$ {}^{w}\mathbf{p}_b = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2, $$

where x and y are measured in metres. The orientation of frame $\{b\}$ relative to frame $\{w\}$ is represented by the rotation matrix $\mathbf{R}_{wb}(\theta) \in \mathbb{R}^{2\times 2}$:

$$\mathbf{R}_{wb}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

The subscript order wb means that $\mathbf{R}_{wb}$ maps vector coordinates from frame $\{b\}$ to frame $\{w\}$. The pose of $\{b\}$ relative to $\{w\}$ can therefore be described by the pair

$$(\mathbf{R}_{wb}, {}^{w}\mathbf{p}_b)\,.$$

This pair separates orientation from position. A homogeneous transformation will combine them into one matrix without confusing their different roles.

6.1.2 Planar rotations and $SO(2)$

The set of valid planar rotation matrices is called the **special orthogonal group in two dimensions**, written $SO(2)$ and defined by

$$SO(2) := \left\{ \mathbf{R} \in \mathbb{R}^{2\times 2} \;\middle|\; \mathbf{R}^{\mathsf{T}}\mathbf{R} = \mathbf{I}_2, \quad \det(\mathbf{R}) = 1 \right\}.$$

The word *orthogonal* refers to $\mathbf{R}^{\mathsf{T}}\mathbf{R} = \mathbf{I}_2$, which means that the columns form an orthonormal basis and that $\mathbf{R}^{-1} = \mathbf{R}^{\mathsf{T}}$. The word *special* adds the condition $\det(\mathbf{R}) = 1$, excluding reflections whose determinant is -1.

The word *group* means that the set is equipped with an operation satisfying four properties: closure, associativity, an identity element, and an inverse for every element. For $SO(2)$, the operation is matrix multiplication.

Closure means that multiplying any two elements of $SO(2)$ produces another element of $SO(2)$. If $\mathbf{R}(\alpha)$ and $\mathbf{R}(\beta)$ represent planar rotations through angles α and β, then

$$\mathbf{R}(\alpha)\mathbf{R}(\beta) = \mathbf{R}(\alpha + \beta) \in SO(2).$$

Geometrically, performing one planar rotation and then another still produces a valid planar rotation; it does not introduce scaling, shear, or reflection.

Associativity means that the placement of parentheses does not change the result of composing three rotations:

$$(\mathbf{R}_1\mathbf{R}_2)\,\mathbf{R}_3 = \mathbf{R}_1\,(\mathbf{R}_2\mathbf{R}_3)\,.$$

Associativity changes only the grouping, not the order, of the matrices. It should not be confused with commutativity, which would permit the order to be exchanged. The identity element is $\mathbf{I}_2$, and every $\mathbf{R} \in SO(2)$ has the inverse $\mathbf{R}^\mathsf{T} \in SO(2)$.

Every planar rotation matrix can be written as $\mathbf{R}(\theta)$ for some $\theta \in \mathbb{R}$. The angle is not unique because

$$\mathbf{R}(\theta + 2\pi n) = \mathbf{R}(\theta), \qquad n \in \mathbb{Z},$$

where $\mathbb{Z}$ is the set of integers. Thus, θ is a **coordinate** used to describe an orientation, while $\mathbf{R}(\theta)$ represents the **orientation** itself. Angles that differ by an integer multiple of 2π describe the same element of $SO(2)$.

6.1.3 Transforming point coordinates by rotation and translation

Let P be a geometric point fixed to the robot. Its position relative to O_b, expressed in body-frame coordinates, is ${}^b\mathbf{p}_P \in \mathbb{R}^2$ and is measured in metres. To obtain its world-frame position, first use $\mathbf{R}_{wb}$ to re-express this body-frame displacement along the world axes and then add the world-frame position of the body origin:

$$\boxed{{}^w\mathbf{p}_P = \mathbf{R}_{wb}\,{}^b\mathbf{p}_P + {}^w\mathbf{p}_b}.$$

This equation follows by tracing the geometric displacement from O_w to P. First travel from O_w to O_b by ${}^w\mathbf{p}_b$. Then travel from O_b to P by the displacement whose body-frame coordinate is ${}^b\mathbf{p}_P$. Multiplication by $\mathbf{R}_{wb}$ expresses that same displacement in world coordinates so that the two terms can be added. This passive coordinate mapping does not physically move P.

A free vector, such as a displacement or velocity vector, has direction and magnitude but no fixed origin. It therefore changes coordinates using rotation alone:

$${}^w\mathbf{v} = \mathbf{R}_{wb}\,{}^b\mathbf{v}.$$

Translation appears in the point equation because a point is located relative to a frame origin. It does not appear in the free-vector equation because a free vector is not attached to an origin.

6.1.4 Homogeneous transformations and $SE(2)$

The point mapping in Section 6.1.3 contains both a matrix multiplication and a vector addition. Homogeneous coordinates package these two operations into one matrix multiplication. Define the body- and world-frame homogeneous coordinate columns of point P by appending a 1:

$$
{}^b\overline{\mathbf{p}}_P := \begin{bmatrix} {}^b\mathbf{p}_P \\ 1 \end{bmatrix} \in \mathbb{R}^3, \qquad {}^w\overline{\mathbf{p}}_P := \begin{bmatrix} {}^w\mathbf{p}_P \\ 1 \end{bmatrix} \in \mathbb{R}^3.
$$

The overline identifies a homogeneous coordinate representation, not a different geometric point. The first two entries have units of metres, while the appended 1 is dimensionless and exists only to make translation part of a matrix multiplication. Define the **homogeneous transformation** from frame $\{b\}$ to frame $\{w\}$ by

$$
\boxed{\mathbf{T}_{wb} := \begin{bmatrix} \mathbf{R}_{wb} & {}^w\mathbf{p}_b \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix} \in \mathbb{R}^{3\times 3}},
$$

where $\mathbf{0}_2 = [0\ 0]^\mathsf{T} \in \mathbb{R}^2$. The subscript order means that $\mathbf{T}_{wb}$ maps homogeneous point coordinates from frame $\{b\}$ to frame $\{w\}$:

$$
\begin{aligned}
{}^w\overline{\mathbf{p}}_P &= \mathbf{T}_{wb}\, {}^b\overline{\mathbf{p}}_P \\
&= \begin{bmatrix} \mathbf{R}_{wb} & {}^w\mathbf{p}_b \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix} \begin{bmatrix} {}^b\mathbf{p}_P \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} \mathbf{R}_{wb}\, {}^b\mathbf{p}_P + {}^w\mathbf{p}_b \\ 1 \end{bmatrix}.
\end{aligned}
$$

The upper two entries reproduce the point-transformation equation. Appending 1 causes the translation column to contribute to the result. Thus, $\mathbf{T}_{wb}$ re-expresses the point's body-relative displacement in world coordinates and then adds the world position of the body origin.

Using ${}^w\mathbf{p}_b = [x\ y]^\mathsf{T}$ and $\mathbf{R}_{wb} = \mathbf{R}_{wb}(\theta)$ gives the component form

$$
\mathbf{T}_{wb} = \begin{bmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{bmatrix}.
$$

The rotation block is dimensionless, while the first two entries of the translation column have units of metres. The final row is an algebraic device that permits rotation and translation to be applied by one matrix multiplication; it does not represent another physical spatial dimension.

To preserve the rotation-only mapping for a free vector, define its homogeneous representation by appending a 0:

$$ {}^{b}\overline{\mathbf{v}} := \begin{bmatrix} {}^{b}\mathbf{v} \\ 0 \end{bmatrix}. $$

Then

$$ \mathbf{T}_{wb} \begin{bmatrix} {}^{b}\mathbf{v} \\ 0 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{wb}\,{}^{b}\mathbf{v} \\ 0 \end{bmatrix}. $$

Appending 0 prevents the translation column from contributing. This algebraic distinction encodes the geometric fact that changing a point's reference frame requires rotation and translation, whereas changing the coordinates of a free vector requires only rotation.

The set of all planar homogeneous transformations is the **special Euclidean group in two dimensions**:

$$ SE(2) := \left\{ \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_2^\top & 1 \end{bmatrix} \,\middle|\, \mathbf{R} \in SO(2), \quad \mathbf{p} \in \mathbb{R}^2 \right\}. $$

Here, *Euclidean* indicates rigid motions that preserve Euclidean distances and angles. Matrix multiplication is the group operation, the 3×3 identity $\mathbf{I}_3$ is the identity motion, and every element has an inverse that reverses the coordinate mapping. Section 6.1.6 derives that inverse explicitly.

6.1.5 Composing transformations

Introduce a third frame $\{c\}$. Let $\mathbf{T}_{ab} \in SE(2)$ map coordinates from frame $\{b\}$ to frame $\{a\}$, and let $\mathbf{T}_{bc} \in SE(2)$ map coordinates from frame $\{c\}$ to frame $\{b\}$. Applying the two mappings to a point gives

$$ {}^{a}\overline{\mathbf{p}}_P = \mathbf{T}_{ab}\left(\mathbf{T}_{bc}\,{}^{c}\overline{\mathbf{p}}_P\right) = \left(\mathbf{T}_{ab}\mathbf{T}_{bc}\right){}^{c}\overline{\mathbf{p}}_P. $$

Therefore, the transform that maps directly from frame $\{c\}$ to frame $\{a\}$ is

$$ \boxed{\mathbf{T}_{ac} = \mathbf{T}_{ab}\mathbf{T}_{bc}}. $$

The adjacent frame label b identifies the intermediate frame. Expanding the block multiplication gives

$$ \begin{aligned} \mathbf{T}_{ab}\mathbf{T}_{bc} &= \begin{bmatrix} \mathbf{R}_{ab} & {}^{a}\mathbf{p}_b \\ \mathbf{0}_2^\top & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_{bc} & {}^{b}\mathbf{p}_c \\ \mathbf{0}_2^\top & 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{R}_{ab}\mathbf{R}_{bc} & \mathbf{R}_{ab}\,{}^{b}\mathbf{p}_c + {}^{a}\mathbf{p}_b \\ \mathbf{0}_2^\top & 1 \end{bmatrix}. \end{aligned} $$

Consequently,

$$\boxed{\mathbf{R}_{ac} = \mathbf{R}_{ab}\mathbf{R}_{bc}, \qquad {}^{a}\mathbf{p}_{c} = \mathbf{R}_{ab}\,{}^{b}\mathbf{p}_{c} + {}^{a}\mathbf{p}_{b}}.$$

The translation vectors cannot be added directly until they are expressed in the same frame. The product first re-expresses ${}^{b}\mathbf{p}_{c}$ in frame $\{a\}$ and only then adds ${}^{a}\mathbf{p}_{b}$.

6.1.6 Inverting a transformation

The inverse transform $\mathbf{T}_{ba} = \mathbf{T}_{ab}^{-1}$ must undo both the translation and the rotation. Starting from the point mapping

$${}^{a}\mathbf{p}_{P} = \mathbf{R}_{ab}\,{}^{b}\mathbf{p}_{P} + {}^{a}\mathbf{p}_{b},$$

subtract ${}^{a}\mathbf{p}_{b}$ and premultiply by $\mathbf{R}_{ab}^{-1} = \mathbf{R}_{ab}^{\mathsf{T}}$:

$${}^{b}\mathbf{p}_{P} = \mathbf{R}_{ab}^{\mathsf{T}}\left({}^{a}\mathbf{p}_{P} - {}^{a}\mathbf{p}_{b}\right).$$

Distributing the rotation gives

$${}^{b}\mathbf{p}_{P} = \mathbf{R}_{ab}^{\mathsf{T}}\,{}^{a}\mathbf{p}_{P} - \mathbf{R}_{ab}^{\mathsf{T}}\,{}^{a}\mathbf{p}_{b}.$$

Therefore,

$$\boxed{\mathbf{T}_{ab}^{-1} = \mathbf{T}_{ba} = \begin{bmatrix} \mathbf{R}_{ab}^{\mathsf{T}} & -\mathbf{R}_{ab}^{\mathsf{T}}\,{}^{a}\mathbf{p}_{b} \\ \mathbf{0}_{2}^{\mathsf{T}} & 1 \end{bmatrix}}.$$

The inverse translation is not generally $-{}^{a}\mathbf{p}_{b}$ because that column is expressed in frame $\{a\}$. It must first be negated and then expressed in frame $\{b\}$ by $\mathbf{R}_{ab}^{\mathsf{T}}$.

6.1.7 Pose coordinates versus the pose itself

The familiar column

$$\mathbf{q} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$$

has the following meaning:

- $x \in \mathbb{R}$ is the x_w-coordinate of the body origin O_b, expressed in the world frame and measured in metres;
- $y \in \mathbb{R}$ is the y_w-coordinate of O_b, expressed in the world frame and measured in metres; and
- $\theta \in \mathbb{R}$ is the counter-clockwise orientation of body frame $\{b\}$ relative to world frame $\{w\}$, measured in radians.

Therefore, the position of the body origin relative to the world origin, expressed in world-frame coordinates, is

$$^{w}\mathbf{p}_b = \begin{bmatrix} x \\ y \end{bmatrix}.$$

The notation $^{w}\mathbf{p}_b$ means "the position of the body origin O_b, expressed in world-frame coordinates." Thus, $\mathbf{q}$ is a coordinate description of the pose of frame $\{b\}$ relative to frame $\{w\}$. The same pose can be represented by $\mathbf{T}_{wb} \in SE(2)$:

$$\boxed{\mathbf{q} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} \quad \longmapsto \quad \mathbf{T}_{wb}(\mathbf{q}) = \begin{bmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{bmatrix}}.$$

The coordinate column is useful for storage, plotting, and differential equations, but it is not an ordinary geometric vector. Its components represent different physical quantities, θ and $\theta + 2\pi n$ represent the same orientation, and componentwise addition does not in general reproduce rigid-motion composition. Transformations should be composed by matrix multiplication with explicit frame labels.

6.1.8 Worked example

Suppose the body origin is at

$$^{w}\mathbf{p}_b = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{m}$$

and the body heading is $\theta = \pi/2\,\text{rad}$. A point P fixed one metre along the body $+x_b$-axis has body-frame coordinates

$$^{b}\mathbf{p}_P = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{m}.$$

The rotation and homogeneous transformation are

$$\mathbf{R}_{wb}\left(\frac{\pi}{2}\right) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{T}_{wb} = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Applying the transform gives

$$\begin{aligned} {}^{w}\overline{\mathbf{p}}_P &= \mathbf{T}_{wb}\,{}^{b}\overline{\mathbf{p}}_P \\ &= \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}. \end{aligned}$$

Thus, ${}^{w}\mathbf{p}_P = [2\ 2]^\mathsf{T}\,\mathrm{m}$. The one-metre body-frame displacement is rotated into the world $+y_w$-direction and then added to the body origin at $[2\ 1]^\mathsf{T}\,\mathrm{m}$.

Quick test A: pose and $SE(2)$

1. What geometric information is contained in the pair $(\mathbf{R}_{wb}, {}^{w}\mathbf{p}_b)$, and what does the subscript order wb mean?
2. Define $SO(2)$. What do its orthogonality and determinant conditions exclude or guarantee?
3. Why do θ and $\theta + 2\pi$ describe the same orientation?
4. Starting from the block definitions of $\mathbf{T}_{ab}$ and $\mathbf{T}_{bc}$, derive the translation column of $\mathbf{T}_{ac} = \mathbf{T}_{ab}\mathbf{T}_{bc}$.
5. Why is the inverse translation $-\mathbf{R}_{ab}^\mathsf{T}\,{}^{a}\mathbf{p}_b$ rather than simply $-{}^{a}\mathbf{p}_b$?
6. A body frame has world position $[3\ 2]^\mathsf{T}\,\mathrm{m}$ and heading $-\pi/2\,\mathrm{rad}$. Find the world coordinates of a point whose body coordinates are $[0.5\ 0]^\mathsf{T}\,\mathrm{m}$.
7. Explain why $\mathbf{q}_1 + \mathbf{q}_2$ is not generally the composition of the two poses represented by $\mathbf{q}_1$ and $\mathbf{q}_2$.
8. Name two transformations that cannot be represented by an element of $SE(2)$ and explain why.

Answers and hints

1. $\mathbf{R}_{wb}$ describes the orientation of frame $\{b\}$ relative to frame $\{w\}$, while ${}^{w}\mathbf{p}_b$ locates O_b relative to O_w using world coordinates. The order wb means that the rotation maps coordinate columns from $\{b\}$ to $\{w\}$.
2. $SO(2) = \{\mathbf{R} \in \mathbb{R}^{2\times 2} \mid \mathbf{R}^\mathsf{T}\mathbf{R} = \mathbf{I}_2, \det \mathbf{R} = 1\}$. Orthogonality preserves lengths and angles and gives $\mathbf{R}^{-1} = \mathbf{R}^\mathsf{T}$; determinant $+1$ preserves orientation and excludes reflections.

3. Sine and cosine are 2π-periodic, so every entry of $\mathbf{R}(\theta + 2\pi)$ equals the corresponding entry of $\mathbf{R}(\theta)$.
4. Block multiplication gives ${}^{a}\mathbf{p}_c = \mathbf{R}_{ab}{}^{b}\mathbf{p}_c + {}^{a}\mathbf{p}_b$. The first term re-expresses the b-frame translation in frame $\{a\}$ before the two translations are added.
5. The reversed displacement must be expressed in frame $\{b\}$, whereas ${}^{a}\mathbf{p}_b$ is expressed in frame $\{a\}$. Premultiplication by $\mathbf{R}_{ab}^{\mathsf{T}}$ performs the necessary coordinate change.
6. $\mathbf{R}_{wb}(-\pi/2)[0.5\ 0]^{\mathsf{T}} = [0\ \ -0.5]^{\mathsf{T}}$ m, so ${}^{w}\mathbf{p}_P = [3\ 1.5]^{\mathsf{T}}$ m.
7. Rigid-motion composition must account for the frame in which each translation is expressed and for the order of rotations and translations. Componentwise addition does neither, and angular coordinates are periodic.
8. Examples include nonuniform scaling, shear, and elastic deformation. They do not preserve every distance and angle, so they are not rigid motions and do not belong to $SE(2)$.

6.2 Passive coordinate changes and active rigid motions

The same numerical rotation or homogeneous-transformation matrix can appear in different geometric questions. Correct interpretation requires identifying what remains fixed, what changes, and the frame in which every coordinate column is expressed.

6.2.1 Passive coordinate change

A **passive coordinate change** keeps the geometric point, vector, or rigid body unchanged and changes only the coordinate frame used to describe it. The word *passive* indicates that nothing physically moves.

For example, let the geometric point P remain fixed while its coordinates are changed from frame $\{b\}$ to frame $\{w\}$. The passive point-coordinate transformation is

$${}^{w}\overline{\mathbf{p}}_P = \mathbf{T}_{wb}\,{}^{b}\overline{\mathbf{p}}_P.$$

The two coordinate columns ${}^{b}\overline{\mathbf{p}}_P$ and ${}^{w}\overline{\mathbf{p}}_P$ contain different numbers in general, but both describe the same geometric point P. The transform $\mathbf{T}_{wb}$ accounts for the position and orientation of frame $\{b\}$ relative to frame $\{w\}$.

For a free vector $\mathbf{v}$, translation is irrelevant and only the rotation block is used:

$${}^{w}\mathbf{v} = \mathbf{R}_{wb}\,{}^{b}\mathbf{v}.$$

Suppose frame $\{b\}$ is rotated counter-clockwise by $\pi/2$ relative to frame $\{w\}$ and the vector lies along $+x_b$. Its world-frame coordinates are

$$
\begin{aligned}
{}^{w}\mathbf{v} &= \mathbf{R}_{wb}\,{}^{b}\mathbf{v} \\
&= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ 1 \end{bmatrix}.
\end{aligned}
$$

The vector has not rotated physically. It is the same geometric vector described once using the axes of $\{b\}$ and once using the axes of $\{w\}$.

6.2.2 Active rigid-body motion

An **active transformation** keeps the reference frame fixed and moves a geometric point, vector, or rigid body. The initial point P and final point P' are different geometric points, but both coordinate columns are expressed in the same frame.

Let $\mathbf{R}_\Delta \in SO(2)$ be an applied planar rotation and let $\mathbf{d}_\Delta \in \mathbb{R}^2$ be an applied translation expressed in world-frame coordinates and measured in metres. Define the active rigid-motion operator

$$
\mathbf{T}_\Delta := \begin{bmatrix} \mathbf{R}_\Delta & \mathbf{d}_\Delta \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix} \in SE(2).
$$

Applied to point P, it produces the new point P' according to

$$
\boxed{{}^{w}\overline{\mathbf{p}}_{P'} = \mathbf{T}_\Delta\,{}^{w}\overline{\mathbf{p}}_{P}},
$$

or, using ordinary two-dimensional coordinates,

$$
{}^{w}\mathbf{p}_{P'} = \mathbf{R}_\Delta\,{}^{w}\mathbf{p}_P + \mathbf{d}_\Delta.
$$

Both sides are expressed in frame $\{w\}$. The point changes from P to P', unlike a passive coordinate change, where the point remains P and the leading frame superscript changes.

For example, let $\mathbf{R}_\Delta = \mathbf{R}(\pi/2)$, $\mathbf{d}_\Delta = [2\ 1]^\mathsf{T}$ m, and ${}^{w}\mathbf{p}_P = [1\ 0]^\mathsf{T}$ m. Then

$$
{}^{w}\mathbf{p}_{P'} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \text{m}.
$$

Here, the coordinates changed because the geometric point moved, not because a different coordinate frame was selected.

6.2.3 One matrix form, three interpretations

A homogeneous transformation may describe a relative pose, perform a passive coordinate change, or act as an active rigid-motion operator. The matrix arithmetic can be identical while the geometric meaning differs.

Interpretation	What changes?	Representative statement
Relative-pose description	Nothing is being operated on; the matrix records the pose of one frame relative to another.	$\mathbf{T}_{wb}$ describes frame $\{b\}$ relative to frame $\{w\}$.
Passive coordinate change	The coordinate column and its frame label change; the geometric point remains fixed.	${}^{w}\overline{\mathbf{p}}_{P} = \mathbf{T}_{wb}{}^{b}\overline{\mathbf{p}}_{P}$.
Active rigid motion	The geometric point changes; the coordinate frame remains fixed.	${}^{w}\overline{\mathbf{p}}_{P'} = \mathbf{T}_{\Delta}{}^{w}\overline{\mathbf{p}}_{P}$.

The project notation reserves labelled transforms such as $\mathbf{T}_{wb}$ for relative poses and coordinate mappings. A symbol such as $\mathbf{T}_{\Delta}$ must be explicitly declared when the same matrix form is used as an active motion operator.

6.2.4 Composition order

Transformations act on column vectors from the left, so the rightmost matrix acts first. For a passive coordinate chain,

$$
{}^{a}\overline{\mathbf{p}}_{P} = \mathbf{T}_{ab}\mathbf{T}_{bc}{}^{c}\overline{\mathbf{p}}_{P},
$$

the rightmost transform $\mathbf{T}_{bc}$ first changes coordinates from frame $\{c\}$ to frame $\{b\}$, and $\mathbf{T}_{ab}$ then changes them from frame $\{b\}$ to frame $\{a\}$. Thus,

$$
\mathbf{T}_{ac} = \mathbf{T}_{ab}\mathbf{T}_{bc}.
$$

For two active motions $\mathbf{T}_{1}$ and $\mathbf{T}_{2}$,

$$
{}^{w}\overline{\mathbf{p}}_{P''} = \mathbf{T}_{2}\mathbf{T}_{1}{}^{w}\overline{\mathbf{p}}_{P},
$$

motion $\mathbf{T}_{1}$ occurs first and motion $\mathbf{T}_{2}$ occurs second. Associativity permits the parentheses in a longer product to be regrouped, but it does not permit the matrix order to be reversed.

6.2.5 Why composition in $SE(2)$ is not commutative

Planar rotations alone commute, but planar rigid transformations generally do not because rotation changes the direction of a translation. Consider a translation of one metre along $+x_w$ and a counter-clockwise rotation of $\pi/2$ about O_w:

$$\mathbf{T}_D = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{T}_R = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Translate first and rotate second:

$$\mathbf{T}_R\mathbf{T}_D = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Rotate first and translate second:

$$\mathbf{T}_D\mathbf{T}_R = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The products differ. Applied to the world origin ${}^{w}\overline{\mathbf{p}}_O = [0\ 0\ 1]^\mathsf{T}$, they give

$$\mathbf{T}_R\mathbf{T}_D\,{}^{w}\overline{\mathbf{p}}_O = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \qquad \mathbf{T}_D\mathbf{T}_R\,{}^{w}\overline{\mathbf{p}}_O = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$\boxed{\mathbf{T}_R\mathbf{T}_D \neq \mathbf{T}_D\mathbf{T}_R}.$$

This noncommutativity is why a transformation sequence must state both the individual motions and their order.

6.2.6 Interpretation failures and engineering checks

- **Missing frame labels:** An equation such as $\mathbf{p}' = \mathbf{T}\mathbf{p}$ does not reveal whether $\mathbf{T}$ changes coordinates or moves the point. Declare the interpretation and frames.
- **Applying transformations in the wrong order:** For column-vector notation, transformations act from right to left. In $\mathbf{T}_2\mathbf{T}_1\mathbf{p}$, $\mathbf{T}_1$ acts first and $\mathbf{T}_2$ acts second. Verify an implementation using a point whose expected final location is easy to calculate.
- **Mixing coordinate frames:** Translation columns may be added only after they are expressed in the same frame.
- **Assuming wrong commutativity:** Reordering a rotation and translation generally changes the resulting pose.

Quick test B: interpretation and composition

1. Define a passive coordinate change. What changes, and what remains fixed?
2. Define an active rigid-body motion. What changes, and what remains fixed?
3. Explain the three interpretations of a homogeneous transformation used in this chapter.
4. In $\mathbf{T}_{ab}\mathbf{T}_{bc}{}^{c}\overline{\mathbf{p}}_P$, which matrix acts first, and what frame sequence does the product represent?
5. For active motions, what sequence is represented by $\mathbf{T}_2\mathbf{T}_1{}^{w}\overline{\mathbf{p}}_P$?
6. Using $\mathbf{T}_D$ and $\mathbf{T}_R$ from Section 6.2.5, explain geometrically why applying them to the origin in opposite orders gives different results.
7. Does the commutativity of planar rotations imply that all elements of $SE(2)$ commute? Explain.
8. A program receives an unlabelled matrix $\mathbf{T}$ and an unlabelled coordinate column $\mathbf{p}$. What information is missing before the multiplication can be interpreted safely?

Answers and hints

1. A passive coordinate change preserves the geometric object and changes only the frame used to express its coordinates. The coordinate values and frame label may change.
2. An active rigid-body motion preserves the reference frame and moves the geometric object. The initial and final coordinates describe different points or poses in the same frame.
3. A homogeneous transformation can describe the pose of one frame relative to another, map coordinates of an unchanged object between frames, or actively move an object while its coordinates remain expressed in one frame.
4. $\mathbf{T}_{bc}$ acts first, mapping from $\{c\}$ to $\{b\}$; $\mathbf{T}_{ab}$ then maps from $\{b\}$ to $\{a\}$. The complete product maps from $\{c\}$ to $\{a\}$.
5. $\mathbf{T}_1$ moves the point first, and $\mathbf{T}_2$ moves the resulting point second.
6. Translating the origin first places it at $[1\ 0]^\mathsf{T}$, which the subsequent rotation moves to $[0\ 1]^\mathsf{T}$. Rotating the origin first leaves it at the origin, after which translation places it at $[1\ 0]^\mathsf{T}$.

7. No. Although $SO(2)$ rotations commute with each other, an $SE(2)$ transformation also contains translation. Rotation changes the direction of a translation, so general rotation-translation products do not commute.
8. The program needs to know whether $\mathbf{p}$ represents a point or free vector, the frame in which it is expressed, the source and destination frames or active-motion interpretation of $\mathbf{T}$, the units, and the intended multiplication order.

6.3 Time-varying rotations and rigid-body point velocity

A pose describes where a rigid body is at one instant. A time-varying pose describes motion. This section differentiates the position and orientation components of $\mathbf{T}_{wb}(t)$ and derives the velocity of any point fixed to the body.

6.3.1 Translational and angular velocity

Let the planar body pose vary continuously with time:

$$\mathbf{T}_{wb}(t) = \begin{bmatrix} \mathbf{R}_{wb}(t) & {}^{w}\mathbf{p}_b(t) \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix} \in SE(2),$$

where $t \in \mathbb{R}_{\geq 0}$ is time in seconds. The world-frame position of the body origin is

$${}^{w}\mathbf{p}_b(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} \in \mathbb{R}^2.$$

Its derivative, taken with respect to the fixed world frame and expressed in world-frame coordinates, is the translational velocity of O_b:

$$\boxed{{}^{w}\mathbf{v}_b(t) := \frac{d}{dt}{}^{w}\mathbf{p}_b(t) = \begin{bmatrix} \dot{x}(t) \\ \dot{y}(t) \end{bmatrix}} \in \mathbb{R}^2.$$

The entries of ${}^{w}\mathbf{v}_b$ are measured in $\mathrm{m\,s^{-1}}$. The heading is $\theta(t) \in \mathbb{R}$, measured in radians, and its derivative is the signed yaw rate

$$\boxed{\omega(t) := \dot{\theta}(t)} \in \mathbb{R},$$

measured in $\mathrm{rad\,s^{-1}}$. Positive ω denotes counter-clockwise rotation about the common $+z$-axis of the planar frames.

6.3.2 Deriving the rotation-matrix derivative

The time-varying planar rotation matrix is

$$\mathbf{R}_{wb}(t) = \begin{bmatrix} \cos\theta(t) & -\sin\theta(t) \\ \sin\theta(t) & \cos\theta(t) \end{bmatrix}.$$

Differentiate every entry and apply the chain rule:

$$\begin{aligned} \dot{\mathbf{R}}_{wb}(t) &= \dot{\theta}(t) \begin{bmatrix} -\sin\theta(t) & -\cos\theta(t) \\ \cos\theta(t) & -\sin\theta(t) \end{bmatrix} \\ &= \omega(t) \begin{bmatrix} -\sin\theta(t) & -\cos\theta(t) \\ \cos\theta(t) & -\sin\theta(t) \end{bmatrix}. \end{aligned}$$

A square matrix $\mathbf{A}$ is **skew-symmetric** when its transpose equals its negative:

$$\boxed{\mathbf{A}^\mathsf{T} = -\mathbf{A}}.$$

Transposition reflects the entries across the main diagonal. The condition therefore means that entries mirrored across the diagonal have equal magnitudes and opposite signs; every diagonal entry must be zero. Define the dimensionless planar skew-symmetric generator

$$\boxed{\mathbf{S}_2 := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}}.$$

The matrix $\mathbf{S}_2$ is skew-symmetric because $\mathbf{S}_2^\mathsf{T} = -\mathbf{S}_2$. Premultiplication by $\mathbf{S}_2$ rotates any planar coordinate column counter-clockwise by $\pi/2$:

$$\mathbf{S}_2 \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -b \\ a \end{bmatrix}.$$

Direct multiplication gives

$$\mathbf{S}_2\mathbf{R}_{wb}(\theta) = \mathbf{R}_{wb}(\theta)\mathbf{S}_2 = \begin{bmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{bmatrix}.$$

Therefore,

$$\boxed{\dot{\mathbf{R}}_{wb} = \omega\mathbf{S}_2\mathbf{R}_{wb} = \mathbf{R}_{wb}\omega\mathbf{S}_2}.$$

The two products are equal in the planar case because every element of $SO(2)$ commutes with $\mathbf{S}_2$. This equality is a special simplification of planar rotation; matrix order becomes important for three-dimensional angular velocity.

6.3.3 Why $\dot{\mathrm{R}}_{wb}$ is not a rotation matrix

Although $\mathbf{R}_{wb}(t) \in SO(2)$ at every time t, its derivative $\dot{\mathbf{R}}_{wb}(t)$ is generally not a rotation matrix. A rotation matrix is a dimensionless map that represents a finite change of orientation, whereas $\dot{\mathbf{R}}_{wb}$ measures how rapidly that map is changing and has entries with units of s^{-1}. The derivative therefore describes an instantaneous direction of rotational motion rather than another finite rotation.

6.3.3.1 The rotation constraint restricts its derivative

Every planar rotation matrix satisfies

$$\mathbf{R}_{wb}^{\mathsf{T}}\mathbf{R}_{wb} = \mathbf{I}_2.$$

Differentiating this constraint with respect to time gives

$$\dot{\mathbf{R}}_{wb}^{\mathsf{T}}\mathbf{R}_{wb} + \mathbf{R}_{wb}^{\mathsf{T}}\dot{\mathbf{R}}_{wb} = \mathbf{0}_{2\times 2}.$$

Rearranging shows that

$$\left(\mathbf{R}_{wb}^{\mathsf{T}}\dot{\mathbf{R}}_{wb}\right)^{\mathsf{T}} = -\mathbf{R}_{wb}^{\mathsf{T}}\dot{\mathbf{R}}_{wb}.$$

Thus, $\mathbf{R}_{wb}^{\mathsf{T}}\dot{\mathbf{R}}_{wb}$ is **skew-symmetric**: its transpose equals its negative. The factor $\mathbf{R}_{wb}^{\mathsf{T}}$ expresses the rotational rate relative to the current body orientation; it does not turn $\dot{\mathbf{R}}_{wb}$ into a finite rotation.

The set of all 2×2 real skew-symmetric matrices is

$$\mathfrak{so}(2) := \left\{\mathbf{A} \in \mathbb{R}^{2\times 2} \mid \mathbf{A}^{\mathsf{T}} = -\mathbf{A}\right\} = \left\{\alpha\mathbf{S}_2 \mid \alpha \in \mathbb{R}\right\},$$

where α is a real scalar and

$$\mathbf{S}_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

To justify the second equality, begin with a general matrix

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad a, b, c, d \in \mathbb{R}.$$

The condition $\mathbf{A}^{\mathsf{T}} = -\mathbf{A}$ requires corresponding entries to satisfy

$$a = -a, \qquad d = -d, \qquad c = -b.$$

Therefore, $a = d = 0$. Setting $\alpha = c = -b$ gives

$$\mathbf{A} = \begin{bmatrix} 0 & -\alpha \\ \alpha & 0 \end{bmatrix} = \alpha \mathbf{S}_2.$$

Thus, every 2×2 skew-symmetric matrix is a scalar multiple of $\mathbf{S}_2$. Conversely, for every $\alpha \in \mathbb{R}$,

$$(\alpha \mathbf{S}_2)^\mathsf{T} = \alpha \mathbf{S}_2^\mathsf{T} = -\alpha \mathbf{S}_2,$$

so every scalar multiple of $\mathbf{S}_2$ is skew-symmetric. The two sets are therefore equal. The notation $\mathfrak{so}(2)$ is read "little s o two."

6.3.3.2 Why these matrices form a tangent space

The skew-symmetric matrix $\mathbf{S}_2$ is tangent to $SO(2)$ at the identity rotation $\mathbf{I}_2$. More generally, every scalar multiple $\alpha \mathbf{S}_2$, where $\alpha \in \mathbb{R}$, is a possible tangent direction there. This subsection explains the geometric meaning of that statement and proves that these are all the possible tangent directions.

A **tangent direction** to a curve at a point is the limiting direction of secant lines drawn from that point to nearby points on the curve. Equivalently, it is the derivative at that point of a differentiable path that remains on the curve. This familiar definition applies to $SO(2)$ because every rotation matrix $\mathbf{R}(\theta)$ is determined by the pair $(\cos\theta, \sin\theta)$ on the unit circle. Under this correspondence,

$$\mathbf{R}(0) = \mathbf{I}_2 \quad \longleftrightarrow \quad (\cos 0, \sin 0) = (1, 0).$$

For a nonzero angular increment h measured in radians, the secant direction of the unit circle from $(1, 0)$ to $(\cos h, \sin h)$ is

$$\frac{1}{h}\left(\begin{bmatrix} \cos h \\ \sin h \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right).$$

As $h \to 0$, this secant direction approaches $[0\ 1]^\mathsf{T}$, the usual tangent direction to the circle at $(1, 0)$. The corresponding difference quotient in matrix coordinates is

$$\frac{\mathbf{R}(h) - \mathbf{I}_2}{h} = \begin{bmatrix} \dfrac{\cos h - 1}{h} & -\dfrac{\sin h}{h} \\ \dfrac{\sin h}{h} & \dfrac{\cos h - 1}{h} \end{bmatrix}.$$

The scalar limits

$$\lim_{h\to 0}\frac{\sin h}{h}=1,\qquad \lim_{h\to 0}\frac{\cos h-1}{h}=0$$

give

$$\boxed{\left.\frac{d\mathbf{R}(h)}{dh}\right|_{h=0}=\lim_{h\to 0}\frac{\mathbf{R}(h)-\mathbf{I}_2}{h}=\mathbf{S}_2}.$$

The second scalar limit follows from $\cos h - 1 = -2\sin^2(h/2)$, the first limit, and the continuity of sine at zero. Therefore, $\mathbf{S}_2$ is tangent to the rotation curve at $\mathbf{I}_2$ in exactly the same sense that $[0\ 1]^\mathsf{T}$ is tangent to the unit circle at $(1, 0)$.

To collect every possible tangent direction, let $\mathbf{Q}(s) \in SO(2)$ be any differentiable rotation curve defined for the dimensionless real parameter s near zero, with $\mathbf{Q}(0) = \mathbf{I}_2$. The **tangent space of $SO(2)$ at $\mathbf{I}_2$** is

$$T_{\mathbf{I}_2}SO(2) := \left\{ \left.\frac{d\mathbf{Q}(s)}{ds}\right|_{s=0} \;\middle|\; \mathbf{Q}(s)\in SO(2),\ \mathbf{Q}(0)=\mathbf{I}_2 \right\}.$$

The symbol T denotes a tangent space, and its subscript $\mathbf{I}_2$ identifies the point where the tangent directions are attached. To prove that this space equals $\mathfrak{so}(2)$, both set inclusions must be established: every tangent direction must be shown to be skew-symmetric, and every skew-symmetric matrix must be shown to arise from a valid rotation curve.

First, take any $\mathbf{A} \in T_{\mathbf{I}_2}SO(2)$. By definition, $\mathbf{A} = \mathbf{Q}'(0)$ for some rotation curve $\mathbf{Q}(s)$ through $\mathbf{I}_2$. Differentiating $\mathbf{Q}(s)^\mathsf{T}\mathbf{Q}(s) = \mathbf{I}_2$ gives

$$\mathbf{Q}'(s)^\mathsf{T}\mathbf{Q}(s)+\mathbf{Q}(s)^\mathsf{T}\mathbf{Q}'(s)=\mathbf{0}_{2\times 2}.$$

Now evaluate at (s=0). Because the curve passes through the identity,

$$\mathbf{Q}(0)=\mathbf{I}_2,\qquad \mathbf{Q}'(0)=\mathbf{A}.$$

Substitution gives

$$\mathbf{A}^\mathsf{T}+\mathbf{A}=\mathbf{0}_{2\times 2}.$$

Thus, $\mathbf{A}$ is skew-symmetric and belongs to $\mathfrak{so}(2)$. This proves

$$T_{\mathbf{I}_2}SO(2)\subseteq \mathfrak{so}(2).$$

Second, take any $\mathbf{A} \in \mathfrak{so}(2)$. It has the form $\mathbf{A} = \alpha\mathbf{S}_2$ for some $\alpha \in \mathbb{R}$. The curve $\mathbf{Q}(s) = \mathbf{R}(\alpha s)$ remains in $SO(2)$, passes through $\mathbf{I}_2$ at $s = 0$, and satisfies

$$\mathbf{Q}'(0) = \frac{d}{ds}\mathbf{R}(\alpha s)\bigg|_{s=0} = \alpha\mathbf{S}_2 = \mathbf{A}.$$

Hence, every element of $\mathfrak{so}(2)$ is produced by a valid curve through the identity, which proves

$$\mathfrak{so}(2) \subseteq T_{\mathbf{I}_2}SO(2).$$

The two inclusions establish

$$\boxed{T_{\mathbf{I}_2}SO(2) = \mathfrak{so}(2) = \{\alpha\mathbf{S}_2 \mid \alpha \in \mathbb{R}\}}.$$

This proves that the skew-symmetric matrices in $\mathfrak{so}(2)$ are exactly the instantaneous directions in which a rotation curve can pass through $\mathbf{I}_2$. They describe rotational directions, not finite rotations. The next subsection shows how these directions provide a first-order approximation to nearby rotation matrices.

6.3.3.3 Lie-group terminology and the local linear model

Section 6.1.2 established that $SO(2)$ is a group under matrix multiplication. Its matrices also vary smoothly with the continuous angle θ, and its multiplication and inverse operations are differentiable. A group with such a compatible smooth structure is called a **Lie group**. The tangent space at the identity of a Lie group, together with an operation called the Lie bracket, is its **Lie algebra**. Thus, $\mathfrak{so}(2)$ is the Lie algebra of $SO(2)$.

The Lie algebra is useful because it is linear even though the rotation group is not. For any $\alpha, \beta, c \in \mathbb{R}$,

$$\alpha\mathbf{S}_2 + \beta\mathbf{S}_2 = (\alpha + \beta)\mathbf{S}_2, \qquad c(\alpha\mathbf{S}_2) = (c\alpha)\mathbf{S}_2,$$

and both results remain in $\mathfrak{so}(2)$. Instantaneous rotational directions can therefore be added and scaled using ordinary linear algebra, whereas adding or scaling finite rotation matrices does not generally produce valid rotations.

The tangent space is also a local linear approximation of the group. For a small angle h,

$$\boxed{\mathbf{R}(h) = \mathbf{I}_2 + h\mathbf{S}_2 + O(h^2)}.$$

This result follows more directly from the first-order Taylor expansion of the matrix-valued function $\mathbf{R}(h)$ about $h = 0$:

$$\mathbf{R}(h) = \mathbf{R}(0) + h\left.\frac{d\mathbf{R}(h)}{dh}\right|_{h=0} + O(h^2).$$

The definition of $\mathbf{R}(h)$ and the preceding limit give

$$\mathbf{R}(0) = \mathbf{I}_2, \qquad \left.\frac{d\mathbf{R}(h)}{dh}\right|_{h=0} = \mathbf{S}_2.$$

Substituting these two values immediately gives $\mathbf{R}(h) = \mathbf{I}_2 + h\mathbf{S}_2 + O(h^2)$. The derivative is written $d\mathbf{R}/dh$ rather than $\dot{\mathbf{R}}$ because h is an angular argument, not time. For this matrix equation, $O(h^2)$ means that the norm of the Taylor remainder is bounded by $C|h|^2$ for some constant $C > 0$ when h is sufficiently close to zero; this bound exists because the sine and cosine entries of $\mathbf{R}(h)$ have continuous second derivatives near $h = 0$.

If the same rotation increment is instead parameterised by time as $h(t) = \omega t$ with constant angular velocity ω, then the time-domain Taylor expansion over a short interval Δt is

$$\mathbf{R}(\omega\Delta t) = \mathbf{R}(0) + \Delta t\,\dot{\mathbf{R}}(0) + O(\Delta t^2) = \mathbf{I}_2 + \omega\Delta t\,\mathbf{S}_2 + O(\Delta t^2),$$

because $\dot{\mathbf{R}}(0) = \omega\mathbf{S}_2$. This is the time-parameterised version of the same first-order approximation.

In the angle-domain approximation, $\mathbf{I}_2 + h\mathbf{S}_2$ starts at $\mathbf{I}_2$ and changes by $h\mathbf{S}_2$ along the tangent direction. It is generally not an exact rotation matrix, but its difference from the true rotation is only of order h^2. The approximation therefore matches both the value and the first derivative of the rotation curve at $h = 0$.

6.3.3.4 Returning to the physical rotation trajectory

For the physical trajectory $\mathbf{R}_{wb}(t)$, Section 6.3.2 established

$$\dot{\mathbf{R}}_{wb} = \omega\mathbf{R}_{wb}\mathbf{S}_2.$$

Thus, $\dot{\mathbf{R}}_{wb}$ is tangent to $SO(2)$ at the current rotation $\mathbf{R}_{wb}$, rather than at the identity. Premultiplication by $\mathbf{R}_{wb}^\mathsf{T}$ carries this direction back to the tangent space at the identity:

$$\boxed{\boldsymbol{\Omega}_b := \mathbf{R}_{wb}^\mathsf{T}\dot{\mathbf{R}}_{wb} = \omega\mathbf{S}_2 \in \mathfrak{so}(2)}.$$

Here, $\boldsymbol{\Omega}_b$ is the body-frame angular-velocity matrix, and its entries have units of s^{-1}. The time-domain Taylor expansion above connects this instantaneous generator $\omega\mathbf{S}_2$ to a nearby finite rotation. The derivative is not itself a rotation matrix; it specifies the tangent direction from which the finite rotation develops over time.

Finally, the **Lie bracket** measures whether two infinitesimal matrix motions commute. For matrices it is the commutator

$$[\mathbf{A}, \mathbf{B}] := \mathbf{AB} - \mathbf{BA}, \qquad \mathbf{A}, \mathbf{B} \in \mathfrak{so}(2).$$

For $\mathfrak{so}(2)$, write $\mathbf{A} = \alpha \mathbf{S}_2$ and $\mathbf{B} = \beta \mathbf{S}_2$. Then $\mathbf{AB} = \mathbf{BA}$, so $[\mathbf{A}, \mathbf{B}] = \mathbf{0}_{2\times 2}$. The bracket is trivial for planar rotations because they commute, but it becomes important for rigid motions in $SE(2)$ and rotations in $SO(3)$, where motion order generally matters.

6.3.4 Velocity of a point fixed to the body

Let P be a point rigidly fixed to the body. Its body-frame coordinates ${}^b\mathbf{p}_P \in \mathbb{R}^2$ are constant, so

$$\frac{d}{dt}{}^b\mathbf{p}_P = \mathbf{0}_2.$$

Its world-frame position is

$${}^w\mathbf{p}_P(t) = \mathbf{R}_{wb}(t){}^b\mathbf{p}_P + {}^w\mathbf{p}_b(t).$$

Differentiate both sides:

$$\begin{aligned} {}^w\mathbf{v}_P := \frac{d}{dt}{}^w\mathbf{p}_P &= \dot{\mathbf{R}}_{wb}{}^b\mathbf{p}_P + \mathbf{R}_{wb}\frac{d}{dt}{}^b\mathbf{p}_P + {}^w\mathbf{v}_b \\ &= \dot{\mathbf{R}}_{wb}{}^b\mathbf{p}_P + {}^w\mathbf{v}_b. \end{aligned}$$

Define the relative-position vector from O_b to P, expressed in world-frame coordinates, by

$${}^w\mathbf{r}_{O_bP} := {}^w\mathbf{p}_P - {}^w\mathbf{p}_b = \mathbf{R}_{wb}{}^b\mathbf{p}_P \in \mathbb{R}^2.$$

Using $\dot{\mathbf{R}}_{wb} = \omega \mathbf{S}_2 \mathbf{R}_{wb}$ gives

$$\boxed{{}^w\mathbf{v}_P = {}^w\mathbf{v}_b + \omega \mathbf{S}_2{}^w\mathbf{r}_{O_bP}}.$$

The first term is the translational velocity shared with the body origin. The second term is the additional velocity caused by rotation about the body origin. Because $\mathbf{S}_2$ rotates ${}^w\mathbf{r}_{O_bP}$ by $\pi/2$, the rotational contribution is perpendicular to the relative-position vector.

The two panels in Figure 6.1 separate the position-vector identity from the velocity-vector addition. In the left panel, the two solid vectors add head to tail to produce the dashed world-position vector of P. In the right panel, translated copies of the two velocity contributions form a parallelogram whose diagonal is ${}^{w}\mathbf{v}_P$; the right-angle marker applies specifically to the rotational contribution and ${}^{w}\mathbf{r}_{O_bP}$.

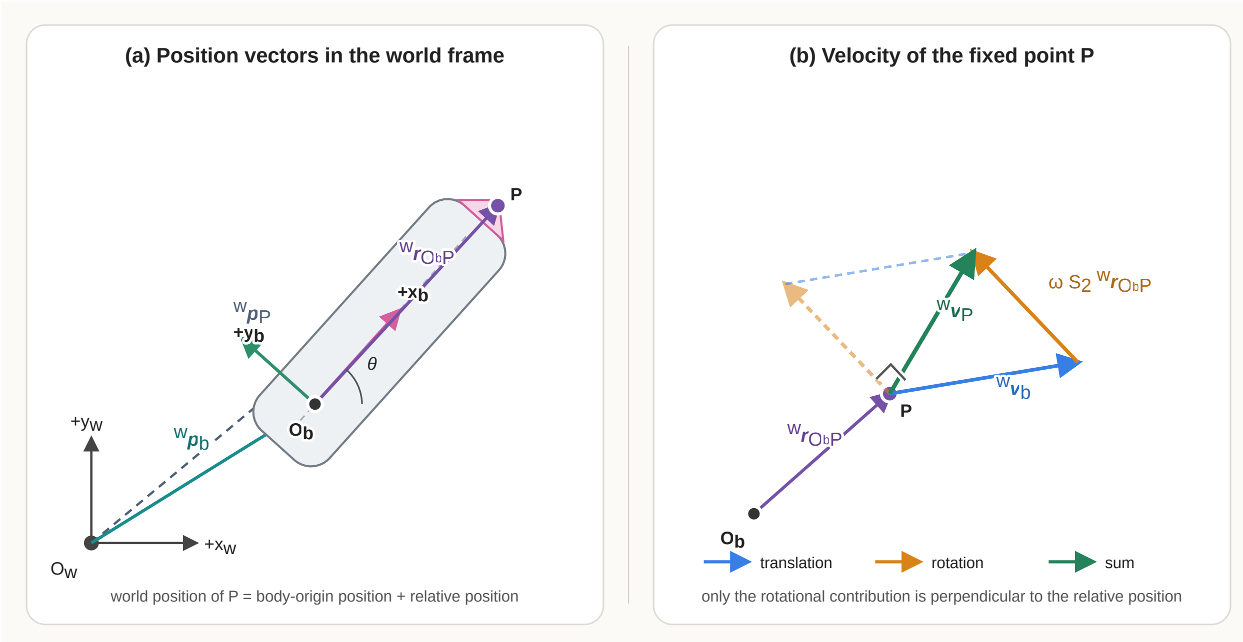

Figure 6.1: Position and velocity relations for a point fixed to a planar rigid body.

The same result can be written with the three-dimensional cross product by embedding the planar quantities in $\mathbb{R}^3$:

$$
{}^{w}\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ \omega \end{bmatrix}, \qquad {}^{w}\mathbf{r}^{(3)}_{O_bP} = \begin{bmatrix} r_x \\ r_y \\ 0 \end{bmatrix}.
$$

Here, r_x and r_y, measured in metres, are the two entries of ${}^{w}\mathbf{r}_{O_bP}$, the superscript (3) marks its three-dimensional embedding, and $\times$ denotes the vector cross product.

Then

$$
{}^{w}\boldsymbol{\omega} \times {}^{w}\mathbf{r}^{(3)}_{O_bP} = \begin{bmatrix} -\omega r_y \\ \omega r_x \\ 0 \end{bmatrix},
$$

whose first two entries equal $\omega \mathbf{S}_2 {}^{w}\mathbf{r}_{O_bP}$. Thus, the planar formula is the two-dimensional form of the rigid-body velocity relation

$$
\mathbf{v}_P = \mathbf{v}_{O_b} + \boldsymbol{\omega} \times \mathbf{r}_{O_bP}.
$$

6.3.5 Worked example

At one instant, suppose the body heading is $\theta = \pi/2\,\mathrm{rad}$, the body-origin velocity is

$$^{w}\mathbf{v}_b = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}},$$

and the yaw rate is $\omega = 2\,\mathrm{rad\,s^{-1}}$. Point P is fixed one metre along the body $+x_b$-axis:

$$^{b}\mathbf{p}_P = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \mathrm{m}.$$

At $\theta = \pi/2$, its relative-position vector in the world frame is

$$^{w}\mathbf{r}_{O_b P} = \mathbf{R}_{wb}\left(\frac{\pi}{2}\right) {}^{b}\mathbf{p}_P = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mathrm{m}.$$

The rotational contribution is

$$\omega \mathbf{S}_2 {}^{w}\mathbf{r}_{O_b P} = 2 \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}.$$

Therefore,

$$\begin{aligned} {}^{w}\mathbf{v}_P &= {}^{w}\mathbf{v}_b + \omega \mathbf{S}_2 {}^{w}\mathbf{r}_{O_b P} \\ &= \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -1.5 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}. \end{aligned}$$

The point lies one metre above the body origin in world coordinates. Counter-clockwise rotation therefore moves it instantaneously toward $-x_w$ at $2\,\mathrm{m\,s^{-1}}$, while translation of the body origin contributes $0.5\,\mathrm{m\,s^{-1}}$ toward $+x_w$.

6.3.6 Assumptions, failure modes, and checks

- **Body-fixed point assumption:** The derivation sets $d({}^{b}\mathbf{p}_P)/dt = \mathbf{0}_2$. If P moves relative to the body, the additional term $\mathbf{R}_{wb}\, d({}^{b}\mathbf{p}_P)/dt$ must be retained.
- **Frame consistency:** ${}^{w}\mathbf{v}_b$ and $\omega \mathbf{S}_2 {}^{w}\mathbf{r}_{O_b P}$ must be expressed in the same frame before they are added.
- **Yaw sign:** With the stated convention, $\omega > 0$ produces counter-clockwise rotation. A sign error reverses the rotational velocity of every off-origin point.

- **Derivative versus rotation:** $\dot{\mathbf{R}}_{wb}$ is not a rotation matrix. Code must not validate or normalise it as though it belonged to $SO(2)$.
- **Numerical differentiation:** Estimating $\dot{\mathbf{R}}$ or velocity from sampled poses amplifies measurement noise and depends on an accurate sampling interval.

Quick test C: rotation derivatives and point velocity

1. Define ${}^{w}\mathbf{v}_b$ and ω, including their frames and units.
2. Differentiate $\mathbf{R}_{wb}(\theta)$ and show that $\dot{\mathbf{R}}_{wb} = \omega\mathbf{S}_2\mathbf{R}_{wb}$.
3. What geometric operation does $\mathbf{S}_2$ perform on a planar vector?
4. Why is $\dot{\mathbf{R}}_{wb}$ generally not an element of $SO(2)$?
5. Differentiate $\mathbf{R}_{wb}^{\mathsf{T}}\mathbf{R}_{wb} = \mathbf{I}_2$ and use the result to show that $\mathbf{R}_{wb}^{\mathsf{T}}\dot{\mathbf{R}}_{wb}$ is skew-symmetric.
6. Starting from ${}^{w}\mathbf{p}_P = \mathbf{R}_{wb}{}^{b}\mathbf{p}_P + {}^{w}\mathbf{p}_b$, derive the velocity of a point fixed to the body.
7. Why is the rotational contribution to point velocity perpendicular to ${}^{w}\mathbf{r}_{O_bP}$?
8. How does the velocity equation change if P moves relative to frame $\{b\}$?

Answers and hints

1. ${}^{w}\mathbf{v}_b = d({}^{w}\mathbf{p}_b)/dt = [\dot{x}\ \dot{y}]^{\mathsf{T}}$ is the velocity of the body origin relative to the fixed world frame and expressed in world coordinates, measured in $\mathrm{m\,s^{-1}}$. The scalar $\omega = \dot{\theta}$ is the signed yaw rate, measured in $\mathrm{rad\,s^{-1}}$.
2. Entrywise differentiation and the chain rule give the matrix displayed in Section 6.3.2. Direct multiplication shows that its coefficient of ω equals $\mathbf{S}_2\mathbf{R}_{wb}$, so $\dot{\mathbf{R}}_{wb} = \omega\mathbf{S}_2\mathbf{R}_{wb}$.
3. $\mathbf{S}_2[a\ b]^{\mathsf{T}} = [-b\ a]^{\mathsf{T}}$, so it rotates the coordinate column counter-clockwise by $\pi/2$ without changing its length.
4. Its entries have units of $\mathrm{s^{-1}}$, and it does not generally satisfy $\dot{\mathbf{R}}_{wb}^{\mathsf{T}}\dot{\mathbf{R}}_{wb} = \mathbf{I}_2$ or have determinant 1.
5. Differentiation gives $\dot{\mathbf{R}}^{\mathsf{T}}\mathbf{R} + \mathbf{R}^{\mathsf{T}}\dot{\mathbf{R}} = \mathbf{0}$. Therefore, $(\mathbf{R}^{\mathsf{T}}\dot{\mathbf{R}})^{\mathsf{T}} = \dot{\mathbf{R}}^{\mathsf{T}}\mathbf{R} = -\mathbf{R}^{\mathsf{T}}\dot{\mathbf{R}}$.
6. Since P is body-fixed, $d({}^{b}\mathbf{p}_P)/dt = \mathbf{0}_2$. Therefore, ${}^{w}\mathbf{v}_P = {}^{w}\mathbf{v}_b + \dot{\mathbf{R}}_{wb}{}^{b}\mathbf{p}_P = {}^{w}\mathbf{v}_b + \omega\mathbf{S}_2{}^{w}\mathbf{r}_{O_bP}$.
7. $\mathbf{S}_2$ rotates the relative-position vector by $\pi/2$, so their dot product is zero. This is the tangential direction of instantaneous circular motion about O_b.
8. Retain the relative-motion term: ${}^{w}\mathbf{v}_P = {}^{w}\mathbf{v}_b + \dot{\mathbf{R}}_{wb}{}^{b}\mathbf{p}_P + \mathbf{R}_{wb}\,d({}^{b}\mathbf{p}_P)/dt$.

6.4 Planar twists and $\mathfrak{se}(2)$

The previous section described translational and angular velocity separately. A **planar twist** combines them into one representation of the instantaneous motion of a rigid body. This representation will connect the pose derivative $\dot{\mathbf{T}}_{wb}$ to the differential-drive inputs v and ω.

6.4.1 Twist coordinates

In this chapter, a planar twist is represented by the coordinate column

$$\boxed{\boldsymbol{\xi} := \begin{bmatrix} \nu_x \\ \nu_y \\ \omega \end{bmatrix} \in \mathbb{R}^3}.$$

Here, ν_x and ν_y use the Greek letter *nu*, ν, not the Latin letter v. They are **generic placeholders** for the x- and y-components of the twist's linear part. Because neither the coordinate frame nor the point chosen as the velocity reference has been specified yet, these symbols do not yet identify a unique physical velocity. Together they form the column $\boldsymbol{\nu} = [\nu_x \ \nu_y]^\mathsf{T} \in \mathbb{R}^2$, whose entries are measured in $\mathrm{m\,s^{-1}}$. The body and spatial cases are defined in Section 6.4.3 and Section 6.4.4.

The third entry $\omega \in \mathbb{R}$ is angular velocity, measured in $\mathrm{rad\,s^{-1}}$. The symbol $\boldsymbol{\xi}$ is the Greek letter *xi*. The word *twist* refers to an instantaneous rigid-body velocity, not to a pose or a finite displacement.

This chapter uses the component order $[\nu_x \ \nu_y \ \omega]^\mathsf{T}$. Some textbooks and software libraries place angular velocity first, so the component order must always be declared. Although $\boldsymbol{\xi}$ is mathematically a three-entry real column, its entries have different physical units. A norm such as $\|\boldsymbol{\xi}\|_2$ therefore has no direct physical meaning unless a length scale is introduced to compare translation with rotation.

6.4.2 The hat operator and $\mathfrak{se}(2)$

Section 6.3.3 showed that $\mathfrak{so}(2)$ contains the tangent directions of the rotation group $SO(2)$ at $\mathbf{I}_2$. A planar rigid motion also includes translation, so its instantaneous direction must combine one rotational component with two translational components.

6.4.2.1 From twist coordinates to a matrix

The **hat operator** maps the twist coordinate column to a 3×3 matrix:

$$\boxed{\widehat{\boldsymbol{\xi}} := \begin{bmatrix} 0 & -\omega & \nu_x \\ \omega & 0 & \nu_y \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \omega \mathbf{S}_2 & \boldsymbol{\nu} \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix}}.$$

The wide hat in $\widehat{\boldsymbol{\xi}}$ denotes this mapping; it does not denote an estimate. The inverse operation, which recovers $\boldsymbol{\xi}$ from its matrix representation, is commonly called the **vee operator** and is written $(\widehat{\boldsymbol{\xi}})^\vee = \boldsymbol{\xi}$.

The upper-left block $\omega\mathbf{S}_2$ represents the instantaneous rotation direction. The upper-right column $\boldsymbol{\nu}$ represents the two-component linear part, and the bottom row is zero because differentiating the constant bottom row [0 0 1] of a homogeneous transformation gives [0 0 0].

6.4.2.2 Why these matrices are tangent to $SE(2)$

Recall that an element of $SE(2)$ is a finite planar homogeneous transformation

$$\mathbf{G} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix}, \qquad \mathbf{R} \in SO(2), \quad \mathbf{p} \in \mathbb{R}^2.$$

It rotates and translates planar point coordinates. The **identity transformation** is

$$\mathbf{I}_3 = \begin{bmatrix} \mathbf{I}_2 & \mathbf{0}_2 \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix}.$$

It represents zero rotation angle and zero translation, so it leaves every homogeneous point coordinate $\overline{\mathbf{p}}_P = [\mathbf{p}_P^\mathsf{T}\ 1]^\mathsf{T} \in \mathbb{R}^3$ unchanged:

$$\mathbf{I}_3\overline{\mathbf{p}}_P = \overline{\mathbf{p}}_P.$$

Let $\mathbf{G}(s) \in SE(2)$ be a differentiable transformation curve with dimensionless real parameter s near zero and $\mathbf{G}(0) = \mathbf{I}_3$. Its derivative $\mathbf{G}'(0)$ is a tangent direction to $SE(2)$ at $\mathbf{I}_3$. The tangent space is the set of all such derivatives:

$$T_{\mathbf{I}_3}SE(2) := \left\{\mathbf{G}'(0) \mid \mathbf{G}(s) \in SE(2),\ \mathbf{G}(0) = \mathbf{I}_3\right\}.$$

Now define the candidate set

$$\mathcal{C} := \left\{ \begin{bmatrix} a\mathbf{S}_2 & \mathbf{u} \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix} \;\middle|\; a \in \mathbb{R},\ \mathbf{u} \in \mathbb{R}^2 \right\},$$

where $\mathcal{C}$ is merely a temporary name for the proposed tangent-matrix structure. The goal is to prove

$$T_{\mathbf{I}_3}SE(2) = \mathcal{C}.$$

Equality requires two set inclusions. The first shows that no other tangent-matrix form is possible. The second shows that every matrix in the proposed set is actually attainable.

First inclusion: every tangent direction has the proposed form. Take any $\mathbf{A} \in T_{\mathbf{I}_3}SE(2)$. By the definition of the tangent space, there is a differentiable curve

$$\mathbf{G}(s) = \begin{bmatrix} \mathbf{R}(s) & \mathbf{p}(s) \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix} \in SE(2)$$

such that $\mathbf{G}(0) = \mathbf{I}_3$ and $\mathbf{A} = \mathbf{G}'(0)$. The identity condition implies $\mathbf{R}(0) = \mathbf{I}_2$ and $\mathbf{p}(0) = \mathbf{0}_2$. Differentiating the block matrix gives

$$\mathbf{A} = \mathbf{G}'(0) = \begin{bmatrix} \mathbf{R}'(0) & \mathbf{p}'(0) \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix}.$$

Because $\mathbf{R}(s) \in SO(2)$, it satisfies $\mathbf{R}(s)^\mathsf{T}\mathbf{R}(s) = \mathbf{I}_2$. Differentiating this constraint with respect to s and evaluating at $s = 0$ gives

$$\mathbf{R}'(0)^\mathsf{T}\mathbf{R}(0) + \mathbf{R}(0)^\mathsf{T}\mathbf{R}'(0) = \mathbf{0}_{2\times 2}.$$

Using $\mathbf{R}(0) = \mathbf{I}_2$ reduces this equation to

$$\mathbf{R}'(0)^\mathsf{T} + \mathbf{R}'(0) = \mathbf{0}_{2\times 2}.$$

Thus, $\mathbf{R}'(0)$ is skew-symmetric and belongs to $\mathfrak{so}(2)$. Section 6.3.3 established that every matrix in $\mathfrak{so}(2)$ has the form $a\mathbf{S}_2$ for some $a \in \mathbb{R}$, so $\mathbf{R}'(0) = a\mathbf{S}_2$. Define $\mathbf{u} := \mathbf{p}'(0) \in \mathbb{R}^2$. Substitution gives

$$\mathbf{A} = \begin{bmatrix} a\mathbf{S}_2 & \mathbf{u} \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix}.$$

Therefore, $\mathbf{A} \in \mathcal{C}$. Because $\mathbf{A}$ was an arbitrary tangent direction, this proves

$$T_{\mathbf{I}_3}SE(2) \subseteq \mathcal{C}.$$

Second inclusion: every matrix of the proposed form is attainable. Take any $\mathbf{A} \in \mathcal{C}$. By the definition of $\mathcal{C}$, there are values $a \in \mathbb{R}$ and $\mathbf{u} \in \mathbb{R}^2$ such that

$$\mathbf{A} = \begin{bmatrix} a\mathbf{S}_2 & \mathbf{u} \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix}.$$

Construct the curve

$$\mathbf{G}_{a,\mathbf{u}}(s) := \begin{bmatrix} \mathbf{R}(as) & s\mathbf{u} \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix}.$$

For every s, $\mathbf{R}(as) \in SO(2)$ and $s\mathbf{u} \in \mathbb{R}^2$, so $\mathbf{G}_{a,\mathbf{u}}(s) \in SE(2)$. At $s = 0$,

$$\mathbf{G}_{a,\mathbf{u}}(0) = \begin{bmatrix} \mathbf{R}(0) & \mathbf{0}_2 \\ \mathbf{0}_2^\top & 1 \end{bmatrix} = \mathbf{I}_3.$$

The chain rule and $d\mathbf{R}(\theta)/d\theta|_{\theta=0} = \mathbf{S}_2$ give its derivative at zero:

$$\begin{aligned} \mathbf{G}'_{a,\mathbf{u}}(0) &= \begin{bmatrix} \dfrac{d}{ds}\mathbf{R}(as)\Big|_{s=0} & \dfrac{d}{ds}(s\mathbf{u})\Big|_{s=0} \\ \mathbf{0}_2^\top & 0 \end{bmatrix} \\ &= \begin{bmatrix} a\mathbf{S}_2 & \mathbf{u} \\ \mathbf{0}_2^\top & 0 \end{bmatrix} = \mathbf{A}. \end{aligned}$$

The curve is valid, passes through $\mathbf{I}_3$, and has the prescribed derivative $\mathbf{A}$. Hence, $\mathbf{A} \in T_{\mathbf{I}_3}SE(2)$. Because $\mathbf{A}$ was an arbitrary member of $\mathcal{C}$, this proves

$$\mathcal{C} \subseteq T_{\mathbf{I}_3}SE(2).$$

Conclusion. The two inclusions prove $T_{\mathbf{I}_3}SE(2) = \mathcal{C}$. This tangent space is denoted $\mathfrak{se}(2)$:

$$\boxed{\mathfrak{se}(2) := T_{\mathbf{I}_3}SE(2) = \mathcal{C} = \left\{ \begin{bmatrix} a\mathbf{S}_2 & \mathbf{u} \\ \mathbf{0}_2^\top & 0 \end{bmatrix} \;\middle|\; a \in \mathbb{R},\ \mathbf{u} \in \mathbb{R}^2 \right\}}.$$

The notation $\mathfrak{se}(2)$ is read "little s e two." Because $SE(2)$ is a Lie group, its tangent space at the identity is called its **Lie algebra**.

This vector space is three-dimensional because each matrix is determined by three independent real numbers: one scalar a for rotation and the two entries of $\mathbf{u}$ for translation. Elements of $SE(2)$ represent finite poses or finite rigid transformations. Elements of $\mathfrak{se}(2)$ represent directions in which such transformations can change instantaneously from the identity.

6.4.2.3 Physical interpretation

When the curve parameter is physical time, the rotational coefficient becomes angular velocity $a = \omega$, and the translation derivative becomes a linear-velocity column $\mathbf{u} = \boldsymbol{\nu}$. The corresponding tangent matrix is exactly the hatted twist $\widehat{\boldsymbol{\xi}}$.

When $\widehat{\boldsymbol{\xi}}$ represents physical velocity, the entries of its 2×2 block have units of s^{-1} and the entries of its right column have units of $\mathrm{m\,s^{-1}}$. The matrix is therefore a structured representation of a physical rate, not a finite homogeneous transformation.

For a sufficiently short time interval $\Delta t > 0$, a smooth transformation curve through the identity with $\mathbf{G}'(0) = \widehat{\boldsymbol{\xi}}$ has the first-order expansion

$$\mathbf{G}(\Delta t) = \mathbf{I}_3 + \Delta t\,\widehat{\boldsymbol{\xi}} + O(\Delta t^2).$$

This equation explains the term **infinitesimal rigid motion**: $\widehat{\boldsymbol{\xi}}$ is not itself a finite transformation in $SE(2)$, but it gives the first-order direction from which a nearby finite transformation develops.

6.4.3 Body twist

6.4.3.1 Plain-language meaning of pose and twist

A **pose** describes where a rigid body is and which direction it is facing relative to a chosen reference frame. In planar motion, the pose contains two position coordinates and one orientation angle. The homogeneous transformation $\mathbf{T}_{wb}$ represents the pose of body frame $\{b\}$ relative to world frame $\{w\}$.

A **twist** describes how a rigid body's pose is changing at one instant. It combines linear velocity and angular velocity in one coordinate column. The word *twist* does not mean that the rigid body bends or deforms; the body remains rigid. In this planar chapter, a twist has two linear-velocity components and one angular-velocity component.

A **body twist** is the motion of the body relative to the world, expressed using the body's instantaneous axes. The phrase *relative to the world* identifies the reference used to measure the motion, whereas *expressed using the body axes* identifies the directions used to report its numerical components. The body frame moves through the world with the rigid body, but it remains fixed relative to the body.

For compactness in this derivation, write

$$\mathbf{T} := \mathbf{T}_{wb} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix}, \qquad \dot{\mathbf{T}} = \begin{bmatrix} \dot{\mathbf{R}} & \dot{\mathbf{p}} \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix},$$

where $\mathbf{R} = \mathbf{R}_{wb} \in SO(2)$ and $\mathbf{p} = {}^{w}\mathbf{p}_b \in \mathbb{R}^2$. The entries of $\mathbf{p}$ are measured in metres, and $\dot{\mathbf{p}} = {}^{w}\mathbf{v}_b$ is the world-frame velocity of the body origin.

Premultiply $\dot{\mathbf{T}}$ by $\mathbf{T}^{-1}$:

$$\begin{aligned} \mathbf{T}^{-1}\dot{\mathbf{T}} &= \begin{bmatrix} \mathbf{R}^\mathsf{T} & -\mathbf{R}^\mathsf{T}\mathbf{p} \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix} \begin{bmatrix} \dot{\mathbf{R}} & \dot{\mathbf{p}} \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{R}^\mathsf{T}\dot{\mathbf{R}} & \mathbf{R}^\mathsf{T}\dot{\mathbf{p}} \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix} \\ &= \begin{bmatrix} \omega\mathbf{S}_2 & \boldsymbol{\nu}_b \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix}. \end{aligned}$$

Here,

$$\boldsymbol{\nu}_b := \mathbf{R}_{wb}^\mathsf{T}\,{}^{w}\mathbf{v}_b = \begin{bmatrix} \nu_{x_b} \\ \nu_{y_b} \end{bmatrix}$$

is the velocity of the body origin expressed along the body axes x_b and y_b. Its entries are measured in $\mathrm{m\,s^{-1}}$. The **body twist** and its matrix representation are therefore

$$\boxed{\boldsymbol{\xi}_b := \begin{bmatrix} \nu_{x_b} \\ \nu_{y_b} \\ \omega \end{bmatrix}, \qquad \widehat{\boldsymbol{\xi}}_b = \mathbf{T}_{wb}^{-1}\dot{\mathbf{T}}_{wb}}.$$

Equivalently,

$$\boxed{\dot{\mathbf{T}}_{wb} = \mathbf{T}_{wb}\widehat{\boldsymbol{\xi}}_b}.$$

The subscript b states that the twist is resolved in the instantaneous body frame. In planar motion, ω has the same scalar value in the world and body descriptions because both frames share the same rotation axis; this simplification does not extend to the three-component angular-velocity coordinates used in spatial motion.

The coordinate conversion

$$\boldsymbol{\nu}_b = \mathbf{R}_{wb}^{\mathsf{T}}\,{}^{w}\mathbf{v}_b$$

does not create a different physical velocity. The column ${}^{w}\mathbf{v}_b = [\dot{x}\ \dot{y}]^{\mathsf{T}}$ describes the body origin's velocity *relative to the world* using the world axes. Multiplication by $\mathbf{R}_{wb}^{\mathsf{T}}$ *expresses* that same velocity using the body's forward and sideways axes:

$$\boldsymbol{\nu}_b = \begin{bmatrix} \nu_{x_b} \\ \nu_{y_b} \end{bmatrix}.$$

Thus, ν_{x_b} is the forward component and ν_{y_b} is the sideways component of the world-relative velocity. For example, a robot facing north and moving north at $1\,\mathrm{m\,s^{-1}}$ has world-frame velocity components $[0\ 1]^{\mathsf{T}}\,\mathrm{m\,s^{-1}}$ but body-frame components $[1\ 0]^{\mathsf{T}}\,\mathrm{m\,s^{-1}}$. These two columns describe the same physical motion.

The body origin does not move relative to its own attached frame. Its body-frame position is always ${}^{b}\mathbf{p}_b = \mathbf{0}_2$, so the derivative of that relative position is also zero. *This zero relative velocity is different from $\boldsymbol{\nu}_b$, which is the body's velocity relative to the world merely expressed using body axes.*

Nor does the coordinate conversion imply that the body directly feels or measures constant linear velocity. A robot estimates its motion relative to the ground or another external reference using information such as wheel encoders, cameras, lidar, satellite positioning, or sensor fusion. The body axes then provide convenient forward and sideways directions in which to express that estimate.

6.4.3.2 Physical meaning and use

The two boxed equations describe the same relationship in opposite directions. The equation

$$\widehat{\boldsymbol{\xi}}_b = \mathbf{T}_{wb}^{-1}\dot{\mathbf{T}}_{wb}$$

extracts motion from a pose trajectory: if $\mathbf{T}_{wb}(t)$ and its derivative are known, it returns the instantaneous angular and linear velocity resolved along the moving body axes. The inverse rotation inside $\mathbf{T}_{wb}^{-1}$ removes the body's current world orientation from the pose rate.

The equation

$$\dot{\mathbf{T}}_{wb} = \mathbf{T}_{wb}\widehat{\boldsymbol{\xi}}_b$$

predicts how the pose changes: given the current pose and a body-frame twist, it produces the corresponding world-frame pose rate. This is the form commonly used to propagate mobile-robot odometry, simulate motion, and predict a future pose from a commanded or estimated body velocity.

For an ideal differential-drive robot, the no-sideways-motion constraint gives $\nu_{y_b} = 0$. The wheel-to-body mapping from Section 3.2.1 therefore supplies

$$\boxed{\boldsymbol{\xi}_b = \begin{bmatrix} v \\ 0 \\ \omega \end{bmatrix}}, \qquad v = \frac{r}{2}\left(\dot{\phi}_L + \dot{\phi}_R\right), \qquad \omega = \frac{r}{L}\left(\dot{\phi}_R - \dot{\phi}_L\right).$$

Here, $r > 0$ is the wheel radius in metres, $L > 0$ is the track width in metres, and $\dot{\phi}_L$ and $\dot{\phi}_R$ are the left and right wheel angular speeds in $\mathrm{rad\,s^{-1}}$. The resulting engineering flow is

$$(\dot{\phi}_L, \dot{\phi}_R) \longrightarrow \boldsymbol{\xi}_b \longrightarrow \dot{\mathbf{T}}_{wb} \longrightarrow \mathbf{T}_{wb}(t + \Delta t).$$

Wheel encoders measure wheel rotation, not the true body twist. Inferring $\boldsymbol{\xi}_b$ from them relies on the wheel geometry and ideal rolling assumptions; slip, deformation, unequal effective radii, and encoder errors make the inferred twist differ from the physical motion.

6.4.4 Spatial twist expressed in the world frame

Postmultiplying $\dot{\mathbf{T}}$ by $\mathbf{T}^{-1}$ produces a different element of $\mathfrak{se}(2)$:

$$\begin{aligned}
\dot{\mathbf{T}}\mathbf{T}^{-1} &= \begin{bmatrix} \dot{\mathbf{R}} & \dot{\mathbf{p}} \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R}^\mathsf{T} & -\mathbf{R}^\mathsf{T}\mathbf{p} \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix} \\
&= \begin{bmatrix} \dot{\mathbf{R}}\mathbf{R}^\mathsf{T} & \dot{\mathbf{p}} - \dot{\mathbf{R}}\mathbf{R}^\mathsf{T}\mathbf{p} \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix} \\
&= \begin{bmatrix} \omega\mathbf{S}_2 & \boldsymbol{\nu}_w \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix},
\end{aligned}$$

where

$$\boxed{\boldsymbol{\nu}_w := \dot{\mathbf{p}} - \omega\mathbf{S}_2\mathbf{p}}.$$

Write $\boldsymbol{\nu}_w = [\nu_{x_w}\ \nu_{y_w}]^\mathsf{T} \in \mathbb{R}^2$, where ν_{x_w} and ν_{y_w} are its components along the world axes and are measured in $\mathrm{m\,s^{-1}}$.

The **spatial twist expressed in the world frame** is

$$\boxed{\boldsymbol{\xi}_w := \begin{bmatrix} \nu_{x_w} \\ \nu_{y_w} \\ \omega \end{bmatrix}, \qquad \widehat{\boldsymbol{\xi}}_w = \dot{\mathbf{T}}_{wb}\mathbf{T}_{wb}^{-1}}.$$

Equivalently,

$$\boxed{\dot{\mathbf{T}}_{wb} = \widehat{\boldsymbol{\xi}}_w\mathbf{T}_{wb}}.$$

Here, *spatial* means that the twist is represented relative to the fixed space or world frame; it does not mean that the motion is three-dimensional.

A crucial distinction is

$$\boldsymbol{\nu}_w \neq \dot{\mathbf{p}} = {}^w\mathbf{v}_b$$

in general. The column $\boldsymbol{\nu}_w$ is the linear part of the spatial twist, not simply the body-origin velocity expressed in world coordinates. Its meaning follows from the rigid-body point-velocity equation derived in Section 6.3.4:

$${}^w\mathbf{v}_P = {}^w\mathbf{v}_b + \omega\mathbf{S}_2\,{}^w\mathbf{r}_{O_bP}.$$

Here, ${}^w\mathbf{v}_P \in \mathbb{R}^2$ is the velocity of a body-fixed point P relative to the world and expressed in world coordinates, ${}^w\mathbf{v}_b = \dot{\mathbf{p}} \in \mathbb{R}^2$ is the corresponding velocity of the body origin O_b, and

${}^{w}\mathbf{r}_{O_bP} \in \mathbb{R}^2$ is the relative-position vector from O_b to P, also expressed in world coordinates. Because

$$ {}^{w}\mathbf{r}_{O_bP} = {}^{w}\mathbf{p}_P - {}^{w}\mathbf{p}_b = {}^{w}\mathbf{p}_P - \mathbf{p}, $$

substitution and regrouping give

$$ \begin{aligned} {}^{w}\mathbf{v}_P &= \dot{\mathbf{p}} + \omega\mathbf{S}_2\left({}^{w}\mathbf{p}_P - \mathbf{p}\right) \\ &= \dot{\mathbf{p}} + \omega\mathbf{S}_2{}^{w}\mathbf{p}_P - \omega\mathbf{S}_2\mathbf{p} \\ &= \underbrace{(\dot{\mathbf{p}} - \omega\mathbf{S}_2\mathbf{p})}_{\boldsymbol{\nu}_w} + \omega\mathbf{S}_2{}^{w}\mathbf{p}_P. \end{aligned} $$

Using the definition $\boldsymbol{\nu}_w = \dot{\mathbf{p}} - \omega\mathbf{S}_2\mathbf{p}$ therefore produces the world-frame velocity field

$$ \boxed{{}^{w}\mathbf{v}_P = \boldsymbol{\nu}_w + \omega\mathbf{S}_2{}^{w}\mathbf{p}_P}. $$

The symbol O_w denotes the fixed origin of world frame $\{w\}$. Its world-coordinate column is $\mathbf{0}_2$. Evaluating the velocity field at that location by setting ${}^{w}\mathbf{p}_P = \mathbf{0}_2$ gives ${}^{w}\mathbf{v}_P = \boldsymbol{\nu}_w$. Thus, $\boldsymbol{\nu}_w$ is the velocity, expressed in world coordinates, assigned by the instantaneous rigid-motion field to the location O_w. *The world origin need not be a physical point on the robot; the rigid-motion velocity field is being extended mathematically to that location.* This dependence on the chosen origin explains why changing the reference origin changes the linear part of a twist.

6.4.4.1 Physical meaning and use

As in the body case, the two boxed equations support two complementary calculations. The equation

$$ \widehat{\boldsymbol{\xi}}_w = \dot{\mathbf{T}}_{wb}\mathbf{T}_{wb}^{-1} $$

extracts the instantaneous rigid-body motion relative to the fixed world axes and the fixed world origin. The equation

$$ \dot{\mathbf{T}}_{wb} = \widehat{\boldsymbol{\xi}}_w\mathbf{T}_{wb} $$

uses that world-referenced motion to calculate the pose rate. The multiplication side records the convention: a body twist acts on $\mathbf{T}_{wb}$ from the right, whereas a spatial twist acts from the left.

The spatial twist describes the velocity of the entire rigid body through the world-frame velocity field derived above. Once $\boldsymbol{\xi}_w$ is known, substituting the world position ${}^{w}\mathbf{p}_P$ of any body-fixed point P gives that point's world-frame velocity. At the body origin, ${}^{w}\mathbf{p}_P = \mathbf{p}$ and $P = O_b$, so $\boldsymbol{\nu}_w = \dot{\mathbf{p}} - \omega \mathbf{S}_2 \mathbf{p}$ gives

$$\boxed{{}^{w}\mathbf{v}_b = \dot{\mathbf{p}} = \boldsymbol{\nu}_w + \omega \mathbf{S}_2 \mathbf{p}}.$$

Therefore,

$$\underbrace{{}^{w}\mathbf{v}_b}_{\text{velocity at } O_b} = \underbrace{\boldsymbol{\nu}_w}_{\text{velocity field at } O_w} + \underbrace{\omega \mathbf{S}_2 \mathbf{p}}_{\text{change caused by the different location}}. \tag{6.1}$$

The linear part $\boldsymbol{\nu}_w$ and the body-origin velocity ${}^{w}\mathbf{v}_b$ refer to different locations. They are equal only when the rotational contribution $\omega \mathbf{S}_2 \mathbf{p}$ is zero, such as when $\omega = 0$ or when O_b currently coincides with O_w.

The spatial representation is useful when an algorithm uses a common fixed world frame: for example, when comparing the motions of several bodies, evaluating point velocities at world-frame locations, or formulating world-referenced estimation and control. Differential-drive wheel kinematics more naturally produces a body twist, so such an application first maps that body twist to the equivalent spatial twist. The next section derives this conversion. *Both twists describe the same physical instantaneous motion; only the reference frame and reference origin used for their coordinates differ.*

6.4.5 Relating body and spatial twists

The body and spatial twists describe the same instantaneous motion, so one must be convertible into the other. Start from their matrix definitions:

$$\widehat{\boldsymbol{\xi}}_b = \mathbf{T}^{-1}\dot{\mathbf{T}}, \qquad \widehat{\boldsymbol{\xi}}_w = \dot{\mathbf{T}}\mathbf{T}^{-1}.$$

The body-twist definition implies $\dot{\mathbf{T}} = \mathbf{T}\widehat{\boldsymbol{\xi}}_b$. Substituting this result into the spatial-twist definition gives

$$\begin{aligned} \widehat{\boldsymbol{\xi}}_w &= \dot{\mathbf{T}}\mathbf{T}^{-1} \\ &= \left(\mathbf{T}\widehat{\boldsymbol{\xi}}_b\right)\mathbf{T}^{-1} \\ &= \mathbf{T}\widehat{\boldsymbol{\xi}}_b\mathbf{T}^{-1}. \end{aligned}$$

The product $\mathbf{T}\widehat{\boldsymbol{\xi}}_b\mathbf{T}^{-1}$ is called **conjugation by $\mathbf{T}$**. To find what it does to the twist coordinates, write the three matrices in block form:

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix}, \qquad \widehat{\boldsymbol{\xi}}_b = \begin{bmatrix} \omega \mathbf{S}_2 & \boldsymbol{\nu}_b \\ \mathbf{0}_2^\mathsf{T} & 0 \end{bmatrix}, \qquad \mathbf{T}^{-1} = \begin{bmatrix} \mathbf{R}^\mathsf{T} & -\mathbf{R}^\mathsf{T}\mathbf{p} \\ \mathbf{0}_2^\mathsf{T} & 1 \end{bmatrix}.$$

First multiply the two matrices on the left:

$$\mathbf{T}\widehat{\boldsymbol{\xi}}_b = \begin{bmatrix} \mathbf{R}\omega\mathbf{S}_2 & \mathbf{R}\boldsymbol{\nu}_b \\ \mathbf{0}_2^\top & 0 \end{bmatrix}.$$

Postmultiplying this result by $\mathbf{T}^{-1}$ gives

$$\mathbf{T}\widehat{\boldsymbol{\xi}}_b\mathbf{T}^{-1} = \begin{bmatrix} \mathbf{R}\omega\mathbf{S}_2\mathbf{R}^\top & -\mathbf{R}\omega\mathbf{S}_2\mathbf{R}^\top\mathbf{p} + \mathbf{R}\boldsymbol{\nu}_b \\ \mathbf{0}_2^\top & 0 \end{bmatrix}.$$

As shown in Section 6.3.2, every planar rotation $\mathbf{R}$ commutes with $\mathbf{S}_2$. Using $\mathbf{R}\mathbf{S}_2 = \mathbf{S}_2\mathbf{R}$ and $\mathbf{R}\mathbf{R}^\top = \mathbf{I}_2$ gives $\mathbf{R}\mathbf{S}_2\mathbf{R}^\top = \mathbf{S}_2$. Therefore,

$$\widehat{\boldsymbol{\xi}}_w = \begin{bmatrix} \omega\mathbf{S}_2 & \mathbf{R}\boldsymbol{\nu}_b - \omega\mathbf{S}_2\mathbf{p} \\ \mathbf{0}_2^\top & 0 \end{bmatrix}.$$

By definition, the same spatial-twist matrix is

$$\widehat{\boldsymbol{\xi}}_w = \begin{bmatrix} \omega\mathbf{S}_2 & \boldsymbol{\nu}_w \\ \mathbf{0}_2^\top & 0 \end{bmatrix}.$$

Comparing the upper-right columns gives

$$\boxed{\boldsymbol{\nu}_w = \mathbf{R}\boldsymbol{\nu}_b - \omega\mathbf{S}_2\mathbf{p}}.$$

This result also follows directly from the definition of $\boldsymbol{\nu}_w$ or ${}^w\mathbf{v}_b = \boldsymbol{\nu}_w + \omega\mathbf{S}_2\mathbf{p}$ in Equation 6.1. The body-origin velocity expressed in world coordinates is ${}^w\mathbf{v}_b = \mathbf{R}\boldsymbol{\nu}_b$, so that equation becomes

$$\mathbf{R}\boldsymbol{\nu}_b = \boldsymbol{\nu}_w + \omega\mathbf{S}_2\mathbf{p}.$$

Rearranging gives $\boldsymbol{\nu}_w = \mathbf{R}\boldsymbol{\nu}_b - \omega\mathbf{S}_2\mathbf{p}$, which agrees with the block-matrix derivation.

The first term rotates the body-axis components of the linear velocity into world-axis components. The second term changes the reference location from the body origin O_b to the world origin O_w. A twist therefore does not transform like a free vector, for which rotation alone would be sufficient.

Now collect the linear and angular coordinates into

$$\boldsymbol{\xi}_b = \begin{bmatrix} \boldsymbol{\nu}_b \\ \omega \end{bmatrix}, \qquad \boldsymbol{\xi}_w = \begin{bmatrix} \boldsymbol{\nu}_w \\ \omega \end{bmatrix}.$$

The preceding component equation can be written as one linear mapping:

$$\begin{aligned}\boldsymbol{\xi}_w &= \begin{bmatrix}\mathbf{R} & -\mathbf{S}_2\mathbf{p}\\ \mathbf{0}_2^\mathsf{T} & 1\end{bmatrix}\begin{bmatrix}\boldsymbol{\nu}_b\\ \omega\end{bmatrix}\\ &= \begin{bmatrix}\mathbf{R}\boldsymbol{\nu}_b - \omega\mathbf{S}_2\mathbf{p}\\ \omega\end{bmatrix}.\end{aligned}$$

This 3×3 coordinate mapping is called the **adjoint representation** of $\mathbf{T}$:

$$\boxed{\mathrm{Ad}_{\mathbf{T}} = \begin{bmatrix}\mathbf{R} & -\mathbf{S}_2\mathbf{p}\\ \mathbf{0}_2^\mathsf{T} & 1\end{bmatrix} \in \mathbb{R}^{3\times 3}}.$$

Equivalently, the adjoint is the linear mapping defined by

$$\widehat{\mathrm{Ad}_{\mathbf{T}}\,\boldsymbol{\xi}} = \mathbf{T}\widehat{\boldsymbol{\xi}}\mathbf{T}^{-1}.$$

For $\mathbf{T} = \mathbf{T}_{wb}$, it converts the body-twist coordinates into the spatial-twist coordinates:

$$\boxed{\boldsymbol{\xi}_w = \mathrm{Ad}_{\mathbf{T}_{wb}}\,\boldsymbol{\xi}_b}.$$

6.4.6 Differential-drive body twist

Under the ideal rolling constraints, a differential-drive robot has no lateral velocity at the body origin. Its body-frame linear velocity is

$$\boldsymbol{\nu}_b = \begin{bmatrix}v\\ 0\end{bmatrix},$$

where v is the signed forward speed in $\mathrm{m\,s^{-1}}$. Its body twist is therefore

$$\boxed{\boldsymbol{\xi}_b = \begin{bmatrix}v\\ 0\\ \omega\end{bmatrix}, \qquad \widehat{\boldsymbol{\xi}}_b = \begin{bmatrix}0 & -\omega & v\\ \omega & 0 & 0\\ 0 & 0 & 0\end{bmatrix}}.$$

Substitute the hatted body twist into $\dot{\mathbf{T}}_{wb} = \mathbf{T}_{wb}\widehat{\boldsymbol{\xi}}_b$. In block form,

$$\mathbf{T}_{wb} = \begin{bmatrix}\mathbf{R}_{wb} & \mathbf{p}\\ \mathbf{0}_2^\mathsf{T} & 1\end{bmatrix}, \qquad \widehat{\boldsymbol{\xi}}_b = \begin{bmatrix}\omega\mathbf{S}_2 & \begin{bmatrix}v\\ 0\end{bmatrix}\\ \mathbf{0}_2^\mathsf{T} & 0\end{bmatrix}.$$

Multiplying these matrices gives

$$\mathbf{T}_{wb}\widehat{\boldsymbol{\xi}}_b = \begin{bmatrix} \mathbf{R}_{wb} & \mathbf{p} \\ \mathbf{0}_2^\top & 1 \end{bmatrix} \begin{bmatrix} \omega \mathbf{S}_2 & \begin{bmatrix} v \\ 0 \end{bmatrix} \\ \mathbf{0}_2^\top & 0 \end{bmatrix}$$
$$= \begin{bmatrix} \mathbf{R}_{wb}\omega \mathbf{S}_2 & \mathbf{R}_{wb}\begin{bmatrix} v \\ 0 \end{bmatrix} \\ \mathbf{0}_2^\top & 0 \end{bmatrix}.$$

The terms involving the translation column $\mathbf{p}$ disappear because they multiply the zero blocks in $\widehat{\boldsymbol{\xi}}_b$. On the other hand,

$$\dot{\mathbf{T}}_{wb} = \begin{bmatrix} \dot{\mathbf{R}}_{wb} & \dot{\mathbf{p}} \\ \mathbf{0}_2^\top & 0 \end{bmatrix}.$$

Because these two block matrices are equal, their corresponding upper-left and upper-right blocks must be equal:

$$\dot{\mathbf{R}}_{wb} = \mathbf{R}_{wb}\omega \mathbf{S}_2, \qquad \dot{\mathbf{p}} = \mathbf{R}_{wb}\begin{bmatrix} v \\ 0 \end{bmatrix}.$$

Since $\dot{\mathbf{p}} = {}^{w}\mathbf{v}_b$ and

$$\mathbf{R}_{wb} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix},$$

the translation block becomes

$${}^{w}\mathbf{v}_b = \mathbf{R}_{wb}\begin{bmatrix} v \\ 0 \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \end{bmatrix}.$$

The rotation derivative derived in Section 6.3.2 is $\dot{\mathbf{R}}_{wb} = \mathbf{R}_{wb}\dot{\theta}\mathbf{S}_2$. Comparing it with $\dot{\mathbf{R}}_{wb} = \mathbf{R}_{wb}\omega\mathbf{S}_2$ gives $\dot{\theta} = \omega$.

Therefore, the familiar differential-drive state equation is recovered:

$$\boxed{\dot{\mathbf{q}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \\ \omega \end{bmatrix}}.$$

The twist formulation has not introduced a different motion model. It has expressed the same model in a form that respects rigid-motion composition and generalises naturally to three-dimensional robotics.

6.4.7 Worked example: the two twist representations

Suppose

$$^{w}\mathbf{p}_b = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \mathrm{m}, \qquad \theta = \frac{\pi}{2}\,\mathrm{rad},$$

and the differential-drive body twist has $v = 0.5\,\mathrm{m\,s^{-1}}$ and $\omega = 2\,\mathrm{rad\,s^{-1}}$. The body-twist column is

$$\boldsymbol{\xi}_b = \begin{bmatrix} 0.5 \\ 0 \\ 2 \end{bmatrix},$$

where the first two entries have units of $\mathrm{m\,s^{-1}}$ and the last has units of $\mathrm{rad\,s^{-1}}$. At $\theta = \pi/2$, the velocity of the body origin expressed in the world frame is

$$\dot{\mathbf{p}} = \mathbf{R}_{wb} \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} \mathrm{m\,s^{-1}}.$$

The spatial twist's linear part is

$$\begin{aligned} \boldsymbol{\nu}_w &= \dot{\mathbf{p}} - \omega \mathbf{S}_2 \mathbf{p} \\ &= \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} - 2 \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -3.5 \end{bmatrix} \mathrm{m\,s^{-1}}. \end{aligned}$$

This does not contradict $^{w}\mathbf{v}_b = [0\ 0.5]^{\mathsf{T}}\,\mathrm{m\,s^{-1}}$. The two columns refer to different points in the same instantaneous velocity field. Substituting the body-origin position into the spatial velocity field recovers its correct velocity:

$$\boldsymbol{\nu}_w + \omega \mathbf{S}_2 {}^{w}\mathbf{p}_b = \begin{bmatrix} 2 \\ -3.5 \end{bmatrix} + \begin{bmatrix} -2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} \mathrm{m\,s^{-1}}.$$

6.4.8 Assumptions, failure modes, and checks

- **Undeclared component order:** A column containing the same three numbers can represent different twists if one convention uses linear velocity first and another uses angular velocity first.
- **Confusing a twist with a pose:** $\boldsymbol{\xi}$ and $\widehat{\boldsymbol{\xi}}$ describe instantaneous motion; neither is an element of $SE(2)$ or a finite pose.

- **Confusing $\boldsymbol{\nu}_w$ with ${}^w\mathbf{v}_b$:** They are equal only when $\omega = 0$, $\mathbf{p} = \mathbf{0}_2$, or their correction term happens to vanish.
- **Rotating a twist like a free vector:** Changing the reference origin introduces the translation-dependent term in $\mathrm{Ad}_{\mathbf{T}}$.
- **Assuming zero lateral velocity for every rigid body:** The condition $\nu_{y_b} = 0$ comes from the ideal differential-drive rolling constraint, not from the definition of a planar twist.
- **Mixing finite and infinitesimal quantities:** In $\mathbf{T}_{wb}$, the rotation block $\mathbf{R}_{wb}$ is dimensionless and the translation column $\mathbf{p}$ has units of metres. In $\widehat{\boldsymbol{\xi}}$, the corresponding blocks have units of s^{-1} and $\mathrm{m\,s}^{-1}$. Therefore, $\mathbf{T}_{wb}$ and $\widehat{\boldsymbol{\xi}}$ cannot be added directly. A twist must first be used as a rate over a time interval through the appropriate pose-evolution equation.

Quick test D: planar twists

1. What physical quantities are stored in $\boldsymbol{\xi} = [\nu_x\ \nu_y\ \omega]^\mathsf{T}$, and what are their units?
2. Define the hat operator for the component order used in this chapter.
3. What is $\mathfrak{se}(2)$, and how does it differ from $SE(2)$?
4. Starting from $\mathbf{T}_{wb}$ and $\dot{\mathbf{T}}_{wb}$, derive $\widehat{\boldsymbol{\xi}}_b = \mathbf{T}_{wb}^{-1}\dot{\mathbf{T}}_{wb}$.
5. Why is the linear part $\boldsymbol{\nu}_w$ of the spatial twist generally different from ${}^w\mathbf{v}_b$?
6. State the two equivalent pose-rate equations involving the body and spatial twists.
7. Derive the differential-drive state equation from $\boldsymbol{\xi}_b = [v\ 0\ \omega]^\mathsf{T}$.
8. A planar twist is $\boldsymbol{\xi} = [\nu_x\ \nu_y\ \omega]^\mathsf{T}$, where ν_x and ν_y are measured in $\mathrm{m\,s}^{-1}$ and ω is measured in $\mathrm{rad\,s}^{-1}$. Why is the ordinary Euclidean expression $\|\boldsymbol{\xi}\|_2 = \sqrt{\nu_x^2 + \nu_y^2 + \omega^2}$ not a physically meaningful twist magnitude? How can a chosen characteristic length $\ell > 0$, measured in metres, make the terms comparable?

Answers and hints

1. ν_x and ν_y are the two components of a linear velocity, measured in $\mathrm{m\,s}^{-1}$; ω is angular velocity, measured in $\mathrm{rad\,s}^{-1}$. Their frame and reference point must also be declared.
2. The hat matrix has $\omega\mathbf{S}_2$ as its upper-left 2×2 block, $\boldsymbol{\nu}$ as its upper-right 2×1 block, and a zero bottom row, as displayed in Section 6.4.2.
3. $SE(2)$ is the group of finite planar rigid transformations. Its Lie algebra $\mathfrak{se}(2)$ is the vector space of structured matrices representing infinitesimal rigid motions and instantaneous velocities.
4. Block multiplication gives upper-left block $\mathbf{R}^\mathsf{T}\dot{\mathbf{R}}$, upper-right block $\mathbf{R}^\mathsf{T}\dot{\mathbf{p}}$, and a zero bottom row. Identify $\mathbf{R}^\mathsf{T}\dot{\mathbf{R}} = \omega\mathbf{S}_2$ and $\boldsymbol{\nu}_b = \mathbf{R}^\mathsf{T}\dot{\mathbf{p}}$.
5. A spatial twist uses the fixed world origin as its velocity reference. Therefore, its linear part is $\boldsymbol{\nu}_w = \dot{\mathbf{p}} - \omega\mathbf{S}_2\mathbf{p}$; the second term accounts for the offset between the world and body origins.
6. $\dot{\mathbf{T}}_{wb} = \mathbf{T}_{wb}\widehat{\boldsymbol{\xi}}_b$ and $\dot{\mathbf{T}}_{wb} = \widehat{\boldsymbol{\xi}}_w\mathbf{T}_{wb}$.
7. Matrix multiplication gives ${}^w\mathbf{v}_b = \mathbf{R}_{wb}[v\ 0]^\mathsf{T} = [v\cos\theta\ v\sin\theta]^\mathsf{T}$ and $\dot{\theta} = \omega$.

8. The expression adds squared linear speeds to a squared angular speed, so its terms represent different physical quantities and its numerical value depends arbitrarily on the chosen units. A characteristic length converts angular speed into the linear-speed scale $\ell\omega$, measured in $\mathrm{m\,s^{-1}}$, giving a weighted magnitude such as $\sqrt{\nu_x^2 + \nu_y^2 + (\ell\omega)^2}$. The choice of ℓ is application-dependent—for example, it might be related to the robot's size—so this is not a unique intrinsic norm for all tasks.

Cumulative chapter review

1. What does the pose $\mathbf{T}_{wb} \in SE(2)$ describe, and how does it transform the body-frame coordinates of a point into world-frame coordinates?
2. Distinguish a passive coordinate change from an active rigid motion. For column-vector notation, which transformation acts first in $\mathbf{T}_2\mathbf{T}_1\mathbf{p}$?
3. Starting from $\mathbf{R}_{wb}^\top \mathbf{R}_{wb} = \mathbf{I}_2$, explain why $\mathbf{R}_{wb}^\top \dot{\mathbf{R}}_{wb}$ is skew-symmetric and identify its planar form.
4. Derive the world-frame velocity of a body-fixed point P from ${}^w\mathbf{p}_P = {}^w\mathbf{p}_b + \mathbf{R}_{wb}{}^b\mathbf{p}_P$.
5. How do $SE(2)$ and $\mathfrak{se}(2)$ differ physically and mathematically? What does the hat operator do?
6. State the physical meanings of the body twist $\boldsymbol{\xi}_b$ and spatial twist $\boldsymbol{\xi}_w$. Why is $\boldsymbol{\nu}_w$ generally not the velocity of the body origin?
7. A robot has $\theta = \pi/2$, $\mathbf{p} = [2\ 0]^\top$ m, and $\boldsymbol{\xi}_b = [1\ 0\ 0.5]^\top$, with linear entries in $\mathrm{m\,s^{-1}}$ and angular entry in $\mathrm{rad\,s^{-1}}$. Find ${}^w\mathbf{v}_b$ and $\boldsymbol{\nu}_w$.
8. Starting from the differential-drive body twist $\boldsymbol{\xi}_b = [v\ 0\ \omega]^\top$, recover the state equation for $\dot{\mathbf{q}}$.

Answers and hints

1. $\mathbf{T}_{wb}$ describes the position and orientation of body frame $\{b\}$ relative to world frame $\{w\}$. It maps a body-frame homogeneous point by ${}^w\overline{\mathbf{p}}_P = \mathbf{T}_{wb}{}^b\overline{\mathbf{p}}_P$, or equivalently ${}^w\mathbf{p}_P = \mathbf{R}_{wb}{}^b\mathbf{p}_P + {}^w\mathbf{p}_b$.
2. A passive change describes the same geometric object in different coordinates; an active motion moves the object while retaining the reference frame. In $\mathbf{T}_2\mathbf{T}_1\mathbf{p}$, the rightmost transformation $\mathbf{T}_1$ acts first.
3. Differentiation gives $\dot{\mathbf{R}}_{wb}^\top \mathbf{R}_{wb} + \mathbf{R}_{wb}^\top \dot{\mathbf{R}}_{wb} = \mathbf{0}_{2\times 2}$. Therefore, $\mathbf{R}_{wb}^\top \dot{\mathbf{R}}_{wb}$ equals the negative of its transpose and has the planar form $\omega\mathbf{S}_2 \in \mathfrak{so}(2)$.
4. Differentiating gives ${}^w\mathbf{v}_P = {}^w\mathbf{v}_b + \dot{\mathbf{R}}_{wb}{}^b\mathbf{p}_P$. Using $\dot{\mathbf{R}}_{wb} = \omega\mathbf{S}_2\mathbf{R}_{wb}$ and ${}^w\mathbf{r}_{O_bP} = \mathbf{R}_{wb}{}^b\mathbf{p}_P$ gives ${}^w\mathbf{v}_P = {}^w\mathbf{v}_b + \omega\mathbf{S}_2{}^w\mathbf{r}_{O_bP}$.
5. $SE(2)$ is the nonlinear group of finite planar poses and rigid transformations. $\mathfrak{se}(2)$ is its three-dimensional tangent vector space at the identity and represents infinitesimal motions or rates. The hat operator maps a twist coordinate column $[\nu_x\ \nu_y\ \omega]^\top$ to its structured 3×3 matrix in $\mathfrak{se}(2)$.

6. $\boldsymbol{\xi}_b$ describes world-relative motion using body axes and the body origin as its linear-velocity reference. $\boldsymbol{\xi}_w$ describes the same motion using world axes and the world origin. Its linear part is the velocity field at O_w, so the velocity at O_b is instead ${}^w\mathbf{v}_b = \boldsymbol{\nu}_w + \omega \mathbf{S}_2 \mathbf{p}$.
7. At $\theta = \pi/2$, $\mathbf{R}_{wb}[1\ 0]^\mathsf{T} = [0\ 1]^\mathsf{T}$, so ${}^w\mathbf{v}_b = [0\ 1]^\mathsf{T}\,\mathrm{m\,s^{-1}}$. Because $0.5\mathbf{S}_2[2\ 0]^\mathsf{T} = [0\ 1]^\mathsf{T}\,\mathrm{m\,s^{-1}}$, $\boldsymbol{\nu}_w = {}^w\mathbf{v}_b - \omega \mathbf{S}_2 \mathbf{p} = \mathbf{0}_2\,\mathrm{m\,s^{-1}}$.
8. Substitution into $\dot{\mathbf{T}}_{wb} = \mathbf{T}_{wb}\widehat{\boldsymbol{\xi}_b}$ gives $\dot{x} = v\cos\theta$, $\dot{y} = v\sin\theta$, and $\dot{\theta} = \omega$, so $\dot{\mathbf{q}} = [v\cos\theta\ v\sin\theta\ \omega]^\mathsf{T}$.

References

- John J. Craig, *Introduction to Robotics: Mechanics and Control*, 4th ed., Chapters 2 and 5, especially frame mappings, transformation interpretations, angular velocity, and rigid-body point velocity.
- Peter Corke, *Robotics, Vision and Control: Fundamental Algorithms in Python*, 3rd ed., Chapters 2-3, especially time-varying pose, angular velocity, and derivatives of orientation and pose.
- Kevin M. Lynch and Frank C. Park, *Modern Robotics: Mechanics, Planning, and Control*, Chapters 2-3, especially configuration spaces, angular velocity, skew-symmetric representations, body and spatial twists, and rigid-body motions. The $SE(3)$ twist definitions are specialised here to planar motion in $SE(2)$.

7 Degrees of Freedom and Spatial Motion: $SO(3)$ and $SE(3)$

Why this chapter matters

The preceding chapter described a rigid body constrained to a plane by three pose coordinates and represented its configuration by an element of $SE(2)$. A robot operating freely in three-dimensional physical space can translate in three independent directions and rotate about three independent directions. This chapter develops the corresponding six-degree-of-freedom model, represents spatial orientation with $SO(3)$, combines orientation and position with $SE(3)$, and extends planar angular velocity and twists to three dimensions.

These ideas are required for robot arms, aerial vehicles, underwater vehicles, cameras, lidar sensors, and any system whose orientation cannot be represented by one planar heading angle.

Prerequisites and assumptions

The reader should understand vectors, matrices, rank, and constraints from Chapter 1; coordinate frames, cross products, and three-dimensional point transformations from Chapter 2; and planar rigid transformations, tangent spaces, and twists from Chapter 6.

Physical space is modelled as three-dimensional Euclidean space. Every body considered here is rigid, so the distance between any two body-fixed points remains constant. Frames are right-handed and orthonormal. Position is measured in metres, time in seconds, and angles in radians.

7.1 Degrees of freedom and configuration space

7.1.1 Configuration is a complete geometric description

A system's **configuration** is a complete geometric specification of the position of every labelled material point of the system at one instant. A *material point* is a particular point fixed in the physical material of the body; its identity does not change as the body moves.

Configuration describes where the system is, not how fast it is moving or what forces act on it.

This definition can be written as a map. Let $\mathcal{B}$ denote the set of labelled material points belonging to one planar rigid body. After attaching frame $\{b\}$ to the body, each point $P \in \mathcal{B}$ has a constant body-frame coordinate ${}^{b}\mathbf{p}_P \in \mathbb{R}^2$. A configuration is a **placement map** χ that assigns a world-frame position to every material point:

$$\chi : \mathcal{B} \to \mathbb{R}^2, \qquad P \longmapsto \chi(P) = {}^{w}\mathbf{p}_P.$$

The symbol χ is the Greek lowercase letter *chi*. Because the body is rigid, a valid placement must preserve the distance between every pair of material points:

$$\|\chi(P) - \chi(Q)\|_2 = \left\|{}^{b}\mathbf{p}_P - {}^{b}\mathbf{p}_Q\right\|_2 \qquad \text{for every } P, Q \in \mathcal{B}.$$

The set of all placement maps allowed by the system's physical constraints is its **configuration space**, abbreviated **C-space** and denoted here by $\mathcal{C}$. One configuration is therefore one element $\chi \in \mathcal{C}$. Calling a configuration a *point in C-space* means one abstract element of this set; it does not mean one physical point of the rigid body or one point in the world.

7.1.1.1 Why one transformation represents the whole rigid body

Listing $\chi(P)$ separately for every $P \in \mathcal{B}$ would be highly redundant. All body-frame point coordinates are fixed by rigidity, so a planar rotation $\mathbf{R}_{wb} \in SO(2)$ and the world position ${}^{w}\mathbf{p}_b \in \mathbb{R}^2$ of the body-frame origin determine every value of the placement map:

$$\boxed{\chi(P) = {}^{w}\mathbf{p}_P = \mathbf{R}_{wb}{}^{b}\mathbf{p}_P + {}^{w}\mathbf{p}_b \qquad \text{for every } P \in \mathcal{B}}.$$

The homogeneous transformation

$$\mathbf{T}_{wb} = \begin{bmatrix} \mathbf{R}_{wb} & {}^{w}\mathbf{p}_b \\ \mathbf{0}_2^{\mathsf{T}} & 1 \end{bmatrix} \in SE(2)$$

packages the same rotation and translation. For one selected point P, the multiplication

$${}^{w}\overline{\mathbf{p}}_P = \mathbf{T}_{wb}{}^{b}\overline{\mathbf{p}}_P$$

evaluates the placement map only at that point and returns its world-frame homogeneous coordinate. Applying the same $\mathbf{T}_{wb}$ to every $P \in \mathcal{B}$ produces the world positions of all the body's material points and therefore specifies the complete configuration. The transformation is not itself the world coordinate of one point; it is the single rule that locates the entire rigid body.

Every configuration uses the same fixed world frame $\{w\}$. Two configurations χ_1 and χ_2 differ when their body-origin positions ${}^w\mathbf{p}_b^{(1)}$ and ${}^w\mathbf{p}_b^{(2)}$ differ, their rotations $\mathbf{R}_{wb}^{(1)}$ and $\mathbf{R}_{wb}^{(2)}$ differ, or both differ. They are the same configuration only when both the translation and rotation agree, equivalently when $\mathbf{T}_{wb}^{(1)} = \mathbf{T}_{wb}^{(2)}$.

Once the body frame has been fixed relative to the body, each orientation-preserving planar placement determines exactly one $\mathbf{T}_{wb} \in SE(2)$, and each $\mathbf{T}_{wb} \in SE(2)$ determines exactly one such placement. The two sets are therefore in one-to-one correspondence, written

$$\boxed{\mathcal{C} \cong SE(2)},$$

where $\cong$ means that the two sets carry the same configuration information through this correspondence. Robotics texts consequently often identify the configuration space of a free planar rigid body with $SE(2)$ and write $\mathcal{C} = SE(2)$. Here, an element of $SE(2)$ is being used as a representation of a complete body placement, even though the same matrix can also act as an operator that transforms point coordinates.

7.1.1.2 Configuration coordinates are not the configuration itself

For the planar rigid body, one convenient coordinate column is

$$\mathbf{q} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}.$$

where x and y are the entries of ${}^w\mathbf{p}_b = [x\ y]^\top$ and θ is the orientation of frame $\{b\}$ relative to frame $\{w\}$. These three **generalised coordinates** construct the transformation representing the configuration:

$$\boxed{\mathbf{q} \longmapsto \mathbf{T}_{wb}(\mathbf{q}) = \begin{bmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{bmatrix} \in SE(2)}.$$

Thus, χ denotes the complete physical placement, $\mathbf{T}_{wb}$ represents that placement as a rigid transformation, and $\mathbf{q}$ is a coordinate description used to construct the transformation. These objects contain equivalent placement information once the world and body frames, axis directions, units, positive rotation direction, and transformation direction have been defined, but they have different mathematical types. In particular, θ and $\theta + 2\pi n$ for any integer n construct the same rotation, so an unrestricted real-valued angle does not give a unique global coordinate for each configuration.

A configuration space is different from a physical **workspace**. A workspace is a region of physical space that a robot or end-effector can reach. A C-space contains complete robot configurations. Different joint arrangements can place an end-effector at the same workspace point while representing different points in C-space.

7.1.2 Definition of degree of freedom

A **degree of freedom**, abbreviated **DOF**, is one independent local direction in which a configuration can change while continuing to satisfy all configuration constraints. The number of degrees of freedom counts these independent allowable directions.

System	Independently chosen coordinates	DOF
Point constrained to a straight rail	Distance s along the rail	1
Point free in a plane	Position (x, y)	2
Door attached by an ideal hinge	Hinge angle θ	1
Free planar rigid body	Position and heading (x, y, θ)	3
Free spatial rigid body	Three position and three orientation coordinates	6

The door illustrates independence. Once its hinge angle is chosen, the positions of all its material points are determined. Recording the world coordinates of its corners would store many scalar values, but the hinge and rigidity constraints prevent those values from being varied independently.

7.1.2.1 Mathematical definition from allowable small changes

Suppose a candidate configuration is represented by a variable column

$$\mathbf{z} = \begin{bmatrix} z_1 & \cdots & z_{n_v} \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^{n_v},$$

where $n_v \in \mathbb{N}_0$ is the number of stored scalar variables. Valid configurations must satisfy $n_c \in \mathbb{N}_0$ scalar equality constraints collected in the mapping

$$\mathbf{g} : \mathbb{R}^{n_v} \to \mathbb{R}^{n_c}, \qquad \mathbf{g}(\mathbf{z}) = \mathbf{0}_{n_c}.$$

The **constraint Jacobian**, or constraint-derivative matrix, is $\mathbf{J}_{\mathbf{g}}(\mathbf{z}) \in \mathbb{R}^{n_c \times n_v}$ with entry

$$[\mathbf{J}_{\mathbf{g}}(\mathbf{z})]_{ij} := \frac{\partial g_i}{\partial z_j}.$$

To describe an allowable local direction at a selected valid configuration $\mathbf{z}_0 \in \mathbb{R}^{n_v}$, let $\mathbf{z}(s)$ be any differentiable curve of valid configurations, where $s \in \mathbb{R}$ is a dimensionless curve parameter, $\mathbf{z}(0) = \mathbf{z}_0$, and $\mathbf{g}(\mathbf{z}(s)) = \mathbf{0}_{n_c}$ for every s near zero. Define the curve's direction at $\mathbf{z}_0$ by $\delta\mathbf{z} := \mathbf{z}'(0) \in \mathbb{R}^{n_v}$. Differentiating the constraint equation with respect to s and applying the multivariable chain rule gives

$$\boxed{\left.\frac{d}{ds}\mathbf{g}(\mathbf{z}(s))\right|_{s=0} = \mathbf{J}_{\mathbf{g}}(\mathbf{z}_0)\,\delta\mathbf{z} = \mathbf{0}_{n_c}}.$$

The set of all vectors satisfying this linear equation is the null space of the Jacobian evaluated at $\mathbf{z}_0$:

$$\boxed{\mathcal{N}(\mathbf{J_g}(\mathbf{z}_0)) = \{\delta\mathbf{z} \in \mathbb{R}^{n_v} \mid \mathbf{J_g}(\mathbf{z}_0)\delta\mathbf{z} = \mathbf{0}_{n_c}\}}.$$

This is not the same set as the configuration constraint set $\{\mathbf{z} \in \mathbb{R}^{n_v} \mid \mathbf{g}(\mathbf{z}) = \mathbf{0}_{n_c}\}$. The constraint set contains valid configurations and may be curved; the null space contains allowable infinitesimal changes at the one selected configuration $\mathbf{z}_0$ and is a linear space.

At a **regular configuration**, meaning that the rank of the constraint Jacobian remains constant in a neighbourhood of $\mathbf{z}_0$, these null-space vectors are exactly the allowable infinitesimal directions. Each independent null-space direction is one independent way to change the configuration locally, so

$$\boxed{n_{\text{dof}}(\mathbf{z}_0) := \dim \mathcal{N}(\mathbf{J_g}(\mathbf{z}_0))}.$$

Applying the rank-nullity theorem from Section 1.2.2.1 gives

$$\boxed{n_{\text{dof}}(\mathbf{z}_0) = n_v - \operatorname{rank}(\mathbf{J_g}(\mathbf{z}_0))}.$$

If the n_c constraints are independent near $\mathbf{z}_0$, then $\operatorname{rank}(\mathbf{J_g}(\mathbf{z}_0)) = n_c$, and the familiar counting rule follows:

$$\boxed{n_{\text{dof}} = n_v - n_c}.$$

7.1.2.2 Example: a point constrained to a circle

Consider a point with Cartesian coordinate column $\mathbf{p} = [x\ y]^\mathsf{T} \in \mathbb{R}^2$ constrained to a circle of fixed radius $r_c > 0$ metres centred at the world origin. Its two stored variables satisfy

$$g(\mathbf{p}) := x^2 + y^2 - r_c^2 = 0.$$

Here, $n_v = 2$ and there is one scalar constraint. Its constraint Jacobian is

$$\mathbf{J}_g(\mathbf{p}) = \begin{bmatrix} 2x & 2y \end{bmatrix} \in \mathbb{R}^{1\times 2}.$$

An allowable change $\delta\mathbf{p} = [\delta x\ \delta y]^\mathsf{T}$ must satisfy

$$\begin{bmatrix} 2x & 2y \end{bmatrix} \begin{bmatrix} \delta x \\ \delta y \end{bmatrix} = 2x\,\delta x + 2y\,\delta y = 0.$$

For this local calculation, $\mathbf{p} = [x\ y]^\mathsf{T}$ is one selected, fixed configuration on the circle. Thus, x and y are given constants, while the allowable change $\delta\mathbf{p}$ is the variable being determined. Because $r_c > 0$, the selected point is not $\mathbf{0}_2$, so the row $\mathbf{J}_g(\mathbf{p}) = [2x\ 2y]$ is nonzero and has rank one. The rank-nullity formula therefore gives

$$\boxed{n_{\mathrm{dof}} = \dim \mathcal{N}(\mathbf{J}_g(\mathbf{p})) = 2 - 1 = 1}.$$

The equation $x\,\delta x + y\,\delta y = 0$ also shows the actual allowable direction: $\delta\mathbf{p}$ must be perpendicular to the fixed radial vector $\mathbf{p}$. Every vector perpendicular to $[x\ y]^\mathsf{T}$ in $\mathbb{R}^2$ is a scalar multiple of $[-y\ x]^\mathsf{T}$, so

$$\mathcal{N}(\mathbf{J}_g(\mathbf{p})) = \left\{ \alpha \begin{bmatrix} -y \\ x \end{bmatrix} \;\middle|\; \alpha \in \mathbb{R} \right\},$$

Here, x and y remain fixed and only the dimensionless scalar $\alpha \in \mathbb{R}$ varies. Choosing α determines both components through $\delta x = -\alpha y$ and $\delta y = \alpha x$. Substitution confirms that every such change satisfies the linearised constraint:

$$\mathbf{J}_g(\mathbf{p})\,\delta\mathbf{p} = \begin{bmatrix} 2x & 2y \end{bmatrix} \alpha \begin{bmatrix} -y \\ x \end{bmatrix} = \alpha(-2xy + 2yx) = 0.$$

The single varying coefficient α explicitly represents the one local degree of freedom. At a different point on the circle, the fixed values of x and y produce a different tangent direction, but the null space still has dimension one.

The number of stored values is consequently not necessarily the number of degrees of freedom. For example, a spatial rotation matrix stores nine scalar entries, but its constraints leave only three independent rotational directions. Likewise, the circle uses two Cartesian entries even though its allowable-change space is one-dimensional.

This counting rule applies directly to geometric, or **holonomic**, equality constraints that restrict configuration. Constraints imposed only on velocity require separate analysis and do not necessarily reduce the dimension of the configuration space.

7.1.3 Why a free planar rigid body has three degrees of freedom

Select three non-collinear points A, B, and C fixed to a planar rigid body. *Non-collinear* means that the points do not all lie on one line, so they form a non-degenerate triangle that visibly represents a two-dimensional rigid shape.

Three points are convenient but not minimal: two distinct labelled points already determine planar position and orientation, whereas one point cannot reveal rotation about itself. The triangle is used here because its fixed pairwise distances make the rigidity constraints especially clear.

If the three points could move independently, their planar coordinates would provide six scalar variables:

$$(x_A, y_A, x_B, y_B, x_C, y_C) \in \mathbb{R}^6.$$

Rigidity fixes the three pairwise distances

$$\|\mathbf{p}_A - \mathbf{p}_B\|_2 = d_{AB}, \qquad \|\mathbf{p}_B - \mathbf{p}_C\|_2 = d_{BC}, \qquad \|\mathbf{p}_A - \mathbf{p}_C\|_2 = d_{AC},$$

where $\mathbf{p}_A, \mathbf{p}_B, \mathbf{p}_C \in \mathbb{R}^2$ are the point-coordinate columns and the positive constants d_{AB}, d_{BC}, and d_{AC} are the fixed distances measured in metres. For a non-degenerate triangle, these are three independent scalar constraints. Therefore,

$$n_{\text{dof}} = 6 - 3 = 3.$$

These three freedoms can be represented more conveniently by two translations and one rotation:

$$\boxed{\text{free planar rigid body:} \qquad (x, y, \theta) \qquad \Longrightarrow \qquad 3 \text{ DOF}}.$$

The fixed distances preserve the body's shape; they do not prevent the complete triangle from translating or rotating as one rigid object.

7.1.4 Why a free spatial rigid body has six degrees of freedom

Allow the same three non-collinear body-fixed points to move in three-dimensional space. Each point has three Cartesian coordinates, so the three points provide nine scalar variables. Rigidity preserves the three pairwise distances d_{AB}, d_{BC}, and d_{AC}, which are three independent constraints for a non-degenerate triangle. Therefore,

$$n_{\text{dof}} = 9 - 3 = 6.$$

The six freedoms are interpreted as three translational and three rotational freedoms:

$$\boxed{\text{free spatial rigid body:} \qquad \underbrace{(x, y, z)}_{\text{3 translations}} \quad + \underbrace{\text{spatial orientation}}_{\text{3 rotational DOF}} \quad \Longrightarrow \quad 6 \text{ DOF}}.$$

The phrase *three rotational degrees of freedom* does not yet prescribe three particular angles. Euler angles, rotation vectors, rotation matrices, and unit quaternions are different representations of the same three-dimensional orientation freedom, each with different coordinate properties and limitations. The next learning block develops the rotation-matrix representation $SO(3)$ before introducing any minimal angle convention.

7.1.5 Constraints, actuators, state variables, and degrees of freedom

Degrees of freedom are often confused with other counts in a robot model. The following quantities answer different questions:

- **Configuration DOF:** How many independent real coordinates are needed locally to specify where the system is?
- **Actuator count:** How many independently commanded physical inputs does the system have?
- **Instantaneous mobility:** How many independent velocity directions are allowed at one configuration?
- **State dimension:** How many variables are required by a chosen dynamic model to predict future behaviour?
- **Representation size:** How many numbers are stored by a particular coordinate representation?

These counts need not be equal. A free spatial rigid body has six configuration degrees of freedom. A dynamic state that stores both its pose and velocity may require twelve local scalar coordinates. A rotation matrix stores nine numbers even though spatial orientation has only three degrees of freedom, because its entries satisfy orthonormality and determinant constraints.

A differential-drive robot provides another important distinction. Its configuration is represented by $\mathbf{T}_{wb} \in SE(2)$, its configuration space is modelled by $SE(2)$, and it has three degrees of freedom described locally by (x, y, θ). Under ideal rolling, however, its instantaneous body velocity has only two independently commanded components, v and ω, because sideways velocity is constrained to zero. The robot cannot move sideways instantaneously, but it can change all three pose coordinates through a sequence of forward and turning motions. Therefore, its two instantaneous control directions do not reduce its configuration space to two dimensions.

7.1.6 Worked examples

7.1.6.1 A hinged door

A free spatial door panel would have six degrees of freedom. An ideal revolute hinge permits only rotation about its hinge axis and prevents the other five independent motions. The constrained door therefore has

$$n_{\text{dof}} = 6 - 5 = 1,$$

which can be represented by one hinge angle $\theta \in \mathbb{R}$ measured in radians over the mechanically allowed interval.

7.1.6.2 A free aerial vehicle

Treat the vehicle as one rigid body with no geometric connection to the environment. Its configuration has six degrees of freedom: three position freedoms and three orientation freedoms. This statement does not imply that the vehicle can independently accelerate in all six directions at one instant. Its actuator arrangement and dynamics determine which accelerations can actually be produced.

7.1.7 Assumptions, limitations, and checks

- **Counting redundant constraints:** Several equations may express the same geometric restriction. Check the rank of the constraint derivatives rather than counting written equations blindly.
- **Confusing coordinates with physical freedom:** A rotation matrix has nine entries but only three rotational degrees of freedom. Identify the independent constraints imposed on a representation.
- **Confusing actuator count with DOF:** A robot may be underactuated or may have redundant actuators. State whether a count refers to configuration, velocity, inputs, or state.
- **Applying $n_v - n_c$ at a singular configuration:** Constraint rank can change at singularities. Verify that the constraints are regular in the region being analysed.
- **Treating a velocity constraint as a configuration constraint:** Under ideal rolling, a wheel moves without sliding sideways at its ground contact. This condition is **nonholonomic** when it restricts the allowable velocities but cannot be replaced by an equivalent equation involving only the configuration. For example, a differential-drive robot cannot move sideways instantaneously, but turning and driving can still change all three coordinates (x, y, θ). The rolling constraint therefore reduces instantaneous mobility without reducing the dimension of its C-space; analyse valid configurations and valid velocities separately.

Quick test A: configuration and degrees of freedom

1. Define configuration and configuration space using the placement map χ.
2. For a planar rigid body, distinguish the placement map χ, the transformation $\mathbf{T}_{wb}$, and the generalised-coordinate column $\mathbf{q}$.
3. Why does applying $\mathbf{T}_{wb}$ to one body-fixed point return only that point's world coordinate, while the same transformation still represents the body's complete configuration?
4. Express the number of degrees of freedom using the null space and rank of a constraint Jacobian. Apply the definition to a point constrained to a circle.
5. Use three non-collinear body-fixed points to explain why a free planar rigid body has three degrees of freedom.
6. Why does an ideal hinged door have one degree of freedom rather than six?

7. Distinguish configuration DOF, instantaneous mobility, actuator count, and state dimension.
8. Why does a differential-drive robot have three configuration degrees of freedom even though its ideal body velocity is specified by only v and ω?
9. Under what condition is the counting rule $n_{\text{dof}} = n_v - n_c$ locally valid?

Answers and hints

1. A configuration is a placement map χ assigning a world position $\chi(P) = {}^{w}\mathbf{p}_P$ to every labelled material point P while satisfying the system's geometric constraints. The configuration space $\mathcal{C}$ is the set of all allowed placement maps.
2. The map χ is the complete placement of all body points. The matrix $\mathbf{T}_{wb} \in SE(2)$ represents that placement through one rotation and translation. The column $\mathbf{q} = [x\ y\ \theta]^\mathsf{T}$ is a set of generalised coordinates used to construct $\mathbf{T}_{wb}$.
3. Multiplication by one point coordinate evaluates the placement rule at that one point. Because every material point has a fixed body-frame coordinate, applying the same transformation to all such coordinates reconstructs the complete placement map; one transformation therefore represents the whole rigid body.
4. At a selected valid configuration $\mathbf{z}_0$, allowable small changes satisfy $\mathbf{J}_\mathbf{g}(\mathbf{z}_0)\delta\mathbf{z} = \mathbf{0}_{n_c}$, so $n_{\text{dof}}(\mathbf{z}_0) = \dim \mathcal{N}(\mathbf{J}_\mathbf{g}(\mathbf{z}_0)) = n_v - \operatorname{rank}(\mathbf{J}_\mathbf{g}(\mathbf{z}_0))$. For $g(x, y) = x^2 + y^2 - r_c^2$, the Jacobian $[2x\ \ 2y]$ has rank one on a circle with $r_c > 0$, and its null space is spanned by $[-y\ x]^\mathsf{T}$, so $n_{\text{dof}} = 2 - 1 = 1$.
5. Three planar points provide six scalar coordinates. Rigidity fixes their three pairwise distances, giving three independent constraints for a non-degenerate triangle. Thus, $6 - 3 = 3$ freedoms remain: two translations and one rotation.
6. The hinge imposes five independent constraints on a free spatial body and leaves only rotation about the hinge axis, so $6 - 5 = 1$.
7. Configuration DOF counts independent pose coordinates; instantaneous mobility counts allowed velocity directions at one configuration; actuator count counts independent command inputs; and state dimension counts the variables retained by a dynamic model.
8. Its configuration still requires (x, y, θ), but the rolling constraint removes instantaneous sideways velocity. Sequences of allowed forward and turning motions can nevertheless change all three configuration coordinates.
9. The rule applies locally when the equality constraints are independent and regular, meaning that their derivative has constant rank in the neighbourhood being considered.

7.2 Spatial orientation and $SO(3)$

7.2.1 Constructing a spatial rotation matrix

The **spatial orientation** of body frame $\{b\}$ relative to world frame $\{w\}$ specifies how the three body axes point when expressed using world-frame coordinates; the locations of the frame origins are irrelevant to orientation. As established in Chapter 2, let $\mathbf{e}_{x,b}$, $\mathbf{e}_{y,b}$, and $\mathbf{e}_{z,b}$

be the geometric unit vectors along the body axes. Their world-frame coordinate columns are

$$
{}^{w}\mathbf{e}_{x,b}, \quad {}^{w}\mathbf{e}_{y,b}, \quad {}^{w}\mathbf{e}_{z,b} \in \mathbb{R}^3.
$$

Each column is dimensionless because it describes a unit direction. Placing the columns side by side defines

$$
\boxed{\mathbf{R}_{wb} := \begin{bmatrix} {}^{w}\mathbf{e}_{x,b} & {}^{w}\mathbf{e}_{y,b} & {}^{w}\mathbf{e}_{z,b} \end{bmatrix} \in \mathbb{R}^{3\times 3}}.
$$

The first column states where the positive x_b-axis points in world coordinates, and the other columns do the same for the positive y_b- and z_b-axes. The subscript order wb means that $\mathbf{R}_{wb}$ maps coordinate columns from frame $\{b\}$ to frame $\{w\}$.

To derive this mapping, let the geometric vector $\mathbf{v}$ have body-frame coordinate column

$$
{}^{b}\mathbf{v} = \begin{bmatrix} v_{x_b} \\ v_{y_b} \\ v_{z_b} \end{bmatrix} \in \mathbb{R}^3,
$$

where the entries are its scalar components along the body axes and carry the physical units of $\mathbf{v}$. Expanding the vector in the body basis and then expressing the result in world coordinates gives

$$
\begin{aligned}
{}^{w}\mathbf{v} &= v_{x_b}\,{}^{w}\mathbf{e}_{x,b} + v_{y_b}\,{}^{w}\mathbf{e}_{y,b} + v_{z_b}\,{}^{w}\mathbf{e}_{z,b} \\
&= \begin{bmatrix} {}^{w}\mathbf{e}_{x,b} & {}^{w}\mathbf{e}_{y,b} & {}^{w}\mathbf{e}_{z,b} \end{bmatrix} \begin{bmatrix} v_{x_b} \\ v_{y_b} \\ v_{z_b} \end{bmatrix}.
\end{aligned}
$$

Therefore,

$$
\boxed{{}^{w}\mathbf{v} = \mathbf{R}_{wb}\,{}^{b}\mathbf{v}}.
$$

This multiplication changes the coordinate description of $\mathbf{v}$; it does not change the geometric vector itself.

7.2.2 Rotation constraints and the definition of $SO(3)$

Write the columns of $\mathbf{R}_{wb}$ as $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3 \in \mathbb{R}^3$. Because they represent the axes of an orthonormal frame, each column has unit length and each distinct pair is perpendicular:

$$\mathbf{r}_i^\mathsf{T}\mathbf{r}_i = 1, \qquad \mathbf{r}_i^\mathsf{T}\mathbf{r}_j = 0 \quad \text{for } i \neq j, \qquad i, j \in \{1, 2, 3\}.$$

Matrix multiplication collects these pairwise dot products:

$$\mathbf{R}_{wb}^\mathsf{T}\mathbf{R}_{wb} = \begin{bmatrix} \mathbf{r}_1^\mathsf{T}\mathbf{r}_1 & \mathbf{r}_1^\mathsf{T}\mathbf{r}_2 & \mathbf{r}_1^\mathsf{T}\mathbf{r}_3 \\ \mathbf{r}_2^\mathsf{T}\mathbf{r}_1 & \mathbf{r}_2^\mathsf{T}\mathbf{r}_2 & \mathbf{r}_2^\mathsf{T}\mathbf{r}_3 \\ \mathbf{r}_3^\mathsf{T}\mathbf{r}_1 & \mathbf{r}_3^\mathsf{T}\mathbf{r}_2 & \mathbf{r}_3^\mathsf{T}\mathbf{r}_3 \end{bmatrix} = \mathbf{I}_3.$$

A real square matrix satisfying $\mathbf{R}^\mathsf{T}\mathbf{R} = \mathbf{I}_3$ is **orthogonal**. This condition permits both right-handed orientations and reflections. For a right-handed body frame, $\mathbf{r}_1 \times \mathbf{r}_2 = \mathbf{r}_3$, so the scalar-triple-product formula gives

$$\det(\mathbf{R}_{wb}) = \mathbf{r}_3^\mathsf{T}\left(\mathbf{r}_1 \times \mathbf{r}_2\right) = \mathbf{r}_3^\mathsf{T}\mathbf{r}_3 = 1.$$

The **special orthogonal group in three dimensions**, read “S O three,” is

$$\boxed{SO(3) := \left\{\mathbf{R} \in \mathbb{R}^{3\times 3} \;\middle|\; \mathbf{R}^\mathsf{T}\mathbf{R} = \mathbf{I}_3, \quad \det(\mathbf{R}) = 1\right\}}.$$

The word *orthogonal* refers to the orthonormal columns, while *special* selects determinant $+1$ and excludes determinant--1 reflections. The group operation is matrix multiplication. For any $\mathbf{R}_1, \mathbf{R}_2 \in SO(3)$,

$$\begin{aligned} \left(\mathbf{R}_1\mathbf{R}_2\right)^\mathsf{T}\left(\mathbf{R}_1\mathbf{R}_2\right) &= \mathbf{R}_2^\mathsf{T}\mathbf{R}_1^\mathsf{T}\mathbf{R}_1\mathbf{R}_2 = \mathbf{I}_3, \\ \det\left(\mathbf{R}_1\mathbf{R}_2\right) &= \det(\mathbf{R}_1)\det(\mathbf{R}_2) = 1. \end{aligned}$$

Thus, $\mathbf{R}_1\mathbf{R}_2 \in SO(3)$, which is the **closure** property. Matrix multiplication is associative, $\mathbf{I}_3$ is the identity orientation, and every element has an inverse in the set. From $\mathbf{R}^\mathsf{T}\mathbf{R} = \mathbf{I}_3$,

$$\boxed{\mathbf{R}^{-1} = \mathbf{R}^\mathsf{T}}.$$

A rotation matrix stores nine scalar entries, but the symmetric equation $\mathbf{R}^\mathsf{T}\mathbf{R} = \mathbf{I}_3$ supplies six independent scalar constraints: three unit-length equations and three pairwise-orthogonality equations. The determinant requirement selects the right-handed component rather than removing another continuous direction. Consequently,

$$\boxed{\dim SO(3) = 9 - 6 = 3}.$$

Here, $\dim SO(3) = 3$ means that three independent continuous coordinates are required near any selected orientation. This is the same local meaning of dimension used in the earlier DOF definition.

7.2.3 Example: a positive quarter-turn about the world z_w-axis

Suppose frame $\{b\}$ starts aligned with frame $\{w\}$ and is rotated through $\pi/2\,\mathrm{rad}$ about the positive z_w-axis according to the right-hand rule. Its body axes, expressed in world coordinates, become

$$ {}^{w}\mathbf{e}_{x,b} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad {}^{w}\mathbf{e}_{y,b} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \qquad {}^{w}\mathbf{e}_{z,b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. $$

Placing these axes into the columns gives

$$ \boxed{\mathbf{R}_{wb} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \in SO(3)}. $$

Direct calculation gives $\mathbf{R}_{wb}^{\mathsf{T}}\mathbf{R}_{wb} = \mathbf{I}_3$ and $\det(\mathbf{R}_{wb}) = 1$. A vector pointing along the positive body x_b-axis has ${}^{b}\mathbf{v} = [1\ 0\ 0]^{\mathsf{T}}$ in arbitrary vector units, so

$$ {}^{w}\mathbf{v} = \mathbf{R}_{wb}\,{}^{b}\mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. $$

The result agrees with the first column of $\mathbf{R}_{wb}$: the positive body x_b-axis points along the positive world y_w-axis.

Quick test B: spatial orientation and $SO(3)$

1. What does column j of $\mathbf{R}_{wb}$ represent?
2. Starting from orthonormal body axes, derive $\mathbf{R}_{wb}^{\mathsf{T}}\mathbf{R}_{wb} = \mathbf{I}_3$.
3. Why is $\det(\mathbf{R}) = 1$ required in addition to $\mathbf{R}^{\mathsf{T}}\mathbf{R} = \mathbf{I}_3$?
4. Why does a rotation matrix store nine entries but represent only three rotational degrees of freedom?
5. For the quarter-turn matrix above, calculate $\mathbf{R}_{bw}$ and use it to express ${}^{w}\mathbf{v} = [0\ 1\ 0]^{\mathsf{T}}$ in body coordinates.

Answers and hints

1. Columns 1, 2, and 3 are respectively the world-frame coordinate columns of the body x_b-, y_b-, and z_b-axes.
2. Entry (i, j) of $\mathbf{R}_{wb}^{\mathsf{T}}\mathbf{R}_{wb}$ is $\mathbf{r}_i^{\mathsf{T}}\mathbf{r}_j$. It equals 1 when $i = j$ and 0 otherwise, producing $\mathbf{I}_3$.
3. Orthogonality alone permits determinant -1 reflections. The determinant-1 condition selects right-handed rigid-body orientations.
4. Orthonormality imposes six independent scalar constraints on the nine entries: three unit-length constraints and three pairwise-orthogonality constraints. Thus, $9 - 6 = 3$ independent continuous values remain.
5. $\mathbf{R}_{bw} = \mathbf{R}_{wb}^{-1} = \mathbf{R}_{wb}^{\mathsf{T}}$. Therefore, ${}^{b}\mathbf{v} = \mathbf{R}_{bw}\,{}^{w}\mathbf{v} = [1\ 0\ 0]^{\mathsf{T}}$.

7.3 Coordinate-axis rotations and composition

7.3.1 Constructing coordinate-axis rotation matrices

Begin with the coordinate-change equation derived in Section 7.2.1:

$$\boxed{{}^{w}\mathbf{v} = \mathbf{R}_{wb}\,{}^{b}\mathbf{v}}.$$

Here, $\mathbf{v}$ is one unchanged geometric vector, while the two columns are its coordinates in body frame $\{b\}$ and world frame $\{w\}$. The same rotation matrix was defined from the world-frame coordinate columns of the body axes:

$$\boxed{\mathbf{R}_{wb} = \begin{bmatrix} {}^{w}\mathbf{e}_{x,b} & {}^{w}\mathbf{e}_{y,b} & {}^{w}\mathbf{e}_{z,b} \end{bmatrix}}.$$

The task is therefore to calculate these three coordinate columns from the geometry and place them into the matrix. Multiplication by standard basis columns is not needed for this construction.

Suppose frames $\{b\}$ and $\{w\}$ are initially aligned, and then frame $\{b\}$ is rotated through $\theta \in \mathbb{R}$ radians about the positive world x_w-axis. A positive rotation follows the right-hand rule. Define

$$c_\theta := \cos\theta, \qquad s_\theta := \sin\theta.$$

The x_b-axis remains aligned with x_w, so its world coordinates are $[1\ 0\ 0]^{\mathsf{T}}$. The y_b- and z_b-axes rotate in the world $y_w z_w$-plane. Their projections onto the world axes give

$${}^{w}\mathbf{e}_{x,b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad {}^{w}\mathbf{e}_{y,b} = \begin{bmatrix} 0 \\ c_\theta \\ s_\theta \end{bmatrix}, \quad {}^{w}\mathbf{e}_{z,b} = \begin{bmatrix} 0 \\ -s_\theta \\ c_\theta \end{bmatrix}.$$

By definition, the first, second, and third columns of the orientation matrix are the world-frame coordinates of the x_b-, y_b-, and z_b-axes, respectively. Substituting the three columns above gives

$$\mathbf{R}_x(\theta) = \begin{bmatrix} {}^w\mathbf{e}_{x,b} & {}^w\mathbf{e}_{y,b} & {}^w\mathbf{e}_{z,b} \end{bmatrix} = \left[\begin{array}{c|c|c} 1 & 0 & 0 \\ 0 & c_\theta & -s_\theta \\ 0 & s_\theta & c_\theta \end{array}\right].$$

Reading the array by columns shows the substitution directly: $[1\ 0\ 0]^\mathsf{T}$ is column one, $[0\ c_\theta\ s_\theta]^\mathsf{T}$ is column two, and $[0\ -s_\theta\ c_\theta]^\mathsf{T}$ is column three. Therefore,

$$\boxed{\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\theta & -s_\theta \\ 0 & s_\theta & c_\theta \end{bmatrix}}.$$

The column placement can be verified by multiplying $\mathbf{R}_x(\theta)$ by each standard basis column:

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \mathbf{R}_x(\theta) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix},$$
$$\begin{bmatrix} 0 \\ c_\theta \\ s_\theta \end{bmatrix} = \mathbf{R}_x(\theta) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix},$$
$$\begin{bmatrix} 0 \\ -s_\theta \\ c_\theta \end{bmatrix} = \mathbf{R}_x(\theta) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The three input columns select columns one, two, and three of $\mathbf{R}_x(\theta)$, respectively. These products verify the assembled matrix; the preceding geometric projection determined the column values.

For a positive rotation about the world y_w-axis, the y_b-axis remains fixed. The right-hand rule sends the positive x_b-axis toward negative z_w and the positive z_b-axis toward positive x_w. Their world-frame coordinate columns are therefore

$${}^w\mathbf{e}_{x,b} = \begin{bmatrix} c_\theta \\ 0 \\ -s_\theta \end{bmatrix}, \quad {}^w\mathbf{e}_{y,b} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad {}^w\mathbf{e}_{z,b} = \begin{bmatrix} s_\theta \\ 0 \\ c_\theta \end{bmatrix}.$$

For a positive rotation about the world z_w-axis, the z_b-axis remains fixed, the positive x_b-axis rotates toward positive y_w, and the positive y_b-axis rotates toward negative x_w. Hence,

$$
{}^{w}\mathbf{e}_{x,b} = \begin{bmatrix} c_\theta \\ s_\theta \\ 0 \end{bmatrix}, \quad {}^{w}\mathbf{e}_{y,b} = \begin{bmatrix} -s_\theta \\ c_\theta \\ 0 \end{bmatrix}, \quad {}^{w}\mathbf{e}_{z,b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.
$$

Placing each set of three world-frame body-axis coordinates into the corresponding matrix columns gives

$$
\boxed{\mathbf{R}_y(\theta) = \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix}, \qquad \mathbf{R}_z(\theta) = \begin{bmatrix} c_\theta & -s_\theta & 0 \\ s_\theta & c_\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}}.
$$

Each matrix has right-handed orthonormal columns and therefore belongs to $SO(3)$.

7.3.1.1 The active-rotation interpretation

The same numerical matrices can also be used as **active rotation operators**. An active rotation changes a geometric vector while the frame used to describe it remains fixed. Let $\mathbf{v}$ denote the vector before rotation and $\mathbf{v}'$ the vector after rotation. Their world-frame coordinate columns satisfy

$$
\boxed{{}^{w}\mathbf{v}' = \mathbf{R}_i(\theta)\,{}^{w}\mathbf{v}, \qquad i \in \{x, y, z\}}.
$$

The prime labels the vector after rotation and does not denote a derivative. The coordinate-change equation describes one geometric vector in two frames; the active-rotation equation describes two geometric vectors in one fixed frame.

Why can the same numerical matrix perform both calculations? Suppose frame $\{b\}$ initially coincides with frame $\{w\}$ and is then physically rotated. The active rotation sends the original world-axis directions to the new body-axis directions:

$$
\mathbf{e}_{x,w} \longmapsto {}^{w}\mathbf{e}_{x,b}, \qquad \mathbf{e}_{y,w} \longmapsto {}^{w}\mathbf{e}_{y,b}, \qquad \mathbf{e}_{z,w} \longmapsto {}^{w}\mathbf{e}_{z,b}.
$$

A linear transformation is determined by the images of its basis vectors, and those images form its matrix columns. Its active-rotation matrix is therefore

$$
\begin{bmatrix} {}^{w}\mathbf{e}_{x,b} & {}^{w}\mathbf{e}_{y,b} & {}^{w}\mathbf{e}_{z,b} \end{bmatrix} = \mathbf{R}_{wb}.
$$

The identical columns also re-express a fixed vector from body coordinates to world coordinates. The numerical operation is the same, but the labels distinguish a changed vector in one frame from an unchanged vector described in two frames.

7.3.2 Composition order: the rightmost rotation acts first

Let $\mathbf{Q}_1, \mathbf{Q}_2 \in SO(3)$ be two active rotation operators whose inputs and outputs are expressed in the same fixed frame. If $\mathbf{Q}_1$ acts first and $\mathbf{Q}_2$ acts second, then

$$\mathbf{v}' = \mathbf{Q}_1\mathbf{v}, \qquad \mathbf{v}'' = \mathbf{Q}_2\mathbf{v}' = \mathbf{Q}_2\mathbf{Q}_1\mathbf{v}.$$

Thus, in the product $\mathbf{Q}_2\mathbf{Q}_1\mathbf{v}$, the rightmost matrix $\mathbf{Q}_1$ acts first. This is a consequence of function composition, not a special convention invented for rotations.

Spatial rotations generally do not commute. To demonstrate this, define

$$\mathbf{R}_x := \mathbf{R}_x\left(\frac{\pi}{2}\right), \qquad \mathbf{R}_y := \mathbf{R}_y\left(\frac{\pi}{2}\right), \qquad \mathbf{e}_z = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Rotating first about x and then about y gives

$$\mathbf{R}_y\mathbf{R}_x\mathbf{e}_z = \mathbf{R}_y \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}.$$

Reversing the order gives

$$\mathbf{R}_x\mathbf{R}_y\mathbf{e}_z = \mathbf{R}_x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

The outputs differ, so

$$\boxed{\mathbf{R}_y\mathbf{R}_x \neq \mathbf{R}_x\mathbf{R}_y}.$$

This calculation assumes that both rotations are active rotations about axes of the same fixed frame. Changing which frame supplies the rotation axes changes the meaning of the sequence.

7.3.3 Fixed-frame and body-frame increments

Consider a rigid body moving while the world frame $\{w\}$ remains fixed. Before the body turns, its three body-fixed axes point in particular world-frame directions. The body then turns, carrying all its body-fixed points and axes with it, and the axes point in new world-frame directions afterward.

An **orientation** describes this rotational state at one instant; it answers how the body axes currently point relative to a reference frame. A **turn** is the physical rotational motion that changes one orientation into another over a time interval. Let frame $\{b\}$ denote the body axes before the turn and frame $\{b'\}$ denote the same body axes afterward. Then $\mathbf{R}_{wb} \in SO(3)$ is the orientation before the turn, whereas $\mathbf{R}_{wb'} \in SO(3)$ is the orientation afterward.

A rotation matrix $\mathbf{Q} \in SO(3)$ that describes only the change from the old orientation to the new orientation is called an **incremental rotation**. The word *incremental* describes the matrix's role as an orientation change; $\mathbf{Q}$ is still an ordinary rotation matrix, and the turn need not be infinitesimally small. Conceptually,

$$\boxed{\text{new orientation} = \text{rotation change} \circ \text{old orientation}},$$

where $\circ$ denotes composition: apply the old orientation first and then apply the rotation change. Matrix multiplication will represent this composition, but the side on which $\mathbf{Q}$ multiplies depends on the coordinate frame used to describe the change.

First describe the change using world-frame coordinates and denote it by $\mathbf{Q}_w \in SO(3)$. The subscript w means that $\mathbf{Q}_w$ acts on world-coordinate columns. For any direction fixed in the body, if ${}^w\mathbf{v}_{\text{old}} \in \mathbb{R}^3$ and ${}^w\mathbf{v}_{\text{new}} \in \mathbb{R}^3$ are its world-coordinate columns before and after the turn, respectively, then the definition of $\mathbf{Q}_w$ is

$${}^w\mathbf{v}_{\text{new}} = \mathbf{Q}_w {}^w\mathbf{v}_{\text{old}}.$$

Write the current orientation as

$$\mathbf{R}_{wb} = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix},$$

where $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3 \in \mathbb{R}^3$ are the current world-frame coordinate columns of the body x_b-, y_b-, and z_b-axes. Applying $\mathbf{Q}_w$ to the physical body rotates each of these three axes, so the updated body-axis columns are

$${}^w\mathbf{e}_{x,b'} = \mathbf{Q}_w\mathbf{r}_1, \qquad {}^w\mathbf{e}_{y,b'} = \mathbf{Q}_w\mathbf{r}_2, \qquad {}^w\mathbf{e}_{z,b'} = \mathbf{Q}_w\mathbf{r}_3.$$

The updated orientation matrix is constructed by placing these new body-axis coordinates into its columns:

$$
\begin{aligned}
\mathbf{R}_{wb'} &= \begin{bmatrix} {}^w\mathbf{e}_{x,b'} & {}^w\mathbf{e}_{y,b'} & {}^w\mathbf{e}_{z,b'} \end{bmatrix} \\
&= \begin{bmatrix} \mathbf{Q}_w\mathbf{r}_1 & \mathbf{Q}_w\mathbf{r}_2 & \mathbf{Q}_w\mathbf{r}_3 \end{bmatrix} \\
&= \mathbf{Q}_w \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix} \\
&= \boxed{\mathbf{Q}_w\mathbf{R}_{wb}}.
\end{aligned}
$$

Thus, an increment described using world-frame axes **premultiplies** the current orientation, where *premultiply* means multiply on the left. This product answers the practical question posed at the start: given the old orientation and an additional world-described rotation, what is the new orientation?

As an alternative, suppose the increment is described relative to the current body axes. Let $\mathbf{Q}_b := \mathbf{R}_{bb'} \in SO(3)$ describe the orientation of the new body frame $\{b'\}$ relative to the old body frame $\{b\}$. Coordinate-frame composition gives

$$
\boxed{\mathbf{R}_{wb'} = \mathbf{R}_{wb}\mathbf{R}_{bb'} = \mathbf{R}_{wb}\mathbf{Q}_b}.
$$

The matrices in this composition have different roles:

- $\mathbf{R}_{wb'}$ is the resulting orientation of the body relative to the world frame.
- $\mathbf{R}_{wb}$ is the body's orientation relative to the world frame before the turn.
- $\mathbf{R}_{bb'} = \mathbf{Q}_b$ is the orientation of the new body frame $\{b'\}$ relative to the old body frame $\{b\}$; it represents the incremental rotation expressed along the old body axes.

Thus, composing the old world-relative orientation with the body-relative orientation change produces the new world-relative orientation.

The body-frame increment therefore **postmultiplies** the current orientation, where *postmultiply* means multiply on the right. These rules can be remembered from their frame labels:

$$
\underbrace{\mathbf{R}_{wb}}_{b\to w}\underbrace{\mathbf{R}_{bb'}}_{b'\to b} = \underbrace{\mathbf{R}_{wb'}}_{b'\to w}.
$$

The two update formulas are not interchangeable. Premultiplication applies a rotation about an axis fixed in the world description; postmultiplication applies a rotation about an axis attached to the current body description.

Quick test C: coordinate-axis rotations and composition

1. Starting from ${}^w\mathbf{v} = \mathbf{R}_{wb}{}^b\mathbf{v}$, derive the columns of $\mathbf{R}_y(\theta)$ from the body axes expressed in world coordinates.
2. In $\mathbf{Q}_3\mathbf{Q}_2\mathbf{Q}_1\mathbf{v}$, which rotation acts first?
3. Use $\mathbf{e}_z$ to show that $\mathbf{R}_y(\pi/2)\mathbf{R}_x(\pi/2) \neq \mathbf{R}_x(\pi/2)\mathbf{R}_y(\pi/2)$.

4. If an orientation increment is expressed about world-frame axes, on which side of $\mathbf{R}_{wb}$ does it multiply?
5. If $\mathbf{Q}_b = \mathbf{R}_{bb'}$ is an increment relative to the current body frame, derive the updated orientation $\mathbf{R}_{wb'}$ from the frame labels.
6. Distinguish an active rotation from a coordinate change.

Answers and hints

1. After a positive rotation about y_w, the world-frame body-axis columns are $[c_\theta \; 0 \; -s_\theta]^\top$, $[0 \; 1 \; 0]^\top$, and $[s_\theta \; 0 \; c_\theta]^\top$. Placing them into the first, second, and third columns gives $\mathbf{R}_y(\theta)$.
2. $\mathbf{Q}_1$ acts first, followed by $\mathbf{Q}_2$ and then $\mathbf{Q}_3$.
3. The first product maps $\mathbf{e}_z$ to $-\mathbf{e}_y$, while the reversed product maps it to $\mathbf{e}_x$; therefore, the operators are unequal.
4. It premultiplies: $\mathbf{R}'_{wb} = \mathbf{Q}_w \mathbf{R}_{wb}$.
5. The labels compose as $(b \to w)(b' \to b) = (b' \to w)$, so $\mathbf{R}_{wb'} = \mathbf{R}_{wb}\mathbf{R}_{bb'} = \mathbf{R}_{wb}\mathbf{Q}_b$.
6. An active rotation changes a geometric vector while retaining one coordinate frame. A coordinate change keeps the geometric vector fixed and changes the frame used to express its coordinate column.

7.4 Angular velocity as the rate of orientation change

7.4.1 From a finite turn to an instantaneous rate

The preceding section compared the body's orientation before and after a finite turn. Motion also requires a rate that describes how rapidly that orientation is changing at one instant. Consider a rigid body whose orientation relative to the fixed world frame is the differentiable function $\mathbf{R}_{wb}(t) \in SO(3)$, where t is time measured in seconds.

Over a time interval from t to $t + \Delta t$, where $\Delta t > 0$ is measured in seconds, the body undergoes a relative rotation from $\mathbf{R}_{wb}(t)$ to $\mathbf{R}_{wb}(t + \Delta t)$. **Euler's rotation theorem**, used here as a stated geometric result, says that every relative orientation change in $SO(3)$ can be represented as one rotation through an angle about an axis direction. Accordingly, let $\mathbf{a}_w(t, \Delta t) \in \mathbb{R}^3$ be a unit rotation-axis column and let $\Delta\theta(t, \Delta t) \in \mathbb{R}$ be the signed rotation angle measured in radians. The subscript w states that the axis components are expressed along the world axes, and

$$\|\mathbf{a}_w(t, \Delta t)\|_2 = 1.$$

The **rotation axis** in an orientation description specifies only a direction $\mathbf{a}_w$. It does not identify a particular line in physical space. To specify a particular line, also choose the world position $\mathbf{q}_w \in \mathbb{R}^3$ of one point on it, measured in metres. The line is then

$$\mathcal{L} := \{\mathbf{q}_w + \lambda \mathbf{a}_w \mid \lambda \in \mathbb{R}\} \subset \mathbb{R}^3,$$

where $\mathcal{L}$ denotes the axis line and λ is a scalar distance measured in metres. Because $\mathbf{R}_{wb}$ describes orientation only, it contains the axis direction associated with the rotation but not the location of a physical pivot line.

The sign of $\Delta\theta$ follows the right-hand rule about $\mathbf{a}_w$. Replacing $\mathbf{a}_w$ by $-\mathbf{a}_w$ and $\Delta\theta$ by $-\Delta\theta$ describes the same turn, so their product is unchanged.

The vector

$$\mathbf{a}_w(t, \Delta t)\frac{\Delta\theta(t, \Delta t)}{\Delta t}$$

describes the rotation-change rate over that interval: its direction gives the rotation axis and its signed magnitude gives the angle change per unit time. Taking the limit as the interval shrinks defines the body's **angular velocity relative to the world frame, expressed in world coordinates**:

$$\boxed{\boldsymbol{\omega}_w(t) := \lim_{\Delta t \to 0^+} \mathbf{a}_w(t, \Delta t)\frac{\Delta\theta(t, \Delta t)}{\Delta t} \in \mathbb{R}^3}.$$

The unit axis $\mathbf{a}_w$ is dimensionless, while $\Delta\theta/\Delta t$ has units of $\mathrm{rad\,s^{-1}}$. Therefore, $\boldsymbol{\omega}_w$ has units of $\mathrm{rad\,s^{-1}}$. The limit is assumed to exist because $\mathbf{R}_{wb}(t)$ is differentiable.

Orientation and angular velocity describe different physical quantities. The matrix $\mathbf{R}_{wb}(t)$ is the rotational state at time t, whereas $\boldsymbol{\omega}_w(t)$ is the instantaneous rate at which that state is changing. Knowing the orientation at one instant does not determine the angular velocity: a stationary body and a turning body can momentarily have the same orientation but different angular velocities.

To connect angular velocity to the orientation matrix, write its three columns explicitly:

$$\boxed{\mathbf{R}_{wb}(t) = \begin{bmatrix}\mathbf{r}_1(t) & \mathbf{r}_2(t) & \mathbf{r}_3(t)\end{bmatrix}}.$$

For each $i \in \{1, 2, 3\}$, the dimensionless column $\mathbf{r}_i(t) \in \mathbb{R}^3$ is the world-coordinate representation of one unit body axis fixed in the rigid body. Thus, $\|\mathbf{r}_i(t)\|_2 = 1$. The geometric construction below selects one column $\mathbf{r}_i$; Figure 7.1 illustrates the choice $i = 1$ and draws the other two columns as dashed vectors.

When $\boldsymbol{\omega}_w \neq \mathbf{0}_3$, define its unit direction by $\mathbf{a}_w := \boldsymbol{\omega}_w/\|\boldsymbol{\omega}_w\|_2$ and let $\varphi_i \in [0, \pi]$ be the angle between $\mathbf{a}_w$ and $\mathbf{r}_i$. Decompose the selected column into components parallel and perpendicular to the angular-velocity direction:

$$\mathbf{r}_{i,\parallel} := (\mathbf{a}_w^\mathsf{T}\mathbf{r}_i)\mathbf{a}_w, \qquad \mathbf{r}_{i,\perp} := \mathbf{r}_i - \mathbf{r}_{i,\parallel}.$$

For the drawing, place the tail of these direction vectors at a common coordinate origin O and call the tip of $\mathbf{r}_i$ point P_i. The point P_i is a graphical endpoint, not necessarily a material point on the body. Let C_i be the tip of $\mathbf{r}_{i,\parallel}$. The segment from C_i to P_i represents $\mathbf{r}_{i,\perp}$ and is perpendicular to the angular-velocity direction.

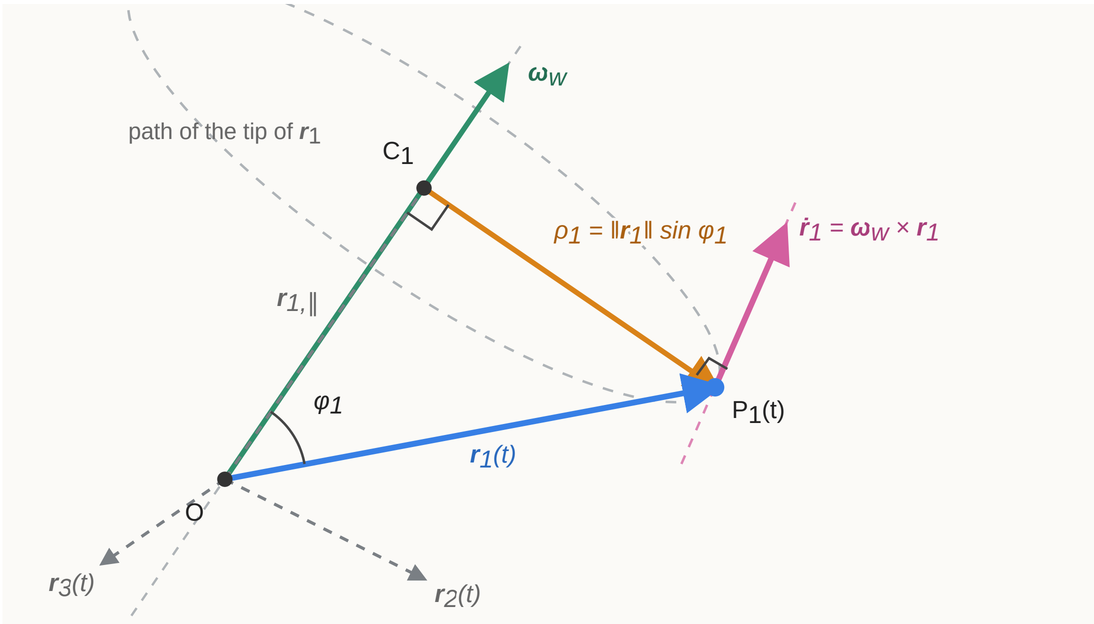

Figure 7.1: Angular velocity and the motion of one orientation-matrix column. The solid vector $\mathbf{r}_1$ is the first column of $\mathbf{R}_{wb}$; the dashed vectors $\mathbf{r}_2$ and $\mathbf{r}_3$ are its other columns.

The figure is a two-dimensional projection of three-dimensional geometry. The orange radius vector $\mathbf{r}_{1,\perp}$, whose length is ρ_1, is perpendicular to $\boldsymbol{\omega}_w$, and $\dot{\mathbf{r}}_1$ is perpendicular to $\mathbf{r}_{1,\perp}$ in three dimensions. The right-angle marker at P_1 therefore appears skewed in the projection even though it denotes a true three-dimensional right angle.

At the selected instant, the tip P_i has the same tangent direction as a point moving on a circle centred at C_i. The circle radius is the length of the perpendicular component:

$$\boxed{\rho_i = \|\mathbf{r}_{i,\perp}\|_2 = \|\mathbf{r}_i\|_2 \sin\varphi_i = \sin\varphi_i}.$$

The last equality uses $\|\mathbf{r}_i\|_2 = 1$. Multiplying the radius by the angular speed gives the speed of the tip:

$$\|\dot{\mathbf{r}}_i\|_2 = \|\boldsymbol{\omega}_w\|_2 \rho_i = \|\boldsymbol{\omega}_w\|_2 \|\mathbf{r}_i\|_2 \sin\varphi_i.$$

This is exactly the magnitude of $\boldsymbol{\omega}_w \times \mathbf{r}_i$, and the cross product's right-hand-rule direction is the tangent direction shown in the figure. When $\boldsymbol{\omega}_w = \mathbf{0}_3$, both the cross product and the column's rate are zero. Therefore, for each of the three columns,

$$\boxed{\dot{\mathbf{r}}_i(t) = \boldsymbol{\omega}_w(t) \times \mathbf{r}_i(t), \qquad i \in \{1,2,3\}}.$$

These equations describe how the three columns of the orientation matrix change. To combine them into one equation for $\dot{\mathbf{R}}_{wb}$, the cross product can be represented as a matrix multiplication.

7.4.2 Cross-product matrices and the orientation derivative

Let

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3.$$

The **cross-product matrix** of $\mathbf{x}$ is defined by

$$\boxed{[\mathbf{x}]_\times := \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix} \in \mathbb{R}^{3\times 3}}.$$

The notation $[\,\cdot\,]_\times$ means "form the matrix that performs a cross product with the enclosed vector." For any $\mathbf{y} = [y_1\ y_2\ y_3]^\mathsf{T} \in \mathbb{R}^3$, direct multiplication gives

$$\begin{aligned} [\mathbf{x}]_\times \mathbf{y} &= \begin{bmatrix} -x_3 y_2 + x_2 y_3 \\ x_3 y_1 - x_1 y_3 \\ -x_2 y_1 + x_1 y_2 \end{bmatrix} \\ &= \boxed{\mathbf{x} \times \mathbf{y}}. \end{aligned}$$

Thus, $[\mathbf{x}]_\times$ is a linear mapping from an input vector $\mathbf{y} \in \mathbb{R}^3$ to the output vector $\mathbf{x} \times \mathbf{y} \in \mathbb{R}^3$. Its entries have the same units as the components of $\mathbf{x}$.

Transposing the matrix changes the sign of every off-diagonal entry:

$$[\mathbf{x}]_\times^\mathsf{T} = -[\mathbf{x}]_\times.$$

A square matrix $\mathbf{A}$ satisfying $\mathbf{A}^\mathsf{T} = -\mathbf{A}$ is called **skew-symmetric**. The set of all real 3×3 skew-symmetric matrices is

$$\boxed{\mathfrak{so}(3) := \left\{\mathbf{A} \in \mathbb{R}^{3\times 3} \;\middle|\; \mathbf{A}^\mathsf{T} = -\mathbf{A}\right\} = \left\{[\mathbf{x}]_\times \;\middle|\; \mathbf{x} \in \mathbb{R}^3\right\}}.$$

The second equality follows from the structure imposed by skew symmetry. The diagonal entries must be zero, and each entry below the diagonal must be the negative of the corresponding entry above it. Hence, every $\mathbf{A} \in \mathfrak{so}(3)$ is determined by three scalars and can be written uniquely as

$$\mathbf{A} = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix} = [\mathbf{x}]_\times.$$

As in the planar case developed in Section 6.3.3, $\mathfrak{so}(3)$ is the **Lie algebra** of $SO(3)$: it is the vector space of instantaneous rotation directions at the identity rotation $\mathbf{I}_3$. A finite rotation matrix belongs to $SO(3)$, whereas an angular-velocity matrix belongs to $\mathfrak{so}(3)$ and describes a rate rather than an orientation.

Apply the cross-product matrix to the three column-rate equations from Section 7.4.1:

$$\begin{aligned} \dot{\mathbf{R}}_{wb} &= \begin{bmatrix} \dot{\mathbf{r}}_1 & \dot{\mathbf{r}}_2 & \dot{\mathbf{r}}_3 \end{bmatrix} \\ &= \begin{bmatrix} [\boldsymbol{\omega}_w]_\times \mathbf{r}_1 & [\boldsymbol{\omega}_w]_\times \mathbf{r}_2 & [\boldsymbol{\omega}_w]_\times \mathbf{r}_3 \end{bmatrix} \\ &= [\boldsymbol{\omega}_w]_\times \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix}. \end{aligned}$$

Because the final matrix of columns is $\mathbf{R}_{wb}$,

$$\boxed{\dot{\mathbf{R}}_{wb} = [\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb}}.$$

Here, $\mathbf{R}_{wb} \in SO(3)$ is dimensionless, $\boldsymbol{\omega}_w \in \mathbb{R}^3$ has units of $\mathrm{rad\,s^{-1}}$, and both $[\boldsymbol{\omega}_w]_\times$ and $\dot{\mathbf{R}}_{wb}$ have units of s^{-1} because radians are dimensionless in SI. This equation says that the world-coordinate rate of every body-axis column is produced by the same world-expressed angular velocity.

Right-multiplying by $\mathbf{R}_{wb}^\top = \mathbf{R}_{wb}^{-1}$ isolates the angular-velocity matrix:

$$\begin{aligned} \dot{\mathbf{R}}_{wb}\mathbf{R}_{wb}^\top &= [\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb}\mathbf{R}_{wb}^\top \\ &= [\boldsymbol{\omega}_w]_\times \mathbf{I}_3. \end{aligned}$$

Therefore,

$$\boxed{[\boldsymbol{\omega}_w]_\times = \dot{\mathbf{R}}_{wb}\mathbf{R}_{wb}^\top}.$$

This formula recovers the angular velocity expressed along the world axes from the orientation and its derivative.

The same physical angular velocity can instead be expressed along the moving body axes. Define

$$\boxed{\boldsymbol{\omega}_b := \mathbf{R}_{wb}^{\mathsf{T}}\boldsymbol{\omega}_w \in \mathbb{R}^3}.$$

The subscript b does not mean that the body rotates relative to itself. It means that the body's angular velocity relative to the world is resolved into components along x_b, y_b, and z_b.

Claim to prove. The world-expressed orientation-rate equation derived above is

$$\dot{\mathbf{R}}_{wb} = [\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb}.$$

We claim that the same orientation rate can be written using the body-expressed angular velocity as

$$\boxed{\dot{\mathbf{R}}_{wb} = \mathbf{R}_{wb}[\boldsymbol{\omega}_b]_\times}.$$

This is the result to be proved. It states that the world-expressed angular-velocity matrix premultiplies $\mathbf{R}_{wb}$, whereas the body-expressed angular-velocity matrix postmultiplies it.

It is sufficient to prove the intermediate coordinate-conversion identity

$$\boxed{[\boldsymbol{\omega}_w]_\times = \mathbf{R}_{wb}[\boldsymbol{\omega}_b]_\times \mathbf{R}_{wb}^{\mathsf{T}}}.$$

Indeed, substituting this identity into $\dot{\mathbf{R}}_{wb} = [\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb}$ will make $\mathbf{R}_{wb}^{\mathsf{T}}\mathbf{R}_{wb} = \mathbf{I}_3$ cancel and produce the claimed body-expressed equation. The remaining task is therefore to prove how a cross-product matrix changes between rotated coordinate frames.

Let $\mathbf{R} \in SO(3)$ map coordinate columns from a source frame to a target frame, and let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ be arbitrary vector-coordinate columns in the source frame. The letters $\mathbf{a}$ and $\mathbf{b}$ are generic placeholders; $\mathbf{a}$ will later be replaced by $\boldsymbol{\omega}_b$. Their target-frame coordinate columns are $\mathbf{Ra}$ and $\mathbf{Rb}$. The general transformation law from Section 1.3.4 is

$$(\mathbf{Aa}) \times (\mathbf{Ab}) = \det(\mathbf{A})\mathbf{A}^{-\mathsf{T}}(\mathbf{a} \times \mathbf{b}).$$

Set $\mathbf{A} = \mathbf{R}$. Because a proper rotation satisfies $\det(\mathbf{R}) = 1$ and $\mathbf{R}^{-\mathsf{T}} = \mathbf{R}$, the general law reduces to

$$\boxed{\mathbf{R}\,(\mathbf{a} \times \mathbf{b}) = (\mathbf{Ra}) \times (\mathbf{Rb})}, \qquad \mathbf{a}, \mathbf{b} \in \mathbb{R}^3.$$

The left-hand side first computes $\mathbf{a} \times \mathbf{b}$ in the source coordinates and then converts the result to the target coordinates. The right-hand side first converts both inputs and then computes their cross product in the target coordinates. The equality says that both procedures describe the same geometric cross-product vector.

We now use this vector identity to convert the cross-product matrix. Keep $\mathbf{a}$ as the source-frame vector that generates $[\mathbf{a}]_\times$, and let $\mathbf{y} \in \mathbb{R}^3$ be an arbitrary target-frame input column. In the product $\mathbf{R}[\mathbf{a}]_\times \mathbf{R}^\mathsf{T}\mathbf{y}$, the operations occur from right to left: $\mathbf{R}^\mathsf{T}$ converts $\mathbf{y}$ to the source coordinates, $[\mathbf{a}]_\times$ forms the cross product there, and $\mathbf{R}$ converts the result back to the target coordinates. Algebraically,

$$\begin{aligned}
\mathbf{R}[\mathbf{a}]_\times \mathbf{R}^\mathsf{T}\mathbf{y} &= \mathbf{R}\left(\mathbf{a} \times (\mathbf{R}^\mathsf{T}\mathbf{y})\right) \\
&= (\mathbf{R}\mathbf{a}) \times (\mathbf{R}\mathbf{R}^\mathsf{T}\mathbf{y}) \\
&= (\mathbf{R}\mathbf{a}) \times \mathbf{y} \\
&= [\mathbf{R}\mathbf{a}]_\times \mathbf{y}.
\end{aligned}$$

Because the two matrices give the same output for every $\mathbf{y}$, they are equal:

$$\boxed{\mathbf{R}[\mathbf{a}]_\times \mathbf{R}^\mathsf{T} = [\mathbf{R}\mathbf{a}]_\times}.$$

Finally, specialize the generic identity to the angular velocity by setting $\mathbf{R} = \mathbf{R}_{wb}$ and $\mathbf{a} = \boldsymbol{\omega}_b$. The definition $\boldsymbol{\omega}_b = \mathbf{R}_{wb}^\mathsf{T}\boldsymbol{\omega}_w$ is equivalent to $\boldsymbol{\omega}_w = \mathbf{R}_{wb}\boldsymbol{\omega}_b$, so

$$\boxed{\mathbf{R}_{wb}[\boldsymbol{\omega}_b]_\times \mathbf{R}_{wb}^\mathsf{T} = [\mathbf{R}_{wb}\boldsymbol{\omega}_b]_\times = [\boldsymbol{\omega}_w]_\times}.$$

This is exactly the required intermediate identity. Substituting it into the known world-expressed orientation-rate equation completes the proof:

$$\begin{aligned}
\dot{\mathbf{R}}_{wb} &= [\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb} \\
&= \mathbf{R}_{wb}[\boldsymbol{\omega}_b]_\times \mathbf{R}_{wb}^\mathsf{T}\mathbf{R}_{wb} \\
&= \boxed{\mathbf{R}_{wb}[\boldsymbol{\omega}_b]_\times}.
\end{aligned}$$

The same intermediate identity can be rearranged to recover the body-expressed cross-product matrix. Premultiplying by $\mathbf{R}_{wb}^\mathsf{T}$ and postmultiplying by $\mathbf{R}_{wb}$ gives

$$[\boldsymbol{\omega}_b]_\times = \mathbf{R}_{wb}^\mathsf{T}[\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb}.$$

Substituting $[\boldsymbol{\omega}_w]_\times = \dot{\mathbf{R}}_{wb}\mathbf{R}_{wb}^\mathsf{T}$ then gives

$$\begin{aligned}
[\boldsymbol{\omega}_b]_\times &= \mathbf{R}_{wb}^\mathsf{T}\dot{\mathbf{R}}_{wb}\mathbf{R}_{wb}^\mathsf{T}\mathbf{R}_{wb} \\
&= \mathbf{R}_{wb}^\mathsf{T}\dot{\mathbf{R}}_{wb}.
\end{aligned}$$

The two coordinate descriptions and the corresponding orientation-rate equations are therefore

$$\boxed{\begin{aligned}[\boldsymbol{\omega}_w]_\times &= \dot{\mathbf{R}}_{wb}\mathbf{R}_{wb}^\top, & \dot{\mathbf{R}}_{wb} &= [\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb},\\ [\boldsymbol{\omega}_b]_\times &= \mathbf{R}_{wb}^\top \dot{\mathbf{R}}_{wb}, & \dot{\mathbf{R}}_{wb} &= \mathbf{R}_{wb}[\boldsymbol{\omega}_b]_\times.\end{aligned}}.$$

7.4.2.1 Connection to finite fixed-frame and body-frame increments

The multiplication sides follow directly from the finite composition rules in Section 7.3.3. To see this, let

$$\mathbf{R} := \mathbf{R}_{wb}(t), \qquad \mathbf{R}^+ := \mathbf{R}_{wb}(t + \Delta t),$$

where $\mathbf{R}$ is the orientation at time t, $\mathbf{R}^+$ is the orientation after a short interval $\Delta t > 0$, and both matrices belong to $SO(3)$.

First express the orientation change using the fixed world axes. The finite composition rule is

$$\mathbf{R}^+ = \mathbf{Q}_w \mathbf{R},$$

where $\mathbf{Q}_w \in SO(3)$ is the incremental rotation expressed using world coordinates. At $\Delta t = 0$, there is no orientation change, so $\mathbf{Q}_w(t, 0) = \mathbf{I}_3$. Its derivative with respect to the interval length at zero is the world-expressed angular-velocity matrix $[\boldsymbol{\omega}_w(t)]_\times$. The first-order Taylor expansion is therefore

$$\mathbf{Q}_w = \mathbf{I}_3 + \Delta t[\boldsymbol{\omega}_w]_\times + O(\Delta t^2).$$

Substituting this expansion into the finite composition rule gives

$$\begin{aligned}\mathbf{R}^+ &= \left(\mathbf{I}_3 + \Delta t[\boldsymbol{\omega}_w]_\times + O(\Delta t^2)\right)\mathbf{R}\\ &= \mathbf{R} + \Delta t[\boldsymbol{\omega}_w]_\times \mathbf{R} + O(\Delta t^2).\end{aligned}$$

Subtract $\mathbf{R}$, divide by Δt, and take the limit $\Delta t \to 0$:

$$\boxed{\dot{\mathbf{R}}_{wb} = [\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb}}.$$

Thus, a rotation change expressed using the fixed world axes premultiplies the finite orientation, and its world-expressed angular-velocity matrix likewise multiplies the orientation on the left.

Now express the same orientation change using the old body axes. The finite composition rule is

$$\mathbf{R}^+ = \mathbf{R}\mathbf{Q}_b,$$

where $\mathbf{Q}_b \in SO(3)$ is the incremental rotation expressed using body coordinates. Similarly, $\mathbf{Q}_b(t, 0) = \mathbf{I}_3$, and its derivative with respect to the interval length at zero is $[\boldsymbol{\omega}_b(t)]_\times$. Its first-order Taylor expansion is

$$\mathbf{Q}_b = \mathbf{I}_3 + \Delta t[\boldsymbol{\omega}_b]_\times + O(\Delta t^2).$$

Substitution gives

$$\begin{aligned}\mathbf{R}^+ &= \mathbf{R}\left(\mathbf{I}_3 + \Delta t[\boldsymbol{\omega}_b]_\times + O(\Delta t^2)\right)\\ &= \mathbf{R} + \Delta t\mathbf{R}[\boldsymbol{\omega}_b]_\times + O(\Delta t^2).\end{aligned}$$

Taking the same difference-quotient limit gives

$$\boxed{\dot{\mathbf{R}}_{wb} = \mathbf{R}_{wb}[\boldsymbol{\omega}_b]_\times}.$$

Thus, a rotation change expressed using the moving body axes postmultiplies the finite orientation, and its body-expressed angular-velocity matrix likewise multiplies the orientation on the right. Both equations describe the same physical turn; only the axes used to express that turn differ.

Coordinates used for the change	Finite composition	Instantaneous orientation rate
Fixed world axes	$\mathbf{R}^+ = \mathbf{Q}_w\mathbf{R}$	$\dot{\mathbf{R}} = [\boldsymbol{\omega}_w]_\times\mathbf{R}$
Moving body axes	$\mathbf{R}^+ = \mathbf{R}\mathbf{Q}_b$	$\dot{\mathbf{R}} = \mathbf{R}[\boldsymbol{\omega}_b]_\times$

Finally, these rate matrices are necessarily skew-symmetric. Differentiating the rotation constraint $\mathbf{R}_{wb}^\mathsf{T}\mathbf{R}_{wb} = \mathbf{I}_3$ gives

$$\dot{\mathbf{R}}_{wb}^\mathsf{T}\mathbf{R}_{wb} + \mathbf{R}_{wb}^\mathsf{T}\dot{\mathbf{R}}_{wb} = \mathbf{0}_{3\times3}.$$

Therefore, $(\mathbf{R}_{wb}^\mathsf{T}\dot{\mathbf{R}}_{wb})^\mathsf{T} = -\mathbf{R}_{wb}^\mathsf{T}\dot{\mathbf{R}}_{wb}$, confirming that $[\boldsymbol{\omega}_b]_\times \in \mathfrak{so}(3)$. Differentiating $\mathbf{R}_{wb}\mathbf{R}_{wb}^\mathsf{T} = \mathbf{I}_3$ similarly confirms that $[\boldsymbol{\omega}_w]_\times = \dot{\mathbf{R}}_{wb}\mathbf{R}_{wb}^\mathsf{T} \in \mathfrak{so}(3)$. These products remove the current finite orientation and express its instantaneous change at the identity rotation.

Quick test D: angular velocity

1. What physical information is contained in the direction and magnitude of $\boldsymbol{\omega}_w$?
2. Why does knowing $\mathbf{R}_{wb}(t)$ at one instant not determine $\boldsymbol{\omega}_w(t)$?
3. Explain geometrically why a body-axis column satisfies $\dot{\mathbf{r}}_i = \boldsymbol{\omega}_w \times \mathbf{r}_i$.
4. Suppose $\boldsymbol{\omega}_w = [0\ 0\ 2]^\mathsf{T}\,\mathrm{rad\,s^{-1}}$ and $\mathbf{r}_i = [1\ 0\ 0]^\mathsf{T}$. Calculate $\dot{\mathbf{r}}_i$ and interpret its direction.
5. Construct $[\mathbf{x}]_\times$ for $\mathbf{x} = [1\ 2\ 3]^\mathsf{T}$ and verify that it is skew-symmetric.

6. Starting from $\dot{\mathbf{R}}_{wb} = [\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb}$, isolate $[\boldsymbol{\omega}_w]_\times$.
7. Do $\boldsymbol{\omega}_w$ and $\boldsymbol{\omega}_b$ describe different physical turns? Explain their relationship.
8. Why is $\mathbf{R}_{wb}^\mathsf{T}\dot{\mathbf{R}}_{wb}$ skew-symmetric rather than a rotation matrix?

Answers and hints

1. Its direction gives the instantaneous rotation axis according to the right-hand rule, and its magnitude gives the angular speed in $\mathrm{rad\,s^{-1}}$.
2. Orientation is a state, whereas angular velocity is its rate of change. Different orientation trajectories can pass through the same $\mathbf{R}_{wb}(t)$ with different derivatives.
3. The tip of $\mathbf{r}_i$ has a tangent velocity about the instantaneous rotation-axis direction. The cross product has that tangent direction and magnitude $\|\boldsymbol{\omega}_w\|_2\|\mathbf{r}_i\|_2 \sin\varphi_i$.
4. $\dot{\mathbf{r}}_i = \boldsymbol{\omega}_w \times \mathbf{r}_i = [0\ 2\ 0]^\mathsf{T}\,\mathrm{s}^{-1}$. The direction initially points toward positive y_w, as expected for a positive turn about z_w.
5. $[\mathbf{x}]_\times = \begin{bmatrix} 0 & -3 & 2 \\ 3 & 0 & -1 \\ -2 & 1 & 0 \end{bmatrix}$, and direct transposition gives $[\mathbf{x}]_\times^\mathsf{T} = -[\mathbf{x}]_\times$.
6. Right-multiply by $\mathbf{R}_{wb}^\mathsf{T}$ and use $\mathbf{R}_{wb}\mathbf{R}_{wb}^\mathsf{T} = \mathbf{I}_3$ to obtain $[\boldsymbol{\omega}_w]_\times = \dot{\mathbf{R}}_{wb}\mathbf{R}_{wb}^\mathsf{T}$.
7. No. They describe the same angular velocity relative to the world but resolve it along different axes: $\boldsymbol{\omega}_b = \mathbf{R}_{wb}^\mathsf{T}\boldsymbol{\omega}_w$.
8. Differentiating $\mathbf{R}_{wb}^\mathsf{T}\mathbf{R}_{wb} = \mathbf{I}_3$ shows that $(\mathbf{R}_{wb}^\mathsf{T}\dot{\mathbf{R}}_{wb})^\mathsf{T} = -\mathbf{R}_{wb}^\mathsf{T}\dot{\mathbf{R}}_{wb}$. It represents an instantaneous rotation rate in $\mathfrak{so}(3)$, not a finite orientation in $SO(3)$.

7.5 Finite rotations from axis–angle coordinates

7.5.1 The problem: converting a rotation rate into a finite turn

The angular-velocity equations in Section 7.4 describe an instantaneous orientation rate. A simulation, estimator, or controller also needs a finite orientation change over a nonzero interval. This section solves the forward problem: given a rotation-axis direction and a signed rotation angle, construct the corresponding matrix in $SO(3)$.

Let $\mathbf{a}_w \in \mathbb{R}^3$ be a unit vector expressed in world coordinates,

$$\|\mathbf{a}_w\|_2 = 1,$$

and let $\alpha \in \mathbb{R}$ be a signed rotation angle measured in radians. The pair $(\mathbf{a}_w, \alpha)$ is an **axis–angle representation** of the intended orientation change: $\mathbf{a}_w$ gives the positive axis direction, and the sign of α selects the direction of rotation by the right-hand rule. This orientation-only description specifies an axis direction through the coordinate origin; locating a physical pivot line would additionally require a point, as explained in Section 7.4.1.

The three-component column

$$\boxed{\boldsymbol{\varphi}_w := \alpha \mathbf{a}_w \in \mathbb{R}^3}$$

is called the **rotation vector**, or the **exponential coordinates of the rotation**. Its direction gives the rotation axis and its norm gives the angle magnitude:

$$\|\boldsymbol{\varphi}_w\|_2 = |\alpha|.$$

The word *exponential* does not refer to the definition $\boldsymbol{\varphi}_w = \alpha \mathbf{a}_w$ itself. It anticipates the construction derived below: the skew-symmetric matrix $[\boldsymbol{\varphi}_w]_\times$ is supplied to the matrix exponential to recover the finite rotation $\mathbf{Q}_w = \exp([\boldsymbol{\varphi}_w]_\times)$.

If the same rotation is produced during a time interval $[t_0, t_1]$ about the fixed axis $\mathbf{a}_w$, let $\alpha(t_0) = 0$ and $\alpha(t_1) = \alpha$. The world-expressed angular velocity and accumulated angle then satisfy

$$\boxed{\boldsymbol{\omega}_w(t) = \dot{\alpha}(t)\mathbf{a}_w}, \qquad \alpha = \int_{t_0}^{t_1} \dot{\alpha}(t)\, dt.$$

Because $\mathbf{a}_w$ is constant over this interval,

$$\boxed{\boldsymbol{\varphi}_w = \alpha \mathbf{a}_w = \int_{t_0}^{t_1} \boldsymbol{\omega}_w(t)\, dt}.$$

The complete fixed-axis relationship is therefore

$$\underbrace{\boldsymbol{\omega}_w(t)}_{\text{angular velocity}} \xrightarrow{\text{integrate over } [t_0, t_1]} \underbrace{\boldsymbol{\varphi}_w = \alpha \mathbf{a}_w}_{\text{accumulated rotation coordinates}} \xrightarrow{\exp([\,\cdot\,]_\times)} \underbrace{\mathbf{Q}_w}_{\text{finite rotation}}.$$

The vector $\boldsymbol{\varphi}_w$ is dimensionless in SI because radians are dimensionless, although its entries are labelled in radians to preserve their physical meaning. It represents accumulated finite rotation coordinates, not angular velocity; angular velocity has units of $\mathrm{rad\,s^{-1}}$.

7.5.2 The matrix exponential

To construct the finite rotation, first consider a differentiable world-coordinate vector $\mathbf{p}(s) \in \mathbb{R}^3$ rotating about $\mathbf{a}_w$, where $s \in \mathbb{R}$ is the accumulated signed rotation angle, measured in radians and dimensionless in SI. The derivation in Section 7.4.1 gave $\dot{\mathbf{r}}_i = \boldsymbol{\omega}_w \times \mathbf{r}_i$ for each body-axis column; the same vector-rate equation applies to any vector rotating with the body:

$$\frac{d\mathbf{p}}{dt} = \boldsymbol{\omega}_w \times \mathbf{p}.$$

If $s = s(t)$, then $\boldsymbol{\omega}_w = \mathbf{a}_w \, ds/dt$. Applying the chain rule gives

$$\frac{d\mathbf{p}}{dt} = \frac{d\mathbf{p}}{ds}\frac{ds}{dt} = (\mathbf{a}_w \times \mathbf{p})\,\frac{ds}{dt}.$$

Cancelling ds/dt when it is nonzero, with the stationary case obtained by continuity, gives the vector-rate equation with angle rather than time as the independent variable:

$$\frac{d\mathbf{p}}{ds} = \mathbf{a}_w \times \mathbf{p}(s) = [\mathbf{a}_w]_\times \mathbf{p}(s).$$

The initial condition is $\mathbf{p}(0) = \mathbf{p}_0$, where $\mathbf{p}_0 \in \mathbb{R}^3$ is the vector before the turn. The coefficient matrix $[\mathbf{a}_w]_\times \in \mathfrak{so}(3)$ is constant because the axis is fixed in world coordinates.

For any square matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, the **matrix exponential** is defined by the convergent power series

$$\boxed{\exp(\mathbf{A}) := \mathbf{I}_n + \mathbf{A} + \frac{\mathbf{A}^2}{2!} + \frac{\mathbf{A}^3}{3!} + \cdots}\,.$$

Here, $\mathbf{A}^0 := \mathbf{I}_n$, $0! := 1$, and $k! := 1 \cdot 2 \cdots k$ for each positive integer k. The dots denote an **infinite series**, defined as the limit of its finite partial sums:

$$\boxed{\exp(\mathbf{A}) := \lim_{N\to\infty} \sum_{k=0}^{N} \frac{\mathbf{A}^k}{k!}}\,.$$

Convergence means that every entry of these partial-sum matrices approaches a finite value as $N \to \infty$. This definition mirrors the scalar series $e^z = \sum_{k=0}^{\infty} z^k/k!$ and converges for every finite square matrix $\mathbf{A}$. For a constant matrix $\mathbf{A}$, term-by-term differentiation of the series gives

$$\frac{d}{ds}\exp(s\mathbf{A}) = \mathbf{A}\exp(s\mathbf{A}).$$

The scalar separation step dp/p cannot be used here because division by a vector and the ordinary logarithm of a vector are not defined. Its matrix counterpart is an **integrating factor**, meaning a matrix chosen so that the differential equation becomes the derivative of one product. Set $\mathbf{A} := [\mathbf{a}_w]_\times \in \mathbb{R}^{3\times 3}$ and left-multiply $d\mathbf{p}/ds = \mathbf{A}\mathbf{p}$ by $\exp(-s\mathbf{A})$. The product rule gives

$$\begin{aligned}
\frac{d}{ds}\left[\exp(-s\mathbf{A})\mathbf{p}(s)\right] &= -\mathbf{A}\exp(-s\mathbf{A})\mathbf{p}(s) + \exp(-s\mathbf{A})\frac{d\mathbf{p}}{ds} \\
&= -\mathbf{A}\exp(-s\mathbf{A})\mathbf{p}(s) + \exp(-s\mathbf{A})\mathbf{A}\mathbf{p}(s) \\
&= \mathbf{0}_3.
\end{aligned}$$

The last equality uses $\mathbf{A}\exp(-s\mathbf{A}) = \exp(-s\mathbf{A})\mathbf{A}$, which holds because the exponential series contains only powers of $\mathbf{A}$. The product is therefore constant. Evaluating it at $s = 0$ and using $\exp(\mathbf{0}_{3\times 3}) = \mathbf{I}_3$ gives

$$\exp(-s\mathbf{A})\mathbf{p}(s) = \mathbf{p}_0.$$

Left-multiplying by $\exp(s\mathbf{A})$ and using $\exp(s\mathbf{A})^{-1} = \exp(-s\mathbf{A})$ yields

$$\boxed{\mathbf{p}(s) = \exp(s[\mathbf{a}_w]_\times)\,\mathbf{p}_0}.$$

This integrating-factor argument derives the solution. Differentiating the result provides a separate verification:

$$\begin{aligned}\frac{d\mathbf{p}}{ds} &= \frac{d}{ds}\left[\exp(s[\mathbf{a}_w]_\times)\right]\mathbf{p}_0\\ &= [\mathbf{a}_w]_\times \exp(s[\mathbf{a}_w]_\times)\,\mathbf{p}_0\\ &= [\mathbf{a}_w]_\times \mathbf{p}(s),\end{aligned}$$

and $\mathbf{p}(0) = \exp(\mathbf{0}_{3\times 3})\mathbf{p}_0 = \mathbf{I}_3\mathbf{p}_0 = \mathbf{p}_0$. The cross-product-matrix construction is linear, so $[\boldsymbol{\varphi}_w]_\times = [\alpha\mathbf{a}_w]_\times = \alpha[\mathbf{a}_w]_\times$. Setting $s = \alpha$ therefore defines the finite rotation

$$\boxed{\mathbf{Q}_w(\mathbf{a}_w, \alpha) := \exp(\alpha[\mathbf{a}_w]_\times) = \exp([\boldsymbol{\varphi}_w]_\times)}.$$

The mapping

$$\exp : \mathfrak{so}(3) \to SO(3)$$

is called the **exponential map** for rotations. The arrow states that the function accepts an element of $\mathfrak{so}(3)$ and returns an element of $SO(3)$. For the rotation vector introduced above,

$$\underbrace{[\boldsymbol{\varphi}_w]_\times}_{\in\mathfrak{so}(3)} \quad\longmapsto\quad \underbrace{\exp([\boldsymbol{\varphi}_w]_\times)}_{\in SO(3)}.$$

Thus, the exponential map converts the dimensionless skew-symmetric matrix form $[\boldsymbol{\varphi}_w]_\times$ of the exponential coordinates into a finite rotation matrix $\mathbf{Q}_w$. Although $[\boldsymbol{\varphi}_w]_\times$ belongs to the tangent space $\mathfrak{so}(3)$, it need not represent an infinitesimal change: its scale is set by the possibly finite angle α. During time-dependent motion, the genuinely infinitesimal rotation coordinate accumulated over an infinitesimal time interval dt is

$$\boxed{[d\boldsymbol{\varphi}_w]_\times = [\boldsymbol{\omega}_w]_\times\,dt},$$

where $\boldsymbol{\omega}_w \in \mathbb{R}^3$ is the world-expressed angular velocity in $\mathrm{rad\,s^{-1}}$.

To verify that the result is a rotation matrix, set $\boldsymbol{\Omega} := [\mathbf{a}_w]_\times$. Because $\boldsymbol{\Omega}^\mathsf{T} = -\boldsymbol{\Omega}$, transposing the exponential series gives

$$\mathbf{Q}_w^\mathsf{T} = \exp(-\alpha\boldsymbol{\Omega}).$$

The inverse property of the matrix exponential gives $\exp(-\alpha\boldsymbol{\Omega}) = \exp(\alpha\boldsymbol{\Omega})^{-1}$. Hence,

$$\mathbf{Q}_w^\mathsf{T}\mathbf{Q}_w = \exp(-\alpha\boldsymbol{\Omega})\exp(\alpha\boldsymbol{\Omega}) = \exp(\alpha\boldsymbol{\Omega})^{-1}\exp(\alpha\boldsymbol{\Omega}) = \mathbf{I}_3.$$

Thus, $\mathbf{Q}_w$ is orthogonal. Taking determinants of $\mathbf{Q}_w^\mathsf{T}\mathbf{Q}_w = \mathbf{I}_3$ gives

$$\det(\mathbf{Q}_w)^2 = 1,$$

so an orthogonal matrix can have determinant only $+1$ or -1. At $\alpha = 0$,

$$\mathbf{Q}_w(0) = \exp(\mathbf{0}_{3\times 3}) = \mathbf{I}_3, \qquad \det(\mathbf{Q}_w(0)) = 1.$$

As α changes continuously, every entry of $\mathbf{Q}_w(\alpha) = \exp(\alpha\boldsymbol{\Omega})$ also changes continuously. The determinant is calculated from those entries using sums and products, so $\det(\mathbf{Q}_w(\alpha))$ is continuous. To change from $+1$ to -1, it would have to pass through intermediate values, including zero. This is impossible while $\mathbf{Q}_w$ remains orthogonal because an orthogonal matrix has determinant only $+1$ or -1 and is never singular. Therefore, $\det(\mathbf{Q}_w) = 1$ for every α, proving $\mathbf{Q}_w \in SO(3)$.

7.5.3 Deriving Rodrigues' rotation formula

Rodrigues' formula is a finite closed form of the matrix exponential derived above. Setting $\boldsymbol{\Omega} := [\mathbf{a}_w]_\times$ in the exponential-series definition gives

$$\mathbf{Q}_w = \exp(\alpha\boldsymbol{\Omega}) = \mathbf{I}_3 + \alpha\boldsymbol{\Omega} + \frac{\alpha^2\boldsymbol{\Omega}^2}{2!} + \frac{\alpha^3\boldsymbol{\Omega}^3}{3!} + \cdots.$$

This expression contains infinitely many powers of $\boldsymbol{\Omega}$. The powers can be reduced to a repeating pattern involving only $\mathbf{I}_3$, $\boldsymbol{\Omega}$, and $\boldsymbol{\Omega}^2$. To find that pattern, apply the vector triple-product identity from Section 1.1.5,

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a}^\mathsf{T}\mathbf{c}) - \mathbf{c}(\mathbf{a}^\mathsf{T}\mathbf{b}),$$

with $\mathbf{a} = \mathbf{b} = \mathbf{a}_w$ and $\mathbf{c} = \mathbf{y}$, where $\mathbf{y} \in \mathbb{R}^3$. This substitution gives

$$\begin{aligned}\mathbf{\Omega}^2\mathbf{y} &= \mathbf{a}_w \times (\mathbf{a}_w \times \mathbf{y}) \\ &= \mathbf{a}_w(\mathbf{a}_w^\mathsf{T}\mathbf{y}) - \mathbf{y}(\mathbf{a}_w^\mathsf{T}\mathbf{a}_w) \\ &= \left(\mathbf{a}_w\mathbf{a}_w^\mathsf{T} - \mathbf{I}_3\right)\mathbf{y}.\end{aligned}$$

The second equality is the triple-product identity after the stated substitution: $\mathbf{a}_w^\mathsf{T}\mathbf{y}$ is the scalar projection factor in the first term, while $\mathbf{a}_w^\mathsf{T}\mathbf{a}_w$ is the squared norm of the axis. The last equality uses $\mathbf{a}_w^\mathsf{T}\mathbf{a}_w = 1$, rewrites $\mathbf{a}_w(\mathbf{a}_w^\mathsf{T}\mathbf{y})$ as $(\mathbf{a}_w\mathbf{a}_w^\mathsf{T})\mathbf{y}$, and rewrites $\mathbf{y}$ as $\mathbf{I}_3\mathbf{y}$. Because the equation holds for every $\mathbf{y}$,

$$\boxed{\mathbf{\Omega}^2 = \mathbf{a}_w\mathbf{a}_w^\mathsf{T} - \mathbf{I}_3}.$$

Furthermore, $\mathbf{\Omega}\mathbf{a}_w = \mathbf{a}_w \times \mathbf{a}_w = \mathbf{0}_3$, so

$$\begin{aligned}\mathbf{\Omega}^3 &= \mathbf{\Omega}\left(\mathbf{a}_w\mathbf{a}_w^\mathsf{T} - \mathbf{I}_3\right) \\ &= (\mathbf{\Omega}\mathbf{a}_w)\mathbf{a}_w^\mathsf{T} - \mathbf{\Omega} \\ &= -\mathbf{\Omega}.\end{aligned}$$

Consequently, the odd powers alternate between $\mathbf{\Omega}$ and $-\mathbf{\Omega}$, while the positive even powers alternate between $\mathbf{\Omega}^2$ and $-\mathbf{\Omega}^2$. Expand the matrix exponential and group those two power sequences:

$$\begin{aligned}\exp(\alpha\mathbf{\Omega}) &= \mathbf{I}_3 + \alpha\mathbf{\Omega} + \frac{\alpha^2}{2!}\mathbf{\Omega}^2 + \frac{\alpha^3}{3!}\mathbf{\Omega}^3 + \cdots \\ &= \mathbf{I}_3 + \left(\alpha - \frac{\alpha^3}{3!} + \frac{\alpha^5}{5!} - \cdots\right)\mathbf{\Omega} \\ &\quad + \left(\frac{\alpha^2}{2!} - \frac{\alpha^4}{4!} + \frac{\alpha^6}{6!} - \cdots\right)\mathbf{\Omega}^2.\end{aligned}$$

The first scalar series is $\sin\alpha$, and the second is $1 - \cos\alpha$. Therefore,

$$\boxed{\mathbf{Q}_w(\mathbf{a}_w, \alpha) = \exp(\alpha[\mathbf{a}_w]_\times) = \mathbf{I}_3 + \sin\alpha[\mathbf{a}_w]_\times + (1 - \cos\alpha)[\mathbf{a}_w]_\times^2}.$$

This closed form is **Rodrigues' rotation formula**. Its geometric meaning becomes clear by decomposing any vector $\mathbf{y} \in \mathbb{R}^3$ into a component parallel to the axis and a component perpendicular to it:

$$\mathbf{y}_\parallel := (\mathbf{a}_w^\mathsf{T}\mathbf{y})\mathbf{a}_w, \qquad \mathbf{y}_\perp := \mathbf{y} - \mathbf{y}_\parallel.$$

Because $\mathbf{a}_w$ is a unit vector, $\mathbf{y} = \mathbf{y}_\parallel + \mathbf{y}_\perp$. Moreover, $\mathbf{a}_w^\mathsf{T}\mathbf{y}_\perp = \mathbf{a}_w^\mathsf{T}\mathbf{y} - (\mathbf{a}_w^\mathsf{T}\mathbf{y})(\mathbf{a}_w^\mathsf{T}\mathbf{a}_w) = 0$, so $\mathbf{y}_\perp$ is perpendicular to the axis. The parallel component is a scalar multiple of $\mathbf{a}_w$, so its cross product with $\mathbf{a}_w$ is zero. Therefore,

$$\begin{aligned}\boldsymbol{\Omega}\mathbf{y} &= \mathbf{a}_w \times \left(\mathbf{y}_\parallel + \mathbf{y}_\perp\right) \\ &= \underbrace{\mathbf{a}_w \times \mathbf{y}_\parallel}_{\mathbf{0}_3} + \mathbf{a}_w \times \mathbf{y}_\perp \\ &= \mathbf{a}_w \times \mathbf{y}_\perp.\end{aligned}$$

The identity $\boldsymbol{\Omega}^2 = \mathbf{a}_w\mathbf{a}_w^\mathsf{T} - \mathbf{I}_3$ derived above gives the second required result:

$$\begin{aligned}\boldsymbol{\Omega}^2\mathbf{y} &= \mathbf{a}_w(\mathbf{a}_w^\mathsf{T}\mathbf{y}) - \mathbf{y} \\ &= \mathbf{y}_\parallel - \left(\mathbf{y}_\parallel + \mathbf{y}_\perp\right) \\ &= -\mathbf{y}_\perp.\end{aligned}$$

Apply Rodrigues' matrix formula to $\mathbf{y}$ and substitute these two results:

$$\begin{aligned}\mathbf{Q}_w\mathbf{y} &= \left[\mathbf{I}_3 + \sin\alpha\,\boldsymbol{\Omega} + (1-\cos\alpha)\boldsymbol{\Omega}^2\right]\mathbf{y} \\ &= \mathbf{y} + \sin\alpha(\mathbf{a}_w \times \mathbf{y}_\perp) - (1-\cos\alpha)\mathbf{y}_\perp \\ &= \mathbf{y}_\parallel + \cos\alpha\,\mathbf{y}_\perp + \sin\alpha(\mathbf{a}_w \times \mathbf{y}_\perp).\end{aligned}$$

Thus,

$$\boxed{\mathbf{Q}_w\mathbf{y} = \mathbf{y}_\parallel + \cos\alpha\,\mathbf{y}_\perp + \sin\alpha(\mathbf{a}_w \times \mathbf{y}_\perp)}.$$

Figure 7.2 illustrates this decomposition for $\mathbf{a}_w = \mathbf{e}_z$. The parallel component remains unchanged. In the plane perpendicular to $\mathbf{a}_w$, the vectors $\mathbf{y}_\perp$ and $\mathbf{a}_w \times \mathbf{y}_\perp$ are perpendicular and have equal length, so the coefficients $\cos\alpha$ and $\sin\alpha$ rotate $\mathbf{y}_\perp$ through angle α according to the right-hand rule.

7.5.4 Check: rotation about the world z_w-axis

Choose

$$\mathbf{a}_w = \mathbf{e}_z = \begin{bmatrix}0\\0\\1\end{bmatrix}, \qquad [\mathbf{a}_w]_\times = \begin{bmatrix}0 & -1 & 0\\1 & 0 & 0\\0 & 0 & 0\end{bmatrix}.$$

Squaring the cross-product matrix gives

$$[\mathbf{a}_w]_\times^2 = \begin{bmatrix}-1 & 0 & 0\\0 & -1 & 0\\0 & 0 & 0\end{bmatrix}.$$

Substitution into Rodrigues' formula produces

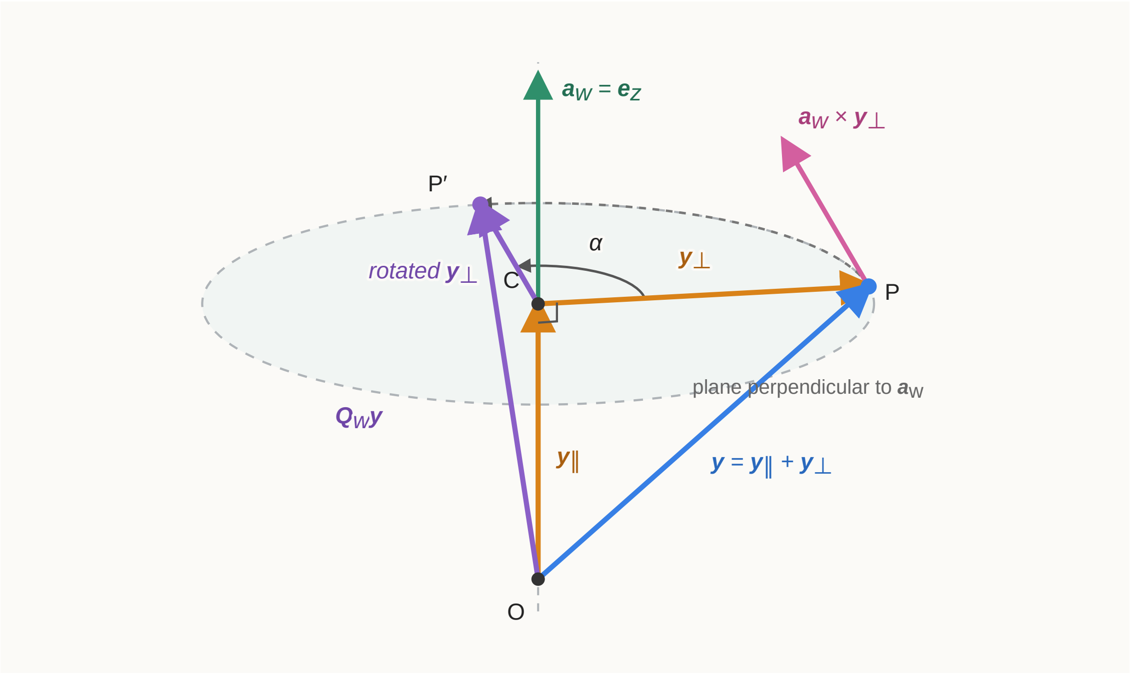

Figure 7.2: Geometric interpretation of Rodrigues' formula for the example axis $\mathbf{a}_w = \mathbf{e}_z$. The vector's parallel component remains fixed while its perpendicular component rotates through α.

$$\mathbf{Q}_w(\mathbf{e}_z, \alpha) = \mathbf{I}_3 + \sin\alpha \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + (1 - \cos\alpha) \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$
$$= \boxed{\begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}}.$$

This is exactly the coordinate-axis rotation $\mathbf{R}_z(\alpha)$ derived in Section 7.3.1.

7.5.5 Applying a constant angular velocity over a time interval

Suppose the world-expressed angular velocity $\boldsymbol{\omega}_w \in \mathbb{R}^3$ remains constant during a time interval $\Delta t > 0$. If $\boldsymbol{\omega}_w \neq \mathbf{0}_3$, define

$$\mathbf{a}_w := \frac{\boldsymbol{\omega}_w}{\|\boldsymbol{\omega}_w\|_2}, \qquad \alpha := \|\boldsymbol{\omega}_w\|_2 \Delta t.$$

The rotation vector over the interval is then

$$\boldsymbol{\varphi}_w = \alpha \mathbf{a}_w = \Delta t \, \boldsymbol{\omega}_w.$$

Because the angular velocity is expressed using the fixed world axes, the finite update premultiplies the current orientation:

$$\boxed{\mathbf{R}_{wb}(t+\Delta t) = \exp(\Delta t[\boldsymbol{\omega}_w]_\times)\,\mathbf{R}_{wb}(t)}.$$

If the constant angular velocity is instead expressed using the body axes, the update postmultiplies:

$$\boxed{\mathbf{R}_{wb}(t+\Delta t) = \mathbf{R}_{wb}(t)\exp(\Delta t[\boldsymbol{\omega}_b]_\times)}.$$

When the angular velocity is zero, define the finite increment as $\mathbf{I}_3$; the orientation then remains unchanged. These finite updates are exact only when the corresponding world- or body-expressed angular velocity remains constant throughout the interval.

Axis–angle coordinates are not globally unique. The pairs $(\mathbf{a}_w, \alpha)$ and $(-\mathbf{a}_w, -\alpha)$ describe the same turn, adding $2\pi k$ to the angle for any $k \in \mathbb{Z}$ produces the same rotation, and when $\alpha = 0$ the axis direction is arbitrary. These facts matter when recovering axis–angle coordinates from a rotation matrix or comparing two numerical orientation representations.

Quick test E: axis–angle coordinates and the exponential map

1. What information is stored in the axis–angle pair $(\mathbf{a}_w, \alpha)$, and how does the rotation vector $\boldsymbol{\varphi}_w$ combine it?
2. State the matrix-exponential definition and explain why $\mathbf{p}(s) = \exp(s[\mathbf{a}_w]_\times)\mathbf{p}_0$ solves $d\mathbf{p}/ds = [\mathbf{a}_w]_\times\mathbf{p}$.
3. Use the vector triple-product identity to derive $[\mathbf{a}_w]_\times^2 = \mathbf{a}_w\mathbf{a}_w^\mathsf{T} - \mathbf{I}_3$ for a unit axis.
4. Starting from the matrix-exponential series and $[\mathbf{a}_w]_\times^3 = -[\mathbf{a}_w]_\times$, derive Rodrigues' rotation formula.
5. Why does Rodrigues' formula leave the vector component parallel to $\mathbf{a}_w$ unchanged?
6. A constant angular velocity $\boldsymbol{\omega}_w = [0\ 0\ 2]^\mathsf{T}\,\mathrm{rad\,s^{-1}}$ acts for $\Delta t = 0.25\,\mathrm{s}$. Find the axis, angle, rotation vector, and finite orientation increment.
7. Why does a world-expressed constant angular velocity premultiply $\mathbf{R}_{wb}$, while a body-expressed constant angular velocity postmultiplies it?
8. Give two reasons why an axis–angle representation of a rotation is not unique.

Answers and hints

1. The unit vector $\mathbf{a}_w$ gives the positive axis direction in world coordinates, and α gives the signed right-hand-rule angle. Their product $\boldsymbol{\varphi}_w = \alpha\mathbf{a}_w$ stores the same information in one three-component column.

2. $\exp(\mathbf{A}) = \mathbf{I} + \mathbf{A} + \mathbf{A}^2/2! + \cdots$. Differentiating $\exp(s\mathbf{A})$ term by term gives $\mathbf{A}\exp(s\mathbf{A})$; setting $\mathbf{A} = [\mathbf{a}_w]_\times$ produces the required differential equation and the identity exponential satisfies the initial condition.
3. For every $\mathbf{y}$, $\mathbf{a}_w \times (\mathbf{a}_w \times \mathbf{y}) = \mathbf{a}_w(\mathbf{a}_w^\mathsf{T}\mathbf{y}) - (\mathbf{a}_w^\mathsf{T}\mathbf{a}_w)\mathbf{y} = (\mathbf{a}_w\mathbf{a}_w^\mathsf{T} - \mathbf{I}_3)\mathbf{y}$. Equality for every $\mathbf{y}$ proves the matrix identity.
4. Replace successive odd powers by alternating $[\mathbf{a}_w]_\times$ and $-[\mathbf{a}_w]_\times$, and successive positive even powers by alternating $[\mathbf{a}_w]_\times^2$ and $-[\mathbf{a}_w]_\times^2$. Their scalar coefficients sum to $\sin\alpha$ and $1 - \cos\alpha$, respectively.
5. The cross product of the axis with a parallel vector is zero, so both $[\mathbf{a}_w]_\times\mathbf{y}_\parallel$ and $[\mathbf{a}_w]_\times^2\mathbf{y}_\parallel$ vanish.
6. $\mathbf{a}_w = [0\ 0\ 1]^\mathsf{T}$, $\alpha = 2(0.25) = 0.5\,\text{rad}$, and $\boldsymbol{\varphi}_w = [0\ 0\ 0.5]^\mathsf{T}\,\text{rad}$. The increment is $\mathbf{Q}_w = \mathbf{R}_z(0.5) = \begin{bmatrix} \cos 0.5 & -\sin 0.5 & 0 \\ \sin 0.5 & \cos 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
7. A finite increment expressed using world axes acts on the world-coordinate columns and therefore premultiplies. An increment expressed using the old body axes composes after the body-to-world mapping and therefore postmultiplies.
8. Examples are $(\mathbf{a}_w, \alpha) \sim (-\mathbf{a}_w, -\alpha)$, angles differing by $2\pi k$ give the same rotation, and the axis is arbitrary for zero rotation.

7.6 Spatial pose and $SE(3)$

7.6.1 From a point transformation to a homogeneous matrix

Orientation alone does not locate a rigid body in three-dimensional space. Suppose frame $\{b\}$ is fixed to a rigid body and frame $\{w\}$ is fixed in the world. A **spatial pose** is the physical placement of frame $\{b\}$ relative to frame $\{w\}$, consisting of:

- the orientation $\mathbf{R}_{wb} \in SO(3)$ of frame $\{b\}$ relative to frame $\{w\}$; and
- the position ${}^w\mathbf{p}_b \in \mathbb{R}^3$ of the body origin O_b, expressed in world coordinates and measured in metres.

Consider a material point P fixed in the body. Its body-frame coordinate ${}^b\mathbf{p}_P \in \mathbb{R}^3$ is constant and measured in metres. The product $\mathbf{R}_{wb}{}^b\mathbf{p}_P$ expresses the displacement from O_b to P along the world axes. It changes the coordinate description of that displacement; it does not physically move P. This displacement is not yet the world position of P. Adding the world position of O_b gives

$$\boxed{{}^w\mathbf{p}_P = \mathbf{R}_{wb}{}^b\mathbf{p}_P + {}^w\mathbf{p}_b}.$$

Every term in this equation belongs to $\mathbb{R}^3$ and has units of metres. The equation first re-expresses the body-relative displacement in world coordinates and then adds the translation from the world origin O_w to the body origin O_b.

The rotation and translation can be packaged into one matrix by appending a final coordinate equal to one. Define the **homogeneous point coordinates**

$$ {}^{b}\overline{\mathbf{p}}_P := \begin{bmatrix} {}^{b}\mathbf{p}_P \\ 1 \end{bmatrix} \in \mathbb{R}^4, \qquad {}^{w}\overline{\mathbf{p}}_P := \begin{bmatrix} {}^{w}\mathbf{p}_P \\ 1 \end{bmatrix} \in \mathbb{R}^4. $$

The overline distinguishes the four-entry homogeneous coordinate from the ordinary three-entry position coordinate. Its first three entries have units of metres, while the appended 1 is dimensionless and exists only to make translation part of a matrix multiplication.

Define the **homogeneous transformation**

$$ \boxed{\mathbf{T}_{wb} := \begin{bmatrix} \mathbf{R}_{wb} & {}^{w}\mathbf{p}_b \\ \mathbf{0}_3^\top & 1 \end{bmatrix} \in \mathbb{R}^{4\times 4}}. $$

Block multiplication now reproduces the physical point-transformation equation:

$$ \begin{aligned} \mathbf{T}_{wb}\,{}^{b}\overline{\mathbf{p}}_P &= \begin{bmatrix} \mathbf{R}_{wb} & {}^{w}\mathbf{p}_b \\ \mathbf{0}_3^\top & 1 \end{bmatrix} \begin{bmatrix} {}^{b}\mathbf{p}_P \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{R}_{wb}\,{}^{b}\mathbf{p}_P + {}^{w}\mathbf{p}_b \\ 1 \end{bmatrix} \\ &= {}^{w}\overline{\mathbf{p}}_P. \end{aligned} $$

For a free vector $\mathbf{v}$, translation is irrelevant because a free vector has no fixed origin. Append a 0 instead of a 1 to define its homogeneous coordinate column:

$$ {}^{b}\overline{\mathbf{v}} := \begin{bmatrix} {}^{b}\mathbf{v} \\ 0 \end{bmatrix} \in \mathbb{R}^4. $$

Applying $\mathbf{T}_{wb}$ then gives

$$ \mathbf{T}_{wb} \begin{bmatrix} {}^{b}\mathbf{v} \\ 0 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{wb}\,{}^{b}\mathbf{v} \\ 0 \end{bmatrix}. $$

The appended 0 prevents the translation column from contributing, so this equation reproduces the rotation-only coordinate mapping ${}^{w}\mathbf{v} = \mathbf{R}_{wb}\,{}^{b}\mathbf{v}$ from Section 7.2.1.

As distinguished for planar transformations in Section 6.2.3, the same homogeneous-matrix form has three possible interpretations:

- As a **relative-pose description**, $\mathbf{T}_{wb}$ stores the orientation and position of frame $\{b\}$ relative to frame $\{w\}$.
- As a **passive coordinate change**, ${}^{w}\overline{\mathbf{p}}_P = \mathbf{T}_{wb}\,{}^{b}\overline{\mathbf{p}}_P$ maps the coordinates of the unchanged geometric point P from frame $\{b\}$ to frame $\{w\}$.

- As an **active rigid-motion operator**, an explicitly declared $\mathbf{T}_\Delta \in SE(3)$ moves a geometric point from P to P' while both coordinate columns remain expressed in the same frame, for example ${}^w\overline{\mathbf{p}}_{P'} = \mathbf{T}_\Delta {}^w\overline{\mathbf{p}}_P$.

The project notation therefore uses a frame-labelled symbol such as $\mathbf{T}_{wb}$ for the first two interpretations and a separately declared symbol such as $\mathbf{T}_\Delta$ for an active motion. A transformation matrix is not itself the position of one point; applying the relative-pose mapping $\mathbf{T}_{wb}$ to every body-fixed point reconstructs the body's complete spatial placement.

The set containing every matrix of this form is the **special Euclidean group in three dimensions**:

$$\boxed{SE(3) := \left\{ \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_3^\mathsf{T} & 1 \end{bmatrix} \in \mathbb{R}^{4\times 4} \;\middle|\; \mathbf{R} \in SO(3),\ \mathbf{p} \in \mathbb{R}^3 \right\}}.$$

Here, *Euclidean* indicates that the transformation preserves Euclidean distances and angles, while *special* indicates that $\det(\mathbf{R}) = 1$, so the transformation preserves orientation and excludes reflections. In a physical pose, the entries of $\mathbf{p}$ are measured in metres even though the abstract set definition writes $\mathbf{p} \in \mathbb{R}^3$.

For a single free rigid body, its spatial pose determines its complete configuration: the body-frame coordinates of all material points are fixed, so the pose's one rotation and one translation locate every point. After the world and body frames have been fixed, each spatial pose produces exactly one matrix $\mathbf{T}_{wb} \in SE(3)$. Conversely, each matrix $\mathbf{T}_{wb} \in SE(3)$ specifies exactly one spatial pose because its rotation block gives $\mathbf{R}_{wb}$ and its translation column gives ${}^w\mathbf{p}_b$. The configuration space of a free spatial rigid body can therefore be identified with $SE(3)$:

$$\boxed{\mathcal{C} \cong SE(3)}.$$

7.6.2 Composing and inverting spatial transformations

To **compose** transformations means to perform two coordinate mappings in sequence and replace them with one direct mapping. Introduce frames $\{a\}$, $\{b\}$, and $\{c\}$. Let $\mathbf{T}_{ab} \in SE(3)$ map point coordinates from frame $\{b\}$ to frame $\{a\}$, and let $\mathbf{T}_{bc} \in SE(3)$ map them from frame $\{c\}$ to frame $\{b\}$. Applying both mappings to a geometric point P gives

$$\begin{aligned} {}^a\overline{\mathbf{p}}_P &= \mathbf{T}_{ab}\left(\mathbf{T}_{bc} {}^c\overline{\mathbf{p}}_P\right) \\ &= \left(\mathbf{T}_{ab}\mathbf{T}_{bc}\right) {}^c\overline{\mathbf{p}}_P. \end{aligned}$$

The rightmost transformation $\mathbf{T}_{bc}$ acts first. The direct mapping from frame $\{c\}$ to frame $\{a\}$ is therefore

$$\boxed{\mathbf{T}_{ac} = \mathbf{T}_{ab}\mathbf{T}_{bc}}.$$

The repeated inner label b identifies the intermediate frame and provides a useful frame-cancellation check. To see how the orientation and position parts combine, write

$$\mathbf{T}_{ab} = \begin{bmatrix} \mathbf{R}_{ab} & {}^{a}\mathbf{p}_{b} \\ \mathbf{0}_3^{\mathsf{T}} & 1 \end{bmatrix}, \qquad \mathbf{T}_{bc} = \begin{bmatrix} \mathbf{R}_{bc} & {}^{b}\mathbf{p}_{c} \\ \mathbf{0}_3^{\mathsf{T}} & 1 \end{bmatrix}.$$

Here, ${}^{a}\mathbf{p}_{b} \in \mathbb{R}^3$ is the position of origin O_b relative to origin O_a, expressed in frame $\{a\}$, and ${}^{b}\mathbf{p}_{c} \in \mathbb{R}^3$ is the position of origin O_c relative to origin O_b, expressed in frame $\{b\}$; both are measured in metres. Block multiplication gives

$$\mathbf{T}_{ab}\mathbf{T}_{bc} = \begin{bmatrix} \mathbf{R}_{ab}\mathbf{R}_{bc} & \mathbf{R}_{ab}{}^{b}\mathbf{p}_{c} + {}^{a}\mathbf{p}_{b} \\ \mathbf{0}_3^{\mathsf{T}} & 1 \end{bmatrix}.$$

Consequently,

$$\boxed{\mathbf{R}_{ac} = \mathbf{R}_{ab}\mathbf{R}_{bc}, \qquad {}^{a}\mathbf{p}_{c} = \mathbf{R}_{ab}{}^{b}\mathbf{p}_{c} + {}^{a}\mathbf{p}_{b}}.$$

The position columns cannot be added until they are expressed along the same axes. Multiplication by $\mathbf{R}_{ab}$ re-expresses ${}^{b}\mathbf{p}_{c}$ in frame $\{a\}$ before it is added to ${}^{a}\mathbf{p}_{b}$. Because $\mathbf{R}_{ab}\mathbf{R}_{bc} \in SO(3)$ and the new translation belongs to $\mathbb{R}^3$, the product remains in $SE(3)$. This proves that $SE(3)$ is closed under matrix multiplication.

The **inverse transformation** reverses a coordinate mapping. Starting from

$${}^{a}\mathbf{p}_{P} = \mathbf{R}_{ab}{}^{b}\mathbf{p}_{P} + {}^{a}\mathbf{p}_{b},$$

subtract ${}^{a}\mathbf{p}_{b}$ and premultiply by $\mathbf{R}_{ab}^{-1} = \mathbf{R}_{ab}^{\mathsf{T}}$:

$$\begin{aligned} {}^{b}\mathbf{p}_{P} &= \mathbf{R}_{ab}^{\mathsf{T}}\left({}^{a}\mathbf{p}_{P} - {}^{a}\mathbf{p}_{b}\right) \\ &= \mathbf{R}_{ab}^{\mathsf{T}}{}^{a}\mathbf{p}_{P} - \mathbf{R}_{ab}^{\mathsf{T}}{}^{a}\mathbf{p}_{b}. \end{aligned}$$

Comparing this result with the standard point-transformation form shows that the reverse rotation and translation are

$$\mathbf{R}_{ba} = \mathbf{R}_{ab}^{\mathsf{T}}, \qquad {}^{b}\mathbf{p}_{a} = -\mathbf{R}_{ab}^{\mathsf{T}}{}^{a}\mathbf{p}_{b}.$$

Therefore,

$$\boxed{\mathbf{T}_{ab}^{-1} = \mathbf{T}_{ba} = \begin{bmatrix} \mathbf{R}_{ab}^{\mathsf{T}} & -\mathbf{R}_{ab}^{\mathsf{T}}{}^{a}\mathbf{p}_{b} \\ \mathbf{0}_3^{\mathsf{T}} & 1 \end{bmatrix} \in SE(3)}.$$

The inverse translation is not generally $-{}^a\mathbf{p}_b$, because that column is expressed in frame $\{a\}$. The factor $\mathbf{R}_{ab}^\top$ re-expresses the negated displacement in frame $\{b\}$, producing the position ${}^b\mathbf{p}_a$ of origin O_a relative to origin O_b.

The identity transformation is

$$\boxed{\mathbf{T}_{aa} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3^\top & 1 \end{bmatrix} = \mathbf{I}_4}.$$

It leaves every homogeneous point coordinate unchanged. Matrix multiplication is associative, the product of two elements remains in $SE(3)$, $\mathbf{I}_4$ is the identity element, and the inverse derived above belongs to $SE(3)$. These four properties establish that $SE(3)$ is a group under matrix multiplication. The operation is generally not commutative: changing the order of spatial rotations and translations can change the final pose.

7.6.3 Worked example: composition and inverse round trip

Consider three coordinate frames $\{a\}$, $\{b\}$, and $\{c\}$. This example uses rotations about two different axes so that the calculation is genuinely three-dimensional. Let frame $\{b\}$ have the following orientation and origin position relative to frame $\{a\}$:

$$\mathbf{R}_{ab} = \mathbf{R}_z\left(\frac{\pi}{2}\right) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad {}^a\mathbf{p}_b = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \text{ m}.$$

The matrix $\mathbf{R}_{ab} \in SO(3)$ maps coordinate columns from frame $\{b\}$ to frame $\{a\}$, while ${}^a\mathbf{p}_b \in \mathbb{R}^3$ is the position of origin O_b relative to origin O_a, expressed along the axes of frame $\{a\}$. Let frame $\{c\}$ have the following orientation and origin position relative to frame $\{b\}$:

$$\mathbf{R}_{bc} = \mathbf{R}_x\left(\frac{\pi}{2}\right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \qquad {}^b\mathbf{p}_c = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \text{ m}.$$

The direct orientation from frame $\{c\}$ to frame $\{a\}$ is found by composition:

$$\begin{aligned} \mathbf{R}_{ac} &= \mathbf{R}_{ab}\mathbf{R}_{bc} \\ &= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}. \end{aligned}$$

To combine the origin positions, ${}^{b}\mathbf{p}_c$ must first be re-expressed in frame $\{a\}$. Therefore,

$$\begin{aligned}
{}^{a}\mathbf{p}_c &= \mathbf{R}_{ab}{}^{b}\mathbf{p}_c + {}^{a}\mathbf{p}_b \\
&= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \text{ m.}
\end{aligned}$$

The composed transformation is consequently

$$\boxed{\mathbf{T}_{ac} = \mathbf{T}_{ab}\mathbf{T}_{bc} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

The first three entries of the last column are measured in metres; the rotation entries and the homogeneous last row are dimensionless.

To check both the numerical result and the multiplication order, choose a geometric point P whose coordinates in frame $\{c\}$ are

$${}^{c}\mathbf{p}_P = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ m.}$$

First map its coordinates from frame $\{c\}$ to frame $\{b\}$:

$$\begin{aligned}
{}^{b}\mathbf{p}_P &= \mathbf{R}_{bc}{}^{c}\mathbf{p}_P + {}^{b}\mathbf{p}_c \\
&= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} \text{ m.}
\end{aligned}$$

Then map the result from frame $\{b\}$ to frame $\{a\}$:

$$
\begin{aligned}
{}^a\mathbf{p}_P &= \mathbf{R}_{ab}\,{}^b\mathbf{p}_P + {}^a\mathbf{p}_b \\
&= \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \text{m}.
\end{aligned}
$$

The direct transformation gives the same coordinates:

$$
\begin{aligned}
{}^a\mathbf{p}_P &= \mathbf{R}_{ac}\,{}^c\mathbf{p}_P + {}^a\mathbf{p}_c \\
&= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \text{m}.
\end{aligned}
$$

This agreement confirms that $\mathbf{T}_{bc}$ must act first and $\mathbf{T}_{ab}$ second in the product $\mathbf{T}_{ab}\mathbf{T}_{bc}$. These equations re-express the coordinates of the same point P; they do not physically move the point.

The inverse should recover the original frame-$\{c\}$ coordinates. Using the inverse formula from Section 7.6.2 gives

$$
\begin{aligned}
\mathbf{R}_{ca} &= \mathbf{R}_{ac}^{\mathsf{T}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \\
{}^c\mathbf{p}_a &= -\mathbf{R}_{ac}^{\mathsf{T}}\,{}^a\mathbf{p}_c \\
&= -\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \\ -1 \end{bmatrix} \text{m}.
\end{aligned}
$$

Therefore,

$$
\mathbf{T}_{ca} = \mathbf{T}_{ac}^{-1} = \begin{bmatrix} 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.
$$

Applying this reverse mapping to the calculated frame-$\{a\}$ coordinates gives

$$
\begin{aligned}
{}^{c}\mathbf{p}_P &= \mathbf{R}_{ca}\,{}^{a}\mathbf{p}_P + {}^{c}\mathbf{p}_a \\
&= \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -3 \\ -1 \\ -1 \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ m.}
\end{aligned}
$$

The round trip returns the original coordinates exactly. Point mapping through both intermediate frames and inverse round trips are therefore useful checks for a spatial-transformation calculation.

Quick test F: spatial pose and $SE(3)$

Try these questions without looking back at the preceding sections.

1. In the transformation

$$
\mathbf{T}_{wb} = \begin{bmatrix} \mathbf{R}_{wb} & {}^{w}\mathbf{p}_b \\ \mathbf{0}_3^{\mathsf{T}} & 1 \end{bmatrix},
$$

   what do $\mathbf{R}_{wb}$ and ${}^{w}\mathbf{p}_b$ describe, and which coordinate mapping does $\mathbf{T}_{wb}$ perform?

2. Why is a point represented by $[\,\mathbf{p}^{\mathsf{T}}\ 1\,]^{\mathsf{T}}$ in homogeneous coordinates, while a free vector is represented by $[\,\mathbf{v}^{\mathsf{T}}\ 0\,]^{\mathsf{T}}$?

3. Let $\mathbf{T}_{ab}$ map coordinates from frame $\{b\}$ to frame $\{a\}$ and let $\mathbf{T}_{bc}$ map coordinates from frame $\{c\}$ to frame $\{b\}$. Derive the rotation and translation blocks of $\mathbf{T}_{ac} = \mathbf{T}_{ab}\mathbf{T}_{bc}$.

4. Why must ${}^{b}\mathbf{p}_c$ be multiplied by $\mathbf{R}_{ab}$ before it can be added to ${}^{a}\mathbf{p}_b$?

5. Derive $\mathbf{T}_{ab}^{-1}$. Why is its translation block generally not equal to $-{}^{a}\mathbf{p}_b$?

6. Suppose

$$
\mathbf{R}_{wb} = \mathbf{R}_z\left(\frac{\pi}{2}\right), \qquad {}^{w}\mathbf{p}_b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ m}, \qquad {}^{b}\mathbf{p}_P = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \text{ m.}
$$

   Calculate ${}^{w}\mathbf{p}_P$.

7. Explain the difference between interpreting $\mathbf{T}_{wb}$ as a passive coordinate mapping and interpreting a separately declared $\mathbf{T}_{\Delta}$ as an active rigid motion.

Answers and hints

1. The rotation $\mathbf{R}_{wb} \in SO(3)$ describes the orientation of frame $\{b\}$ relative to frame $\{w\}$ and maps coordinate columns from body axes to world axes. The position ${}^{w}\mathbf{p}_b \in \mathbb{R}^3$ locates body origin O_b relative to world origin O_w, expressed along world axes and measured in metres. The complete transformation maps body-frame point coordinates to world-frame point coordinates:

$$ {}^{w}\overline{\mathbf{p}}_P = \mathbf{T}_{wb}\, {}^{b}\overline{\mathbf{p}}_P. $$

2. Multiplying a homogeneous point by $\mathbf{T}_{wb}$ gives

$$ \mathbf{T}_{wb} \begin{bmatrix} {}^{b}\mathbf{p}_P \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{wb}\, {}^{b}\mathbf{p}_P + {}^{w}\mathbf{p}_b \\ 1 \end{bmatrix}, $$

so the appended 1 allows the translation column to contribute. A free vector has no fixed origin, so translation must not affect it:

$$ \mathbf{T}_{wb} \begin{bmatrix} {}^{b}\mathbf{v} \\ 0 \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{wb}\, {}^{b}\mathbf{v} \\ 0 \end{bmatrix}. $$

3. Block multiplication gives

$$ \boxed{\mathbf{R}_{ac} = \mathbf{R}_{ab}\mathbf{R}_{bc}, \qquad {}^{a}\mathbf{p}_c = \mathbf{R}_{ab}\, {}^{b}\mathbf{p}_c + {}^{a}\mathbf{p}_b}. $$

4. The columns ${}^{b}\mathbf{p}_c$ and ${}^{a}\mathbf{p}_b$ initially use different coordinate axes. The product $\mathbf{R}_{ab}\, {}^{b}\mathbf{p}_c$ re-expresses the first displacement along frame-$\{a\}$ axes, after which the two frame-$\{a\}$ coordinate columns can be added.

5. Solving ${}^{a}\mathbf{p}_P = \mathbf{R}_{ab}\, {}^{b}\mathbf{p}_P + {}^{a}\mathbf{p}_b$ for ${}^{b}\mathbf{p}_P$ gives

$$ \boxed{\mathbf{T}_{ab}^{-1} = \mathbf{T}_{ba} = \begin{bmatrix} \mathbf{R}_{ab}^{\mathsf{T}} & -\mathbf{R}_{ab}^{\mathsf{T}}\, {}^{a}\mathbf{p}_b \\ \mathbf{0}_3^{\mathsf{T}} & 1 \end{bmatrix}}. $$

   The reverse displacement must be expressed along frame-$\{b\}$ axes. The factor $\mathbf{R}_{ab}^{\mathsf{T}}$ performs this coordinate change, so merely negating the frame-$\{a\}$ column is insufficient.

6. A positive quarter-turn about z_w maps $[\,2\ 0\ 1\,]^{\mathsf{T}}$ to $[\,0\ 2\ 1\,]^{\mathsf{T}}$. Therefore,

$$\begin{aligned}
{}^{w}\mathbf{p}_P &= \mathbf{R}_{wb}\,{}^{b}\mathbf{p}_P + {}^{w}\mathbf{p}_b \\
&= \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \\
&= \boxed{\begin{bmatrix} 1 \\ 4 \\ 4 \end{bmatrix} \mathrm{m}}.
\end{aligned}$$

7. In the passive interpretation, the geometric point remains unchanged and $\mathbf{T}_{wb}$ changes only its coordinate description from frame $\{b\}$ to frame $\{w\}$. In the active interpretation, $\mathbf{T}_{\Delta}$ physically moves a point or rigid body while the input and output coordinates are expressed in the same declared frame. The same matrix multiplication can implement either interpretation, so the frames and intended meaning must be stated.

7.7 Embedding planar poses in $SE(3)$

A mobile robot may be modelled as moving only in a plane even though its software exchanges three-dimensional poses with other robots, sensors, and maps. A planar pose must therefore be represented inside the spatial-pose space without changing its geometric meaning.

Choose the world plane $z_w = 0$ as the plane containing the body origin. Restrict the body to translation along x_w and y_w and rotation about z_w. The body cannot translate along z_w or rotate about the horizontal x_w and y_w axes under this model. Its planar pose coordinates are

$$\mathbf{q} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix},$$

Here:

- x is the x_w-coordinate of body origin O_b relative to world origin O_w, measured in metres;
- y is the y_w-coordinate of O_b relative to O_w, measured in metres;
- θ is the orientation of body frame $\{b\}$ relative to world frame $\{w\}$, measured as a positive right-hand rotation about the shared z-axis in radians.

The corresponding world-frame position of the body origin is

$${}^{w}\mathbf{p}_b = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} \mathrm{m}.$$

To distinguish matrices of different sizes, let the superscripts (2) and (3) label planar and spatial representations; they are dimension labels, not matrix powers. The planar transformation from Section 6.2.3 is

$$\mathbf{T}_{wb}^{(2)} = \begin{bmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{bmatrix} \in SE(2).$$

The same physical placement is represented spatially by

$$\boxed{\mathbf{T}_{wb}^{(3)} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & x \\ \sin\theta & \cos\theta & 0 & y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \in SE(3)}.$$

The upper-left 3×3 block is the rotation $\mathbf{R}_z(\theta) \in SO(3)$ derived in Section 7.3.1. Its third column $[\,0\ 0\ 1\,]^\mathsf{T}$ states that z_b remains aligned with z_w. The translation column $[\,x\ y\ 0\,]^\mathsf{T}$ states that the body origin remains in the chosen world plane.

Write a generic planar transformation using dimensionless rotation entries r_{ij}, where $i, j \in \{1, 2\}$ are row and column indices, and translation entries p_1 and p_2, measured in metres. The conversion can then be defined by the mapping

$$\boxed{\begin{gathered} \iota : SE(2) \longrightarrow SE(3), \\ \begin{bmatrix} r_{11} & r_{12} & p_1 \\ r_{21} & r_{22} & p_2 \\ 0 & 0 & 1 \end{bmatrix} \longmapsto \begin{bmatrix} r_{11} & r_{12} & 0 & p_1 \\ r_{21} & r_{22} & 0 & p_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \end{gathered}}$$

The Greek letter ι names this **group embedding**. A group embedding is a one-to-one mapping that places one group inside another while preserving the group operation. This mapping is one-to-one because all four planar rotation entries and both planar translation entries can be read back uniquely from the spatial matrix.

To verify that point transformations are preserved, let a point P in the body plane have body-frame coordinates

$${}^{b}\mathbf{p}_P^{(2)} = \begin{bmatrix} p_x \\ p_y \end{bmatrix} \in \mathbb{R}^2, \qquad {}^{b}\mathbf{p}_P^{(3)} = \begin{bmatrix} p_x \\ p_y \\ 0 \end{bmatrix} \in \mathbb{R}^3,$$

where p_x and p_y are measured in metres. The spatial transformation gives

$$
\begin{aligned}
{}^{w}\mathbf{p}_P^{(3)} &= \begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}p_x\\ p_y\\ 0\end{bmatrix} + \begin{bmatrix}x\\ y\\ 0\end{bmatrix}\\
&= \begin{bmatrix}\cos\theta\, p_x - \sin\theta\, p_y + x\\ \sin\theta\, p_x + \cos\theta\, p_y + y\\ 0\end{bmatrix}.
\end{aligned}
$$

The first two entries are exactly the result of the planar transformation $\mathbf{T}_{wb}^{(2)}$, while the third entry remains zero. The spatial matrix therefore reproduces the planar point mapping rather than approximating it.

The planar motion model does not flatten the robot into a two-dimensional object. If a body-fixed point has height p_z relative to the body origin, then

$$
{}^{b}\mathbf{p}_P^{(3)} = \begin{bmatrix}p_x\\ p_y\\ p_z\end{bmatrix} \quad\Longrightarrow\quad {}^{w}\mathbf{p}_P^{(3)} = \begin{bmatrix}\cos\theta\, p_x - \sin\theta\, p_y + x\\ \sin\theta\, p_x + \cos\theta\, p_y + y\\ p_z\end{bmatrix}.
$$

Here, p_z is measured in metres. The body can therefore have three-dimensional geometry; the restriction is that its origin has no vertical motion and its orientation has no rotation away from the shared z-axis.

The embedding also preserves composition. Let

$$
\mathbf{T}_1^{(2)} = \begin{bmatrix}\mathbf{R}_1 & \mathbf{p}_1\\ \mathbf{0}_2^{\mathsf{T}} & 1\end{bmatrix}, \qquad \mathbf{T}_2^{(2)} = \begin{bmatrix}\mathbf{R}_2 & \mathbf{p}_2\\ \mathbf{0}_2^{\mathsf{T}} & 1\end{bmatrix} \in SE(2),
$$

where $\mathbf{R}_1, \mathbf{R}_2 \in SO(2)$ and $\mathbf{p}_1, \mathbf{p}_2 \in \mathbb{R}^2$, with the position entries measured in metres. Their planar product is

$$
\mathbf{T}_1^{(2)}\mathbf{T}_2^{(2)} = \begin{bmatrix}\mathbf{R}_1\mathbf{R}_2 & \mathbf{R}_1\mathbf{p}_2 + \mathbf{p}_1\\ \mathbf{0}_2^{\mathsf{T}} & 1\end{bmatrix}.
$$

In each embedded matrix, the planar rotation and translation occupy the same entries, while the added z-axis row and column remain unchanged. Multiplying the embedded matrices therefore produces the embedded planar product:

$$
\boxed{\iota\left(\mathbf{T}_1^{(2)}\mathbf{T}_2^{(2)}\right) = \iota\left(\mathbf{T}_1^{(2)}\right)\iota\left(\mathbf{T}_2^{(2)}\right)}.
$$

The planar identity maps to the spatial identity:

$$
\iota(\mathbf{I}_3) = \mathbf{I}_4.
$$

Applying the composition identity to $\mathbf{T}^{(2)}(\mathbf{T}^{(2)})^{-1} = \mathbf{I}_3$ therefore gives

$$\boxed{\iota\big((\mathbf{T}^{(2)})^{-1}\big) = \iota\big(\mathbf{T}^{(2)}\big)^{-1}}.$$

Thus, planar composition and inversion can be performed before or after embedding with the same result.

For example, take $x = 2\,\mathrm{m}$, $y = -1\,\mathrm{m}$, and $\theta = \pi/2$. Then

$$\mathbf{T}_{wb}^{(3)} = \begin{bmatrix} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

A body-frame point ${}^{b}\mathbf{p}_P^{(3)} = [\,1\ 0\ 0\,]^\mathsf{T}\,\mathrm{m}$ is mapped to

$${}^{w}\mathbf{p}_P^{(3)} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}\ \mathrm{m}.$$

7.7.1 Recovering and checking a planar pose

The reverse conversion is defined only for spatial poses that satisfy the planar restrictions. Let

$$\mathbf{T}^{(3)} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_3^\mathsf{T} & 1 \end{bmatrix} \in SE(3), \qquad \mathbf{e}_z := \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Here, $\mathbf{p} = [\,p_1\ p_2\ p_3\,]^\mathsf{T} \in \mathbb{R}^3$ is the translation column, measured in metres. When $\mathbf{T}^{(3)} = \mathbf{T}_{wb}^{(3)}$, it is the world-frame position ${}^{w}\mathbf{p}_b$ of body origin O_b.

The spatial pose belongs to the embedded copy $\iota(SE(2)) \subset SE(3)$ exactly when

$$\boxed{\mathbf{R}\mathbf{e}_z = \mathbf{e}_z, \qquad \mathbf{e}_z^\mathsf{T}\mathbf{p} = 0}.$$

The second condition selects the third component of the translation column:

$$\mathbf{e}_z^\mathsf{T}\mathbf{p} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = p_3.$$

The first equation says that rotation leaves the common z-axis unchanged. Because $\mathbf{R} \in SO(3)$, its remaining two columns must then be a right-handed orthonormal basis of the x–y plane. The second equation says that the body origin has zero world-frame z-coordinate. The transformation must consequently have the structure

$$\mathbf{T}^{(3)} = \begin{bmatrix} r_{11} & r_{12} & 0 & p_1 \\ r_{21} & r_{22} & 0 & p_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix} \in SO(2).$$

The planar coordinates can therefore be recovered as

$$\boxed{x = p_1, \qquad y = p_2, \qquad \theta = \operatorname{atan2}(r_{21}, r_{11})}.$$

The function $\operatorname{atan2}(u, v)$ returns the signed angle α satisfying $\sin\alpha = u/\sqrt{u^2+v^2}$ and $\cos\alpha = v/\sqrt{u^2+v^2}$. Here, $r_{21} = \sin\theta$ and $r_{11} = \cos\theta$, so using both entries identifies the correct quadrant. Its usual output interval is $(-\pi, \pi]$; angles differing by an integer multiple of 2π represent the same orientation.

An exact planar matrix uses zeros and ones in the restricted entries, but computed matrices contain rounding error. Let $\widetilde{\mathbf{R}}$ and $\widetilde{\mathbf{p}}$ denote computed rotation and translation blocks after the complete $SE(3)$ transformation has first been validated. A numerical planar check can use the residuals

$$\delta_R := \left\| \widetilde{\mathbf{R}}\mathbf{e}_z - \mathbf{e}_z \right\|_2, \qquad \delta_p := \left| \mathbf{e}_z^{\mathsf{T}} \widetilde{\mathbf{p}} \right|.$$

The orientation residual δ_R is dimensionless, while the position residual δ_p is measured in metres. With declared tolerances $\varepsilon_R > 0$ and $\varepsilon_p > 0$ having the same respective units, the pose may be treated as planar only when

$$\boxed{\delta_R \le \varepsilon_R, \qquad \delta_p \le \varepsilon_p}.$$

The tolerances should reflect the numerical precision and the application's geometric accuracy rather than being hidden constants. A pose that fails either check should not be silently converted to (x, y, θ). It must either be rejected as non-planar or passed to an explicitly declared projection that acknowledges the discarded information.

The embedding represents only the three-degree-of-freedom subset of $SE(3)$ satisfying the planar restrictions. It is not a projection of an arbitrary spatial pose onto a plane: such a projection would discard vertical position and rotations about horizontal axes and could not be inverted uniquely.

Quick test G: planar poses inside $SE(3)$

Try these questions without looking back at the section.

1. Write the $SE(3)$ matrix that embeds the planar pose coordinates (x, y, θ).
2. Which entries of the embedded matrix enforce the planar-motion restrictions?
3. Does embedding a planar pose require every point of the robot to have zero z-coordinate? Explain with a body-fixed point $[\, p_x\; p_y\; p_z \,]^\mathsf{T}$.
4. For a validated $\mathbf{T}^{(3)} \in SE(3)$ with rotation block $\mathbf{R}$ and translation block $\mathbf{p}$, state the two equations that determine whether it belongs to $\iota(SE(2))$, and show how to recover x, y, and θ.
5. Why should software use separate orientation and position tolerances when checking whether a computed spatial pose is planar?

Answers and hints

1. The embedding is

$$\boxed{\mathbf{T}_{wb}^{(3)} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & x \\ \sin\theta & \cos\theta & 0 & y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}.$$

2. The third rotation column and row keep z_b aligned with z_w, and the zero third translation entry keeps the body origin in the plane:

$$\mathbf{R}\mathbf{e}_z = \mathbf{e}_z, \qquad \mathbf{e}_z^\mathsf{T}\mathbf{p} = 0.$$

3. No. The pose restricts the motion of the body frame, not the shape of the rigid body. An embedded pose maps the point to

$${}^{w}\mathbf{p}_P = \begin{bmatrix} \cos\theta\, p_x - \sin\theta\, p_y + x \\ \sin\theta\, p_x + \cos\theta\, p_y + y \\ p_z \end{bmatrix},$$

so the point retains its height p_z relative to the body origin.

4. For $\mathbf{e}_z = [\, 0\; 0\; 1 \,]^\mathsf{T}$, the exact conditions and recovered coordinates are

$$\boxed{\mathbf{R}\mathbf{e}_z = \mathbf{e}_z, \qquad \mathbf{e}_z^\mathsf{T}\mathbf{p} = 0, \qquad x = p_1, \quad y = p_2, \quad \theta = \operatorname{atan2}(r_{21}, r_{11})}.$$

5. The rotation residual is dimensionless, whereas the vertical-position residual is measured in metres. One numerical threshold cannot express meaningful limits for both quantities. The tolerances must also be declared so that the accepted geometric error is visible and testable.

7.8 Unit quaternions and planar yaw

A rotation matrix is convenient for geometric calculations, but it stores nine entries even though a spatial orientation has only three degrees of freedom. Many robotics interfaces therefore store the same orientation using four quaternion components constrained to have unit norm. Unit quaternions also provide a compact rule for composing rotations. Specialised quaternion interpolation methods can construct intermediate orientations between two given orientations, but interpolation is outside this chapter's scope. The immediate problem in this section is to understand what the four quaternion components mean, derive how they represent and compose finite rotations, and then convert the planar yaw angle θ into a unit quaternion representing the same orientation as $\mathbf{R}_z(\theta)$.

7.8.1 Why introduce unit quaternions

The axis–angle representation from Section 7.5 describes a finite rotation using a unit axis direction $\mathbf{a} \in \mathbb{R}^3$, expressed in a declared coordinate frame, and a signed rotation angle $\alpha \in \mathbb{R}$ measured in radians. A unit quaternion packages this same rotation information into the ordered scalar–vector pair

$$\boxed{\mathsf{q} := \left(\underbrace{\cos\frac{\alpha}{2}}_{q_w}, \quad \underbrace{\mathbf{a}\sin\frac{\alpha}{2}}_{\mathbf{q}_v} \right), \qquad \mathbf{q}_v := \begin{bmatrix} q_x \\ q_y \\ q_z \end{bmatrix} \in \mathbb{R}^3},$$

where q denotes the quaternion, $q_w \in \mathbb{R}$ is its scalar part, and $\mathbf{q}_v \in \mathbb{R}^3$ is its vector part. The vector part is the rotation axis $\mathbf{a}$ scaled by $\sin(\alpha/2)$, while the scalar part is $\cos(\alpha/2)$; together they encode the axis and signed angle. The half-angle does not mean that the physical rotation is only $\alpha/2$; the quaternion rotation operation derived below produces the full rotation angle α.

The quaternion is therefore a representation of a three-dimensional orientation, not a position or a physical vector in four-dimensional space. Its components can be collected into a column in $\mathbb{R}^4$ for storage and computation, but the individual values q_x, q_y, and q_z are not three independent rotation angles. They are the components of the scaled axis $\mathbf{a}\sin(\alpha/2)$. All four components are dimensionless.

Because $\mathbf{a}$ is a unit vector, $\mathbf{a}^\mathsf{T}\mathbf{a} = 1$, the four components satisfy

$$\boxed{\|\mathsf{q}\|_2^2 := q_w^2 + \mathbf{q}_v^\mathsf{T}\mathbf{q}_v = q_w^2 + q_x^2 + q_y^2 + q_z^2 = \cos^2\frac{\alpha}{2} + \sin^2\frac{\alpha}{2}\mathbf{a}^\mathsf{T}\mathbf{a} = 1}.$$

A quaternion satisfying this equation is called a **unit quaternion**. Although it stores four numbers, the unit-norm equation imposes one scalar constraint, leaving three independent local degrees of freedom, matching a spatial orientation in $SO(3)$. For zero rotation, $\alpha = 0$ gives $\mathsf{q} = (1, \mathbf{0}_3)$; for a rotation through π radians, it gives $\mathsf{q} = (0, \mathbf{a})$.

7.8.2 The Hamilton product and quaternion rotation

Quaternion multiplication is not componentwise multiplication. For $\mathsf{q}_1 = (q_{w1}, \mathbf{q}_{v1})$ and $\mathsf{q}_2 = (q_{w2}, \mathbf{q}_{v2})$, the **Hamilton product** $\otimes$ is defined by

$$\boxed{\mathsf{q}_1 \otimes \mathsf{q}_2 := \left(q_{w1}q_{w2} - \mathbf{q}_{v1}^\mathsf{T}\mathbf{q}_{v2}, \quad q_{w1}\mathbf{q}_{v2} + q_{w2}\mathbf{q}_{v1} + \mathbf{q}_{v1} \times \mathbf{q}_{v2}\right)}.$$

The first output is the scalar part and the second output is the three-dimensional vector part. The cross-product term makes the Hamilton product generally noncommutative: changing the order can change the result.

The **Hamilton-product identity** is the quaternion

$$\mathbf{1}_q := (1, \mathbf{0}_3),$$

because substituting it into the Hamilton-product definition gives

$$\mathsf{q} \otimes \mathbf{1}_q = \mathbf{1}_q \otimes \mathsf{q} = \mathsf{q}.$$

The inverse q^{-1} of a nonzero quaternion is defined by the requirement

$$\boxed{\mathsf{q} \otimes \mathsf{q}^{-1} = \mathsf{q}^{-1} \otimes \mathsf{q} = \mathbf{1}_q}.$$

Both multiplication orders are stated because the Hamilton product is generally noncommutative. To calculate this inverse, define the **quaternion conjugate** by reversing the sign of the vector part:

$$\mathsf{q}^* := (q_w, -\mathbf{q}_v).$$

Substitution into the Hamilton-product definition gives the same result in both orders:

$$\mathsf{q} \otimes \mathsf{q}^* = \mathsf{q}^* \otimes \mathsf{q} = \left(q_w^2 + \mathbf{q}_v^\mathsf{T}\mathbf{q}_v, \mathbf{0}_3\right).$$

The scalar component is the squared quaternion norm $\|\mathsf{q}\|_2^2$. Therefore, if $\|\mathsf{q}\|_2 > 0$, dividing the conjugate by this scalar gives

$$\boxed{\mathsf{q}^{-1} = \frac{\mathsf{q}^*}{\|\mathsf{q}\|_2^2}}.$$

Indeed, multiplying q by this expression from either side gives $\mathbf{1}_q$. For a unit quaternion, $\|\mathsf{q}\|_2^2 = 1$, so the inverse simplifies to

$$\boxed{\mathsf{q}^{-1} = \mathsf{q}^*}.$$

Thus, the identity defines what an inverse means, whereas the calculation proves that the conjugate is the inverse specifically when the quaternion has unit norm.

To rotate a coordinate column $\mathbf{y} \in \mathbb{R}^3$, first represent it as the **pure quaternion** $\mathsf{y} := (0, \mathbf{y})$, where *pure* means that the scalar part is zero. Quaternion rotation is defined by

$$\boxed{\mathsf{y}' = \mathsf{q} \otimes \mathsf{y} \otimes \mathsf{q}^*},$$

where $\mathsf{y}' = (0, \mathbf{y}')$ contains the rotated coordinate column $\mathbf{y}'$. To connect this definition to rotation matrices, first apply one Hamilton product:

$$\mathsf{q} \otimes \mathsf{y} = \left(-\mathbf{q}_v^\mathsf{T}\mathbf{y}, \quad q_w\mathbf{y} + \mathbf{q}_v \times \mathbf{y}\right).$$

Multiplying this result by $\mathsf{q}^* = (q_w, -\mathbf{q}_v)$ makes the scalar part zero because

$$\begin{aligned}
&- q_w\mathbf{q}_v^\mathsf{T}\mathbf{y} + (q_w\mathbf{y} + \mathbf{q}_v \times \mathbf{y})^\mathsf{T}\mathbf{q}_v \\
&= -q_w\mathbf{q}_v^\mathsf{T}\mathbf{y} + q_w\mathbf{y}^\mathsf{T}\mathbf{q}_v + (\mathbf{q}_v \times \mathbf{y})^\mathsf{T}\mathbf{q}_v \\
&= -q_w\mathbf{q}_v^\mathsf{T}\mathbf{y} + q_w\mathbf{q}_v^\mathsf{T}\mathbf{y} + 0 \\
&= 0.
\end{aligned}$$

The second equality distributes the transpose over the sum. The third equality uses the symmetry $\mathbf{y}^\mathsf{T}\mathbf{q}_v = \mathbf{q}_v^\mathsf{T}\mathbf{y}$ of the dot product and the fact that $\mathbf{q}_v \times \mathbf{y}$ is perpendicular to $\mathbf{q}_v$. Thus, the two dot-product terms cancel and the result remains a pure quaternion.

The vector part follows from the same Hamilton product. First distribute the scalar multiplication and cross product:

$$\begin{aligned}
\mathbf{y}' &= (\mathbf{q}_v^\mathsf{T}\mathbf{y})\mathbf{q}_v + q_w(q_w\mathbf{y} + \mathbf{q}_v \times \mathbf{y}) - (q_w\mathbf{y} + \mathbf{q}_v \times \mathbf{y}) \times \mathbf{q}_v \\
&= (\mathbf{q}_v^\mathsf{T}\mathbf{y})\mathbf{q}_v + q_w^2\mathbf{y} + q_w(\mathbf{q}_v \times \mathbf{y}) - q_w(\mathbf{y} \times \mathbf{q}_v) - (\mathbf{q}_v \times \mathbf{y}) \times \mathbf{q}_v.
\end{aligned}$$

Anticommutativity gives $\mathbf{y} \times \mathbf{q}_v = -\mathbf{q}_v \times \mathbf{y}$. The vector triple-product identity from Section 1.1.5 gives

$$(\mathbf{q}_v \times \mathbf{y}) \times \mathbf{q}_v = (\mathbf{q}_v^\mathsf{T}\mathbf{q}_v)\mathbf{y} - (\mathbf{q}_v^\mathsf{T}\mathbf{y})\mathbf{q}_v.$$

Substituting these two identities and collecting like terms gives

$$\begin{aligned}\mathbf{y}' &= (\mathbf{q}_v^\mathsf{T}\mathbf{y})\mathbf{q}_v + q_w^2\mathbf{y} + 2q_w(\mathbf{q}_v \times \mathbf{y}) - (\mathbf{q}_v^\mathsf{T}\mathbf{q}_v)\mathbf{y} + (\mathbf{q}_v^\mathsf{T}\mathbf{y})\mathbf{q}_v \\ &= \left(q_w^2 - \mathbf{q}_v^\mathsf{T}\mathbf{q}_v\right)\mathbf{y} + 2\mathbf{q}_v(\mathbf{q}_v^\mathsf{T}\mathbf{y}) + 2q_w(\mathbf{q}_v \times \mathbf{y}).\end{aligned}$$

The final expression separates the rotated result into a term parallel to $\mathbf{y}$, a term parallel to $\mathbf{q}_v$, and a term perpendicular to both vectors. It describes the active rotation of $\mathbf{y}$ while its coordinate frame remains fixed.

7.8.3 Why the half-angle formula produces the full rotation

It remains to verify that the unit quaternion defined above represents rotation through the full angle α. The required half-angle identities are

$$\boxed{\begin{aligned}\cos^2\frac{\alpha}{2} - \sin^2\frac{\alpha}{2} &= \cos\alpha, \\ 2\sin^2\frac{\alpha}{2} &= 1 - \cos\alpha, \\ 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2} &= \sin\alpha\end{aligned}}.$$

Substitute $q_w = \cos(\alpha/2)$ and $\mathbf{q}_v = \mathbf{a}\sin(\alpha/2)$ into each coefficient of the quaternion-rotation result. Because $\mathbf{a}$ is a unit vector, $\mathbf{a}^\mathsf{T}\mathbf{a} = 1$, the first coefficient becomes

$$\begin{aligned}q_w^2 - \mathbf{q}_v^\mathsf{T}\mathbf{q}_v &= \cos^2\frac{\alpha}{2} - \sin^2\frac{\alpha}{2}(\mathbf{a}^\mathsf{T}\mathbf{a}) \\ &= \cos^2\frac{\alpha}{2} - \sin^2\frac{\alpha}{2} \\ &= \cos\alpha.\end{aligned}$$

For the term parallel to the rotation axis,

$$\begin{aligned}2\mathbf{q}_v(\mathbf{q}_v^\mathsf{T}\mathbf{y}) &= 2\left(\mathbf{a}\sin\frac{\alpha}{2}\right)\left(\sin\frac{\alpha}{2}\,\mathbf{a}^\mathsf{T}\mathbf{y}\right) \\ &= 2\sin^2\frac{\alpha}{2}\,\mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}) \\ &= (1 - \cos\alpha)\mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}).\end{aligned}$$

For the cross-product term, bilinearity of the cross product allows the scalar $\sin(\alpha/2)$ to be taken outside:

$$
\begin{aligned}
2q_w(\mathbf{q}_v \times \mathbf{y}) &= 2\cos\frac{\alpha}{2}\left[\left(\mathbf{a}\sin\frac{\alpha}{2}\right) \times \mathbf{y}\right] \\
&= 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}(\mathbf{a} \times \mathbf{y}) \\
&= \sin\alpha(\mathbf{a} \times \mathbf{y}).
\end{aligned}
$$

Therefore,

$$
\boxed{\mathbf{y}' = \cos\alpha\,\mathbf{y} + (1-\cos\alpha)\mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}) + \sin\alpha(\mathbf{a} \times \mathbf{y})}.
$$

To make the connection with the rotation matrix explicit, Section 7.5 derived Rodrigues' formula for the rotation matrix $\mathbf{Q}(\mathbf{a}, \alpha) \in SO(3)$:

$$
\mathbf{Q}(\mathbf{a}, \alpha) = \mathbf{I}_3 + \sin\alpha[\mathbf{a}]_\times + (1-\cos\alpha)[\mathbf{a}]_\times^2.
$$

Applying this matrix to $\mathbf{y}$ and using $[\mathbf{a}]_\times\mathbf{y} = \mathbf{a} \times \mathbf{y}$ gives

$$
\mathbf{Q}(\mathbf{a}, \alpha)\mathbf{y} = \mathbf{y} + \sin\alpha(\mathbf{a} \times \mathbf{y}) + (1-\cos\alpha)\mathbf{a} \times (\mathbf{a} \times \mathbf{y}).
$$

The vector triple-product identity and $\mathbf{a}^\mathsf{T}\mathbf{a} = 1$ give

$$
\mathbf{a} \times (\mathbf{a} \times \mathbf{y}) = \mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}) - \mathbf{y}.
$$

Substitution therefore produces

$$
\begin{aligned}
\mathbf{Q}(\mathbf{a}, \alpha)\mathbf{y} &= \mathbf{y} + \sin\alpha(\mathbf{a} \times \mathbf{y}) + (1-\cos\alpha)\left[\mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}) - \mathbf{y}\right] \\
&= \cos\alpha\,\mathbf{y} + (1-\cos\alpha)\mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}) + \sin\alpha(\mathbf{a} \times \mathbf{y}) \\
&= \mathbf{y}'.
\end{aligned}
$$

Thus, quaternion multiplication and Rodrigues' matrix transform every input vector $\mathbf{y}$ in the same way. They are two mathematical representations of the same finite rotation through angle α about axis $\mathbf{a}$.

The geometry is clearer after decomposing $\mathbf{y}$ into components parallel and perpendicular to the axis:

$$
\mathbf{y}_\parallel := \mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}), \qquad \mathbf{y}_\perp := \mathbf{y} - \mathbf{y}_\parallel.
$$

By definition, $\mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}) = \mathbf{y}_\parallel$. Moreover, $\mathbf{a} \times \mathbf{y}_\parallel = \mathbf{0}_3$, so $\mathbf{a} \times \mathbf{y} = \mathbf{a} \times \mathbf{y}_\perp$. Substituting these relations and $\mathbf{y} = \mathbf{y}_\parallel + \mathbf{y}_\perp$ into the quaternion result gives

$$
\begin{aligned}
\mathbf{y}' &= \cos\alpha\left(\mathbf{y}_{\parallel} + \mathbf{y}_{\perp}\right) + (1-\cos\alpha)\mathbf{y}_{\parallel} + \sin\alpha(\mathbf{a}\times\mathbf{y}_{\perp}) \\
&= \left[\cos\alpha + (1-\cos\alpha)\right]\mathbf{y}_{\parallel} + \cos\alpha\,\mathbf{y}_{\perp} + \sin\alpha(\mathbf{a}\times\mathbf{y}_{\perp}) \\
&= \boxed{\mathbf{y}_{\parallel} + \cos\alpha\,\mathbf{y}_{\perp} + \sin\alpha(\mathbf{a}\times\mathbf{y}_{\perp})}.
\end{aligned}
$$

This is exactly the parallel–perpendicular geometric form derived in Section 7.5. The parallel component $\mathbf{y}_{\parallel}$ remains unchanged. In the plane perpendicular to $\mathbf{a}$, the two perpendicular directions $\mathbf{y}_{\perp}$ and $\mathbf{a}\times\mathbf{y}_{\perp}$ combine through $\cos\alpha$ and $\sin\alpha$, which rotates $\mathbf{y}_{\perp}$ through angle α according to the right-hand rule.

7.8.4 Composing active rotations

Suppose the unit quaternion q_1 actively rotates a geometric vector first and the unit quaternion q_2 actively rotates the result second, while all coordinate columns remain expressed in the same fixed frame. The question is which single quaternion produces the combined rotation.

Let $\mathsf{y} = (0, \mathbf{y})$ be the pure quaternion formed from the original coordinate column $\mathbf{y} \in \mathbb{R}^3$. After the first rotation,

$$
\mathsf{y}_1 = \mathsf{q}_1 \otimes \mathsf{y} \otimes \mathsf{q}_1^*.
$$

Applying the second rotation gives

$$
\begin{aligned}
\mathsf{y}_2 &= \mathsf{q}_2 \otimes \mathsf{y}_1 \otimes \mathsf{q}_2^* \\
&= \mathsf{q}_2 \otimes (\mathsf{q}_1 \otimes \mathsf{y} \otimes \mathsf{q}_1^*) \otimes \mathsf{q}_2^*.
\end{aligned}
$$

The Hamilton product is **associative**, meaning that regrouping consecutive products does not change their value:

$$
(\mathsf{q}_a \otimes \mathsf{q}_b) \otimes \mathsf{q}_c = \mathsf{q}_a \otimes (\mathsf{q}_b \otimes \mathsf{q}_c).
$$

It is generally not commutative, so regrouping is allowed but reversing the factors is not. Regrouping the two-rotation expression gives

$$
\mathsf{y}_2 = (\mathsf{q}_2 \otimes \mathsf{q}_1) \otimes \mathsf{y} \otimes (\mathsf{q}_1^* \otimes \mathsf{q}_2^*).
$$

To identify the final factor as a conjugate, write $\mathsf{q}_i = (q_{wi}, \mathbf{q}_{vi})$ for $i \in \{1, 2\}$. The Hamilton-product definition gives

$$
\mathsf{q}_2 \otimes \mathsf{q}_1 = \left(q_{w2}q_{w1} - \mathbf{q}_{v2}^{\mathsf{T}}\mathbf{q}_{v1}, \quad q_{w2}\mathbf{q}_{v1} + q_{w1}\mathbf{q}_{v2} + \mathbf{q}_{v2}\times\mathbf{q}_{v1}\right).
$$

Conjugating this result negates its vector part. Conversely, multiplying the two conjugates in reversed order gives the same scalar part and the vector part

$$
\begin{aligned}
q_{w1}(-\mathbf{q}_{v2}) + q_{w2}(-\mathbf{q}_{v1}) + (-\mathbf{q}_{v1}) \times (-\mathbf{q}_{v2}) \\
&= -q_{w1}\mathbf{q}_{v2} - q_{w2}\mathbf{q}_{v1} + \mathbf{q}_{v1} \times \mathbf{q}_{v2} \\
&= -\left(q_{w2}\mathbf{q}_{v1} + q_{w1}\mathbf{q}_{v2} + \mathbf{q}_{v2} \times \mathbf{q}_{v1}\right),
\end{aligned}
$$

where the second equality uses $\mathbf{q}_{v1} \times \mathbf{q}_{v2} = -\mathbf{q}_{v2} \times \mathbf{q}_{v1}$. Thus, conjugating a product reverses the factor order:

$$
\boxed{(\mathsf{q}_2 \otimes \mathsf{q}_1)^* = \mathsf{q}_1^* \otimes \mathsf{q}_2^*}.
$$

The expression for y_2 therefore has the standard quaternion-rotation form

$$
\mathsf{y}_2 = \underbrace{(\mathsf{q}_2 \otimes \mathsf{q}_1)}_{\mathsf{q}_{\text{total}}} \otimes \mathsf{y} \otimes \underbrace{(\mathsf{q}_2 \otimes \mathsf{q}_1)^*}_{\mathsf{q}^*_{\text{total}}}.
$$

Hence, the single quaternion representing the two active rotations is

$$
\boxed{\mathsf{q}_{\text{total}} = \mathsf{q}_2 \otimes \mathsf{q}_1}, \qquad \mathsf{q}_1 \text{ acts first and } \mathsf{q}_2 \text{ acts second.}
$$

The rightmost rotation therefore acts first, just as in the matrix product $\mathbf{R}_{\text{total}} = \mathbf{R}_2\mathbf{R}_1$. Reversing the quaternion order generally produces a different orientation because spatial rotations are not generally commutative.

7.8.5 Planar yaw and ROS 2 component order

For planar yaw, the rotation axis is $\mathbf{a} = \mathbf{e}_z = [0\ 0\ 1]^\mathsf{T}$ and the rotation angle is $\alpha = \theta$. The scalar–vector quaternion is consequently

$$
\mathsf{q}_{wb}^{\text{yaw}} = \left(\cos\frac{\theta}{2}, \quad \begin{bmatrix} 0 \\ 0 \\ \sin(\theta/2) \end{bmatrix}\right).
$$

The ROS 2 `geometry_msgs/msg/Quaternion` message stores the vector components first and the scalar component last. Its field-order column is therefore

$$
\boxed{\begin{bmatrix} q_x \\ q_y \\ q_z \\ q_w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \sin(\theta/2) \\ \cos(\theta/2) \end{bmatrix}}.
$$

The quaternion is already normalized because

$$q_x^2 + q_y^2 + q_z^2 + q_w^2 = \sin^2\frac{\theta}{2} + \cos^2\frac{\theta}{2} = 1.$$

For example, a positive yaw of $\theta = \pi/2$ gives

$$\begin{bmatrix} q_x \\ q_y \\ q_z \\ q_w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}.$$

The quaternion components q_x, q_y, and q_z must not be confused with the position coordinates x, y, and z in a pose message. A spatial pose stores position and orientation as separate fields.

7.8.6 Equivalent signs and numerical normalization

The unit quaternions q and $-\mathsf{q}$ represent the same orientation. Their rotation actions are identical because

$$(-\mathsf{q}) \otimes \mathsf{y} \otimes (-\mathsf{q})^* = \mathsf{q} \otimes \mathsf{y} \otimes \mathsf{q}^*.$$

The two minus signs cancel because the Hamilton product is linear in each input. For the yaw formula, increasing θ by 2π changes both half-angle terms' signs and therefore changes q to $-\mathsf{q}$, while leaving the physical orientation unchanged. Componentwise quaternion equality is consequently too strict for testing orientation equality unless a sign convention has first been imposed.

Floating-point computations may produce a finite quaternion component column $\widetilde{\mathbf{c}}_{xyzw} := [\,\widetilde{q}_x\ \widetilde{q}_y\ \widetilde{q}_z\ \widetilde{q}_w\,]^\mathsf{T} \in \mathbb{R}^4$ whose norm is close to, but not exactly, one. Define its dimensionless unit-norm residual by

$$e_q := \left| \|\widetilde{\mathbf{c}}_{xyzw}\|_2 - 1 \right|.$$

Let $\varepsilon_{\text{norm}} > 0$ be a declared dimensionless normalization tolerance. If all components are finite, the norm is safely nonzero, and $e_q \le \varepsilon_{\text{norm}}$, the deviation is classified as small numerical drift and normalization gives

$$\boxed{\mathbf{c}_{xyzw} = \frac{\widetilde{\mathbf{c}}_{xyzw}}{\|\widetilde{\mathbf{c}}_{xyzw}\|_2}}.$$

The tolerance does not allow the norm error to remain: normalization reduces the accepted residual to floating-point precision. Instead, the tolerance decides whether the input is close enough to a unit quaternion to repair. If $e_q > \varepsilon_{\text{norm}}$, silently normalizing could hide corrupted data or an incorrect calculation, so the input should be rejected unless the interface explicitly specifies a different projection policy. Normalization also cannot repair non-finite components, a zero or near-zero norm, an incorrect frame convention, or an incorrect rotation.

Quick test H: unit quaternions and planar yaw

Try these questions without looking back at the section.

1. For a unit rotation axis $\mathbf{a} \in \mathbb{R}^3$ and signed angle $\alpha \in \mathbb{R}$, write the corresponding unit quaternion as a scalar–vector pair. What do its two parts mean?
2. Why does a unit quaternion store four real components but describe only three local degrees of freedom?
3. Starting from the Hamilton-product definition, state the identity quaternion and derive the inverse of a nonzero quaternion. How does the result simplify for a unit quaternion?
4. Let q_1 act first and q_2 act second under the active-rotation rule $\mathsf{y}' = \mathsf{q} \otimes \mathsf{y} \otimes \mathsf{q}^*$. Derive the quaternion representing the combined rotation and explain why its order matters.
5. Why does the half-angle in $\mathsf{q} = (\cos(\alpha/2), \mathbf{a}\sin(\alpha/2))$ still produce a physical rotation through the full angle α?
6. Convert the planar yaw angle $\theta = \pi/2$ into the ROS 2 component order $[\, q_x \; q_y \; q_z \; q_w \,]^\mathsf{T}$.
7. Why do q and $-\mathsf{q}$ represent the same orientation, and what problem does this cause for direct componentwise comparisons?
8. A computed component column is finite and has norm $1 + 10^{-8}$. What declared check determines whether normalization is appropriate? Why would normalization be unsafe for a zero-norm or non-finite input?

Answers and hints

1. The axis–angle quaternion is

   $$\boxed{\mathsf{q} = \left(\cos\frac{\alpha}{2}, \mathbf{a}\sin\frac{\alpha}{2}\right)}.$$

   The scalar part stores $\cos(\alpha/2)$, while the vector part points along the rotation axis and has magnitude $|\sin(\alpha/2)|$. All components are dimensionless.

2. The four components satisfy the independent unit-norm constraint $q_w^2 + q_x^2 + q_y^2 + q_z^2 = 1$. Therefore, only three components can vary independently in a local coordinate description: $4 - 1 = 3$ degrees of freedom.

3. The Hamilton-product identity is $\mathbf{1}_q = (1, \mathbf{0}_3)$. For $\mathsf{q} \neq (0, \mathbf{0}_3)$,

   $$\mathsf{q} \otimes \mathsf{q}^* = \left(\|\mathsf{q}\|_2^2, \mathbf{0}_3\right),$$

   so

$$\boxed{\mathsf{q}^{-1} = \frac{\mathsf{q}^*}{\|\mathsf{q}\|_2^2}}.$$

If $\|\mathsf{q}\|_2 = 1$, then $\mathsf{q}^{-1} = \mathsf{q}^*$.

4. Two successive rotations give

$$\begin{aligned}\mathsf{y}_2 &= \mathsf{q}_2 \otimes (\mathsf{q}_1 \otimes \mathsf{y} \otimes \mathsf{q}_1^*) \otimes \mathsf{q}_2^* \\ &= (\mathsf{q}_2 \otimes \mathsf{q}_1) \otimes \mathsf{y} \otimes (\mathsf{q}_2 \otimes \mathsf{q}_1)^*.\end{aligned}$$

Therefore, $\mathsf{q}_{\text{total}} = \mathsf{q}_2 \otimes \mathsf{q}_1$. The Hamilton product is generally not commutative, so reversing the order generally changes the final orientation.

5. Substitution into the quaternion rotation action produces

$$\mathbf{y}' = \cos\alpha\,\mathbf{y} + (1-\cos\alpha)\mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}) + \sin\alpha(\mathbf{a}\times\mathbf{y}),$$

Because $\mathbf{a}(\mathbf{a}^\mathsf{T}\mathbf{y}) = \mathbf{y}_\parallel$, $\mathbf{y} = \mathbf{y}_\parallel + \mathbf{y}_\perp$, and $\mathbf{a}\times\mathbf{y} = \mathbf{a}\times\mathbf{y}_\perp$, the same equation becomes

$$\mathbf{y}' = \mathbf{y}_\parallel + \cos\alpha\,\mathbf{y}_\perp + \sin\alpha(\mathbf{a}\times\mathbf{y}_\perp),$$

which is the geometric Rodrigues formula from Section 7.5. The full-angle terms arise from the half-angle identities used when the quaternion appears twice in $\mathsf{q}\otimes\mathsf{y}\otimes\mathsf{q}^*$.

6. Substituting $\theta = \pi/2$ gives

$$\boxed{\begin{bmatrix} q_x \\ q_y \\ q_z \\ q_w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}}.$$

7. Both signs produce the same sandwich product because the two negative factors cancel. Direct componentwise comparison can therefore report two equivalent orientations as unequal; a comparison must account for the sign ambiguity or compare their rotation actions.

8. Compute the residual $e_q = |\|\widetilde{\mathbf{c}}_{xyzw}\|_2 - 1| = 10^{-8}$ and compare it with the declared normalization tolerance $\varepsilon_{\text{norm}}$. If $e_q \le \varepsilon_{\text{norm}}$ and the norm is safely nonzero, divide the column by its norm; the tolerance classifies the error as repairable rather than allowing it to remain. A zero or near-zero norm makes the division undefined or unstable, while non-finite components cannot be repaired by scaling; such inputs must be rejected.

7.9 Spatial rigid-body velocity and twists

A pose $\mathbf{T}_{wb}(t) \in SE(3)$ locates a rigid body at time t, but control and estimation also require its instantaneous motion. The body has one angular velocity and one translational velocity at a chosen reference point, yet different body-fixed points generally have different linear velocities when the body rotates. Before defining a spatial twist, this section derives the velocity of any body-fixed point from those two quantities.

7.9.1 Velocity of a body-fixed point

Let $t \in \mathbb{R}$ denote time in seconds and let

$$\mathbf{T}_{wb}(t) = \begin{bmatrix} \mathbf{R}_{wb}(t) & {}^{w}\mathbf{p}_b(t) \\ \mathbf{0}_3^{\mathsf{T}} & 1 \end{bmatrix} \in SE(3)$$

represent the pose of body frame $\{b\}$ relative to world frame $\{w\}$. Here, $\mathbf{R}_{wb}(t) \in SO(3)$ describes the body's orientation relative to the world, and ${}^{w}\mathbf{p}_b(t) \in \mathbb{R}^3$ is the position of the body origin O_b, expressed in world coordinates and measured in metres.

Choose a material point P fixed in the rigid body. Its body-frame coordinate ${}^{b}\mathbf{p}_P \in \mathbb{R}^3$, measured in metres, is constant because both P and frame $\{b\}$ move with the same rigid body. The point transformation from Section 7.6.1 gives its world position:

$$\boxed{{}^{w}\mathbf{p}_P(t) = \mathbf{R}_{wb}(t)\,{}^{b}\mathbf{p}_P + {}^{w}\mathbf{p}_b(t)}.$$

Differentiate this equation with respect to time. Because ${}^{b}\mathbf{p}_P$ is constant, its derivative is zero, so

$$\begin{aligned} {}^{w}\mathbf{v}_P := \frac{d}{dt}\,{}^{w}\mathbf{p}_P &= \dot{\mathbf{R}}_{wb}\,{}^{b}\mathbf{p}_P + \frac{d}{dt}\,{}^{w}\mathbf{p}_b \\ &= \dot{\mathbf{R}}_{wb}\,{}^{b}\mathbf{p}_P + {}^{w}\mathbf{v}_b, \end{aligned}$$

where ${}^{w}\mathbf{v}_P \in \mathbb{R}^3$ is the velocity of point P expressed in world coordinates and ${}^{w}\mathbf{v}_b := d({}^{w}\mathbf{p}_b)/dt \in \mathbb{R}^3$ is the velocity of the body origin expressed in world coordinates. Both velocities are measured in $\mathrm{m\,s^{-1}}$.

The angular-velocity derivation in Section 7.4 gave the world-expressed orientation-rate equation

$$\dot{\mathbf{R}}_{wb} = [\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb},$$

where $\boldsymbol{\omega}_w \in \mathbb{R}^3$ is the body's angular velocity relative to the world, expressed along the world axes and measured in $\mathrm{rad\,s^{-1}}$. Define the relative-position vector from O_b to P, expressed in world coordinates, by

$$^{w}\mathbf{r}_{O_bP} := {}^{w}\mathbf{p}_P - {}^{w}\mathbf{p}_b = \mathbf{R}_{wb}{}^{b}\mathbf{p}_P \in \mathbb{R}^3.$$

The vector $^{w}\mathbf{r}_{O_bP}$ is measured in metres. It describes the displacement from the body origin to P at the selected instant; it is not the world position of either endpoint by itself.

Substituting these two relations into the point-velocity equation gives

$$\begin{aligned} {}^{w}\mathbf{v}_P &= {}^{w}\mathbf{v}_b + [\boldsymbol{\omega}_w]_\times \mathbf{R}_{wb}{}^{b}\mathbf{p}_P \\ &= {}^{w}\mathbf{v}_b + [\boldsymbol{\omega}_w]_\times {}^{w}\mathbf{r}_{O_bP} \\ &= \boxed{{}^{w}\mathbf{v}_b + \boldsymbol{\omega}_w \times {}^{w}\mathbf{r}_{O_bP}}. \end{aligned}$$

The first term is the translational velocity shared with the body origin. The second term is the additional velocity caused by rotation about the body origin. It is perpendicular to both $\boldsymbol{\omega}_w$ and $^{w}\mathbf{r}_{O_bP}$, and its magnitude is

$$\|\boldsymbol{\omega}_w \times {}^{w}\mathbf{r}_{O_bP}\|_2 = \|\boldsymbol{\omega}_w\|_2 \|{}^{w}\mathbf{r}_{O_bP}\|_2 \sin\varphi,$$

where $\varphi \in [0, \pi]$ is the angle between the angular-velocity vector and the relative-position vector. A point farther from the rotation axis therefore has a larger rotational speed. If $P = O_b$, if $\boldsymbol{\omega}_w = \mathbf{0}_3$, or if $^{w}\mathbf{r}_{O_bP}$ is parallel to $\boldsymbol{\omega}_w$, the rotational contribution is zero.

For example, suppose that, at one selected instant, a sensor is located $0.2\,\mathrm{m}$ from O_b along the positive world direction x_w, while

$$^{w}\mathbf{r}_{O_bP} = \begin{bmatrix} 0.2 \\ 0 \\ 0 \end{bmatrix} \mathrm{m}, \qquad {}^{w}\mathbf{v}_b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}, \qquad \boldsymbol{\omega}_w = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mathrm{rad\,s^{-1}}.$$

The rotational contribution is

$$\boldsymbol{\omega}_w \times {}^{w}\mathbf{r}_{O_bP} = \begin{bmatrix} 0 \\ 0.2 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}},$$

so the sensor velocity is

$$\boxed{{}^{w}\mathbf{v}_P = \begin{bmatrix} 1 \\ 0.2 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}}.$$

The body origin moves only along x_w at this instant, but the offset sensor also moves along y_w because the body is rotating about z_w.

The planar point-velocity equation from Section 6.3.4 is the special case

$$\boldsymbol{\omega}_w = \begin{bmatrix} 0 \\ 0 \\ \omega \end{bmatrix}, \qquad {}^w\mathbf{r}_{O_bP} = \begin{bmatrix} r_x \\ r_y \\ 0 \end{bmatrix}.$$

Indeed,

$$\boldsymbol{\omega}_w \times {}^w\mathbf{r}_{O_bP} = \begin{bmatrix} -\omega r_y \\ \omega r_x \\ 0 \end{bmatrix},$$

whose first two components equal $\omega\mathbf{S}_2[\, r_x \; r_y \,]^\mathsf{T}$. The three-dimensional equation is therefore a direct extension of the planar rigid-body velocity relation.

7.9.2 Body twist and $\mathfrak{se}(3)$

The point-velocity equation requires two quantities: the velocity of the body origin and the angular velocity of the body. A **spatial rigid-body twist** packages these two parts into one six-component representation of the body's instantaneous motion. This subsection first expresses both parts along the moving body axes; the resulting representation is called a **body twist**.

The word *body* describes the coordinate axes used for the components. It does not mean that the body origin moves relative to its own attached frame. The physical motion is still measured relative to the world; only its numerical components are expressed along the instantaneous body axes.

Expressing the two velocity parts along the body axes

For compactness, write the time-varying pose and its derivative as

$$\mathbf{T} := \mathbf{T}_{wb} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_3^\mathsf{T} & 1 \end{bmatrix}, \qquad \dot{\mathbf{T}} = \begin{bmatrix} \dot{\mathbf{R}} & \dot{\mathbf{p}} \\ \mathbf{0}_3^\mathsf{T} & 0 \end{bmatrix},$$

where $\mathbf{R} = \mathbf{R}_{wb} \in SO(3)$ is the orientation of frame $\{b\}$ relative to frame $\{w\}$, $\mathbf{p} = {}^w\mathbf{p}_b \in \mathbb{R}^3$ is the world position of the body origin measured in metres, and $\dot{\mathbf{p}} = {}^w\mathbf{v}_b \in \mathbb{R}^3$ is that origin's velocity relative to the world, expressed in world coordinates and measured in $\mathrm{m\,s^{-1}}$.

The inverse pose from Section 7.6.2 is

$$\mathbf{T}^{-1} = \begin{bmatrix} \mathbf{R}^\mathsf{T} & -\mathbf{R}^\mathsf{T}\mathbf{p} \\ \mathbf{0}_3^\mathsf{T} & 1 \end{bmatrix}.$$

Premultiplying the pose derivative by this inverse gives

$$\begin{aligned}\mathbf{T}^{-1}\dot{\mathbf{T}} &= \begin{bmatrix}\mathbf{R}^\top & -\mathbf{R}^\top\mathbf{p}\\ \mathbf{0}_3^\top & 1\end{bmatrix}\begin{bmatrix}\dot{\mathbf{R}} & \dot{\mathbf{p}}\\ \mathbf{0}_3^\top & 0\end{bmatrix}\\ &= \begin{bmatrix}\mathbf{R}^\top\dot{\mathbf{R}} & \mathbf{R}^\top\dot{\mathbf{p}}\\ \mathbf{0}_3^\top & 0\end{bmatrix}.\end{aligned}$$

The translation column $-\mathbf{R}^\top\mathbf{p}$ contributes zero because it multiplies the zero bottom row of $\dot{\mathbf{T}}$. The angular-velocity relations from Section 7.4 are

$$\mathbf{R}_{wb}^\top\dot{\mathbf{R}}_{wb} = [\boldsymbol{\omega}_b]_\times, \qquad \boldsymbol{\omega}_b = \mathbf{R}_{wb}^\top\boldsymbol{\omega}_w \in \mathbb{R}^3,$$

where $\boldsymbol{\omega}_b$ is the body's angular velocity relative to the world, expressed along the body axes and measured in $\mathrm{rad\,s^{-1}}$. Define the body-expressed linear velocity of the body origin by

$$\boxed{\boldsymbol{\nu}_b := \mathbf{R}_{wb}^\top\,{}^w\mathbf{v}_b = \begin{bmatrix}\nu_{x_b}\\ \nu_{y_b}\\ \nu_{z_b}\end{bmatrix} \in \mathbb{R}^3}.$$

The entries ν_{x_b}, ν_{y_b}, and ν_{z_b} are the forward, lateral, and vertical components of the world-relative body-origin velocity along the body axes x_b, y_b, and z_b, respectively. They are measured in $\mathrm{m\,s^{-1}}$. Multiplication by $\mathbf{R}_{wb}^\top$ changes the coordinate description; it does not create a different physical velocity.

Because $\mathbf{R} = \mathbf{R}_{wb}$ and $\dot{\mathbf{p}} = {}^w\mathbf{v}_b$, the two nonzero blocks in the product are therefore

$$\mathbf{R}^\top\dot{\mathbf{R}} = [\boldsymbol{\omega}_b]_\times, \qquad \mathbf{R}^\top\dot{\mathbf{p}} = \mathbf{R}_{wb}^\top\,{}^w\mathbf{v}_b = \boldsymbol{\nu}_b.$$

Substituting both results back into the block product completes the calculation:

$$\boxed{\mathbf{T}^{-1}\dot{\mathbf{T}} = \begin{bmatrix}[\boldsymbol{\omega}_b]_\times & \boldsymbol{\nu}_b\\ \mathbf{0}_3^\top & 0\end{bmatrix}}.$$

Thus, premultiplication by $\mathbf{T}^{-1}$ uses the current pose to express the pose rate in body coordinates. The terms containing the current position $\mathbf{p}$ vanish because they multiply zeros, while $\mathbf{R}^\top$ converts the remaining orientation and position rates into angular and linear velocity components expressed along the body axes. The body's motion is re-expressed, not removed.

Twist coordinates and the hat operator

Using the component order established for planar twists in Section 6.4.3, define the six-component **body-twist column** by

$$\boxed{\boldsymbol{\xi}_b := \begin{bmatrix} \boldsymbol{\nu}_b \\ \boldsymbol{\omega}_b \end{bmatrix} = \begin{bmatrix} \nu_{x_b} \\ \nu_{y_b} \\ \nu_{z_b} \\ \omega_{x_b} \\ \omega_{y_b} \\ \omega_{z_b} \end{bmatrix} \in \mathbb{R}^6}.$$

The first three entries are linear-velocity components in $\mathrm{m\,s^{-1}}$, and the final three entries are angular-velocity components in $\mathrm{rad\,s^{-1}}$. Thus, $\boldsymbol{\xi}_b$ is a six-entry real coordinate column, but its entries do not all have the same physical units. Some references place angular velocity before linear velocity; a twist's component order must therefore always be declared.

The three-dimensional **hat operator** maps this coordinate column to a structured 4×4 matrix:

$$\boxed{\widehat{\boldsymbol{\xi}}_b := \begin{bmatrix} [\boldsymbol{\omega}_b]_\times & \boldsymbol{\nu}_b \\ \mathbf{0}_3^\mathsf{T} & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\omega_{z_b} & \omega_{y_b} & \nu_{x_b} \\ \omega_{z_b} & 0 & -\omega_{x_b} & \nu_{y_b} \\ -\omega_{y_b} & \omega_{x_b} & 0 & \nu_{z_b} \\ 0 & 0 & 0 & 0 \end{bmatrix}}.$$

The upper-left block represents the angular-velocity operation $\boldsymbol{\omega}_b \times (\cdot)$, the upper-right column stores the body-origin linear velocity, and the bottom row is zero because the bottom row of every homogeneous transformation is the constant row $[\,0\ 0\ 0\ 1\,]$. The inverse mapping from the structured matrix to the six-component column is called the **vee operator** and is written $(\widehat{\boldsymbol{\xi}}_b)^\vee = \boldsymbol{\xi}_b$.

Comparing the hat-operator definition with the completed block product above gives

$$\boxed{\widehat{\boldsymbol{\xi}}_b = \mathbf{T}_{wb}^{-1}\dot{\mathbf{T}}_{wb}}.$$

Multiplying both sides on the left by $\mathbf{T}_{wb}$ gives the equivalent pose-rate equation

$$\boxed{\dot{\mathbf{T}}_{wb} = \mathbf{T}_{wb}\widehat{\boldsymbol{\xi}}_b}.$$

The first equation extracts the body twist from a differentiable pose trajectory. The second equation computes the world-frame pose rate from the current pose and a known body twist, making it suitable for kinematic prediction and numerical integration.

Pose rate and body twist: same motion, different forms

The pose rate and body twist do not describe two different physical motions. Given the current pose $\mathbf{T}_{wb}$, they contain equivalent information about the same instantaneous rigid-body motion, and either one can be calculated from the other:

$$\boxed{\dot{\mathbf{T}}_{wb} \quad \longleftrightarrow \quad \widehat{\boldsymbol{\xi}}_b, \qquad \widehat{\boldsymbol{\xi}}_b = \mathbf{T}_{wb}^{-1}\dot{\mathbf{T}}_{wb}, \qquad \dot{\mathbf{T}}_{wb} = \mathbf{T}_{wb}\widehat{\boldsymbol{\xi}}_b}.$$

They differ in mathematical form and coordinate meaning. The pose rate $\dot{\mathbf{T}}_{wb} \in \mathbb{R}^{4\times 4}$ is the derivative of the homogeneous pose representation at its current value. Its blocks $\dot{\mathbf{R}}_{wb}$ and $\dot{\mathbf{p}} = {}^{w}\mathbf{v}_b$ describe how the body axes and body-origin position change relative to the world. It is not an element of $SE(3)$ because $\dot{\mathbf{R}}_{wb}$ is not a finite rotation matrix and its bottom-right entry is zero.

The body twist $\boldsymbol{\xi}_b \in \mathbb{R}^6$ instead packages the motion into three linear-velocity and three angular-velocity components expressed along the instantaneous body axes. Its linear part $\boldsymbol{\nu}_b = \mathbf{R}_{wb}^{\mathsf{T}}\dot{\mathbf{p}}$ is not the derivative of the body origin's position relative to its own frame: that relative position is always ${}^{b}\mathbf{p}_b = \mathbf{0}_3$. Rather, $\boldsymbol{\nu}_b$ is the body origin's velocity relative to the world, expressed using body-axis coordinates.

For example, consider the body twist

$$\boldsymbol{\xi}_b = \begin{bmatrix}1 & 0 & 0 & 0 & 0 & 0\end{bmatrix}^{\mathsf{T}},$$

whose first three entries are measured in $\mathrm{m\,s^{-1}}$ and final three entries are measured in $\mathrm{rad\,s^{-1}}$. It describes motion at $1\,\mathrm{m\,s^{-1}}$ along positive x_b with no rotation. If $\mathbf{R}_{wb} = \mathbf{I}_3$, then

$$\dot{\mathbf{p}} = \mathbf{R}_{wb}\boldsymbol{\nu}_b = \begin{bmatrix}1\\0\\0\end{bmatrix}\mathrm{m\,s^{-1}}.$$

If the same body is instead oriented by $\mathbf{R}_{wb} = \mathbf{R}_z(\pi/2)$, the same body twist gives

$$\dot{\mathbf{p}} = \mathbf{R}_{wb}\boldsymbol{\nu}_b = \begin{bmatrix}0\\1\\0\end{bmatrix}\mathrm{m\,s^{-1}}.$$

The body-frame instruction "move along positive x_b" is unchanged, but its world-coordinate direction changes with the current orientation. Consequently, the body twist is the same while the world-frame pose rate is different. The current pose supplies the conversion between them.

The useful mental model is therefore

$$\boxed{\text{pose rate and body twist encode the same instantaneous motion}}$$

but

$$\boxed{\text{they express it in different mathematical forms and coordinates.}}$$

Why the hat matrix belongs to $\mathfrak{se}(3)$

The set $SE(3)$ contains finite spatial rigid transformations. Its instantaneous directions at the identity transformation $\mathbf{I}_4$ form the tangent space $T_{\mathbf{I}_4}SE(3)$, denoted $\mathfrak{se}(3)$ and read "little s e three." This statement can be established directly from the block structure of a curve in $SE(3)$.

Let

$$\mathbf{G}(s) = \begin{bmatrix} \mathbf{Q}(s) & \mathbf{d}(s) \\ \mathbf{0}_3^\mathsf{T} & 1 \end{bmatrix} \in SE(3), \qquad \mathbf{G}(0) = \mathbf{I}_4,$$

be any differentiable curve with dimensionless parameter $s \in \mathbb{R}$. The identity condition gives $\mathbf{Q}(0) = \mathbf{I}_3$ and $\mathbf{d}(0) = \mathbf{0}_3$. Differentiation at $s = 0$ gives

$$\mathbf{G}'(0) = \begin{bmatrix} \mathbf{Q}'(0) & \mathbf{d}'(0) \\ \mathbf{0}_3^\mathsf{T} & 0 \end{bmatrix}.$$

Because $\mathbf{Q}(s) \in SO(3)$, differentiating $\mathbf{Q}(s)^\mathsf{T}\mathbf{Q}(s) = \mathbf{I}_3$ at $s = 0$ gives

$$\mathbf{Q}'(0)^\mathsf{T} + \mathbf{Q}'(0) = \mathbf{0}_{3\times 3}.$$

Therefore, $\mathbf{Q}'(0)$ is skew-symmetric and has the unique form $[\mathbf{a}]_\times$ for some $\mathbf{a} \in \mathbb{R}^3$. The translation derivative $\mathbf{d}'(0)$ may be any column $\mathbf{u} \in \mathbb{R}^3$. Hence, every tangent direction must have the form

$$\begin{bmatrix} [\mathbf{a}]_\times & \mathbf{u} \\ \mathbf{0}_3^\mathsf{T} & 0 \end{bmatrix}.$$

The converse is also true. Given any $\mathbf{a}, \mathbf{u} \in \mathbb{R}^3$, the curve

$$\mathbf{G}_{\mathbf{a},\mathbf{u}}(s) := \begin{bmatrix} \exp(s[\mathbf{a}]_\times) & s\mathbf{u} \\ \mathbf{0}_3^\mathsf{T} & 1 \end{bmatrix}$$

belongs to $SE(3)$, passes through $\mathbf{I}_4$ at $s = 0$, and has derivative

$$\mathbf{G}'_{\mathbf{a},\mathbf{u}}(0) = \begin{bmatrix} [\mathbf{a}]_\times & \mathbf{u} \\ \mathbf{0}_3^\mathsf{T} & 0 \end{bmatrix}.$$

The first direction shows that no other tangent-matrix structure is possible; the converse shows that every matrix with the stated structure is attainable by a valid curve through the identity. Therefore,

$$\boxed{\mathfrak{se}(3) := T_{\mathbf{I}_4} SE(3) = \left\{ \begin{bmatrix} [\mathbf{a}]_\times & \mathbf{u} \\ \mathbf{0}_3^\mathsf{T} & 0 \end{bmatrix} \,\middle|\, \mathbf{a}, \mathbf{u} \in \mathbb{R}^3 \right\}}.$$

This is a six-dimensional vector space because three independent numbers determine **a** and three determine **u**. When the curve parameter is physical time, **a** becomes angular velocity, **u** becomes linear velocity, and the resulting matrix is the hatted twist. An element of $SE(3)$ describes a finite pose or coordinate transformation; an element of $\mathfrak{se}(3)$ describes an instantaneous direction or rate and is not itself a finite pose.

Worked coordinate check

Suppose that, at one instant, the body orientation and world-expressed velocities are

$$\mathbf{R}_{wb} = \mathbf{R}_z\left(\frac{\pi}{2}\right), \qquad {}^w\mathbf{v}_b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}, \qquad \boldsymbol{\omega}_w = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \mathrm{rad\,s^{-1}}.$$

The body is turned by $\pi/2$ about z_w, so its positive y_b-axis points along negative x_w. Expressing the same velocities along the body axes gives

$$\begin{aligned} \boldsymbol{\nu}_b &= \mathbf{R}_{wb}^\mathsf{T}\, {}^w\mathbf{v}_b = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}, \\ \boldsymbol{\omega}_b &= \mathbf{R}_{wb}^\mathsf{T} \boldsymbol{\omega}_w = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \mathrm{rad\,s^{-1}}. \end{aligned}$$

Thus,

$$\boldsymbol{\xi}_b = \begin{bmatrix} 0 & -1 & 0 & 0 & 0 & 2 \end{bmatrix}^\mathsf{T},$$

with the first three entries measured in $\mathrm{m\,s^{-1}}$ and the final three measured in $\mathrm{rad\,s^{-1}}$. The negative second component does not indicate a different motion: it states that motion along positive x_w points along negative y_b at this orientation.

7.9.3 Spatial twist expressed in the world frame

The body twist expresses instantaneous motion using the moving body axes and the body origin as its reference. Some calculations instead need one fixed reference shared by every body, such as comparing several robots in a world map or evaluating velocities at world-coordinate locations. The corresponding representation is called the **spatial twist expressed in the world frame**. Here, *spatial* means that the fixed space, or world, frame supplies both the coordinate axes and the reference origin.

Retain the compact pose notation from Section 7.9.2:

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_3^\top & 1 \end{bmatrix}, \qquad \dot{\mathbf{T}} = \begin{bmatrix} \dot{\mathbf{R}} & \dot{\mathbf{p}} \\ \mathbf{0}_3^\top & 0 \end{bmatrix},$$

where $\mathbf{R} = \mathbf{R}_{wb} \in SO(3)$, $\mathbf{p} = {}^w\mathbf{p}_b \in \mathbb{R}^3$, and $\dot{\mathbf{p}} = {}^w\mathbf{v}_b \in \mathbb{R}^3$. To express the pose rate relative to the fixed world frame, postmultiply $\dot{\mathbf{T}}$ by the inverse pose:

$$\begin{aligned} \dot{\mathbf{T}}\mathbf{T}^{-1} &= \begin{bmatrix} \dot{\mathbf{R}} & \dot{\mathbf{p}} \\ \mathbf{0}_3^\top & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R}^\top & -\mathbf{R}^\top\mathbf{p} \\ \mathbf{0}_3^\top & 1 \end{bmatrix} \\ &= \begin{bmatrix} \dot{\mathbf{R}}\mathbf{R}^\top & \dot{\mathbf{p}} - \dot{\mathbf{R}}\mathbf{R}^\top\mathbf{p} \\ \mathbf{0}_3^\top & 0 \end{bmatrix}. \end{aligned}$$

The upper-left block is the world-expressed angular-velocity matrix derived in Section 7.4:

$$\dot{\mathbf{R}}\mathbf{R}^\top = [\boldsymbol{\omega}_w]_\times,$$

where $\boldsymbol{\omega}_w \in \mathbb{R}^3$ is the body's angular velocity relative to the world, expressed along the world axes and measured in $\mathrm{rad\,s^{-1}}$. The upper-right block is therefore

$$\dot{\mathbf{p}} - [\boldsymbol{\omega}_w]_\times \mathbf{p} = \dot{\mathbf{p}} - \boldsymbol{\omega}_w \times \mathbf{p}.$$

Define this column as

$$\boxed{\boldsymbol{\nu}_w := \dot{\mathbf{p}} - \boldsymbol{\omega}_w \times \mathbf{p} = \begin{bmatrix} \nu_{x_w} \\ \nu_{y_w} \\ \nu_{z_w} \end{bmatrix} \in \mathbb{R}^3}.$$

Its entries are components along x_w, y_w, and z_w and are measured in $\mathrm{m\,s^{-1}}$. Radians are dimensionless, so $(\mathrm{rad\,s^{-1}})(\mathrm{m}) = \mathrm{m\,s^{-1}}$ for the cross-product term.

The physical meaning of $\boldsymbol{\nu}_w$ can be stated immediately. The relative-position vector from the body origin O_b, currently at $\mathbf{p}$, to the world origin O_w, currently at $\mathbf{0}_3$, is

$${}^w\mathbf{r}_{O_bO_w} = \mathbf{0}_3 - \mathbf{p} = -\mathbf{p}.$$

The rigid-body point-velocity relation therefore gives

$$\boxed{\underbrace{\boldsymbol{\nu}_w}_{\text{velocity field at } O_w} = \underbrace{\dot{\mathbf{p}}}_{\text{velocity at } O_b} + \underbrace{\boldsymbol{\omega}_w \times (-\mathbf{p})}_{\text{velocity change caused by the different location}}}.$$

Thus, $\boldsymbol{\nu}_w$ is the velocity that the body's instantaneous rigid-motion field assigns to the world-origin location, expressed along the world axes. It is generally not the actual velocity $\dot{\mathbf{p}}$ of the body origin. The detailed point-velocity derivation below verifies this interpretation. Substituting both named blocks back into the matrix product now completes the calculation:

$$\boxed{\dot{\mathbf{T}}\mathbf{T}^{-1} = \begin{bmatrix} [\boldsymbol{\omega}_w]_\times & \boldsymbol{\nu}_w \\ \mathbf{0}_3^\mathsf{T} & 0 \end{bmatrix}}.$$

Using the same linear-first component order as the body twist, define the **spatial-twist column** and its hat matrix by

$$\boxed{\boldsymbol{\xi}_w := \begin{bmatrix} \boldsymbol{\nu}_w \\ \boldsymbol{\omega}_w \end{bmatrix} \in \mathbb{R}^6, \qquad \widehat{\boldsymbol{\xi}}_w := \begin{bmatrix} [\boldsymbol{\omega}_w]_\times & \boldsymbol{\nu}_w \\ \mathbf{0}_3^\mathsf{T} & 0 \end{bmatrix} \in \mathfrak{se}(3)}.$$

Comparison with the completed block product gives

$$\boxed{\widehat{\boldsymbol{\xi}}_w = \dot{\mathbf{T}}_{wb}\mathbf{T}_{wb}^{-1}}, \qquad \boxed{\dot{\mathbf{T}}_{wb} = \widehat{\boldsymbol{\xi}}_w \mathbf{T}_{wb}}.$$

The first equation extracts the world-referenced spatial twist from a pose trajectory. The second reconstructs the pose rate from the current pose and that twist. The multiplication side distinguishes the two conventions: a body-twist matrix multiplies the pose on the right, whereas a spatial-twist matrix multiplies it on the left.

Physical meaning of $\boldsymbol{\nu}_w$

The column $\boldsymbol{\nu}_w$ is generally not the body-origin velocity $\dot{\mathbf{p}}$. Its physical meaning follows from the body-fixed point-velocity equation in Section 7.9.1:

$${}^w\mathbf{v}_P = {}^w\mathbf{v}_b + \boldsymbol{\omega}_w \times {}^w\mathbf{r}_{O_bP},$$

where ${}^w\mathbf{v}_P \in \mathbb{R}^3$ is the world-relative velocity of a body-fixed point P, expressed in world coordinates, and ${}^w\mathbf{r}_{O_bP} \in \mathbb{R}^3$ is the relative-position vector from the body origin O_b to P, also expressed in world coordinates. Because

$${}^w\mathbf{v}_b = \dot{\mathbf{p}}, \qquad {}^w\mathbf{r}_{O_bP} = {}^w\mathbf{p}_P - \mathbf{p},$$

substitution and regrouping give

$$\begin{aligned} {}^{w}\mathbf{v}_P &= \dot{\mathbf{p}} + \boldsymbol{\omega}_w \times ({}^{w}\mathbf{p}_P - \mathbf{p}) \\ &= \dot{\mathbf{p}} - \boldsymbol{\omega}_w \times \mathbf{p} + \boldsymbol{\omega}_w \times {}^{w}\mathbf{p}_P \\ &= \boldsymbol{\nu}_w + \boldsymbol{\omega}_w \times {}^{w}\mathbf{p}_P. \end{aligned}$$

Thus, the spatial twist defines the world-frame **rigid-motion velocity field**—a rule that calculates the instantaneous velocity of any body-fixed point from that point's world-coordinate location:

$$\boxed{{}^{w}\mathbf{v}_P = \boldsymbol{\nu}_w + \boldsymbol{\omega}_w \times {}^{w}\mathbf{p}_P}.$$

Let O_w denote the fixed world origin, whose world-coordinate column is $\mathbf{0}_3$. Evaluating the field at that location gives

$${}^{w}\mathbf{v}_P\big|_{{}^{w}\mathbf{p}_P=\mathbf{0}_3} = \boldsymbol{\nu}_w.$$

Therefore, $\boldsymbol{\nu}_w$ is the velocity that the instantaneous rigid-motion field assigns to the location O_w, expressed along the world axes. The world origin does not need to be a physical point on the robot; the same rigid-motion rule can be evaluated mathematically at that location.

At the body origin, ${}^{w}\mathbf{p}_P = \mathbf{p}$ and $P = O_b$, so the same field gives

$$\boxed{\underbrace{{}^{w}\mathbf{v}_b}_{\text{velocity at } O_b} = \underbrace{\boldsymbol{\nu}_w}_{\text{velocity field at } O_w} + \underbrace{\boldsymbol{\omega}_w \times \mathbf{p}}_{\text{change caused by the different location}}}. \tag{7.1}$$

This equation also follows immediately by rearranging $\boldsymbol{\nu}_w = \dot{\mathbf{p}} - \boldsymbol{\omega}_w \times \mathbf{p}$. It shows why $\boldsymbol{\nu}_w$ alone must not be called the body-origin velocity: the two quantities refer to different locations. They are equal when $\boldsymbol{\omega}_w \times \mathbf{p} = \mathbf{0}_3$, including pure translation or a body origin lying on the instantaneous rotation-axis line through O_w.

Worked reference-origin check

Suppose that, at one instant,

$$\mathbf{p} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \mathrm{m}, \qquad \dot{\mathbf{p}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}, \qquad \boldsymbol{\omega}_w = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \mathrm{rad\,s^{-1}}.$$

The velocity difference caused by rotation between the world-origin location and the body origin is

$$\boldsymbol{\omega}_w \times \mathbf{p} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \times \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}.$$

Therefore, the linear part of the spatial twist is

$$\boldsymbol{\nu}_w = \dot{\mathbf{p}} - \boldsymbol{\omega}_w \times \mathbf{p} = \begin{bmatrix} 3 \\ -4 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}.$$

Because $\boldsymbol{\omega}_w \times \mathbf{p} \neq \mathbf{0}_3$, the spatial-twist component $\boldsymbol{\nu}_w = [\,3\ \ -4\ 0\,]^\mathsf{T}\,\mathrm{m\,s^{-1}}$ differs from the given body-origin velocity $\dot{\mathbf{p}} = [\,1\ 0\ 0\,]^\mathsf{T}\,\mathrm{m\,s^{-1}}$. The first is the velocity field evaluated at O_w; the second is its value at O_b. Their difference is exactly the rotational term $\boldsymbol{\omega}_w \times \mathbf{p}$. The next subsection will derive how the current pose converts between the complete body and spatial twists.

7.9.4 Relating body and spatial twists through the adjoint

The body twist $\boldsymbol{\xi}_b$ and spatial twist $\boldsymbol{\xi}_w$ describe the same instantaneous rigid-body motion, but their components use different axes and different reference origins. A conversion between them must therefore account for both the orientation $\mathbf{R}_{wb}$ and position $\mathbf{p} = {}^w\mathbf{p}_b$ in the current pose $\mathbf{T}_{wb}$.

Convert the angular and linear parts

Write $\mathbf{R} := \mathbf{R}_{wb}$. The angular-velocity coordinate relation was already established in Section 7.4.2:

$$\boxed{\boldsymbol{\omega}_w = \mathbf{R}\boldsymbol{\omega}_b}.$$

This equation rotates the coordinate column of the same physical angular velocity from the body axes into the world axes. The present derivation uses this previously established relation.

For the linear part, $\boldsymbol{\nu}_b$ is the velocity of the body origin O_b expressed along the body axes. Its world-coordinate column is therefore

$${}^w\mathbf{v}_b = \mathbf{R}\boldsymbol{\nu}_b.$$

The spatial-twist velocity field in Equation 7.1 gives the same body-origin velocity as

$${}^w\mathbf{v}_b = \boldsymbol{\nu}_w + \boldsymbol{\omega}_w \times \mathbf{p}.$$

Equating these two descriptions of ${}^w\mathbf{v}_b$ and solving for $\boldsymbol{\nu}_w$ gives

$$\begin{aligned}\boldsymbol{\nu}_w &= \mathbf{R}\boldsymbol{\nu}_b - \boldsymbol{\omega}_w \times \mathbf{p} \\ &= \mathbf{R}\boldsymbol{\nu}_b + \mathbf{p} \times \boldsymbol{\omega}_w \\ &= \boxed{\mathbf{R}\boldsymbol{\nu}_b + [\mathbf{p}]_\times \mathbf{R}\boldsymbol{\omega}_b}.\end{aligned}$$

The second equality uses cross-product anticommutativity, $-\boldsymbol{\omega}_w \times \mathbf{p} = \mathbf{p} \times \boldsymbol{\omega}_w$. The final equality uses $\mathbf{p} \times \boldsymbol{\omega}_w = [\mathbf{p}]_\times \boldsymbol{\omega}_w$ and the already-established relation $\boldsymbol{\omega}_w = \mathbf{R}\boldsymbol{\omega}_b$.

The two terms have different purposes. The term $\mathbf{R}\boldsymbol{\nu}_b$ changes the linear-velocity components from body axes to world axes. The term $[\mathbf{p}]_\times \mathbf{R}\boldsymbol{\omega}_b$ changes the reference location for the linear part from O_b to O_w. Angular velocity needs no reference-location correction because every point of a rigid body shares the same angular velocity.

Confirm the conversion using the hat matrices

The same result follows from the two hat-matrix definitions:

$$\widehat{\boldsymbol{\xi}}_b = \mathbf{T}^{-1}\dot{\mathbf{T}}, \qquad \widehat{\boldsymbol{\xi}}_w = \dot{\mathbf{T}}\mathbf{T}^{-1}.$$

The body-twist definition gives $\dot{\mathbf{T}} = \mathbf{T}\widehat{\boldsymbol{\xi}}_b$. Substitution into the spatial-twist definition gives

$$\begin{aligned}\widehat{\boldsymbol{\xi}}_w &= \dot{\mathbf{T}}\mathbf{T}^{-1} \\ &= \left(\mathbf{T}\widehat{\boldsymbol{\xi}}_b\right)\mathbf{T}^{-1} \\ &= \boxed{\mathbf{T}\widehat{\boldsymbol{\xi}}_b\mathbf{T}^{-1}}.\end{aligned}$$

This three-matrix product is called **conjugation by $\mathbf{T}$**. Here, conjugation uses the current pose and its inverse to change the reference used by the hatted twist while preserving the represented physical motion. The following block multiplication shows exactly how its six coordinates change.

Write the matrices as

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_3^\top & 1 \end{bmatrix}, \qquad \widehat{\boldsymbol{\xi}}_b = \begin{bmatrix} [\boldsymbol{\omega}_b]_\times & \boldsymbol{\nu}_b \\ \mathbf{0}_3^\top & 0 \end{bmatrix}, \qquad \mathbf{T}^{-1} = \begin{bmatrix} \mathbf{R}^\top & -\mathbf{R}^\top\mathbf{p} \\ \mathbf{0}_3^\top & 1 \end{bmatrix},$$

where $\mathbf{R} = \mathbf{R}_{wb} \in SO(3)$ and $\mathbf{p} = {}^w\mathbf{p}_b \in \mathbb{R}^3$. First multiply the two matrices on the left:

$$\mathbf{T}\widehat{\boldsymbol{\xi}}_b = \begin{bmatrix} \mathbf{R}[\boldsymbol{\omega}_b]_\times & \mathbf{R}\boldsymbol{\nu}_b \\ \mathbf{0}_3^\top & 0 \end{bmatrix}.$$

Multiplying this result by $\mathbf{T}^{-1}$ gives

$$\begin{aligned}\mathbf{T}\widehat{\boldsymbol{\xi}}_b\mathbf{T}^{-1} &= \begin{bmatrix}\mathbf{R}[\boldsymbol{\omega}_b]_\times & \mathbf{R}\boldsymbol{\nu}_b \\ \mathbf{0}_3^\top & 0\end{bmatrix}\begin{bmatrix}\mathbf{R}^\top & -\mathbf{R}^\top\mathbf{p} \\ \mathbf{0}_3^\top & 1\end{bmatrix} \\ &= \begin{bmatrix}\mathbf{R}[\boldsymbol{\omega}_b]_\times\mathbf{R}^\top & -\mathbf{R}[\boldsymbol{\omega}_b]_\times\mathbf{R}^\top\mathbf{p} + \mathbf{R}\boldsymbol{\nu}_b \\ \mathbf{0}_3^\top & 0\end{bmatrix}.\end{aligned}$$

The proper-rotation cross-product identity established in Section 7.4.2 gives

$$\mathbf{R}[\boldsymbol{\omega}_b]_\times\mathbf{R}^\top = [\mathbf{R}\boldsymbol{\omega}_b]_\times.$$

The previously established coordinate relation $\boldsymbol{\omega}_w = \mathbf{R}\boldsymbol{\omega}_b$ then gives

$$\boxed{\mathbf{R}[\boldsymbol{\omega}_b]_\times\mathbf{R}^\top = [\boldsymbol{\omega}_w]_\times}.$$

No new angular-velocity relation is being derived in this step. The proper-rotation identity only confirms that conjugating the body-coordinate skew block produces the angular block required by $\widehat{\boldsymbol{\xi}}_w$.

The conjugated upper-right block becomes

$$\begin{aligned}&-\mathbf{R}[\boldsymbol{\omega}_b]_\times\mathbf{R}^\top\mathbf{p} + \mathbf{R}\boldsymbol{\nu}_b \\ &\qquad = -[\boldsymbol{\omega}_w]_\times\mathbf{p} + \mathbf{R}\boldsymbol{\nu}_b \\ &\qquad = \mathbf{R}\boldsymbol{\nu}_b - \boldsymbol{\omega}_w \times \mathbf{p} \\ &\qquad = \boldsymbol{\nu}_w,\end{aligned}$$

where the final equality is the direct linear relation derived at the start of this subsection. Therefore, the complete conjugated matrix is

$$\mathbf{T}\widehat{\boldsymbol{\xi}}_b\mathbf{T}^{-1} = \begin{bmatrix}[\boldsymbol{\omega}_w]_\times & \boldsymbol{\nu}_w \\ \mathbf{0}_3^\top & 0\end{bmatrix} = \widehat{\boldsymbol{\xi}}_w.$$

The component calculation and the conjugation calculation therefore agree: both convert the body-referenced description of one instantaneous motion into its world-referenced description.

The adjoint matrix

Collect the linear and angular relations using the declared linear-first twist order:

$$\begin{aligned}\boldsymbol{\xi}_w &= \begin{bmatrix}\boldsymbol{\nu}_w \\ \boldsymbol{\omega}_w\end{bmatrix} \\ &= \begin{bmatrix}\mathbf{R} & [\mathbf{p}]_\times \mathbf{R} \\ \mathbf{0}_{3\times 3} & \mathbf{R}\end{bmatrix}\begin{bmatrix}\boldsymbol{\nu}_b \\ \boldsymbol{\omega}_b\end{bmatrix}.\end{aligned}$$

The 6×6 matrix in this equation is called the **adjoint representation** of $\mathbf{T}$. The word *representation* means that the finite pose $\mathbf{T} \in SE(3)$ is represented by a linear operator acting on six-component twist coordinates. For the component order $[\,\boldsymbol{\nu}^\mathsf{T}\ \boldsymbol{\omega}^\mathsf{T}\,]^\mathsf{T}$ used in this book, define

$$\boxed{\mathrm{Ad}_{\mathbf{T}} := \begin{bmatrix}\mathbf{R} & [\mathbf{p}]_\times \mathbf{R} \\ \mathbf{0}_{3\times 3} & \mathbf{R}\end{bmatrix} \in \mathbb{R}^{6\times 6}}.$$

Each block is 3×3. The rotation blocks are dimensionless, whereas $[\mathbf{p}]_\times \mathbf{R}$ has units of metres. Multiplying that block by $\boldsymbol{\omega}_b$ in $\mathrm{rad\,s^{-1}}$ produces the required $\mathrm{m\,s^{-1}}$ contribution to $\boldsymbol{\nu}_w$. The zero block shows that linear velocity does not contribute to angular velocity, while the upper-right block shows that angular velocity contributes to the linear part whenever the two reference origins differ.

For $\mathbf{T} = \mathbf{T}_{wb}$, the complete conversion is

$$\boxed{\boldsymbol{\xi}_w = \mathrm{Ad}_{\mathbf{T}_{wb}}\, \boldsymbol{\xi}_b}.$$

Equivalently, the adjoint can be characterised directly by its action on any twist column $\boldsymbol{\xi} \in \mathbb{R}^6$:

$$\boxed{\widehat{\mathrm{Ad}_{\mathbf{T}}\, \boldsymbol{\xi}} = \mathbf{T}\widehat{\boldsymbol{\xi}}\mathbf{T}^{-1}}.$$

Inverse adjoint operator

The forward operator $\mathrm{Ad}_{\mathbf{T}_{wb}} : \mathbb{R}^6 \to \mathbb{R}^6$ converts body-twist coordinates into spatial-twist coordinates. Recovering the body twist requires the inverse of this conversion. Two similar-looking operators appear:

- $\mathrm{Ad}_{\mathbf{T}_{wb}^{-1}}$ means first form the inverse pose $\mathbf{T}_{wb}^{-1} \in SE(3)$ and then construct the adjoint associated with that pose.
- $\mathrm{Ad}_{\mathbf{T}_{wb}}^{-1}$ means first construct the 6×6 adjoint matrix $\mathrm{Ad}_{\mathbf{T}_{wb}}$ and then take its ordinary matrix inverse.

These procedures produce the same 6×6 linear operator. To prove this, let $\boldsymbol{\xi} \in \mathbb{R}^6$ be arbitrary and first apply $\mathrm{Ad}_{\mathbf{T}}$, followed by $\mathrm{Ad}_{\mathbf{T}^{-1}}$. Applying the hat definition twice gives

$$\begin{aligned}
\widehat{\mathrm{Ad}_{\mathbf{T}^{-1}} (\mathrm{Ad}_{\mathbf{T}}\, \boldsymbol{\xi})} &= \mathbf{T}^{-1} \widehat{\mathrm{Ad}_{\mathbf{T}}\, \boldsymbol{\xi}} \mathbf{T} \\
&= \mathbf{T}^{-1} \left(\mathbf{T} \widehat{\boldsymbol{\xi}} \mathbf{T}^{-1} \right) \mathbf{T} \\
&= \widehat{\boldsymbol{\xi}}.
\end{aligned}$$

The hat map is one-to-one: each twist column has exactly one hatted matrix in $\mathfrak{se}(3)$, and each matrix in $\mathfrak{se}(3)$ determines exactly one twist column. The equality of the hatted matrices therefore implies

$$\mathrm{Ad}_{\mathbf{T}^{-1}} (\mathrm{Ad}_{\mathbf{T}}\, \boldsymbol{\xi}) = \boldsymbol{\xi}$$

for every $\boldsymbol{\xi} \in \mathbb{R}^6$. Hence, the product of the two adjoint matrices is the identity operator on $\mathbb{R}^6$:

$$\mathrm{Ad}_{\mathbf{T}^{-1}}\, \mathrm{Ad}_{\mathbf{T}} = \mathbf{I}_6.$$

Repeating the same calculation in the opposite order, equivalently replacing $\mathbf{T}$ by $\mathbf{T}^{-1}$, gives

$$\mathrm{Ad}_{\mathbf{T}}\, \mathrm{Ad}_{\mathbf{T}^{-1}} = \mathbf{I}_6.$$

Thus, $\mathrm{Ad}_{\mathbf{T}^{-1}}$ is both a left and a right inverse of $\mathrm{Ad}_{\mathbf{T}}$. By the definition of a matrix inverse,

$$\boxed{\mathrm{Ad}_{\mathbf{T}^{-1}} = \mathrm{Ad}_{\mathbf{T}}^{-1}}.$$

Setting $\mathbf{T} = \mathbf{T}_{wb}$ and applying this inverse operator to $\boldsymbol{\xi}_w$ gives the inverse twist-coordinate conversion:

$$\boxed{\boldsymbol{\xi}_b = \mathrm{Ad}_{\mathbf{T}_{wb}^{-1}}\, \boldsymbol{\xi}_w = \mathrm{Ad}_{\mathbf{T}_{wb}}^{-1}\, \boldsymbol{\xi}_w}.$$

Physically, $\mathbf{T}_{wb}^{-1}$ reverses the pose-based change of reference from the body frame to the world frame. Its adjoint therefore reverses the corresponding change from body-twist coordinates to spatial-twist coordinates.

If the two frame origins coincide, then $\mathbf{p} = \mathbf{0}_3$ and the cross-product block vanishes. In that special case, both the linear and angular parts change only by rotation. In general, a twist is not a pair of free vectors: changing its reference frame requires both rotation and the position-dependent correction encoded by the adjoint.

Quick test I: spatial rigid-body velocity and twists

Try these questions without looking back at the section.

1. **Recall:** State the world-frame velocity of a body-fixed point P in terms of the body-origin velocity, angular velocity, and relative-position vector.
2. **Contrast:** Define the body- and spatial-twist hat matrices from $\mathbf{T}_{wb}$ and $\dot{\mathbf{T}}_{wb}$. Which side of the pose does each twist matrix multiply when reconstructing $\dot{\mathbf{T}}_{wb}$?
3. **Explanation:** What physical quantity does $\boldsymbol{\nu}_b$ describe? What does $\boldsymbol{\nu}_w$ describe, and why is $\boldsymbol{\nu}_w$ generally not the velocity of the body origin?
4. **Application:** At one instant, let $\mathbf{p} = [1\ 0\ 0]^\mathsf{T}\,\mathrm{m}$, $\dot{\mathbf{p}} = [0\ 1\ 0]^\mathsf{T}\,\mathrm{m\,s^{-1}}$, and $\boldsymbol{\omega}_w = [0\ 0\ 2]^\mathsf{T}\,\mathrm{rad\,s^{-1}}$. Calculate $\boldsymbol{\nu}_w$. Then use the spatial-twist velocity field to calculate the velocity of a body-fixed point currently at ${}^w\mathbf{p}_P = [1\ 1\ 0]^\mathsf{T}\,\mathrm{m}$.
5. **Derivation:** Starting from $\boldsymbol{\omega}_w = \mathbf{R}\boldsymbol{\omega}_b$ and $\boldsymbol{\nu}_w = \mathbf{R}\boldsymbol{\nu}_b + [\mathbf{p}]_\times \mathbf{R}\boldsymbol{\omega}_b$, stack the equations to derive $\boldsymbol{\xi}_w = \mathrm{Ad}_\mathbf{T}\,\boldsymbol{\xi}_b$ for the linear-first component order.
6. **Reference origin:** Under what condition does the adjoint reduce to rotating the linear and angular parts separately? Why does the position-dependent block vanish in that case?
7. **Inverse conversion:** Explain the difference between $\mathrm{Ad}_{\mathbf{T}^{-1}}$ and $\mathrm{Ad}_\mathbf{T}^{-1}$. Why are they equal?
8. **Common error:** Why is the rule $\boldsymbol{\nu}_w = \mathbf{R}\boldsymbol{\nu}_b$ generally insufficient for converting the linear part of a twist, even though $\boldsymbol{\omega}_w = \mathbf{R}\boldsymbol{\omega}_b$ is sufficient for the angular part?

Answers and hints

1. With all quantities expressed in world coordinates,

$$\boxed{{}^w\mathbf{v}_P = {}^w\mathbf{v}_b + \boldsymbol{\omega}_w \times {}^w\mathbf{r}_{O_bP}}.$$

   The first term is the translational velocity at the body origin. The second is the additional velocity caused by the point's displacement from that origin while the body rotates.

2. The definitions and reconstruction equations are

$$\boxed{\widehat{\boldsymbol{\xi}}_b = \mathbf{T}_{wb}^{-1}\dot{\mathbf{T}}_{wb}, \qquad \dot{\mathbf{T}}_{wb} = \mathbf{T}_{wb}\widehat{\boldsymbol{\xi}}_b},$$

   and

$$\boxed{\widehat{\boldsymbol{\xi}}_w = \dot{\mathbf{T}}_{wb}\mathbf{T}_{wb}^{-1}, \qquad \dot{\mathbf{T}}_{wb} = \widehat{\boldsymbol{\xi}}_w\mathbf{T}_{wb}}.$$

   The body-twist matrix multiplies the pose on the right, whereas the spatial-twist matrix multiplies it on the left.

3. The column $\boldsymbol{\nu}_b = \mathbf{R}_{wb}^{\mathsf{T}}\,{}^{w}\mathbf{v}_b$ is the velocity of the body origin relative to the world, expressed along the body axes. The column $\boldsymbol{\nu}_w$ is the value of the instantaneous rigid-motion velocity field at the world-origin location, expressed along the world axes. They satisfy

$$ {}^{w}\mathbf{v}_b = \boldsymbol{\nu}_w + \boldsymbol{\omega}_w \times \mathbf{p}, $$

so they differ whenever the rotational contribution between O_w and O_b is nonzero.

4. First,

$$ \boldsymbol{\omega}_w \times \mathbf{p} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}. $$

Therefore,

$$ \boldsymbol{\nu}_w = \dot{\mathbf{p}} - \boldsymbol{\omega}_w \times \mathbf{p} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}. $$

At the declared point,

$$ \begin{aligned} {}^{w}\mathbf{v}_P &= \boldsymbol{\nu}_w + \boldsymbol{\omega}_w \times {}^{w}\mathbf{p}_P \\ &= \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \\ 0 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}}. \end{aligned} $$

5. Using $\boldsymbol{\xi} = [\,\boldsymbol{\nu}^{\mathsf{T}}\ \boldsymbol{\omega}^{\mathsf{T}}\,]^{\mathsf{T}}$,

$$ \begin{aligned} \boldsymbol{\xi}_w &= \begin{bmatrix} \boldsymbol{\nu}_w \\ \boldsymbol{\omega}_w \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{R}\boldsymbol{\nu}_b + [\mathbf{p}]_\times \mathbf{R}\boldsymbol{\omega}_b \\ \mathbf{R}\boldsymbol{\omega}_b \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} \mathbf{R} & [\mathbf{p}]_\times \mathbf{R} \\ \mathbf{0}_{3\times 3} & \mathbf{R} \end{bmatrix}}_{\mathrm{Ad}_{\mathbf{T}}} \begin{bmatrix} \boldsymbol{\nu}_b \\ \boldsymbol{\omega}_b \end{bmatrix}. \end{aligned} $$

6. If the source and target frame origins coincide, then $\mathbf{p} = \mathbf{0}_3$. Consequently, $[\mathbf{p}]_\times = \mathbf{0}_{3\times 3}$ and

$$\mathrm{Ad}_{\mathbf{T}} = \begin{bmatrix} \mathbf{R} & \mathbf{0}_{3\times 3} \\ \mathbf{0}_{3\times 3} & \mathbf{R} \end{bmatrix}.$$

   No reference-origin shift remains, so both parts require only a change of coordinate axes.

7. The expression $\mathrm{Ad}_{\mathbf{T}^{-1}}$ means invert the pose first and then construct its adjoint. The expression $\mathrm{Ad}_{\mathbf{T}}^{-1}$ means construct the 6×6 adjoint matrix first and then invert that matrix. Applying the two maps in succession gives

$$\mathrm{Ad}_{\mathbf{T}^{-1}} \, \mathrm{Ad}_{\mathbf{T}} = \mathrm{Ad}_{\mathbf{T}} \, \mathrm{Ad}_{\mathbf{T}^{-1}} = \mathbf{I}_6,$$

   because conjugation by $\mathbf{T}$ followed by conjugation by $\mathbf{T}^{-1}$ restores the original hatted twist. Therefore, $\mathrm{Ad}_{\mathbf{T}^{-1}} = \mathrm{Ad}_{\mathbf{T}}^{-1}$.

8. Angular velocity is the same at every point of a rigid body, so changing its coordinate frame requires only $\boldsymbol{\omega}_w = \mathbf{R}\boldsymbol{\omega}_b$. The linear part of a twist is tied to a reference location as well as a coordinate basis. Changing from the body origin to the world origin contributes

$$[\mathbf{p}]_\times \mathbf{R}\boldsymbol{\omega}_b = \mathbf{p} \times \boldsymbol{\omega}_w,$$

   so the complete relation is $\boldsymbol{\nu}_w = \mathbf{R}\boldsymbol{\nu}_b + [\mathbf{p}]_\times \mathbf{R}\boldsymbol{\omega}_b$.

7.10 Representation limitations and numerical checks

The preceding sections describe exact mathematical objects. Software stores their components as finite-precision numbers, so round-off, an incorrect convention, or an invalid input can produce an array that no longer represents the intended geometry. Validation must therefore check the defining constraints of each representation rather than merely checking its array shape.

7.10.1 Equivalent information does not make representations interchangeable

A physical orientation can be represented by a rotation matrix, an axis–angle pair, or a unit quaternion. A physical pose can likewise be represented by a homogeneous transformation or by a chosen coordinate column. These representations may encode equivalent geometric information, but they have different mathematical types, constraints, and composition rules.

For example, $\mathbf{R}_{wb} \in SO(3)$ contains nine entries constrained by orthogonality and determinant conditions, whereas a unit quaternion contains four components constrained by one unit-norm equation. The two unit quaternions q and $-\mathsf{q}$ represent the same rotation, so comparing their component columns directly can report a large difference even when their represented orientations are identical.

Frame and storage conventions are also part of the data's meaning. An implementation must state whether a rotation maps body coordinates to world coordinates or the reverse, whether an operation is active or passive, whether a quaternion is stored in scalar-first or scalar-last order, and whether a twist is body or spatial. This chapter uses the twist component order

$$\boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\nu} \\ \boldsymbol{\omega} \end{bmatrix} \in \mathbb{R}^6,$$

with the three linear components first and the three angular components second. Reordering these six numbers without changing the associated formulas changes the represented motion.

7.10.2 Checking rotation matrices, transformations, and quaternions

Before checking a geometric constraint, first verify that every stored component is finite. A component equal to NaN or $\pm\infty$ does not represent a valid physical quantity and makes later residual tests unreliable.

For a real matrix $\mathbf{M} = [m_{ij}] \in \mathbb{R}^{m\times n}$, define the **Frobenius norm**

$$\|\mathbf{M}\|_{\mathrm{F}} := \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} m_{ij}^2}.$$

It measures the combined size of all matrix entries. For a candidate rotation matrix $\widetilde{\mathbf{R}} \in \mathbb{R}^{3\times 3}$, useful dimensionless residuals are

$$\boxed{e_{\mathrm{orth}} := \left\|\widetilde{\mathbf{R}}^{\mathsf{T}}\widetilde{\mathbf{R}} - \mathbf{I}_3\right\|_{\mathrm{F}}, \qquad e_{\mathrm{det}} := \left|\det(\widetilde{\mathbf{R}}) - 1\right|}.$$

The first residual checks whether the columns are mutually perpendicular unit vectors. The second distinguishes a proper rotation from a reflection. For example, $\mathrm{diag}(1, 1, -1)$ has orthonormal columns but determinant -1, so its orthogonality residual is zero while its determinant residual is two. An implementation accepts the matrix only when both residuals are below declared dimensionless tolerances.

A candidate homogeneous transformation $\widetilde{\mathbf{T}} \in \mathbb{R}^{4\times 4}$ must have the structure

$$\widetilde{\mathbf{T}} = \begin{bmatrix} \widetilde{\mathbf{R}} & \widetilde{\mathbf{p}} \\ \mathbf{0}_3^{\mathsf{T}} & 1 \end{bmatrix},$$

where $\widetilde{\mathbf{R}}$ passes the rotation checks, $\widetilde{\mathbf{p}} \in \mathbb{R}^3$ is a finite translation column measured in metres, and the bottom row equals $[\,0\ 0\ 0\ 1\,]$. Rotation residuals are dimensionless, whereas translation errors have units of metres; they should not be combined into one unscaled residual.

For a quaternion component column $\widetilde{\mathbf{c}} = [\,q_w\ q_x\ q_y\ q_z\,]^{\mathsf{T}} \in \mathbb{R}^4$, the unit-constraint residual is

$$\boxed{e_q := \left|\|\widetilde{\mathbf{c}}\|_2 - 1\right|}.$$

The quaternion section in Section 7.8 explains when a small norm error may be corrected by normalisation. Validation and repair are different operations: validation reports whether the input satisfies a requirement, while repair changes the input according to an explicit policy. A zero, near-zero, or non-finite quaternion cannot be repaired safely by division, and an arbitrary invalid matrix should not be silently treated as a rotation.

7.10.3 Checking twists and their units

A twist must have six finite components, the declared component order, a declared expression frame, and a declared reference origin. Its linear part $\boldsymbol{\nu} \in \mathbb{R}^3$ is measured in $\mathrm{m\,s^{-1}}$, while its angular part $\boldsymbol{\omega} \in \mathbb{R}^3$ is measured in $\mathrm{rad\,s^{-1}}$. The associated matrix must also have the structure

$$\widehat{\boldsymbol{\xi}} = \begin{bmatrix} [\boldsymbol{\omega}]_\times & \boldsymbol{\nu} \\ \mathbf{0}_3^{\mathsf{T}} & 0 \end{bmatrix},$$

so its upper-left block is skew-symmetric and its bottom row is zero.

An ordinary Euclidean expression such as

$$\|\boldsymbol{\xi}\|_2^2 = \|\boldsymbol{\nu}\|_2^2 + \|\boldsymbol{\omega}\|_2^2$$

adds squared quantities with different physical units and therefore is **not**, by itself, a meaningful physical magnitude. If an application needs one scalar for comparison, it must choose a characteristic length $\ell > 0$ measured in metres and may define

$$\boxed{\|\boldsymbol{\xi}\|_\ell := \sqrt{\|\boldsymbol{\nu}\|_2^2 + \ell^2\,\|\boldsymbol{\omega}\|_2^2}}.$$

This quantity has units of $\mathrm{m\,s^{-1}}$. The product $\ell\|\boldsymbol{\omega}\|_2$ is the tangential-speed scale associated with rotation at distance ℓ from the reference axis. Different choices of ℓ express different engineering priorities, so this weighted magnitude is not a universal property of the rigid-body motion.

7.10.4 Round-trip checks and implementation consequences

For numerical tests, write $\mathbf{A} \approx \mathbf{B}$ when a declared residual between the two arrays is below a chosen tolerance. The tolerance must account for the quantity's units, expected numerical scale, and floating-point precision. Useful round-trip and consistency checks include

$$\begin{gathered}\mathbf{R}^{\mathsf{T}}\mathbf{R} \approx \mathbf{I}_3, \quad \mathbf{T}\mathbf{T}^{-1} \approx \mathbf{I}_4, \\ \mathrm{Ad}_{\mathbf{T}^{-1}}\,\mathrm{Ad}_{\mathbf{T}} \approx \mathbf{I}_6, \quad \widehat{\mathrm{Ad}_{\mathbf{T}}\,\boldsymbol{\xi}} \approx \mathbf{T}\widehat{\boldsymbol{\xi}}\mathbf{T}^{-1}.\end{gathered}$$

The first check tests a rotation constraint. The second tests the transformation inverse. The third tests the forward and inverse twist-coordinate mappings. The fourth checks that the coordinate adjoint and matrix-conjugation definitions describe the same operation. A point transformation can also be checked by mapping body coordinates to world coordinates and then applying $\mathbf{T}^{-1}$ to recover the original body coordinates.

The body-expressed orientation-rate equation derived in Section 7.4 is

$$\boxed{\dot{\mathbf{R}}_{wb} = \mathbf{R}_{wb}[\boldsymbol{\omega}_b]_\times}.$$

For a numerical step beginning at time t_k, define

$$\mathbf{R}_k := \mathbf{R}_{wb}(t_k), \qquad \boldsymbol{\omega}_{b,k} := \boldsymbol{\omega}_b(t_k), \qquad \boldsymbol{\Omega}_k := [\boldsymbol{\omega}_{b,k}]_\times.$$

Evaluating the orientation-rate equation at t_k gives

$$\dot{\mathbf{R}}_{wb}(t_k) = \mathbf{R}_k\boldsymbol{\Omega}_k.$$

A direct Forward Euler update illustrates why preserving the representation constraints matters. Forward Euler replaces the next orientation by the current orientation plus the time step multiplied by the current orientation rate:

$$\widetilde{\mathbf{R}}_{k+1} := \mathbf{R}_k + \Delta t\,\dot{\mathbf{R}}_{wb}(t_k) = \mathbf{R}_k + \Delta t\,\mathbf{R}_k\boldsymbol{\Omega}_k = \mathbf{R}_k\left(\mathbf{I}_3 + \Delta t\,\boldsymbol{\Omega}_k\right).$$

Because $\boldsymbol{\Omega}_k^{\mathsf{T}} = -\boldsymbol{\Omega}_k$ and $\mathbf{R}_k^{\mathsf{T}}\mathbf{R}_k = \mathbf{I}_3$,

$$\begin{aligned}\widetilde{\mathbf{R}}_{k+1}^{\mathsf{T}}\widetilde{\mathbf{R}}_{k+1} &= \left(\mathbf{I}_3 - \Delta t\,\boldsymbol{\Omega}_k\right)\left(\mathbf{I}_3 + \Delta t\,\boldsymbol{\Omega}_k\right) \\ &= \mathbf{I}_3 - \Delta t^2\boldsymbol{\Omega}_k^2,\end{aligned}$$

which is generally not $\mathbf{I}_3$. The one-step orthogonality error is of order $O(\Delta t^2)$, so repeated raw matrix updates can drift away from $SO(3)$.

The exact finite-rotation factor is available when the body-expressed angular velocity remains constant throughout the interval. The body-expressed update from Section 7.5.5 becomes

$$\boxed{\mathbf{R}_{k+1} = \mathbf{R}_k \underbrace{\exp(\Delta t\, \boldsymbol{\Omega}_k)}_{\text{exact finite-rotation factor}}}.$$

The matrix exponential is a valid element of $SO(3)$. Its series begins

$$\exp(\Delta t\, \boldsymbol{\Omega}_k) = \mathbf{I}_3 + \Delta t\, \boldsymbol{\Omega}_k + O(\Delta t^2).$$

Forward Euler keeps only the first two terms and therefore replaces the exact factor $\exp(\Delta t\, \boldsymbol{\Omega}_k)$ by $\mathbf{I}_3 + \Delta t\, \boldsymbol{\Omega}_k$. A production implementation must use a declared structure-preserving update or a documented correction policy. The following chapter develops finite rigid-body motion and integration methods; the present result establishes the requirement they must satisfy.

The principal implementation rules from this chapter are therefore:

- store frames, units, component order, and active or passive meaning with the interface contract;
- use the transpose as a validated rotation's inverse and the block formula from Section 7.6.2 as a validated homogeneous transformation's inverse;
- compose transformations in the physical order of the motions rather than adding pose components;
- reject non-finite values and geometric-constraint violations deliberately;
- normalise quaternions only under a stated tolerance and failure policy; and
- test representation invariants, inverse round trips, and equivalent conversion paths.

Cumulative chapter review

Try these questions without looking back. They test definitions, derivations, applications, and limitations from the whole chapter.

1. What is the difference among a rigid body's physical configuration, a homogeneous transformation $\mathbf{T}_{wb}$, and a coordinate column such as $[\,x\ y\ \theta\,]^\mathsf{T}$?
2. For constraints $\mathbf{g}(\mathbf{z}) = \mathbf{0}$ at a regular configuration $\mathbf{z}_0 \in \mathbb{R}^{n_v}$, state the Jacobian null-space description of allowable infinitesimal changes and the corresponding DOF formula. Why does a free spatial rigid body have six DOF?
3. State the defining equations of $SO(3)$ and explain the geometric meaning of the columns of $\mathbf{R}_{wb}$.
4. If $\mathbf{Q}_w$ is an orientation change expressed along fixed world axes and $\mathbf{Q}_b$ is an orientation change expressed along the old body axes, how does each update $\mathbf{R}_{wb}$?
5. Give the world- and body-expressed angular-velocity equations for $\dot{\mathbf{R}}_{wb}$. Why do the two angular-velocity columns describe the same physical turning?
6. Let $\mathbf{a} \in \mathbb{R}^3$ be a unit rotation axis and $\alpha \in \mathbb{R}$ an angle. State the exponential-coordinate vector, its matrix exponential, and Rodrigues' formula.

7. State how $\mathbf{T}_{wb} \in SE(3)$ transforms a body-frame point coordinate into world coordinates, and derive the inverse transformation.
8. What constraints identify the planar-motion subgroup embedded in $SE(3)$, and how is planar yaw recovered from its rotation matrix?
9. Write the unit quaternion for an axis–angle rotation. Why do q and $-\mathsf{q}$ represent the same orientation, and what is the Hamilton-product order when q_1 acts first and q_2 acts second?
10. Distinguish a pose rate $\dot{\mathbf{T}}_{wb}$, a body twist $\boldsymbol{\xi}_b$, and a spatial twist $\boldsymbol{\xi}_w$. What physical quantities do their linear parts represent?
11. Let $\mathbf{R} = \mathbf{R}_z(\pi/2)$, $\mathbf{p} = [\,1\ 2\ 3\,]^\mathsf{T}\,\mathrm{m}$, $\boldsymbol{\nu}_b = [\,1\ 0\ 0\,]^\mathsf{T}\,\mathrm{m\,s^{-1}}$, and $\boldsymbol{\omega}_b = [\,0\ 0\ 2\,]^\mathsf{T}\,\mathrm{rad\,s^{-1}}$. Use the adjoint relation to calculate $\boldsymbol{\omega}_w$ and $\boldsymbol{\nu}_w$.
12. Why must software check both orthogonality and determinant for a candidate rotation matrix? Why is an ordinary Euclidean norm of a twist not a physically meaningful magnitude?
13. Show why the Forward Euler orientation update $\widetilde{\mathbf{R}}_{k+1} = \mathbf{R}_k(\mathbf{I}_3 + \Delta t[\boldsymbol{\omega}_b]_\times)$ generally leaves $SO(3)$.

Answers and hints

1. The configuration is the body's physical placement relative to the fixed world. The matrix $\mathbf{T}_{wb}$ represents that placement as an element of $SE(3)$ and maps body-frame point coordinates to world-frame coordinates. A coordinate column is a chosen list of numbers used to construct a representation; it is not itself the physical body or an ordinary free vector.

2. Linearising the constraints gives

$$\mathbf{J}_\mathbf{g}(\mathbf{z}_0)\delta\mathbf{z} = \mathbf{0},$$

so the allowable infinitesimal changes form

$$\mathcal{N}(\mathbf{J}_\mathbf{g}(\mathbf{z}_0)) = \{\delta\mathbf{z} \in \mathbb{R}^{n_v} \mid \mathbf{J}_\mathbf{g}(\mathbf{z}_0)\delta\mathbf{z} = \mathbf{0}\}.$$

Therefore,

$$n_{\mathrm{dof}} = \dim \mathcal{N}(\mathbf{J}_\mathbf{g}(\mathbf{z}_0)) = n_v - \operatorname{rank}(\mathbf{J}_\mathbf{g}(\mathbf{z}_0)).$$

A free spatial rigid body has three independent translation coordinates and three independent local orientation coordinates, giving six DOF.

3. A proper rotation satisfies

$$\mathbf{R}^\top\mathbf{R} = \mathbf{I}_3, \qquad \det(\mathbf{R}) = 1.$$

   The columns of $\mathbf{R}_{wb}$ are the world-coordinate columns of the body axes x_b, y_b, and z_b. They form a right-handed orthonormal basis.

4. A world-expressed change premultiplies, whereas a body-expressed change postmultiplies:

$$\boxed{\mathbf{R}_{wb'} = \mathbf{Q}_w\mathbf{R}_{wb}}, \qquad \boxed{\mathbf{R}_{wb'} = \mathbf{R}_{wb}\mathbf{Q}_b}.$$

5. The two rate equations are

$$\boxed{\dot{\mathbf{R}}_{wb} = [\boldsymbol{\omega}_w]_\times\mathbf{R}_{wb} = \mathbf{R}_{wb}[\boldsymbol{\omega}_b]_\times}, \qquad \boldsymbol{\omega}_b = \mathbf{R}_{wb}^\top\boldsymbol{\omega}_w.$$

   The physical turning is the same; multiplication by $\mathbf{R}_{wb}^\top$ only resolves its components along the body axes instead of the world axes.

6. Define

$$\boldsymbol{\varphi} := \alpha\mathbf{a}, \qquad [\boldsymbol{\varphi}]_\times = \alpha[\mathbf{a}]_\times.$$

   The finite rotation is

$$\mathbf{Q} = \exp([\boldsymbol{\varphi}]_\times) = \mathbf{I}_3 + \sin\alpha[\mathbf{a}]_\times + (1-\cos\alpha)[\mathbf{a}]_\times^2.$$

   Applied to $\mathbf{y} \in \mathbb{R}^3$, Rodrigues' formula is

$$\mathbf{Q}\mathbf{y} = \cos\alpha\,\mathbf{y} + (1-\cos\alpha)\mathbf{a}(\mathbf{a}^\top\mathbf{y}) + \sin\alpha(\mathbf{a}\times\mathbf{y}).$$

7. With

$$\mathbf{T}_{wb} = \begin{bmatrix} \mathbf{R}_{wb} & {}^w\mathbf{p}_b \\ \mathbf{0}_3^\top & 1 \end{bmatrix},$$

   point coordinates satisfy

$${}^w\mathbf{p}_P = \mathbf{R}_{wb}{}^b\mathbf{p}_P + {}^w\mathbf{p}_b.$$

   Rearranging and using $\mathbf{R}_{wb}^{-1} = \mathbf{R}_{wb}^\top$ gives

$${}^b\mathbf{p}_P = \mathbf{R}_{wb}^\top\left({}^w\mathbf{p}_P - {}^w\mathbf{p}_b\right),$$

and therefore

$$\mathbf{T}_{wb}^{-1} = \begin{bmatrix} \mathbf{R}_{wb}^{\mathsf{T}} & -\mathbf{R}_{wb}^{\mathsf{T}}\,{}^{w}\mathbf{p}_b \\ \mathbf{0}_3^{\mathsf{T}} & 1 \end{bmatrix}.$$

8. Using $\mathbf{e}_z = [\,0\;0\;1\,]^{\mathsf{T}}$, planar motion satisfies

$$\mathbf{R}\mathbf{e}_z = \mathbf{e}_z, \qquad \mathbf{e}_z^{\mathsf{T}}\mathbf{p} = 0.$$

For the embedded yaw rotation, $\theta = \operatorname{atan2}(R_{21}, R_{11})$.

9. For a unit axis $\mathbf{a}$,

$$\mathsf{q} = \left(\cos\frac{\alpha}{2}, \mathbf{a}\sin\frac{\alpha}{2}\right).$$

Negating both parts does not change the quaternion sandwich product $\mathsf{q} \otimes \mathsf{y} \otimes \mathsf{q}^*$, so q and $-\mathsf{q}$ represent the same rotation. If q_1 acts first and q_2 acts second, then $\mathsf{q}_{\text{total}} = \mathsf{q}_2 \otimes \mathsf{q}_1$.

10. The pose rate $\dot{\mathbf{T}}_{wb}$ is the entrywise derivative of a pose matrix and depends on the current pose. The body and spatial twists are obtained by

$$\widehat{\boldsymbol{\xi}}_b = \mathbf{T}_{wb}^{-1}\dot{\mathbf{T}}_{wb}, \qquad \widehat{\boldsymbol{\xi}}_w = \dot{\mathbf{T}}_{wb}\mathbf{T}_{wb}^{-1}.$$

The body linear part is the world-relative velocity of the body origin expressed along body axes:

$$\boldsymbol{\nu}_b = \mathbf{R}_{wb}^{\mathsf{T}}\,{}^{w}\mathbf{v}_b.$$

The spatial linear part is the value of the rigid-motion velocity field at the world-origin location:

$$\boldsymbol{\nu}_w = {}^{w}\mathbf{v}_b - \boldsymbol{\omega}_w \times \mathbf{p}.$$

11. Rotation about z leaves the angular-velocity column unchanged and maps the body x-axis to the world y-axis:

$$\boldsymbol{\omega}_w = \mathbf{R}\boldsymbol{\omega}_b = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \mathrm{rad\,s^{-1}}, \qquad \mathbf{R}\boldsymbol{\nu}_b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}.$$

The origin-shift term is

$$[\mathbf{p}]_\times \boldsymbol{\omega}_w = \mathbf{p} \times \boldsymbol{\omega}_w = \begin{bmatrix} 4 \\ -2 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}.$$

Hence,

$$\boxed{\boldsymbol{\nu}_w = \begin{bmatrix} 4 \\ -1 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}}.$$

12. Orthogonality alone admits both proper rotations and reflections. The determinant check rejects the reflection case by requiring $\det(\mathbf{R}) = 1$. A twist's linear and angular components have different units, so their squared Euclidean norms cannot be added as a physical magnitude without first choosing a length scale that converts angular rate to a linear-speed scale.

13. Let $\boldsymbol{\Omega} = [\boldsymbol{\omega}_b]_\times$. Then

$$\begin{aligned} \widetilde{\mathbf{R}}_{k+1}^\top \widetilde{\mathbf{R}}_{k+1} &= (\mathbf{I}_3 - \Delta t\, \boldsymbol{\Omega})\, \mathbf{R}_k^\top \mathbf{R}_k\, (\mathbf{I}_3 + \Delta t\, \boldsymbol{\Omega}) \\ &= \mathbf{I}_3 - \Delta t^2 \boldsymbol{\Omega}^2. \end{aligned}$$

This is generally not $\mathbf{I}_3$, so the updated matrix is generally not in $SO(3)$ even though its one-step constraint error is $O(\Delta t^2)$.

References

- Kevin M. Lynch and Frank C. Park, *Modern Robotics: Mechanics, Planning, and Control*, Chapter 2 for configuration space and degrees of freedom, and Chapter 3, especially Sections 3.2.1-3.2.3 on spatial rotation matrices, angular velocity, exponential coordinates, and Rodrigues' formula, Section 3.3.1 on homogeneous transformation matrices, and Section 3.3.2 on spatial rigid-body velocities and twists.
- John J. Craig, *Introduction to Robotics: Mechanics and Control*, 4th ed., Chapters 1-2, especially degrees of freedom, body-attached frames, proper orthonormal rotation matrices, spatial orientation, homogeneous transformations, equivalent angle–axis representations, and Euler parameters.
- Peter Corke, *Robotics, Vision and Control: Fundamental Algorithms in Python*, 3rd ed., Chapters 2-3, especially rotation matrices, the matrix exponential, Rodrigues' formula, unit quaternions, rotational velocity, and body- and world-frame expressions of spatial motion.
- ROS 2 Jazzy `geometry_msgs`, Quaternion message definition, for the `x`, `y`, `z`, and `w` field order.

8 General Robot Kinematics and Jacobians

Why this chapter matters

A robot joint permits a constrained relative motion between two rigid links. To predict where a serial robot will move, the geometry of each joint must be expressed in a common frame, composed into a configuration-to-pose map, and differentiated to obtain rigid-body and task velocities. This chapter develops that chain from screw axes through forward kinematics and Jacobians.

The chapter treats open serial chains with ideal rigid links and one-degree-of-freedom revolute or prismatic joints. It does not yet model joint compliance, backlash, closed kinematic loops, collision, forces, actuator dynamics, or feedback control.

Prerequisites and assumptions

The reader should understand vectors, matrices, rank, null spaces, and constraints from Chapter 1; coordinate frames and cross products from Chapter 2; and spatial poses, angular velocity, twists, the adjoint, and the project's linear-first twist order from Chapter 7.

Physical space is modelled as three-dimensional Euclidean space. Frames are right-handed and orthonormal. Position and length are measured in metres, time in seconds, and angles in radians. Unless stated otherwise, every screw-axis coordinate column is referred to one declared fixed frame: that frame supplies both the coordinate axes and the reference origin of the associated spatial-twist velocity field.

8.1 Screw axes and one-degree-of-freedom joint motion

8.1.1 The physical problem: one joint, one allowed motion

A door hinge and a straight linear slide each have one degree of freedom, so one scalar joint coordinate is sufficient to describe the allowed relative configuration. Their motions are nevertheless different. The hinge permits rotation about a fixed line, whereas the slide permits translation along a fixed direction. A useful joint model must describe both cases, locate the permitted motion in physical space, and produce the instantaneous twist caused by a joint rate.

A **screw motion** is rotation about an oriented line combined with translation along the same line. A **screw axis** specifies the line, its positive direction, and the translation per unit rotation. Pure rotation and pure translation are limiting cases of this model. Most robot manipulators use these two cases as **revolute joints** and **prismatic joints**, respectively.

The physical motion must be distinguished from its representation. A joint axis is a geometric feature of the mechanism. Its coordinate column changes when the reference frame changes, even though the physical joint does not. The current joint angle or displacement is also separate from the axis: the axis states which motion is allowed, while the joint coordinate states how far the joint has moved.

8.1.2 An oriented line in a declared frame

Let $\{a\}$ be a fixed reference frame with origin O_a. Let ${}^a\mathbf{p}_\ell \in \mathbb{R}^3$ be the coordinate column of any point on a joint-axis line, expressed along the axes of $\{a\}$ and measured in metres. Let ${}^a\mathbf{s} \in \mathbb{R}^3$ be a dimensionless unit column pointing in the chosen positive direction of that axis, so

$$\|{}^a\mathbf{s}\|_2 = 1.$$

The coordinate representation of the oriented line is the set

$$\boxed{{}^a\mathcal{L} = \{{}^a\mathbf{p}_\ell + \lambda\,{}^a\mathbf{s} \mid \lambda \in \mathbb{R}\}},$$

where the scalar λ is measured in metres. The point ${}^a\mathbf{p}_\ell$ is not unique: adding any distance along the line gives another valid point. The unit direction fixes an orientation for the line, so replacing ${}^a\mathbf{s}$ by $-{}^a\mathbf{s}$ reverses the positive joint direction.

This line description contains only geometry. It does not contain a joint position, a joint rate, or a finite rigid displacement.

8.1.3 Revolute motion about the line

A **revolute joint** permits pure rotation about its joint-axis line. Let $t \in \mathbb{R}$ denote time in seconds, let j denote the joint index, let $\theta_j(t) \in \mathbb{R}$ be joint j's angle in radians, and let $\dot{\theta}_j(t)$ be its angular rate in $\mathrm{rad\,s^{-1}}$. The positive direction ${}^a\mathbf{s}$ and the right-hand rule determine the sign of θ_j.

The joint's angular velocity relative to the fixed frame, expressed along the axes of $\{a\}$, is

$${}^a\boldsymbol{\omega} = {}^a\mathbf{s}\,\dot{\theta}_j.$$

The spatial-twist definition in Section 7.9.3 showed that a twist referred to a fixed frame defines the rigid-motion velocity field

$$ {}^{a}\mathbf{v}_P = {}^{a}\boldsymbol{\nu} + {}^{a}\boldsymbol{\omega} \times {}^{a}\mathbf{p}_P, $$

where ${}^{a}\mathbf{p}_P \in \mathbb{R}^3$ is the current position of a body-fixed point P in metres, ${}^{a}\mathbf{v}_P \in \mathbb{R}^3$ is its velocity in $\mathrm{m\,s^{-1}}$, and ${}^{a}\boldsymbol{\nu} \in \mathbb{R}^3$ is the linear part of the spatial twist referred to O_a, also in $\mathrm{m\,s^{-1}}$. The quantity ${}^{a}\boldsymbol{\nu}$ is the velocity field evaluated at O_a; it is not generally the velocity of a link-frame origin.

For pure rotation, every point on the joint axis is instantaneously stationary. Evaluate the velocity field at the chosen axis point ${}^{a}\mathbf{p}_\ell$ and set that velocity to zero:

$$ \mathbf{0}_3 = {}^{a}\boldsymbol{\nu} + \left({}^{a}\mathbf{s}\,\dot{\theta}_j\right) \times {}^{a}\mathbf{p}_\ell. $$

Solving for the linear part gives

$$ {}^{a}\boldsymbol{\nu} = -\left({}^{a}\mathbf{s} \times {}^{a}\mathbf{p}_\ell\right)\dot{\theta}_j. $$

The project's twist convention places the linear component before the angular component. The **normalised revolute screw-axis coordinates** referred to frame $\{a\}$ are therefore

$$ \boxed{ {}^{a}\mathbf{S}_R := \begin{bmatrix} -{}^{a}\mathbf{s} \times {}^{a}\mathbf{p}_\ell \\ {}^{a}\mathbf{s} \end{bmatrix} \in \mathbb{R}^6 }. $$

The subscript R labels the revolute case. Stacking the two parts of the joint-induced spatial twist now gives

$$ \boxed{ {}^{a}\boldsymbol{\xi} = \begin{bmatrix} {}^{a}\boldsymbol{\nu} \\ {}^{a}\boldsymbol{\omega} \end{bmatrix} = {}^{a}\mathbf{S}_R\dot{\theta}_j }. $$

The first three entries of ${}^{a}\mathbf{S}_R$ have units of metres per radian, conventionally written as metres because radians are dimensionless, and the final three entries are dimensionless. Multiplication by $\dot{\theta}_j$ produces linear velocity in $\mathrm{m\,s^{-1}}$ and angular velocity in $\mathrm{rad\,s^{-1}}$.

8.1.4 Helical motion and screw pitch

A **helical joint** permits rotation about an axis and translation along that same axis in a fixed proportion. Its **pitch** $h \in \mathbb{R}$ is the signed axial distance travelled per positive radian of rotation, measured in $\mathrm{m\,rad^{-1}}$. The axial speed is therefore $h\dot{\theta}_j$, and its direction is ${}^a\mathbf{s}$.

For a helical motion, a point on the axis is no longer stationary. Its velocity is $h{}^a\mathbf{s}\dot{\theta}_j$. Evaluate the spatial velocity field at ${}^a\mathbf{p}_\ell$:

$$h{}^a\mathbf{s}\dot{\theta}_j = {}^a\boldsymbol{\nu} + \left({}^a\mathbf{s}\dot{\theta}_j\right) \times {}^a\mathbf{p}_\ell.$$

Rearranging separates the geometry from the angular rate:

$${}^a\boldsymbol{\nu} = \left(-{}^a\mathbf{s} \times {}^a\mathbf{p}_\ell + h{}^a\mathbf{s}\right)\dot{\theta}_j.$$

Hence the normalised helical screw-axis coordinates are

$$\boxed{{}^a\mathbf{S}_H := \begin{bmatrix} -{}^a\mathbf{s} \times {}^a\mathbf{p}_\ell + h{}^a\mathbf{s} \\ {}^a\mathbf{s} \end{bmatrix} \in \mathbb{R}^6, \qquad {}^a\boldsymbol{\xi} = {}^a\mathbf{S}_H\dot{\theta}_j}.$$

The subscript H labels the helical case. Setting $h = 0$ recovers the revolute screw axis. A positive pitch translates in the ${}^a\mathbf{s}$ direction during positive rotation; a negative pitch translates in the opposite direction.

The choice of point on the line does not change the screw-axis coordinates. If ${}^a\mathbf{p}'_\ell = {}^a\mathbf{p}_\ell + \lambda{}^a\mathbf{s}$ for any $\lambda \in \mathbb{R}$ measured in metres, then

$$\begin{aligned} -{}^a\mathbf{s} \times {}^a\mathbf{p}'_\ell + h{}^a\mathbf{s} &= -{}^a\mathbf{s} \times \left({}^a\mathbf{p}_\ell + \lambda{}^a\mathbf{s}\right) + h{}^a\mathbf{s} \\ &= -{}^a\mathbf{s} \times {}^a\mathbf{p}_\ell - \lambda\underbrace{{}^a\mathbf{s} \times {}^a\mathbf{s}}_{\mathbf{0}_3} + h{}^a\mathbf{s} \\ &= -{}^a\mathbf{s} \times {}^a\mathbf{p}_\ell + h{}^a\mathbf{s}. \end{aligned}$$

The coordinate column therefore depends on the oriented line, pitch, and reference frame, but not on which point is selected along that line.

8.1.5 Prismatic motion along a direction

A **prismatic joint** permits pure translation along a fixed direction. Let $d_j(t) \in \mathbb{R}$ be the joint displacement in metres and let $\dot{d}_j(t)$ be its rate in $\mathrm{m\,s^{-1}}$. There is no angular velocity, and every point of the moving link has the same instantaneous velocity:

$$^{a}\boldsymbol{\omega} = \mathbf{0}_3, \qquad ^{a}\mathbf{v}_P = {}^{a}\mathbf{s}\dot{d}_j.$$

The normalised prismatic screw-axis coordinates and resulting twist are therefore

$$\boxed{^{a}\mathbf{S}_P := \begin{bmatrix} ^{a}\mathbf{s} \\ \mathbf{0}_3 \end{bmatrix} \in \mathbb{R}^6, \qquad ^{a}\boldsymbol{\xi} = {}^{a}\mathbf{S}_P \dot{d}_j}.$$

The subscript P labels the prismatic case. The first three entries of $^{a}\mathbf{S}_P$ are dimensionless direction components, and multiplication by $\dot{d}_j$ produces linear velocity in $\mathrm{m\,s^{-1}}$. The angular entries remain zero. Unlike a revolute or helical motion, a pure translation has no unique finite axis location; its direction is sufficient.

Pure translation is sometimes described as screw motion with infinite pitch. That phrase is a geometric limiting interpretation, not a numerical instruction. A software model should use the finite prismatic form $^{a}\mathbf{S}_P$ and should not store $h = \infty$.

8.1.6 The normalisation rule and the declared component order

Write any normalised screw-axis coordinate column in the project's linear-first order as

$$^{a}\mathbf{S} = \begin{bmatrix} ^{a}\mathbf{s}_v \\ ^{a}\mathbf{s}_\omega \end{bmatrix} \in \mathbb{R}^6.$$

The subscripts v and ω identify the coefficients that generate the linear and angular parts of the twist after multiplication by the appropriate scalar joint rate. The two valid normalisation cases are

$$\boxed{\begin{array}{lll} \|^{a}\mathbf{s}_\omega\|_2 = 1, & ^{a}\mathbf{s}_v = -{}^{a}\mathbf{s}_\omega \times {}^{a}\mathbf{p}_\ell + h\,{}^{a}\mathbf{s}_\omega & \text{for revolute or helical motion,} \\ ^{a}\mathbf{s}_\omega = \mathbf{0}_3, & \|^{a}\mathbf{s}_v\|_2 = 1 & \text{for prismatic motion.} \end{array}}$$

These are separate normalisation rules because the coordinate column has mixed physical units. Taking one ordinary Euclidean norm of all six entries would add squared lengths to squared dimensionless quantities and would not define a physical magnitude.

Modern Robotics writes twists and screw axes in angular-first order, $[\boldsymbol{\omega}^\mathsf{T}\ \boldsymbol{\nu}^\mathsf{T}]^\mathsf{T}$. This project uses the linear-first order established in Section 7.9.2 and Section 7.9.3, $[\boldsymbol{\nu}^\mathsf{T}\ \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}$. The formulas above have already been reordered; copying an angular-first six-vector directly into this convention would describe the wrong motion.

If the same physical screw is represented in another fixed frame $\{b\}$, its coordinate column changes through the adjoint already defined in Section 7.9.4. If $\mathbf{T}_{ab} \in SE(3)$ transforms point coordinates from $\{b\}$ to $\{a\}$, then

$$\boxed{{}^{a}\mathbf{S} = \mathrm{Ad}_{\mathbf{T}_{ab}}\, {}^{b}\mathbf{S}}.$$

The physical joint axis has not moved in this equation. Only the reference axes and origin used to represent it have changed.

8.1.7 Worked checks

Consider a pure revolute joint whose axis points along positive z_a and passes through

$${}^{a}\mathbf{p}_{\ell} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \mathrm{m}, \qquad {}^{a}\mathbf{s} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

The linear coefficient is

$$-{}^{a}\mathbf{s} \times {}^{a}\mathbf{p}_{\ell} = -\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \mathrm{m},$$

so the linear-first screw-axis coordinates are

$${}^{a}\mathbf{S}_{R} = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 1 \end{bmatrix}^{\mathsf{T}}.$$

At $\dot{\theta}_j = 3\,\mathrm{rad\,s^{-1}}$, the twist parts are

$${}^{a}\boldsymbol{\nu} = \begin{bmatrix} 6 \\ -3 \\ 0 \end{bmatrix} \mathrm{m\,s^{-1}}, \qquad {}^{a}\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \mathrm{rad\,s^{-1}}.$$

Let ${}^{a}\mathbf{v}_{\ell} \in \mathbb{R}^3$ denote the velocity field evaluated at the selected axis point, expressed in frame $\{a\}$ and measured in $\mathrm{m\,s^{-1}}$. Its value verifies the required zero velocity:

$$
\begin{aligned}
{}^{a}\mathbf{v}_{\ell} &= {}^{a}\boldsymbol{\nu} + {}^{a}\boldsymbol{\omega} \times {}^{a}\mathbf{p}_{\ell} \\
&= \begin{bmatrix} 6 \\ -3 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} 6 \\ -3 \\ 0 \end{bmatrix} + \begin{bmatrix} -6 \\ 3 \\ 0 \end{bmatrix} = \mathbf{0}_3 \,\mathrm{m\,s^{-1}}.
\end{aligned}
$$

As a contrasting prismatic example, take the unit direction

$$
{}^{a}\mathbf{s} = \frac{1}{3} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.
$$

Its norm is one because $\sqrt{1^2 + 2^2 + 2^2}/3 = 1$. At $\dot{d}_j = 0.3\,\mathrm{m\,s^{-1}}$, every moving-link point has velocity

$$
{}^{a}\mathbf{v}_P = {}^{a}\mathbf{s}\dot{d}_j = \begin{bmatrix} 0.1 \\ 0.2 \\ 0.2 \end{bmatrix} \mathrm{m\,s^{-1}},
$$

and the angular velocity is zero.

8.1.8 Contrasting the geometric and motion quantities

The terms introduced above describe related but different objects:

Quantity	What it describes	What it depends on
Oriented axis line $\mathcal{L}$	A physical line and its positive direction	Joint geometry; its coordinate representation depends on the frame
Screw pitch h	Signed axial travel per radian of rotation	Joint geometry, not the current joint rate
Normalised screw-axis coordinates ${}^{a}\mathbf{S}$	A six-entry coordinate representation of the permitted instantaneous motion per unit joint rate	Axis, pitch, component order, and reference frame
Twist ${}^{a}\boldsymbol{\xi}$	The actual instantaneous rigid-motion velocity field at one instant	Screw-axis coordinates and the current joint rate

Quantity	What it describes	What it depends on
Joint coordinate θ_j or d_j	The current amount of rotation or translation from a declared zero configuration	The current joint configuration and zero convention

A finite relative pose is a further object. It will be obtained from a screw axis and a finite joint-coordinate change through the $SE(3)$ exponential in Section 8.2. A screw-axis column alone is neither the current pose nor a finite transformation.

Quick test A: screw axes and joint motion

1. **Recall:** What three geometric ingredients describe a finite-pitch screw axis before it is converted to a six-entry coordinate column?
2. **Contrast:** Distinguish an oriented axis line, normalised screw-axis coordinates, a twist, and a joint coordinate.
3. **Derivation:** A pure revolute axis has unit direction ${}^a\mathbf{s}$ and passes through ${}^a\mathbf{p}_\ell$. Starting from the spatial velocity field, derive the linear coefficient $-{}^a\mathbf{s} \times {}^a\mathbf{p}_\ell$.
4. **Application:** Find the linear-first screw-axis coordinates of a pure revolute joint about positive z_a through $[1\ 2\ 0]^\mathsf{T}$ m.
5. **Application:** A prismatic joint translates along negative y_a with $\dot{d}_j = 0.4\,\mathrm{m\,s^{-1}}$. Write its normalised screw-axis coordinates and the resulting twist.
6. **Pitch:** A helical joint has $h = 0.01\,\mathrm{m\,rad^{-1}}$ and $\dot{\theta}_j = 5\,\mathrm{rad\,s^{-1}}$. What is its signed axial speed?
7. **Proof:** Show that replacing ${}^a\mathbf{p}_\ell$ by another point on the same line leaves the revolute or helical screw-axis coordinates unchanged.
8. **Limitation:** Why should software use the prismatic screw form instead of storing an infinite pitch? Why should it not normalise a general six-entry screw column with one ordinary Euclidean norm?

Answers and hints

1. Any point on the axis, a unit direction along the axis, and the signed pitch.
2. The line is physical joint geometry; ${}^a\mathbf{S}$ is its frame-dependent six-entry motion coordinate; ${}^a\boldsymbol{\xi}$ is the instantaneous motion after multiplication by the current joint rate; and θ_j or d_j records the joint's current displacement from a declared zero.
3. For pure rotation, a point on the axis is stationary. Substitute ${}^a\boldsymbol{\omega} = {}^a\mathbf{s}\dot{\theta}_j$ and ${}^a\mathbf{p}_P = {}^a\mathbf{p}_\ell$ into ${}^a\mathbf{v}_P = {}^a\boldsymbol{\nu} + {}^a\boldsymbol{\omega} \times {}^a\mathbf{p}_P$, set the result to zero, and collect the coefficient of $\dot{\theta}_j$.
4. The result is $[2\ -1\ 0\ 0\ 0\ 1]^\mathsf{T}$ in the project's linear-first order.
5. The unit direction is $[0\ -1\ 0]^\mathsf{T}$, so ${}^a\mathbf{S}_P = [0\ -1\ 0\ 0\ 0\ 0]^\mathsf{T}$ and ${}^a\boldsymbol{\xi} = [0\ -0.4\ 0\ 0\ 0\ 0]^\mathsf{T}$, with linear entries in $\mathrm{m\,s^{-1}}$ and angular entries in $\mathrm{rad\,s^{-1}}$.
6. The signed axial speed is $h\dot{\theta}_j = 0.05\,\mathrm{m\,s^{-1}}$ in the positive axis direction.

7. Write ${}^{a}\mathbf{p}'_{\ell} = {}^{a}\mathbf{p}_{\ell} + \lambda {}^{a}\mathbf{s}$ and use ${}^{a}\mathbf{s} \times {}^{a}\mathbf{s} = \mathbf{0}_3$.
8. Infinity is not needed to represent pure translation and can create non-finite arithmetic. The six entries have mixed physical units, so their squared values cannot be added into one physically meaningful norm. Normalise the angular part for revolute or helical motion and the linear part for prismatic motion.

8.2 Rigid motion from the $SE(3)$ exponential

8.2.1 From a constant twist to a finite pose

8.2.2 Closed forms for rotational and translational screws

8.2.3 Recovering screw coordinates with the $SE(3)$ logarithm

8.3 Constant-twist pose integration

8.3.1 Space-twist pose updates

8.3.2 Body-twist pose updates

8.3.3 Multiplication side and physical interpretation

8.4 Numerical branches and validation

8.4.1 Small angular displacements

8.4.2 Rotations near π and logarithm branches

8.4.3 Invalid inputs and deterministic checks

8.4.4 Worked spatial cases

8.5 Configuration coordinates and forward kinematics

8.5.1 Configuration space and coordinate vectors

8.5.2 Task variables and the forward map

8.5.3 The total derivative and multivariable chain rule

8.6 Serial-chain pose through products of exponentials

8.6.1 Home configuration and joint screw axes

8.6.2 Space-form product of exponentials

8.6.3 Body-form product of exponentials

normalised screw axes.

- John J. Craig, *Introduction to Robotics: Mechanics and Control*, Chapter 2 on spatial descriptions, transformations, and rotations about general axes, and Chapter 3 on one-degree-of-freedom revolute and prismatic joints.
- Peter Corke, *Robotics, Vision and Control: Fundamental Algorithms in Python*, Chapters 2–3, especially the three-dimensional twist representation and its revolute, prismatic, and finite-pitch screw cases.